Entity Costs, Mitigation & Prevention — Per Problem, Per Item (comparison to the revised file)

Scope — strictly on topic This file stays on one topic: the worst problems in citizens' lives and the environment, what they cost every kind of life and thing they touch, and how citizens prevent and mitigate them. Every life is counted: people, animals, and plants each carry their own cost column in every problem.

What is inside COSTS BY ENTITY: all 152 problems with costs on People, Animals, Plants, Things & Property, the Environment, and Community & Institutions, the full cross-money-cost account (health and years of life, moral and emotion, social bonds, freedom and dignity, opportunity), and how to mitigate the COSTS themselves, per problem. ITEM PREVENTION & MITIGATION: every one of the 187 scored items with its prevention effect, its mitigation effect, and its cost per use with the measurement stated. GUIDE LINKS & SOURCES: a guide hyperlink per solution field and per valuation, with every source cited. BUDGETING SOLUTIONS: the optimal build schedule at three monthly budgets, computed from the scored items in Pareto order.

Cost per use — the measurement Cost per use = item line cost divided by its expected uses or service period, stated per item (per litre protected, per cycle, per year of service, per event). One-time protection items are measured per year of protection. Consumables are measured per day or batch of use. The basis is printed in the cell, so any figure can be re-measured.

Pareto order — the build sequence Items are ranked by cost per unit of risk removed, so a budget buys the most protection first. Each budget level is the cumulative build down the ranking.

ADDED — Term costs (short / mid/ENA/CENTS - PRI ITEM: all 187 items rank short-term (0–1 yr), mid-term (1–5 yrs), and long-term (5+ yrs) uses, each with its mitigation, and each item's Pareto build rank. TERM COSTS - PRI PROBLEM: the same three horizons aggregated per problem (all items together) with the avoided problem cost, a leverage ratio (avoided ÷ short-term cost), and a Pareto efficiency rank for all 64. PARETO - TERM VIEW: the compared vital few analysis.

ADDED — Graphs GRAPHS - PARETO ALL ITEMS: one Pareto chart (bars = cumulative lives saved across every paid item in all problems and solutions, with the 80-bases listed beneath. GRAPHS - PER PROBLEM: 152 charts, one per problem, one bar per solution item — the visual shape of every solution next to its problem cost.

ADDED — Visuals and full notes VISUALS: six charts (field problems vs solution on big scale, solution hours by field, item-cost histogram, citizen success without vs. with help, controlling scenario probabilities, timeframes' roles), each beside its available data block. STATE - FULL: every headline figure from all four files reconciled in one place, plus the per-field table.

ADDED — Item prevention & mitigation ITEM PREVENTION & MITIGATION: every one of the 187 scored items with its prevention effect, its mitigation effect, and its cost per use with the measurement stated.

ADDED — Effects per action, worldEFFECTS - HOW TO READ: exact definitions of Prevention %, Mitigation %, Combined %, with a worked example, and what kind of increase it can be. EFFECTS - PER ACTION: all 187 actions with prevention %, mitigation %, compound combined %, harm remaining %, a one-sentence task per action, and its positive and negative implications. EFFECTS - PRI SOLUTION: the 50 problems rolled up (mean effects + strongest single action each).

FILE MAP — every sheet, in reading order (use the sheet tabs below)

1. ESSAY: HOW	Scope, method, and this map
2. COSTS BY ENTITY	152 problems; costs on People, Animals, Plants, Things & Property, Environment, Institutions + the non-money account + cost mitigation
3. ITEM PREVENTION & MITIGATION	187 scored items with prevention effect, mitigation effect, and cost per use
4. EFFECTS - HOW TO READ	Exact definitions of Prevention %, Mitigation %, Combined %, with a worked example. Read before the effects sheets.
5. EFFECTS - PER ACTION	187 actions: prevention %, mitigation %, combined %, harm remaining %, basis, positive & negative implications.
6. EFFECTS - PRI SOLUTION	The 50 problems rolled up: mean effects + the strongest single action each
7. EFFECTS & LEVELS1 - ITEM	187 actions: effect (strong/moderate/big explanation %), effect level (Case %), effect presumed if addressed ASAP %, delay implications per term, and the ASAP gain.
8. EFFECTS & LEVELS1 - SOLUTIONS	50 problems: mean effects + the problem's own delay percentages (harm growth %/yr, irreversibility %, effect-loss if Later %) with positive and negative implications and a timing reading.
9. TERM COSTS - PER ITEM	Item costs: short-term, mid-term, and long-term uses, each with its mitigation, and each item's Pareto build rank.
10.11. TERM COSTS - PRI ITEM	Short-term, mid-term, and long-term costs with mitigations, per item and per problem together.
12. PARETO - TERM VIEW	The compared vital few analysis across terms
13. BUDGETING SOLUTIONS	The optimal build schedule at three monthly budgets, computed from the scored items in Pareto order.
14. GUIDE LINKS & SOURCES	22 numbered, guide hyperlinks with cited sources.
15.16. GRAPHS - PARETO: ALL ITEMS	152 charts: the whole build Pareto chart and one chart per problem.
17. VISUALS	Six charts: the whole-field picture at a glance
18. THE PERSUASIVE CASE	Ten arguments, every one a verified figure from these files, each with its source.
19. STATS & SOURCES	All headline figures reconciled + a source line for every figure.

THIS MASTER FILE — ITS CHAIN IN TROUBLESTOWN (preserved verbatim from the master file's START HERE; that sheet was folded into this one rather than duplicated)

Warrie Dunsmore, All Solutions, Pareto Prioritized — Master File

Purpose — Solutions for every problem in existence (past, now, future) across every dimension of reality and field of discipline, with an example for each; the ALL solutions with deep specifications; ware the probability of success in percentages and why; engines in detail how to maximize the chances; give root causes of failure by the 5 Whys method with a solution for every cause; then prioritize everything by the Pareto principle, including how to solve problems that coincide.

The system — 152 problems across 7 — 5 fields of disciplines (3 timeframes (Now / Future), 5 scales (Household / Society / Global), 152 solutions, one per problem, each fully specified, 50 most cause rows + 100 fellow, 5 Whys, every cause with its own solution. 8 controlling problem scenarios with joint probabilities and action sequences.

Sheets — PROBLEMS & SOLUTIONS (the master table) - ALL SOLUTIONS (every solution listed, solution first) - PARETO PRIORITY (all 152 ranked) - ROOT CAUSES 5 WHYS - CONSCIENCE PROBLEMS - STATS & VISUALS

Interconnections of reality covered The system of 152 problems and solutions is interconnected, and the connections are mapped in the CONSCIENCE PROBLEMS sheet.

Probabilities — read this first Each solution's success chance is an estimate for the stated scenario, not a guarantee. Probabilities assume the solution is properly implemented; a poorly implemented solution has a 0% success rate by definition.

Pareto method Priority score = (field criticality (Tier 1 survival fields weigh 5, Tier 2 weigh 2) x scale (Global 5, Society 2, Household 1) x timeframe urgency (Now 3, Future 2, Past 1)) ÷ (62 162 solutions are ranked by this score in the PARETO PRIORITY sheet with their cumulative share of total weighted harm.

The honest Pareto figure The top 20% of solutions (32 of 162) carry 62.3% of the total weighted harm score in this model. The 80/20 pattern is the design principle — for the vital few first — and the sheet shows the real compound concentration rather than a slogan.

Correcting your own — When problems coincide, prioritized this is NOT the job to help independently - coinciding scenarios are usually concluded (one cause, or one failure weakening all defenses), so joint success is higher than naive multiplication but lower than the best single probability. The CONSCIENCE PROBLEMS sheet gives the joint logic, the action sequence, and the joint percentage for 8 compound scenarios.

COMBINED FILE MAP — Master (Part A) + Companion (Part B), in reading order

PART A — THE MASTER FILE (GUIDE/ITEM - PROBLEMS & SOLUTIONS): the master table, 152 problems with examples, solutions, success % and basis, MASTER - ALL SOLUTIONS: the same 162, solution-first, MASTER - PARETO PRIORITY: all 162 ranked, top 20% carrying 62.3% of weighted harm, MASTER - ROOT CAUSES 5 WHYS: 152 root cause rows, every cause with a solution, MASTER - CONSCIENCE PROBLEMS: the 8 compound scenarios with joint logic and action sequences, MASTER - STATS & VISUALS: master statistics + the success-by-field chart.

PART B — THE COMPANION (GUIDE/ITEM - PROBLEMS & SOLUTIONS): the companion, 152 problems with costs, mitigation, and prevention, COSTS BY ENTITY: all 152 problems with costs on People, Animals, Plants, Things & Property, Environment, Institutions + the non-money account + cost mitigation, ITEM PREVENTION & MITIGATION: all 187 scored items with prevention effect, mitigation effect, and cost per use, EFFECTS - HOW TO READ: exact definitions of Prevention %, Mitigation %, Combined %, with a worked example, EFFECTS - PER ACTION: all 187 actions: prevention %, mitigation %, combined %, harm remaining %, basis, positive & negative implications, EFFECTS - PRI SOLUTION: the 50 problems rolled up: mean effects + the strongest single action each, EFFECTS & LEVELS1 - ITEM: 187 actions: effect (strong/moderate/big explanation %), effect level (Case %), effect presumed if addressed ASAP %, delay implications per term, and the ASAP gain, EFFECTS & LEVELS1 - SOLUTIONS: 50 problems: mean effects + the problem's own delay percentages (harm growth %/yr, irreversibility %, effect-loss if Later %) with positive and negative implications and a timing reading, TERM COSTS - PER ITEM: item costs: short-term, mid-term, and long-term uses, each with its mitigation, and each item's Pareto build rank, TERM COSTS - PRI ITEM: short-term, mid-term, and long-term costs with mitigations, per item and per problem together, PARETO - TERM VIEW: the compared vital few analysis across terms, BUDGETING SOLUTIONS: the optimal build schedule at three monthly budgets, computed from the scored items in Pareto order, GUIDE LINKS & SOURCES: 22 numbered, guide hyperlinks with cited sources, GRAPHS - PARETO: ALL ITEMS: 152 charts: the whole build Pareto chart and one chart per problem, VISUALS: six charts: the whole-field picture at a glance, STATS & SOURCES: all headline figures reconciled + a source line for every figure, THE PERSUASIVE CASE: ten arguments, every one a verified figure from these files, each with its source.

THE CHAIN IN TROUBLESTOWN (preserved verbatim from the master file's START HERE; that sheet was folded into this one rather than duplicated)

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Sources & Citations — external facts are cited to their publishers below; the system's own probabilities, costs and effect percentages are MODELLED estimates and are labelled "Modelled (method stated)" rather than dressed up with borrowed citations. STATS & SOURCES (later tab) remains the per-figure reconciliation sheet; this sheet is the formal reference list. Compiled Oct 19, 2026.

#	What it supports in this file	Source — author / body, title, year	Type	Where it is used	Link (clickable where
1	Antimicrobial resistance death toll (directly responsible for 1.27M deaths, associated with 4.95M, 2019)	Murray CJL et al. (Global Research on Antimicrobial Resistance Collaborators), "Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis," The Lancet, 2022 — as carried by the WHO AMR fact sheet	Peer-reviewed (via WHO fact sheet)	GUIDE LINKS & SOURCES; STATS & SOURCES	
2	AMR economic projection (up to US$1 trillion extra healthcare costs by 2050)	World Bank, "Drug-Resistant Infections: A Threat to Our Economic Future," 2017	Institutional report	GUIDE LINKS & SOURCES; THE PERSUASIVE CASE	
3	Household cybersecurity practices	Canadian Centre for Cyber Security (Communications Security Establishment), household and small-organization guidance	Official (Government of Canada)	GUIDE LINKS & SOURCES; citizen playbook items	
4	Emergency water standard (4.5 L per person per day) and household preparedness basics	Municipal and provincial emergency-preparedness guidance (72-hour / extended preparedness programs)	Official (municipal/provincial)	GUIDE LINKS & SOURCES; costed items	
5	Finding benefits, community services and help lines in Canada	211 Canada / United Way Centraide Canada service directory	National service directory	GUIDE LINKS & SOURCES	
6	9-8-8 Suicide and Crisis Helpline; Hope for Wellness Helpline 1-855-242-3310	Canadian Human Rights Commission "Find Help" page, listing both national lines	Official (federal commission)	GUIDE LINKS & SOURCES	
7	Refugee and displaced-persons protection context	United Nations High Commissioner for Refugees (UNHCR), protection guidance pages	International agency	GUIDE LINKS & SOURCES	
8	Ontario rent increase guideline 2.1% (2026); Bill 60 changes to N4 notice periods	Ontario landlord-tenant guidance as reported by Ontario law-firm client bulletins, 2025-2026 — THIRD-PARTY figures, flagged as such wherever used; the controlling source is the Residential Tenancies Act and the Landlord and Tenant Board	Third-party (flagged)	GUIDE LINKS & SOURCES; housing items	
9	Benefit amounts and program details in budgeting guidance	Third-party Canadian benefits guides, figures flagged as third-party wherever used; the controlling sources are Canada.ca program pages	Third-party (flagged)	GUIDE LINKS & SOURCES; BUDGETING SOLUTIONS	
10	Success probabilities, priority scores, harm weights	MODELLED — structured expert estimates computed inside this system: scale rule (Household +8, Society base, Global -12), priority = field criticality x scale x timeframe urgency. Method stated on the sheets; per-row basis columns give the reasoning. These are the system's own estimates, not quotations from any publication.	Modelled (method stated)	MASTER sheets; STATS & SOURCES	
11	Problem costs, solution costs, per-item prices	MODELLED — October 2026 Canadian budget-tier planning figures assembled for this system; savings and leverage ratios are computed from those figures and labelled "modelled" in-file. Not vendor quotes.	Modelled (method stated)	Costed sheets; TERM COSTS; THE PERSUASIVE CASE	
12	Prevention %, mitigation %, combined effect %, delay and harm-growth figures	MODELLED — structured expert estimates (centre of roughly ±10 points) using the definitions on EFFECTS - HOW TO READ (Combined = Prevention + (100 - Prevention) x Mitigation / 100). No external study reports these exact percentages; the method is the citation.	Modelled (method stated)	EFFECTS sheets; DELAY sheets	
13	Root-cause chains (5 Whys)	Analytical method: the 5 Whys technique from the Toyota Production System tradition (Sakichi Toyoda / Taiichi Ohno). The chains themselves are this system's analysis.	Method citation	MASTER - ROOT CAUSES 5 WHYS	
14	Coinciding-problem scenarios and joint probabilities	MODELLED — scenario construction and probability estimates by this system, from the documented historical co-occurrences noted per scenario (e.g., heatwave + grid failure + hospital surge).	Modelled (method stated)	MASTER - COINCIDING PROBLEMS; DELAY - COINCIDING PROBLEMS	
15	Danger-awareness content (companion volume)	Fully cited in Danger_Awareness_Worldwide.xlsx, SOURCES & CITATIONS sheet (official guides, legislation, databases, peer-reviewed literature).	See companion file	START HERE note	

#	Dimension of Reality	Field of Discipline	Pareto Tier of Impact	Timeframe	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Success	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
1	Physical / Material	Water	Tier 1	Past	Household / Personal	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable.	The Aral Sea shrank to a fraction of its size after river diversion, salting farmland and collapsing fishing towns.	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	86% (base range 70–85%)	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
2	Physical / Material	Water	Tier 1	Past	Society / National	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable.	The Aral Sea shrank to a fraction of its size after river diversion, salting farmland and collapsing fishing towns.	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	78% (base range 70–85%)	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
3	Physical / Material	Water	Tier 1	Past	Global / Civilizational	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable.	The Aral Sea shrank to a fraction of its size after river diversion, salting farmland and collapsing fishing towns.	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	66% (base range 70–85%)	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
4	Physical / Material	Water	Tier 1	Now	Household / Personal	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	A single treatment-plant failure or pipe contamination event forces boil-water advisories for hundreds of thousands of people at once.	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	93% (base range 78–92%)	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
5	Physical / Material	Water	Tier 1	Now	Society / National	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	A single treatment-plant failure or pipe contamination event forces boil-water advisories for hundreds of thousands of people at once.	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	85% (base range 78–92%)	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
6	Physical / Material	Water	Tier 1	Now	Global / Civilizational	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	A single treatment-plant failure or pipe contamination event forces boil-water advisories for hundreds of thousands of people at once.	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	73% (base range 78–92%)	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
7	Physical / Material	Water	Tier 1	Future	Household / Personal	Multi-year megadrought plus population growth makes current cities structurally water-short.	A metropolitan region of millions faces day-zero conditions as reservoirs fall below dead-storage levels for consecutive years.	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	63% (base range 45–65%)	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
8	Physical / Material	Water	Tier 1	Future	Society / National	Multi-year megadrought plus population growth makes current cities structurally water-short.	A metropolitan region of millions faces day-zero conditions as reservoirs fall below dead-storage levels for consecutive years.	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	55% (base range 45–65%)	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
9	Physical / Material	Water	Tier 1	Future	Global / Civilizational	Multi-year megadrought plus population growth makes current cities structurally water-short.	A metropolitan region of millions faces day-zero conditions as reservoirs fall below dead-storage levels for consecutive years.	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	43% (base range 45–65%)	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
10	Biological / Life	Food & Agriculture	Tier 1	Past	Household / Personal	Single-crop dependence plus blight or drought produces famine at national scale.	The Irish potato famine: one crop variety, one pathogen, a million dead and a million emigrated.	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	90% (base range 75–88%)	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.

11	Biological / Life	Food & Agriculture	Tier 1	Past	Society / National	Single-crop dependence plus blight or drought produces famine at national scale.	The Irish potato famine: one crop variety, one pathogen, a million dead and a million emigrated.	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain – 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	82% (base range 75–88%)	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
12	Biological / Life	Food & Agriculture	Tier 1	Past	Global / Civilizational	Single-crop dependence plus blight or drought produces famine at national scale.	The Irish potato famine: one crop variety, one pathogen, a million dead and a million emigrated.	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain – 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	70% (base range 75–88%)	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
13	Biological / Life	Food & Agriculture	Tier 1	Now	Household / Personal	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	A fertilizer-export ban plus a blocked shipping lane doubled bread prices in import-dependent countries within one season.	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples = 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	80% (base range 64–80%)	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
14	Biological / Life	Food & Agriculture	Tier 1	Now	Society / National	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	A fertilizer-export ban plus a blocked shipping lane doubled bread prices in import-dependent countries within one season.	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples = 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	72% (base range 64–80%)	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
15	Biological / Life	Food & Agriculture	Tier 1	Now	Global / Civilizational	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	A fertilizer-export ban plus a blocked shipping lane doubled bread prices in import-dependent countries within one season.	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples = 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	60% (base range 64–80%)	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
16	Biological / Life	Food & Agriculture	Tier 1	Future	Household / Personal	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Simultaneous heat domes over multiple grain regions in one summer cut global wheat output while soils, already degraded, cannot compensate.	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	66% (base range 48–68%)	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
17	Biological / Life	Food & Agriculture	Tier 1	Future	Society / National	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Simultaneous heat domes over multiple grain regions in one summer cut global wheat output while soils, already degraded, cannot compensate.	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	58% (base range 48–68%)	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
18	Biological / Life	Food & Agriculture	Tier 1	Future	Global / Civilizational	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Simultaneous heat domes over multiple grain regions in one summer cut global wheat output while soils, already degraded, cannot compensate.	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	46% (base range 48–68%)	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Why 1 — shelves empty fast; Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope. | Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity. | Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract. | Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it. | Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
19	Biological / Life	Health & Medicine	Tier 1	Past	Household / Personal	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	The 1918 influenza killed tens of millions in months; cities ran out of coffins and grave-diggers.	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	88% (base range 72–88%)	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
20	Biological / Life	Health & Medicine	Tier 1	Past	Society / National	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	The 1918 influenza killed tens of millions in months; cities ran out of coffins and grave-diggers.	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	80% (base range 72–88%)	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
21	Biological / Life	Health & Medicine	Tier 1	Past	Global / Civilizational	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	The 1918 influenza killed tens of millions in months; cities ran out of coffins and grave-diggers.	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	68% (base range 72–88%)	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.

22	Biological / Life	Health & Medicine	Tier 1	Now	Household / Personal	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Drug-resistant infections already kill over a million people a year globally; common procedures like hip replacements become gambles.	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	70% (base range 54–70%)	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
23	Biological / Life	Health & Medicine	Tier 1	Now	Society / National	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Drug-resistant infections already kill over a million people a year globally; common procedures like hip replacements become gambles.	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	62% (base range 54–70%)	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
24	Biological / Life	Health & Medicine	Tier 1	Now	Global / Civilizational	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Drug-resistant infections already kill over a million people a year globally; common procedures like hip replacements become gambles.	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	50% (base range 54–70%)	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
25	Biological / Life	Health & Medicine	Tier 1	Future	Household / Personal	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicine as an institution.	A high-fatality airborne pathogen emerges while hospitals are already at 95% occupancy from ordinary demand; elective and emergency care collapse together.	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	56% (base range 38–58%)	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
26	Biological / Life	Health & Medicine	Tier 1	Future	Society / National	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicine as an institution.	A high-fatality airborne pathogen emerges while hospitals are already at 95% occupancy from ordinary demand; elective and emergency care collapse together.	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	48% (base range 38–58%)	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
27	Biological / Life	Health & Medicine	Tier 1	Future	Global / Civilizational	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicine as an institution.	A high-fatality airborne pathogen emerges while hospitals are already at 95% occupancy from ordinary demand; elective and emergency care collapse together.	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	36% (base range 38–58%)	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
28	Physical / Material	Energy	Tier 1	Past	Household / Personal	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	The 1973 oil embargo quadrupled prices, idled factories, and forced rationing across importing countries.	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	92% (base range 76–90%)	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
29	Physical / Material	Energy	Tier 1	Past	Society / National	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	The 1973 oil embargo quadrupled prices, idled factories, and forced rationing across importing countries.	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	84% (base range 76–90%)	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
30	Physical / Material	Energy	Tier 1	Past	Global / Civilizational	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	The 1973 oil embargo quadrupled prices, idled factories, and forced rationing across importing countries.	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	72% (base range 76–90%)	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
31	Physical / Material	Energy	Tier 1	Now	Household / Personal	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fails with it.	continental-scale blackouts have left tens of millions without power; water pumps, payment systems, hospitals, and communications degrade within hours.	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	84% (base range 68–84%)	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
32	Physical / Material	Energy	Tier 1	Now	Society / National	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fails with it.	continental-scale blackouts have left tens of millions without power; water pumps, payment systems, hospitals, and communications degrade within hours.	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	76% (base range 68–84%)	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.

33	Physical / Material	Energy	Tier 1	Now	Global / Civilizational	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fails with it.	continental-scale blackouts have left tens of millions without power; water pumps, payment systems, hospitals, and communications degrade within hours.	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	64% (base range 68–84%)	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
34	Physical / Material	Energy	Tier 1	Future	Household / Personal	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes it down continent-wide for months.	A Carrington-class geomagnetic storm destroys large transformers that take 12–24 months to manufacture, leaving regions dark for seasons.	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	58% (base range 40–60%)	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
35	Physical / Material	Energy	Tier 1	Future	Society / National	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes it down continent-wide for months.	A Carrington-class geomagnetic storm destroys large transformers that take 12–24 months to manufacture, leaving regions dark for seasons.	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	50% (base range 40–60%)	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
36	Physical / Material	Energy	Tier 1	Future	Global / Civilizational	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes it down continent-wide for months.	A Carrington-class geomagnetic storm destroys large transformers that take 12–24 months to manufacture, leaving regions dark for seasons.	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	38% (base range 40–60%)	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
37	Physical / Material	Shelter & Housing	Tier 1	Past	Household / Personal	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	The 1906 San Francisco earthquake and fire destroyed 80% of the city; the 1940s bombing campaigns displaced tens of millions across Europe.	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	89% (base range 74–88%)	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
38	Physical / Material	Shelter & Housing	Tier 1	Past	Society / National	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	The 1906 San Francisco earthquake and fire destroyed 80% of the city; the 1940s bombing campaigns displaced tens of millions across Europe.	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	81% (base range 74–88%)	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
39	Physical / Material	Shelter & Housing	Tier 1	Past	Global / Civilizational	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	The 1906 San Francisco earthquake and fire destroyed 80% of the city; the 1940s bombing campaigns displaced tens of millions across Europe.	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	69% (base range 74–88%)	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
40	Physical / Material	Shelter & Housing	Tier 1	Now	Household / Personal	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	In major cities, median homes cost 10–13x median income; full-time workers sleep in vehicles while vacancies are held as investments.	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	72% (base range 55–72%)	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
41	Physical / Material	Shelter & Housing	Tier 1	Now	Society / National	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	In major cities, median homes cost 10–13x median income; full-time workers sleep in vehicles while vacancies are held as investments.	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	64% (base range 55–72%)	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
42	Physical / Material	Shelter & Housing	Tier 1	Now	Global / Civilizational	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	In major cities, median homes cost 10–13x median income; full-time workers sleep in vehicles while vacancies are held as investments.	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	52% (base range 55–72%)	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
43	Physical / Material	Shelter & Housing	Tier 1	Future	Household / Personal	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safely occupied.	Coastal and fire-zone properties lose insurance and value simultaneously; millions hold worthless assets in places becoming unlivable.	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	60% (base range 42–62%)	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).

44	Physical / Material	Shelter & Housing	Tier 1	Future	Society / National	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safely occupied.	Coastal and fire-zone properties lose insurance and value simultaneously; millions hold worthless assets in places becoming unlivable.	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	52% (base range 42–62%)	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
45	Physical / Material	Shelter & Housing	Tier 1	Future	Global / Civilizational	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safely occupied.	Coastal and fire-zone properties lose insurance and value simultaneously; millions hold worthless assets in places becoming unlivable.	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	40% (base range 42–62%)	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
46	Social / Order	Security & Defense	Tier 1	Past	Household / Personal	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	The 1990s Balkan wars and the Rwandan genocide showed how fast neighbours, police, and armies can become the threat itself.	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	53% (base range 35–55%)	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
47	Social / Order	Security & Defense	Tier 1	Past	Society / National	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	The 1990s Balkan wars and the Rwandan genocide showed how fast neighbours, police, and armies can become the threat itself.	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	45% (base range 35–55%)	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
48	Social / Order	Security & Defense	Tier 1	Past	Global / Civilizational	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	The 1990s Balkan wars and the Rwandan genocide showed how fast neighbours, police, and armies can become the threat itself.	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	33% (base range 35–55%)	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
49	Social / Order	Security & Defense	Tier 1	Now	Household / Personal	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legitimacy.	Cities where homicide concentrates in a few postal codes: a small fraction of places and people account for most violence.	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	82% (base range 66–82%)	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
50	Social / Order	Security & Defense	Tier 1	Now	Society / National	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legitimacy.	Cities where homicide concentrates in a few postal codes: a small fraction of places and people account for most violence.	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	74% (base range 66–82%)	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
51	Social / Order	Security & Defense	Tier 1	Now	Global / Civilizational	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legitimacy.	Cities where homicide concentrates in a few postal codes: a small fraction of places and people account for most violence.	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	62% (base range 66–82%)	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
52	Social / Order	Security & Defense	Tier 1	Future	Household / Personal	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built for human-speed threats.	A single actor with a swarm of consumer drones attacks a stadium, substation, or water plant; defence budgets built for jets and tanks are irrelevant.	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	64% (base range 46–66%)	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
53	Social / Order	Security & Defense	Tier 1	Future	Society / National	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built for human-speed threats.	A single actor with a swarm of consumer drones attacks a stadium, substation, or water plant; defence budgets built for jets and tanks are irrelevant.	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	56% (base range 46–66%)	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
54	Social / Order	Security & Defense	Tier 1	Future	Global / Civilizational	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built for human-speed threats.	A single actor with a swarm of consumer drones attacks a stadium, substation, or water plant; defence budgets built for jets and tanks are irrelevant.	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	44% (base range 46–66%)	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.

55	Social / Order	Governance & Law	Tier 1	Past	Household / Personal	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Weimar Germany's institutions were dismantled largely by legal instruments — an enabling act, packed courts, and emergency rule that never ended.	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	68% (base range 50–70%)	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
56	Social / Order	Governance & Law	Tier 1	Past	Society / National	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Weimar Germany's institutions were dismantled largely by legal instruments — an enabling act, packed courts, and emergency rule that never ended.	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	60% (base range 50–70%)	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
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58	Social / Order	Governance & Law	Tier 1	Now	Household / Personal	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	In captured states, procurement, courts, and media serve a network; citizens pay bribes for what law already entitles them to.	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	76% (base range 60–76%)	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
59	Social / Order	Governance & Law	Tier 1	Now	Society / National	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	In captured states, procurement, courts, and media serve a network; citizens pay bribes for what law already entitles them to.	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	68% (base range 60–76%)	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
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61	Social / Order	Governance & Law	Tier 1	Future	Household / Personal	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a dictator.	A future where elections, courts, money, and news are all distrusted simultaneously; every collective action — vaccination, evacuation, taxation — fails for lack of belief.	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	55% (base range 37–57%)	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
62	Social / Order	Governance & Law	Tier 1	Future	Society / National	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a dictator.	A future where elections, courts, money, and news are all distrusted simultaneously; every collective action — vaccination, evacuation, taxation — fails for lack of belief.	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	47% (base range 37–57%)	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
63	Social / Order	Governance & Law	Tier 1	Future	Global / Civilizational	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a dictator.	A future where elections, courts, money, and news are all distrusted simultaneously; every collective action — vaccination, evacuation, taxation — fails for lack of belief.	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	35% (base range 37–57%)	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
64	Social / Order	Economy & Finance	Tier 1	Past	Household / Personal	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	After 1929, bank runs destroyed savings, credit froze, and unemployment reached 25% in the United States and higher elsewhere.	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	91% (base range 76–89%)	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
65	Social / Order	Economy & Finance	Tier 1	Past	Society / National	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	After 1929, bank runs destroyed savings, credit froze, and unemployment reached 25% in the United States and higher elsewhere.	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	83% (base range 76–89%)	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.

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67	Social / Order	Economy & Finance	Tier 1	Now	Household / Personal	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Households carrying record debt-to-income ratios meet higher rates; defaults cluster; whole suburbs' collateral values fall together.	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.	71% (base range 55–71%)	Debt-service caps at origination demonstrably reduce default cascades; the obstacle is the credit boom's political popularity.	Apply the stressed-rate test (+2%) to every loan; open refinancing windows automatically when rates fall; enforce the 30% personal debt-payment ceiling.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
68	Social / Order	Economy & Finance	Tier 1	Now	Society / National	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Households carrying record debt-to-income ratios meet higher rates; defaults cluster; whole suburbs' collateral values fall together.	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.	63% (base range 55–71%)	Debt-service caps at origination demonstrably reduce default cascades; the obstacle is the credit boom's political popularity.	Apply the stressed-rate test (+2%) to every loan; open refinancing windows automatically when rates fall; enforce the 30% personal debt-payment ceiling.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
69	Social / Order	Economy & Finance	Tier 1	Now	Global / Civilizational	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Households carrying record debt-to-income ratios meet higher rates; defaults cluster; whole suburbs' collateral values fall together.	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.	51% (base range 55–71%)	Debt-service caps at origination demonstrably reduce default cascades; the obstacle is the credit boom's political popularity.	Apply the stressed-rate test (+2%) to every loan; open refinancing windows automatically when rates fall; enforce the 30% personal debt-payment ceiling.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
70	Social / Order	Economy & Finance	Tier 1	Future	Household / Personal	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines and models.	Whole occupations — driving, clerical work, routine analysis — shrink within a decade; wages decouple from productivity entirely; politics radicalizes.	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	52% (base range 34–54%)	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
71	Social / Order	Economy & Finance	Tier 1	Future	Society / National	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines and models.	Whole occupations — driving, clerical work, routine analysis — shrink within a decade; wages decouple from productivity entirely; politics radicalizes.	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	44% (base range 34–54%)	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
72	Social / Order	Economy & Finance	Tier 1	Future	Global / Civilizational	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines and models.	Whole occupations — driving, clerical work, routine analysis — shrink within a decade; wages decouple from productivity entirely; politics radicalizes.	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	32% (base range 34–54%)	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
73	Technological	Technology & AI	Tier 1	Past	Household / Personal	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before correction.	Leaded gasoline and asbestos were known hazards for decades while use expanded; lead exposure lowered population IQ and asbestos deaths continue a century later.	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	74% (base range 58–74%)	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
74	Technological	Technology & AI	Tier 1	Past	Society / National	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before correction.	Leaded gasoline and asbestos were known hazards for decades while use expanded; lead exposure lowered population IQ and asbestos deaths continue a century later.	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	66% (base range 58–74%)	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
75	Technological	Technology & AI	Tier 1	Past	Global / Civilizational	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before correction.	Leaded gasoline and asbestos were known hazards for decades while use expanded; lead exposure lowered population IQ and asbestos deaths continue a century later.	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	54% (base range 58–74%)	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
76	Technological	Technology & AI	Tier 1	Now	Household / Personal	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragile.	A single ransomware incident closed a major fuel pipeline and triggered panic buying across a seaboard; hospitals routinely divert ambulances during attacks.	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	86% (base range 70–85%)	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.

77	Technological	Technology & AI	Tier 1	Now	Society / National	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragile.	A single ransomware incident closed a major fuel pipeline and triggered panic buying across a seaboard; hospitals routinely divert ambulances during attacks.	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	78% (base range 70–85%)	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
78	Technological	Technology & AI	Tier 1	Now	Global / Civilizational	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragile.	A single ransomware incident closed a major fuel pipeline and triggered panic buying across a seaboard; hospitals routinely divert ambulances during attacks.	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	66% (base range 70–85%)	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
79	Technological	Technology & AI	Tier 1	Future	Household / Personal	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale failure or harm.	Frontier systems embedded in finance, logistics, medicine, and weapons make correlated errors at machine speed, or are deliberately turned to bioweapon design and mass manipulation.	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	50% (base range 32–52%)	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
80	Technological	Technology & AI	Tier 1	Future	Society / National	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale failure or harm.	Frontier systems embedded in finance, logistics, medicine, and weapons make correlated errors at machine speed, or are deliberately turned to bioweapon design and mass manipulation.	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	42% (base range 32–52%)	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
81	Technological	Technology & AI	Tier 1	Future	Global / Civilizational	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale failure or harm.	Frontier systems embedded in finance, logistics, medicine, and weapons make correlated errors at machine speed, or are deliberately turned to bioweapon design and mass manipulation.	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	30% (base range 32–52%)	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
82	Physical / Material	Environment & Climate	Tier 1	Past	Household / Personal	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society follows.	Classic cases from Mesopotamian salinization to the Dust Bowl: farming and extraction practices consumed the resource they depended on.	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	79% (base range 63–79%)	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
83	Physical / Material	Environment & Climate	Tier 1	Past	Society / National	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society follows.	Classic cases from Mesopotamian salinization to the Dust Bowl: farming and extraction practices consumed the resource they depended on.	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	71% (base range 63–79%)	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
84	Physical / Material	Environment & Climate	Tier 1	Past	Global / Civilizational	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society follows.	Classic cases from Mesopotamian salinization to the Dust Bowl: farming and extraction practices consumed the resource they depended on.	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	59% (base range 63–79%)	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
85	Physical / Material	Environment & Climate	Tier 1	Now	Household / Personal	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Heat domes killing hundreds in a weekend in northern cities; floods overwhelming drains sized for historical rainfall; wildfire smoke closing schools across a continent.	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	77% (base range 61–77%)	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
86	Physical / Material	Environment & Climate	Tier 1	Now	Society / National	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Heat domes killing hundreds in a weekend in northern cities; floods overwhelming drains sized for historical rainfall; wildfire smoke closing schools across a continent.	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	69% (base range 61–77%)	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
87	Physical / Material	Environment & Climate	Tier 1	Now	Global / Civilizational	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Heat domes killing hundreds in a weekend in northern cities; floods overwhelming drains sized for historical rainfall; wildfire smoke closing schools across a continent.	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	57% (base range 61–77%)	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.

88	Physical / Material	Environment & Climate	Tier 1	Future	Household / Personal	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level rise and uninhabitable zones.	Compound worst case: several tipping elements interact, raising seas 2+ metres over centuries, shifting monsoons that feed billions, and rendering equatorial regions seasonally unsurvivable outdoors.	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	46% (base range 28–48%)	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
89	Physical / Material	Environment & Climate	Tier 1	Future	Society / National	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level rise and uninhabitable zones.	Compound worst case: several tipping elements interact, raising seas 2+ metres over centuries, shifting monsoons that feed billions, and rendering equatorial regions seasonally unsurvivable outdoors.	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	38% (base range 28–48%)	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
90	Physical / Material	Environment & Climate	Tier 1	Future	Global / Civilizational	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level rise and uninhabitable zones.	Compound worst case: several tipping elements interact, raising seas 2+ metres over centuries, shifting monsoons that feed billions, and rendering equatorial regions seasonally unsurvivable outdoors.	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	26% (base range 28–48%)	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
91	Informational	Information & Media	Tier 2	Past	Household / Personal	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Radio propaganda in 1930s Germany and 1994 Rwanda turned neighbours into killers; the medium outran every correction.	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	66% (base range 48–68%)	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
92	Informational	Information & Media	Tier 2	Past	Society / National	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Radio propaganda in 1930s Germany and 1994 Rwanda turned neighbours into killers; the medium outran every correction.	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	58% (base range 48–68%)	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
93	Informational	Information & Media	Tier 2	Past	Global / Civilizational	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Radio propaganda in 1930s Germany and 1994 Rwanda turned neighbours into killers; the medium outran every correction.	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	46% (base range 48–68%)	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
94	Informational	Information & Media	Tier 2	Now	Household / Personal	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic facts.	Two citizens in the same city now inhabit different factual worlds; elections, health measures, and emergencies are fought inside incompatible realities.	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	65% (base range 49–65%)	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
95	Informational	Information & Media	Tier 2	Now	Society / National	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic facts.	Two citizens in the same city now inhabit different factual worlds; elections, health measures, and emergencies are fought inside incompatible realities.	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	57% (base range 49–65%)	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
96	Informational	Information & Media	Tier 2	Now	Global / Civilizational	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic facts.	Two citizens in the same city now inhabit different factual worlds; elections, health measures, and emergencies are fought inside incompatible realities.	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	45% (base range 49–65%)	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
97	Informational	Information & Media	Tier 2	Future	Household / Personal	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, and all records become deniable.	A fabricated video of a leader announcing surrender, or of a candidate committing a crime, lands hours before a vote; authentic evidence of real crimes is dismissed as fake — the liar's dividend pays out everywhere.	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	69% (base range 52–70%)	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
98	Informational	Information & Media	Tier 2	Future	Society / National	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, and all records become deniable.	A fabricated video of a leader announcing surrender, or of a candidate committing a crime, lands hours before a vote; authentic evidence of real crimes is dismissed as fake — the liar's dividend pays out everywhere.	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	61% (base range 52–70%)	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.

99	Informational	Information & Media	Tier 2	Future	Global / Civilizational	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, and all records become deniable.	A fabricated video of a leader announcing surrender, or of a candidate committing a crime, lands hours before a vote; authentic evidence of real crimes is dismissed as fake — the liar's dividend pays out everywhere.	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	49% (base range 52–70%)	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
100	Psychological	Psychology & Mind	Tier 2	Past	Household / Personal	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Post-war generations raised by traumatized parents show elevated anxiety, violence, and illness decades after the events ended.	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	73% (base range 57–73%)	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
101	Psychological	Psychology & Mind	Tier 2	Past	Society / National	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Post-war generations raised by traumatized parents show elevated anxiety, violence, and illness decades after the events ended.	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	65% (base range 57–73%)	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
102	Psychological	Psychology & Mind	Tier 2	Past	Global / Civilizational	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Post-war generations raised by traumatized parents show elevated anxiety, violence, and illness decades after the events ended.	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	53% (base range 57–73%)	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
103	Psychological	Psychology & Mind	Tier 2	Now	Household / Personal	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolated.	Adolescent mental-health indicators worsened sharply in the smartphone era across many countries simultaneously; loneliness at scale matches smoking as a mortality risk factor.	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	75% (base range 59–75%)	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
104	Psychological	Psychology & Mind	Tier 2	Now	Society / National	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolated.	Adolescent mental-health indicators worsened sharply in the smartphone era across many countries simultaneously; loneliness at scale matches smoking as a mortality risk factor.	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	67% (base range 59–75%)	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
105	Psychological	Psychology & Mind	Tier 2	Now	Global / Civilizational	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolated.	Adolescent mental-health indicators worsened sharply in the smartphone era across many countries simultaneously; loneliness at scale matches smoking as a mortality risk factor.	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	55% (base range 59–75%)	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
106	Psychological	Psychology & Mind	Tier 2	Future	Household / Personal	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human contact, work, and reproduction.	AI companions that never tire, never judge, and always agree become the primary relationship for a large minority; birth rates and civic participation fall off a cliff.	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	54% (base range 36–56%)	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
107	Psychological	Psychology & Mind	Tier 2	Future	Society / National	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human contact, work, and reproduction.	AI companions that never tire, never judge, and always agree become the primary relationship for a large minority; birth rates and civic participation fall off a cliff.	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	46% (base range 36–56%)	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
108	Psychological	Psychology & Mind	Tier 2	Future	Global / Civilizational	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human contact, work, and reproduction.	AI companions that never tire, never judge, and always agree become the primary relationship for a large minority; birth rates and civic participation fall off a cliff.	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	34% (base range 36–56%)	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.

109	Informational	Education & Knowledge	Tier 2	Past	Household / Personal	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies with a generation.	The destruction of libraries from Alexandria to Baghdad, and colonial bans on teaching the enslaved to read, erased centuries of accumulated knowledge.	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	94% (base range 79–92%)	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
110	Informational	Education & Knowledge	Tier 2	Past	Society / National	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies with a generation.	The destruction of libraries from Alexandria to Baghdad, and colonial bans on teaching the enslaved to read, erased centuries of accumulated knowledge.	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	86% (base range 79–92%)	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
111	Informational	Education & Knowledge	Tier 2	Past	Global / Civilizational	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies with a generation.	The destruction of libraries from Alexandria to Baghdad, and colonial bans on teaching the enslaved to read, erased centuries of accumulated knowledge.	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	74% (base range 79–92%)	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
112	Informational	Education & Knowledge	Tier 2	Now	Household / Personal	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinformation outruns curricula.	Large shares of 10-year-olds in many countries cannot read a simple text, even after years enrolled in school; graduates report their degrees did not teach the skills their jobs require.	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	78% (base range 62–78%)	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
113	Informational	Education & Knowledge	Tier 2	Now	Society / National	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinformation outruns curricula.	Large shares of 10-year-olds in many countries cannot read a simple text, even after years enrolled in school; graduates report their degrees did not teach the skills their jobs require.	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	70% (base range 62–78%)	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
114	Informational	Education & Knowledge	Tier 2	Now	Global / Civilizational	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinformation outruns curricula.	Large shares of 10-year-olds in many countries cannot read a simple text, even after years enrolled in school; graduates report their degrees did not teach the skills their jobs require.	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	58% (base range 62–78%)	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
115	Informational	Education & Knowledge	Tier 2	Future	Household / Personal	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calculate, or reason unaided.	A generation that has never written, navigated, or calculated without machine assistance cannot detect machine error — or function when systems fail.	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	67% (base range 50–68%)	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
116	Informational	Education & Knowledge	Tier 2	Future	Society / National	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calculate, or reason unaided.	A generation that has never written, navigated, or calculated without machine assistance cannot detect machine error — or function when systems fail.	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	59% (base range 50–68%)	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
117	Informational	Education & Knowledge	Tier 2	Future	Global / Civilizational	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calculate, or reason unaided.	A generation that has never written, navigated, or calculated without machine assistance cannot detect machine error — or function when systems fail.	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	47% (base range 50–68%)	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Why 1 — systems measure attendance, not learning: Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
118	Social / Order	Work & Livelihood	Tier 2	Past	Household / Personal	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housing, food, and law.	Company towns paid in scrip redeemable only at the company store; workers owed their souls and their rent to the same ledger.	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of (job, housing, food retail, credit) to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	85% (base range 69–84%)	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
119	Social / Order	Work & Livelihood	Tier 2	Past	Society / National	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housing, food, and law.	Company towns paid in scrip redeemable only at the company store; workers owed their souls and their rent to the same ledger.	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of (job, housing, food retail, credit) to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	77% (base range 69–84%)	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.

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121 Social / Order	Work & Livelihood	Tier 2	Now	Household / Personal	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheque from crisis.	Delivery and care workers classified as contractors without benefits, working full-time hours in poverty; towns that lose one plant lose their whole economy.	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	74% (base range 58–74%)	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
122 Social / Order	Work & Livelihood	Tier 2	Now	Society / National	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheque from crisis.	Delivery and care workers classified as contractors without benefits, working full-time hours in poverty; towns that lose one plant lose their whole economy.	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	66% (base range 58–74%)	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
123 Social / Order	Work & Livelihood	Tier 2	Now	Global / Civilizational	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheque from crisis.	Delivery and care workers classified as contractors without benefits, working full-time hours in poverty; towns that lose one plant lose their whole economy.	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	54% (base range 58–74%)	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
124 Social / Order	Work & Livelihood	Tier 2	Future	Household / Personal	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	If AI and robotics perform most current jobs at near-zero marginal cost, wages as the main distribution mechanism fail for most of the population simultaneously.	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	51% (base range 33–53%)	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
125 Social / Order	Work & Livelihood	Tier 2	Future	Society / National	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	If AI and robotics perform most current jobs at near-zero marginal cost, wages as the main distribution mechanism fail for most of the population simultaneously.	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	43% (base range 33–53%)	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
126 Social / Order	Work & Livelihood	Tier 2	Future	Global / Civilizational	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	If AI and robotics perform most current jobs at near-zero marginal cost, wages as the main distribution mechanism fail for most of the population simultaneously.	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	31% (base range 33–53%)	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
127 Physical / Material	Transport & Mobility	Tier 2	Past	Household / Personal	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or blockade could sever in days.	Sieges from Leningrad to Sarajevo: when roads and rails close, urban populations face starvation within weeks regardless of money.	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	87% (base range 71–86%)	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
128 Physical / Material	Transport & Mobility	Tier 2	Past	Society / National	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or blockade could sever in days.	Sieges from Leningrad to Sarajevo: when roads and rails close, urban populations face starvation within weeks regardless of money.	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	79% (base range 71–86%)	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
129 Physical / Material	Transport & Mobility	Tier 2	Past	Global / Civilizational	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or blockade could sever in days.	Sieges from Leningrad to Sarajevo: when roads and rails close, urban populations face starvation within weeks regardless of money.	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	67% (base range 71–86%)	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
130 Physical / Material	Transport & Mobility	Tier 2	Now	Household / Personal	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike strands millions.	A single blocked canal halted 12% of world trade; domestically, households without cars in sprawled regions cannot reach work, food, or care at all.	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	76% (base range 60–76%)	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.

131 Physical / Material	Transport & Mobility	Tier 2	Now	Society / National	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike strands millions.	A single blocked canal halted 12% of world trade; domestically, households without cars in sprawled regions cannot reach work, food, or care at all.	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	68% (base range 60–76%)	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
132 Physical / Material	Transport & Mobility	Tier 2	Now	Global / Civilizational	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike strands millions.	A single blocked canal halted 12% of world trade; domestically, households without cars in sprawled regions cannot reach work, food, or care at all.	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	56% (base range 60–76%)	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
133 Physical / Material	Transport & Mobility	Tier 2	Future	Household / Personal	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or battery-material cutoff grounds a continent.	A corrupted fleet-wide update or a compromised dispatch platform immobilizes autonomous fleets mid-journey, everywhere, at the same hour.	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	62% (base range 45–63%)	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
134 Physical / Material	Transport & Mobility	Tier 2	Future	Society / National	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or battery-material cutoff grounds a continent.	A corrupted fleet-wide update or a compromised dispatch platform immobilizes autonomous fleets mid-journey, everywhere, at the same hour.	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	54% (base range 45–63%)	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
135 Physical / Material	Transport & Mobility	Tier 2	Future	Global / Civilizational	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or battery-material cutoff grounds a continent.	A corrupted fleet-wide update or a compromised dispatch platform immobilizes autonomous fleets mid-journey, everywhere, at the same hour.	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	42% (base range 45–63%)	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
136 Informational	Communication	Tier 2	Past	Household / Personal	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumour governed.	In the 1857 Indian rebellion, cut telegraph lines and rumour cascades moved faster than any verified information, with catastrophic results on all sides.	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	83% (base range 67–82%)	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
137 Informational	Communication	Tier 2	Past	Society / National	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumour governed.	In the 1857 Indian rebellion, cut telegraph lines and rumour cascades moved faster than any verified information, with catastrophic results on all sides.	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	75% (base range 67–82%)	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
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139 Informational	Communication	Tier 2	Now	Household / Personal	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; outages and bans silence millions instantly.	A single platform outage in 2021 knocked out businesses, emergency lines, and family contact across continents for hours; account bans erase people's records, audiences, and livelihoods without appeal.	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	77% (base range 61–77%)	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
140 Informational	Communication	Tier 2	Now	Society / National	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; outages and bans silence millions instantly.	A single platform outage in 2021 knocked out businesses, emergency lines, and family contact across continents for hours; account bans erase people's records, audiences, and livelihoods without appeal.	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	69% (base range 61–77%)	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
141 Informational	Communication	Tier 2	Now	Global / Civilizational	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; outages and bans silence millions instantly.	A single platform outage in 2021 knocked out businesses, emergency lines, and family contact across continents for hours; account bans erase people's records, audiences, and livelihoods without appeal.	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	57% (base range 61–77%)	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.

142	Informational	Communication	Tier 2	Future	Household / Personal	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable coordination makes law enforcement blind to real conspiracies.	A state that reads everything chills all dissent and error-correction; a world with no lawful access anywhere shelters atrocity planning equally.	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	60% (base range 43–61%)	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity. | Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right. | Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans. | Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited. | Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
143	Informational	Communication	Tier 2	Future	Society / National	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable coordination makes law enforcement blind to real conspiracies.	A state that reads everything chills all dissent and error-correction; a world with no lawful access anywhere shelters atrocity planning equally.	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	52% (base range 43–61%)	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity. | Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right. | Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans. | Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited. | Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
144	Informational	Communication	Tier 2	Future	Global / Civilizational	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable coordination makes law enforcement blind to real conspiracies.	A state that reads everything chills all dissent and error-correction; a world with no lawful access anywhere shelters atrocity planning equally.	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	40% (base range 43–61%)	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity. | Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right. | Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans. | Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited. | Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
145	Biological / Life	Demographics & Population	Tier 2	Past	Household / Personal	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowledge lineages cut.	The Black Death killed a third to a half of Europe's population; some regions took two centuries to recover their numbers and skills.	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	72% (base range 55–72%)	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
146	Biological / Life	Demographics & Population	Tier 2	Past	Society / National	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowledge lineages cut.	The Black Death killed a third to a half of Europe's population; some regions took two centuries to recover their numbers and skills.	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	64% (base range 55–72%)	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
147	Biological / Life	Demographics & Population	Tier 2	Past	Global / Civilizational	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowledge lineages cut.	The Black Death killed a third to a half of Europe's population; some regions took two centuries to recover their numbers and skills.	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	52% (base range 55–72%)	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
148	Biological / Life	Demographics & Population	Tier 2	Now	Household / Personal	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns hollow out.	Countries at fertility rates of 0.7–1.3 face shrinking school cohorts, closing villages, and pension systems whose math no longer closes.	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	65% (base range 48–66%)	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
149	Biological / Life	Demographics & Population	Tier 2	Now	Society / National	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns hollow out.	Countries at fertility rates of 0.7–1.3 face shrinking school cohorts, closing villages, and pension systems whose math no longer closes.	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	57% (base range 48–66%)	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
150	Biological / Life	Demographics & Population	Tier 2	Now	Global / Civilizational	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns hollow out.	Countries at fertility rates of 0.7–1.3 face shrinking school cohorts, closing villages, and pension systems whose math no longer closes.	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	45% (base range 48–66%)	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
151	Biological / Life	Demographics & Population	Tier 2	Future	Household / Personal	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows destabilize both.	By 2100 projections, parts of East Asia and Europe halve in population while parts of Africa triple; infrastructure, states, and cultures built for the opposite trend fail.	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	57% (base range 40–58%)	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
152	Biological / Life	Demographics & Population	Tier 2	Future	Society / National	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows destabilize both.	By 2100 projections, parts of East Asia and Europe halve in population while parts of Africa triple; infrastructure, states, and cultures built for the opposite trend fail.	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	49% (base range 40–58%)	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.

153	Biological / Life	Demographics & Population	Tier 2	Future	Global / Civilizational	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows destabilize both.	By 2100 projections, parts of East Asia and Europe halve in population while parts of Africa triple; infrastructure, states, and cultures built for the opposite trend fail.	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	37% (base range 40–58%)	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
154	Social / Order	Justice & Ethics	Tier 2	Past	Household / Personal	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	From Stalin's purges to lynching regimes, courts and mobs delivered punishment by loyalty and identity, not by evidence.	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	80% (base range 64–80%)	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
155	Social / Order	Justice & Ethics	Tier 2	Past	Society / National	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	From Stalin's purges to lynching regimes, courts and mobs delivered punishment by loyalty and identity, not by evidence.	Evidence as the *only* currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	72% (base range 64–80%)	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
156	Social / Order	Justice & Ethics	Tier 2	Past	Global / Civilizational	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	From Stalin's purges to lynching regimes, courts and mobs delivered punishment by loyalty and identity, not by evidence.	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	60% (base range 64–80%)	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
157	Social / Order	Justice & Ethics	Tier 2	Now	Household / Personal	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make 'justice delayed' the norm.	Defendants held pre-trial longer than the sentence for the charge itself; civil cases priced beyond ordinary people's reach; trust in courts falls across democracies.	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	73% (base range 57–73%)	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
158	Social / Order	Justice & Ethics	Tier 2	Now	Society / National	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make 'justice delayed' the norm.	Defendants held pre-trial longer than the sentence for the charge itself; civil cases priced beyond ordinary people's reach; trust in courts falls across democracies.	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	65% (base range 57–73%)	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
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160	Social / Order	Justice & Ethics	Tier 2	Future	Household / Personal	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and no human is answerable.	Risk scores trained on historical enforcement data send patrols and sentences in self-confirming loops; defendants cannot cross-examine a model.	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	66% (base range 49–67%)	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
161	Social / Order	Justice & Ethics	Tier 2	Future	Society / National	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and no human is answerable.	Risk scores trained on historical enforcement data send patrols and sentences in self-confirming loops; defendants cannot cross-examine a model.	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	58% (base range 49–67%)	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
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Solution #	Field of Discipline	Timeframe	Scale	Solution — Full Specification	Probability of Su	How to Maximize the Chances (detailed)	Why This Probability
1	Water	Past	Household / Personal	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	86%	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.
2	Water	Past	Society / National	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	78%	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.
3	Water	Past	Global / Civilizational	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	66%	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.
4	Water	Now	Household / Personal	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	93%	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.
5	Water	Now	Society / National	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	85%	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.
6	Water	Now	Global / Civilizational	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	73%	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.
7	Water	Future	Household / Personal	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	63%	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.
8	Water	Future	Society / National	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	55%	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.
9	Water	Future	Global / Civilizational	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	43%	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.
10	Food & Agriculture	Past	Household / Personal	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	90%	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.
11	Food & Agriculture	Past	Society / National	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	82%	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.
12	Food & Agriculture	Past	Global / Civilizational	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	70%	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.
13	Food & Agriculture	Now	Household / Personal	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	80%	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.
14	Food & Agriculture	Now	Society / National	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	72%	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.
15	Food & Agriculture	Now	Global / Civilizational	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	60%	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.
16	Food & Agriculture	Future	Household / Personal	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	66%	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.
17	Food & Agriculture	Future	Society / National	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	58%	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.
18	Food & Agriculture	Future	Global / Civilizational	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	46%	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.
19	Health & Medicine	Past	Household / Personal	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	88%	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.
20	Health & Medicine	Past	Society / National	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	80%	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.
21	Health & Medicine	Past	Global / Civilizational	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	68%	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.

22	Health & Medicine	Now	Household / Personal	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	70%	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.
23	Health & Medicine	Now	Society / National	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	62%	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.
24	Health & Medicine	Now	Global / Civilizational	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	50%	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.
25	Health & Medicine	Future	Household / Personal	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	56%	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.
26	Health & Medicine	Future	Society / National	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	48%	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.
27	Health & Medicine	Future	Global / Civilizational	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	36%	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.
28	Energy	Past	Household / Personal	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	92%	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).
29	Energy	Past	Society / National	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	84%	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).
30	Energy	Past	Global / Civilizational	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	72%	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).
31	Energy	Now	Household / Personal	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	84%	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.
32	Energy	Now	Society / National	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	76%	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.
33	Energy	Now	Global / Civilizational	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	64%	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.
34	Energy	Future	Household / Personal	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	58%	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.
35	Energy	Future	Society / National	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	50%	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.
36	Energy	Future	Global / Civilizational	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	38%	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.
37	Shelter & Housing	Past	Household / Personal	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	89%	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.
38	Shelter & Housing	Past	Society / National	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	81%	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.
39	Shelter & Housing	Past	Global / Civilizational	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	69%	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.
40	Shelter & Housing	Now	Household / Personal	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	72%	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.
41	Shelter & Housing	Now	Society / National	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	64%	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.

42	Shelter & Housing	Now	Global / Civilizational	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	52%	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.
43	Shelter & Housing	Future	Household / Personal	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	60%	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.
44	Shelter & Housing	Future	Society / National	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	52%	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.
45	Shelter & Housing	Future	Global / Civilizational	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	40%	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.
46	Security & Defense	Past	Household / Personal	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	53%	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.
47	Security & Defense	Past	Society / National	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	45%	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.
48	Security & Defense	Past	Global / Civilizational	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	33%	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.
49	Security & Defense	Now	Household / Personal	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	82%	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.
50	Security & Defense	Now	Society / National	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	74%	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.
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52	Security & Defense	Future	Household / Personal	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	64%	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.
53	Security & Defense	Future	Society / National	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	56%	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.
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55	Governance & Law	Past	Household / Personal	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	68%	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.
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58	Governance & Law	Now	Household / Personal	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	76%	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.

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61	Governance & Law	Future	Household / Personal	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	55%	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.
62	Governance & Law	Future	Society / National	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	47%	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.
63	Governance & Law	Future	Global / Civilizational	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	35%	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.
64	Economy & Finance	Past	Household / Personal	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	91%	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.
65	Economy & Finance	Past	Society / National	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	83%	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.
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67	Economy & Finance	Now	Household / Personal	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.	71%	Apply the stressed-rate test (+2%) to every loan; open refinancing windows automatically when rates fall; enforce the 30% personal debt-payment ceiling.	Debt-service caps at origination demonstrably reduce default cascades; the obstacle is the credit boom's political popularity.
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70	Economy & Finance	Future	Household / Personal	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	52%	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.
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73	Technology & AI	Past	Household / Personal	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	74%	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.
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76	Technology & AI	Now	Household / Personal	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	86%	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.

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79	Technology & AI	Future	Household / Personal	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	50%	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.
80	Technology & AI	Future	Society / National	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	42%	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.
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82	Environment & Climate	Past	Household / Personal	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	79%	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.
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85	Environment & Climate	Now	Household / Personal	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	77%	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.
86	Environment & Climate	Now	Society / National	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	69%	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.
87	Environment & Climate	Now	Global / Civilizational	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	57%	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.
88	Environment & Climate	Future	Household / Personal	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	46%	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.
89	Environment & Climate	Future	Society / National	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	38%	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.
90	Environment & Climate	Future	Global / Civilizational	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	26%	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.
91	Information & Media	Past	Household / Personal	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	66%	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.
92	Information & Media	Past	Society / National	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	58%	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.
93	Information & Media	Past	Global / Civilizational	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	46%	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.

94	Information & Media	Now	Household / Personal	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	65%	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.
95	Information & Media	Now	Society / National	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	57%	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.
96	Information & Media	Now	Global / Civilizational	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	45%	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.
97	Information & Media	Future	Household / Personal	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	69%	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.
98	Information & Media	Future	Society / National	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	61%	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.
99	Information & Media	Future	Global / Civilizational	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	49%	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.
100	Psychology & Mind	Past	Household / Personal	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	73%	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.
101	Psychology & Mind	Past	Society / National	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	65%	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.
102	Psychology & Mind	Past	Global / Civilizational	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	53%	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.
103	Psychology & Mind	Now	Household / Personal	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	75%	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.
104	Psychology & Mind	Now	Society / National	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	67%	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.
105	Psychology & Mind	Now	Global / Civilizational	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	55%	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.
106	Psychology & Mind	Future	Household / Personal	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	54%	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.
107	Psychology & Mind	Future	Society / National	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	46%	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.
108	Psychology & Mind	Future	Global / Civilizational	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	34%	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.
109	Education & Knowledge	Past	Household / Personal	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	94%	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.
110	Education & Knowledge	Past	Society / National	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	86%	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.

111	Education & Knowledge	Past	Global / Civilizational	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	74%	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.
112	Education & Knowledge	Now	Household / Personal	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	78%	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.
113	Education & Knowledge	Now	Society / National	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	70%	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.
114	Education & Knowledge	Now	Global / Civilizational	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	58%	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.
115	Education & Knowledge	Future	Household / Personal	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	67%	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.
116	Education & Knowledge	Future	Society / National	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	59%	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.
117	Education & Knowledge	Future	Global / Civilizational	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	47%	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.
118	Work & Livelihood	Past	Household / Personal	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	85%	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.
119	Work & Livelihood	Past	Society / National	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	77%	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.
120	Work & Livelihood	Past	Global / Civilizational	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	65%	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.
121	Work & Livelihood	Now	Household / Personal	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	74%	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.
122	Work & Livelihood	Now	Society / National	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	66%	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.
123	Work & Livelihood	Now	Global / Civilizational	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	54%	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.
124	Work & Livelihood	Future	Household / Personal	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	51%	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.
125	Work & Livelihood	Future	Society / National	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	43%	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.
126	Work & Livelihood	Future	Global / Civilizational	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	31%	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.
127	Transport & Mobility	Past	Household / Personal	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	87%	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.
128	Transport & Mobility	Past	Society / National	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	79%	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.
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130	Transport & Mobility	Now	Household / Personal	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	76%	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.
131	Transport & Mobility	Now	Society / National	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	68%	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.
132	Transport & Mobility	Now	Global / Civilizational	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	56%	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.
133	Transport & Mobility	Future	Household / Personal	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	62%	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.
134	Transport & Mobility	Future	Society / National	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	54%	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.
135	Transport & Mobility	Future	Global / Civilizational	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	42%	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.
136	Communication	Past	Household / Personal	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	83%	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.
137	Communication	Past	Society / National	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	75%	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.
138	Communication	Past	Global / Civilizational	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	63%	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.
139	Communication	Now	Household / Personal	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	77%	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.
140	Communication	Now	Society / National	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	69%	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.
141	Communication	Now	Global / Civilizational	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	57%	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.
142	Communication	Future	Household / Personal	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	60%	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.
143	Communication	Future	Society / National	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	52%	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.
144	Communication	Future	Global / Civilizational	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	40%	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.
145	Demographics & Population	Past	Household / Personal	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	72%	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.
146	Demographics & Population	Past	Society / National	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	64%	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.
147	Demographics & Population	Past	Global / Civilizational	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	52%	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.

148	Demographics & Population	Now	Household / Personal	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	65%	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.
149	Demographics & Population	Now	Society / National	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	57%	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.
150	Demographics & Population	Now	Global / Civilizational	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	45%	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.
151	Demographics & Population	Future	Household / Personal	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	57%	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.
152	Demographics & Population	Future	Society / National	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	49%	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.
153	Demographics & Population	Future	Global / Civilizational	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	37%	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.
154	Justice & Ethics	Past	Household / Personal	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	80%	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.
155	Justice & Ethics	Past	Society / National	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	72%	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.
156	Justice & Ethics	Past	Global / Civilizational	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	60%	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.
157	Justice & Ethics	Now	Household / Personal	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	73%	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.
158	Justice & Ethics	Now	Society / National	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	65%	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.
159	Justice & Ethics	Now	Global / Civilizational	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	53%	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.
160	Justice & Ethics	Future	Household / Personal	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	66%	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.
161	Justice & Ethics	Future	Society / National	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	58%	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.
162	Justice & Ethics	Future	Global / Civilizational	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	46%	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of S	Full Solution
1	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	Water	Now	Global / Civilizational	27	1.6%	73%	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.
2	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragi	Technology & AI	Now	Global / Civilizational	27	3.3%	66%	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.
3	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fa	Energy	Now	Global / Civilizational	27	4.9%	64%	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).
4	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legit	Security & Defense	Now	Global / Civilizational	27	6.5%	62%	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.
5	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	Food & Agriculture	Now	Global / Civilizational	27	8.2%	60%	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.
6	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Environment & Climate	Now	Global / Civilizational	27	9.8%	57%	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.
7	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	Governance & Law	Now	Global / Civilizational	27	11.4%	56%	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.
8	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	Shelter & Housing	Now	Global / Civilizational	27	13.0%	52%	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.
9	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Economy & Finance	Now	Global / Civilizational	27	14.7%	51%	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.
10	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Health & Medicine	Now	Global / Civilizational	27	16.3%	50%	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.
11	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	Water	Now	Society / National	18	17.4%	85%	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.
12	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragi	Technology & AI	Now	Society / National	18	18.5%	78%	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.
13	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fa	Energy	Now	Society / National	18	19.6%	76%	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).
14	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legit	Security & Defense	Now	Society / National	18	20.7%	74%	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.
15	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	Food & Agriculture	Now	Society / National	18	21.7%	72%	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.

16	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Environment & Climate	Now	Society / National	18	22.8%	69%	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.
17	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	Governance & Law	Now	Society / National	18	23.9%	68%	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.
18	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	Shelter & Housing	Now	Society / National	18	25.0%	64%	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.
19	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Economy & Finance	Now	Society / National	18	26.1%	63%	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.
20	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Health & Medicine	Now	Society / National	18	27.2%	62%	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.
21	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinform	Education & Knowledge	Now	Global / Civilizational	18	28.3%	58%	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.
22	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; o	Communication	Now	Global / Civilizational	18	29.3%	57%	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.
23	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike stran	Transport & Mobility	Now	Global / Civilizational	18	30.4%	56%	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.
24	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolat	Psychology & Mind	Now	Global / Civilizational	18	31.5%	55%	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.
25	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheq	Work & Livelihood	Now	Global / Civilizational	18	32.6%	54%	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.
26	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make	Justice & Ethics	Now	Global / Civilizational	18	33.7%	53%	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.
27	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Food & Agriculture	Future	Global / Civilizational	18	34.8%	46%	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.
28	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic	Information & Media	Now	Global / Civilizational	18	35.9%	45%	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.
29	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns h	Demographics & Population	Now	Global / Civilizational	18	37.0%	45%	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.

30	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built f	Security & Defense	Future	Global / Civilizational	18	38.0%	44%	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.
31	Multi-year megadrought plus population growth makes current cities structurally water-short.	Water	Future	Global / Civilizational	18	39.1%	43%	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.
32	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safe	Shelter & Housing	Future	Global / Civilizational	18	40.2%	40%	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.
33	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes	Energy	Future	Global / Civilizational	18	41.3%	38%	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.
34	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicin	Health & Medicine	Future	Global / Civilizational	18	42.4%	36%	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.
35	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a	Governance & Law	Future	Global / Civilizational	18	43.5%	35%	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.
36	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines	Economy & Finance	Future	Global / Civilizational	18	44.6%	32%	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.
37	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale fail	Technology & AI	Future	Global / Civilizational	18	45.7%	30%	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.
38	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level ris	Environment & Climate	Future	Global / Civilizational	18	46.7%	26%	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.
39	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinform	Education & Knowledge	Now	Society / National	12	47.5%	70%	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.
40	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; o	Communication	Now	Society / National	12	48.2%	69%	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.
41	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike stran	Transport & Mobility	Now	Society / National	12	48.9%	68%	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.
42	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolat	Psychology & Mind	Now	Society / National	12	49.6%	67%	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.
43	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheq	Work & Livelihood	Now	Society / National	12	50.4%	66%	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.

44	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make	Justice & Ethics	Now	Society / National	12	51.1%	65%	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.
45	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Food & Agriculture	Future	Society / National	12	51.8%	58%	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.
46	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic	Information & Media	Now	Society / National	12	52.5%	57%	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.
47	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns h	Demographics & Population	Now	Society / National	12	53.3%	57%	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.
48	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built f	Security & Defense	Future	Society / National	12	54.0%	56%	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.
49	Multi-year megadrought plus population growth makes current cities structurally water-short.	Water	Future	Society / National	12	54.7%	55%	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.
50	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safe	Shelter & Housing	Future	Society / National	12	55.4%	52%	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.
51	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes	Energy	Future	Society / National	12	56.2%	50%	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.
52	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, an	Information & Media	Future	Global / Civilizational	12	56.9%	49%	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.
53	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicin	Health & Medicine	Future	Society / National	12	57.6%	48%	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.
54	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a	Governance & Law	Future	Society / National	12	58.3%	47%	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.
55	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calcula	Education & Knowledge	Future	Global / Civilizational	12	59.1%	47%	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.
56	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and	Justice & Ethics	Future	Global / Civilizational	12	59.8%	46%	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.
57	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines	Economy & Finance	Future	Society / National	12	60.5%	44%	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.
58	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale fail	Technology & AI	Future	Society / National	12	61.2%	42%	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.

59	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or b	Transport & Mobility	Future	Global / Civilizational	12	62.0%	42%	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.
60	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable	Communication	Future	Global / Civilizational	12	62.7%	40%	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.
61	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level ris	Environment & Climate	Future	Society / National	12	63.4%	38%	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.
62	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows	Demographics & Population	Future	Global / Civilizational	12	64.1%	37%	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.
63	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human conta	Psychology & Mind	Future	Global / Civilizational	12	64.9%	34%	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.
64	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	Work & Livelihood	Future	Global / Civilizational	12	65.6%	31%	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.
65	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	Water	Now	Household / Personal	9	66.1%	93%	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.
66	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragi	Technology & AI	Now	Household / Personal	9	66.7%	86%	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.
67	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fa	Energy	Now	Household / Personal	9	67.2%	84%	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).
68	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legit	Security & Defense	Now	Household / Personal	9	67.8%	82%	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.
69	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	Food & Agriculture	Now	Household / Personal	9	68.3%	80%	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.
70	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Environment & Climate	Now	Household / Personal	9	68.8%	77%	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.
71	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	Governance & Law	Now	Household / Personal	9	69.4%	76%	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.
72	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	Energy	Past	Global / Civilizational	9	69.9%	72%	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.

73	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	Shelter & Housing	Now	Household / Personal	9	70.5%	72%	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.
74	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	Economy & Finance	Past	Global / Civilizational	9	71.0%	71%	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.
75	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Economy & Finance	Now	Household / Personal	9	71.6%	71%	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.
76	Single-crop dependence plus blight or drought produces famine at national scale.	Food & Agriculture	Past	Global / Civilizational	9	72.1%	70%	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.
77	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Health & Medicine	Now	Household / Personal	9	72.6%	70%	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.
78	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	Shelter & Housing	Past	Global / Civilizational	9	73.2%	69%	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.
79	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	Health & Medicine	Past	Global / Civilizational	9	73.7%	68%	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.
80	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable	Water	Past	Global / Civilizational	9	74.3%	66%	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.
81	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society	Environment & Climate	Past	Global / Civilizational	9	74.8%	59%	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.
82	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before corre	Technology & AI	Past	Global / Civilizational	9	75.4%	54%	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.
83	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Governance & Law	Past	Global / Civilizational	9	75.9%	48%	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.
84	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	Security & Defense	Past	Global / Civilizational	9	76.4%	33%	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.
85	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, an	Information & Media	Future	Society / National	8	76.9%	61%	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.
86	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calcula	Education & Knowledge	Future	Society / National	8	77.4%	59%	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.
87	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and	Justice & Ethics	Future	Society / National	8	77.9%	58%	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.

88	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or b	Transport & Mobility	Future	Society / National	8	78.4%	54%	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.
89	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable	Communication	Future	Society / National	8	78.9%	52%	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.
90	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows	Demographics & Population	Future	Society / National	8	79.3%	49%	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.
91	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human conta	Psychology & Mind	Future	Society / National	8	79.8%	46%	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.
92	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	Work & Livelihood	Future	Society / National	8	80.3%	43%	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.
93	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	Energy	Past	Society / National	6	80.7%	84%	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.
94	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	Economy & Finance	Past	Society / National	6	81.0%	83%	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.
95	Single-crop dependence plus blight or drought produces famine at national scale.	Food & Agriculture	Past	Society / National	6	81.4%	82%	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.
96	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	Shelter & Housing	Past	Society / National	6	81.8%	81%	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.
97	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	Health & Medicine	Past	Society / National	6	82.1%	80%	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.
98	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable	Water	Past	Society / National	6	82.5%	78%	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.
99	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinform	Education & Knowledge	Now	Household / Personal	6	82.9%	78%	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.
100	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; o	Communication	Now	Household / Personal	6	83.2%	77%	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.
101	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike stran	Transport & Mobility	Now	Household / Personal	6	83.6%	76%	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.
102	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolat	Psychology & Mind	Now	Household / Personal	6	83.9%	75%	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.

103	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies	Education & Knowledge	Past	Global / Civilizational	6	84.3%	74%	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.
104	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheq	Work & Livelihood	Now	Household / Personal	6	84.7%	74%	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.
105	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make	Justice & Ethics	Now	Household / Personal	6	85.0%	73%	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.
106	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society	Environment & Climate	Past	Society / National	6	85.4%	71%	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.
107	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or	Transport & Mobility	Past	Global / Civilizational	6	85.7%	67%	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.
108	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Food & Agriculture	Future	Household / Personal	6	86.1%	66%	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.
109	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before corre	Technology & AI	Past	Society / National	6	86.5%	66%	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.
110	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic	Information & Media	Now	Household / Personal	6	86.8%	65%	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.
111	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housi	Work & Livelihood	Past	Global / Civilizational	6	87.2%	65%	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.
112	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns h	Demographics & Population	Now	Household / Personal	6	87.6%	65%	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.
113	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built f	Security & Defense	Future	Household / Personal	6	87.9%	64%	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.
114	Multi-year megadrought plus population growth makes current cities structurally water-short.	Water	Future	Household / Personal	6	88.3%	63%	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.
115	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumou	Communication	Past	Global / Civilizational	6	88.6%	63%	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.
116	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safe	Shelter & Housing	Future	Household / Personal	6	89.0%	60%	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.
117	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Governance & Law	Past	Society / National	6	89.4%	60%	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.

118	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	Justice & Ethics	Past	Global / Civilizational	6	89.7%	60%	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.
119	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes	Energy	Future	Household / Personal	6	90.1%	58%	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.
120	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicin	Health & Medicine	Future	Household / Personal	6	90.5%	56%	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.
121	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a	Governance & Law	Future	Household / Personal	6	90.8%	55%	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.
122	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Psychology & Mind	Past	Global / Civilizational	6	91.2%	53%	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.
123	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines	Economy & Finance	Future	Household / Personal	6	91.5%	52%	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.
124	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowled	Demographics & Population	Past	Global / Civilizational	6	91.9%	52%	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.
125	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale fail	Technology & AI	Future	Household / Personal	6	92.3%	50%	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks andban fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.
126	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level ris	Environment & Climate	Future	Household / Personal	6	92.6%	46%	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.
127	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Information & Media	Past	Global / Civilizational	6	93.0%	46%	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.
128	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	Security & Defense	Past	Society / National	6	93.4%	45%	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.
129	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies	Education & Knowledge	Past	Society / National	4	93.6%	86%	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.
130	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or	Transport & Mobility	Past	Society / National	4	93.8%	79%	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.
131	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housi	Work & Livelihood	Past	Society / National	4	94.1%	77%	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.
132	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumou	Communication	Past	Society / National	4	94.3%	75%	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.

133	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	Justice & Ethics	Past	Society / National	4	94.6%	72%	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.
134	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, an	Information & Media	Future	Household / Personal	4	94.8%	69%	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.
135	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calcula	Education & Knowledge	Future	Household / Personal	4	95.0%	67%	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.
136	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and	Justice & Ethics	Future	Household / Personal	4	95.3%	66%	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.
137	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Psychology & Mind	Past	Society / National	4	95.5%	65%	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.
138	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowled	Demographics & Population	Past	Society / National	4	95.8%	64%	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.
139	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or b	Transport & Mobility	Future	Household / Personal	4	96.0%	62%	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.
140	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable	Communication	Future	Household / Personal	4	96.3%	60%	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.
141	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Information & Media	Past	Society / National	4	96.5%	58%	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.
142	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows	Demographics & Population	Future	Household / Personal	4	96.7%	57%	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.
143	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human conta	Psychology & Mind	Future	Household / Personal	4	97.0%	54%	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.
144	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	Work & Livelihood	Future	Household / Personal	4	97.2%	51%	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.
145	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	Energy	Past	Household / Personal	3	97.4%	92%	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.
146	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	Economy & Finance	Past	Household / Personal	3	97.6%	91%	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.
147	Single-crop dependence plus blight or drought produces famine at national scale.	Food & Agriculture	Past	Household / Personal	3	97.8%	90%	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.

148	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	Shelter & Housing	Past	Household / Personal	397.9%	89%	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.
149	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	Health & Medicine	Past	Household / Personal	398.1%	88%	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.
150	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable	Water	Past	Household / Personal	398.3%	86%	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.
151	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society	Environment & Climate	Past	Household / Personal	398.5%	79%	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.
152	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before corre	Technology & AI	Past	Household / Personal	398.7%	74%	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.
153	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Governance & Law	Past	Household / Personal	398.9%	68%	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.
154	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	Security & Defense	Past	Household / Personal	399.0%	53%	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.
155	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies	Education & Knowledge	Past	Household / Personal	299.2%	94%	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.
156	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or	Transport & Mobility	Past	Household / Personal	299.3%	87%	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.
157	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housi	Work & Livelihood	Past	Household / Personal	299.4%	85%	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.
158	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumou	Communication	Past	Household / Personal	299.5%	83%	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.
159	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	Justice & Ethics	Past	Household / Personal	299.6%	80%	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.
160	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Psychology & Mind	Past	Household / Personal	299.8%	73%	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.
161	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowled	Demographics & Population	Past	Household / Personal	299.9%	72%	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.
162	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Information & Media	Past	Household / Personal	2100.0%	66%	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.

Field of Discipline	Why Level	Root Cause of Possible Failure	Solution to This Cause
Water	Why 1 — supplies fail	Infrastructure ages and maintenance is deferred because water is priced below its cost.	Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum.
Water	Why 2 — deferral is invisible	Underground assets fail out of sight; politicians gain nothing from pipes nobody sees.	Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades.
Water	Why 3 — no one owns the number	Responsibility is split across municipalities, regions, and private operators.	One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature.
Water	Why 4 — basins cross borders	Upstream and downstream jurisdictions optimize against each other.	Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap.
Water	Why 5 — root	Water is treated as a free gift of nature rather than a managed stock with a measurable balance.	Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
Food & Agriculture	Why 1 — shelves empty fast	Just-in-time logistics hold almost no buffer by design.	Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.
Food & Agriculture	Why 2 — buffers cost money	Private actors are punished by markets for holding 'inefficient' stock.	Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.
Food & Agriculture	Why 3 — efficiency outbids resilience	Quarterly metrics reward cost per unit, never survival per decade.	Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.
Food & Agriculture	Why 4 — soil is mined, not farmed	Yield this year is priced; soil next decade is not.	Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.
Food & Agriculture	Why 5 — root	Food is treated as a commodity first and a survival system second.	Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
Health & Medicine	Why 1 — systems run at 95%+ occupancy	Efficiency targets eliminate the slack that crises consume.	Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness.
Health & Medicine	Why 2 — readiness looks like waste	Idle beds and stockpiles photograph badly in budget debates.	Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines.
Health & Medicine	Why 3 — prevention is underpaid	Payment follows procedures performed, not illness avoided.	Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results.
Health & Medicine	Why 4 — workforce pipelines are long	Clinicians take a decade to train; shortages cannot be improvised.	Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months.
Health & Medicine	Why 5 — root	Health is budgeted as a consumption expense instead of survival infrastructure.	Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
Energy	Why 1 — grids fail in cascade	Systems are optimized for average load and cost, with single points of failure.	Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first.
Energy	Why 2 — redundancy is cut in procurement	Lowest-bid rules select the design with the least slack.	Score bids on failure behaviour (tested black-start, islanding) with the same weight as price.
Energy	Why 3 — long-lead parts are not stocked	Transformers and breakers are treated as order-when-needed items.	A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses.
Energy	Why 4 — demand only grows	Every new load (vehicles, data centres, heating) is connected first and planned later.	Connection approvals require a supply plan: new large loads bring generation or storage with them by rule.
Energy	Why 5 — root	Energy is sold as a commodity while being used as a life-support system.	Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
Shelter & Housing	Why 1 — not enough homes near work	Zoning forbids the density the population needs where it needs it.	Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.
Shelter & Housing	Why 2 — incumbents block supply	Homeowners gain from scarcity and hold local veto power.	Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.
Shelter & Housing	Why 3 — risk is mispriced	Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones.	Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.
Shelter & Housing	Why 4 — construction capacity is thin	Trades pipelines and factory-built capacity were allowed to shrink.	Fund trades training against the housing target and permit factory-built housing at national standard.
Shelter & Housing	Why 5 — root	Housing is treated primarily as an investment asset and only secondarily as shelter.	Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
Security & Defense	Why 1 — violence concentrates	A tiny network and a few places generate most harm, while responses spread thin everywhere.	Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly.
Security & Defense	Why 2 — legitimacy collapses first	Abusive or absent policing destroys the cooperation that produces information.	Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers.
Security & Defense	Why 3 — weapons get cheaper	Each decade lowers the cost of precision harm.	Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after.
Security & Defense	Why 4 — early warnings are ignored	Acting early looks like overreaction until it looks like negligence.	Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait.
Security & Defense	Why 5 — root	Security is funded as reaction — after events — rather than as prevention with measured returns.	Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
Governance & Law	Why 1 — institutions are captured	Appointment, money, and media levers concentrate in one network.	Split every lever across independent bodies; publish ownership, contracts, and assets by default.
Governance & Law	Why 2 — capture is gradual and legal	Each step is small enough to seem normal.	Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness).

Governance & Law	Why 3 — citizens cannot verify	Records are secret or unusable, so accountability is theatrical.	Open data as the default state of every public record; secrecy justified line by line, expiring automatically.
Governance & Law	Why 4 — emergencies become permanent	Temporary powers are never handed back voluntarily.	Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public.
Governance & Law	Why 5 — root	Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time.	Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
Economy & Finance	Why 1 — crises cascade through debt	Leverage links every balance sheet to every other.	Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest.
Economy & Finance	Why 2 — booms silence regulators	Warnings during bubbles are punished as pessimism.	Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment.
Economy & Finance	Why 3 — gains privatize, losses socialize	Institutions too big to fail keep the upside.	Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually.
Economy & Finance	Why 4 — households carry the fragility	Wages stagnate while housing and credit costs compound.	Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing.
Economy & Finance	Why 5 — root	Finance is treated as an end in itself rather than as plumbing for the real economy.	Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
Technology & AI	Why 1 — harms arrive before rules	Development cycles are months; legislation is years.	Staged gates with automatic review dates, so oversight moves at the technology's speed by default.
Technology & AI	Why 2 — builders don't pay for harm	Latent and externalized costs fall on the public.	Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.
Technology & AI	Why 3 — basics are skipped	Backups, segmentation, and MFA are known but unglamorous.	Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.
Technology & AI	Why 4 — dependence concentrates	One platform, one model, one vendor becomes a single point of civilizational failure.	Interoperability, portability, and manual fallbacks as certification requirements for essential services.
Technology & AI	Why 5 — root	Speed is rewarded and safety is unpriced, so the race always outruns the guardrails.	Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
Environment & Climate	Why 1 — emissions continue	The cost of carbon falls on everyone except the emitter.	Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics.
Environment & Climate	Why 2 — adaptation lags	Infrastructure is built to historical weather by code and habit.	Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections.
Environment & Climate	Why 3 — risk is hidden	Buyers and builders cannot see address-level future risk.	Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit.
Environment & Climate	Why 4 — feedback is slow	Damage accumulates silently for decades before it is undeniable.	Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions.
Environment & Climate	Why 5 — root	Nature is accounted as an externality — a free input and a free dump.	Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
Information & Media	Why 1 — outrage outcompetes fact	Engagement optimization pays for emotion, not accuracy.	Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public.
Information & Media	Why 2 — corrections don't travel	A lie reaches millions before its correction reaches thousands.	Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days.
Information & Media	Why 3 — ownership concentrates	A few owners set a population's information diet.	Cross-media ownership caps enforced before capture; arm's-length public media funded by formula.
Information & Media	Why 4 — citizens lack verification habits	Schools teach content, not source-tracing.	Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin.
Information & Media	Why 5 — root	Attention is treated as a free resource to be mined rather than a commons to be governed.	Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
Psychology & Mind	Why 1 — distress is treated late	Help arrives at crisis, years after onset.	Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months.
Psychology & Mind	Why 2 — environments manufacture distress	Sleep loss, isolation, and engineered attention damage are built into daily design.	Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure.
Psychology & Mind	Why 3 — belonging collapsed	Clubs, congregations, leagues, and unions thinned for fifty years.	Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics.
Psychology & Mind	Why 4 — synthetic comfort outcompetes people	Machines offer frictionless company; humans are comparatively costly.	Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier.
Psychology & Mind	Why 5 — root	Mental health is framed as individual chemistry rather than as the output of a designed environment.	Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
Education & Knowledge	Why 1 — systems measure attendance, not learning	Enrolment and graduation are countable; learning is not, so it is not managed.	Independent annual learning assessments published per school; funding follows measured gains.
Education & Knowledge	Why 2 — teachers are undervalued	Pay and status lose the competition for strong graduates.	Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession.
Education & Knowledge	Why 3 — knowledge is centralized	Single publishers, platforms, and archives are single points of failure.	Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula.
Education & Knowledge	Why 4 — verification is untaught	Curricula deliver content while the environment demands checking.	Verification (source-tracing, estimation, consistency) as a tested core skill at every level.

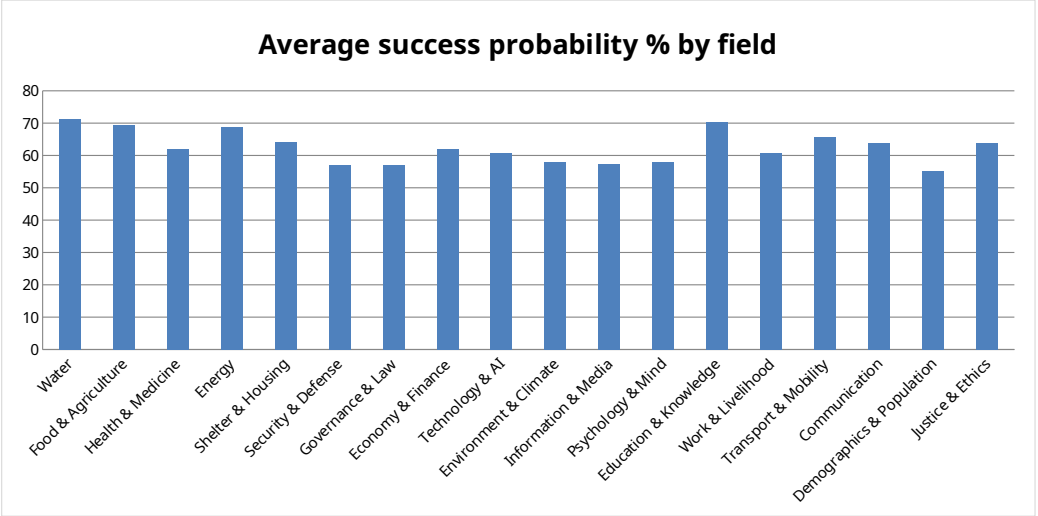
Education & Knowledge	Why 5 — root	Education is run as a sorting machine for employers rather than as the formation of capable citizens.	Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
Work & Livelihood	Why 1 — protections attach to job titles	Benefits were designed for one employer, one career, one contract type.	Attach all social insurance to the person, pro-rated per hour worked in any form.
Work & Livelihood	Why 2 — employers arbitrage the categories	Misclassification is cheaper than compliance where penalties are rare.	Per-task standards with automatic penalties and joint liability up the contracting chain.
Work & Livelihood	Why 3 — regions depend on single employers	Local economies were allowed to become monocultures.	Automatic diversification triggers at the 20% threshold, with funds released by formula.
Work & Livelihood	Why 4 — transitions are unfunded	Retraining is offered after displacement, from general revenue, too late.	Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.
Work & Livelihood	Why 5 — root	Income is tied exclusively to selling labour in a world where labour's share is structurally falling.	Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
Transport & Mobility	Why 1 — single threads everywhere	Cost optimization removed parallel routes and modes.	Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.
Transport & Mobility	Why 2 — sprawl locks in fragility	Land use separated homes from needs by design for seventy years.	The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.
Transport & Mobility	Why 3 — fleets homogenize	One vendor, one software stack, one battery chemistry per fleet.	Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.
Transport & Mobility	Why 4 — fallbacks are never drilled	Manual procedures exist on paper only.	Annual no-network drill days with published results for transit, logistics, and emergency services.
Transport & Mobility	Why 5 — root	Transport is optimized for cost per trip in normal times, not for continuity in bad times.	Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
Communication	Why 1 — speech moved onto private platforms	Public communication infrastructure was allowed to wither.	Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.
Communication	Why 2 — exit is costly	Audiences, records, and identities are locked in by design.	Portability and interoperability as law for dominant platforms; monthly export as a consumer right.
Communication	Why 3 — emergency channels are single-thread	Alerts ride the same networks that fail in disasters.	Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.
Communication	Why 4 — privacy and safety are traded falsely	Policy frames encryption and lawful access as absolutes at war.	Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.
Communication	Why 5 — root	Communication is treated as a market commodity rather than as civic infrastructure.	Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
Demographics & Population	Why 1 — family formation is priced out	Housing and childcare costs peak exactly at family-forming ages.	Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans.
Demographics & Population	Why 2 — pensions promise the unprojected	Formulas were set for pyramids that no longer exist.	Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic.
Demographics & Population	Why 3 — migration is improvised	Flows arrive whether planned or not; only the reception differs.	Permanent, competent administration sized to the measured gap, with 12-month credential and language completion.
Demographics & Population	Why 4 — regions hollow silently	Depopulation has no constituency until services collapse.	Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards.
Demographics & Population	Why 5 — root	Policy is made for the current electorate's age structure while the future population is already born and countable.	A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
Justice & Ethics	Why 1 — justice is priced	Expertise, delay, and bail convert wealth into outcomes.	Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review.
Justice & Ethics	Why 2 — backlogs normalize delay	Court capacity is funded below caseload by habit.	Published time standards with automatic capacity triggers (90% of cases within 12 months).
Justice & Ethics	Why 3 — error is unmeasured	Systems that don't count wrongful convictions can't reduce them.	Conviction-integrity audits and exoneration statistics published annually; compensation automatic.
Justice & Ethics	Why 4 — algorithms inherit the bias	Historical enforcement data encodes historical disparity.	Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone.
Justice & Ethics	Why 5 — root	Justice is administered as a service to the state rather than as a right held against it.	Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.

#	Coinciding Scenario	Why They Amplify Each Other	Action Sequence — First 24 Hours	First Week	First Month	Joint Probabilit	Why This Joint Probability	How to Maximize (detailed)
1	Grid failure + extreme heatwave + hospital surge	Power loss kills cooling and medical devices exactly when heat casualties peak; hospitals lose power as admissions triple; water pressure falls as pumps stop.	Open cooling centres on generator power first (before restoring general load); hospitals to emergency generation with fuel convoys escorted; publish the restoration order publicly: hospitals, water, cooling, then neighbourhoods.	Rolling restoration by survival-core priority; mutual-aid crews from unaffected regions; daily public accounting of deaths, capacity, and restoration percentages.	After-action rebuild: islanding cells, transformer spares, canopy and cooling standards — the scenario becomes a funded program, not a report.	62%	Each element alone scores 69–76%, but they share one cause (the grid) and one defence (generation), so fixing generation lifts all three at once; the joint figure sits below the best single because heat casualties already incurred cannot be undone.	Pre-position generation at hospitals and cooling sites BEFORE heat season; drill the compound scenario (not single failures) annually; hold 72 hours of fuel at every critical site; check in on isolated elderly by roster within the first 6 hours — most heat deaths are solitary and early.
2	Pandemic + supply-chain break + misinformation wave	Disease raises demand for medicine and food exactly as imports fail; false cures and denial delay every effective behaviour; panic buying empties shelves faster than the virus empties beds.	Publish the verified facts with signed provenance and the raw data; release strategic reserves visibly to break panic buying; protect the two highest-risk settings (care homes, hospitals) with air and testing barriers first.	Stand up local production of the top shortage items; daily briefings with error corrections stated openly; targeted support (delivery, income) so isolation is economically possible.	Institutionalize: wastewater surveillance permanent, platform provenance duties enforced, reserve levels restored to 90 days.	55%	Health measures alone score ~80% in the best case, but the coinciding information failure degrades compliance — the binding constraint is behaviour, not medicine; joint success is set by the weakest layer.	Pre-build the trust infrastructure in peacetime (citizen assemblies, published service metrics) — it cannot be manufactured mid-crisis; pre-sign local production contracts; communicate uncertainty honestly from hour one, because the first correction costs more trust than the first admission.
3	Household triple: job loss + serious illness + housing loss	Income stops as medical costs start; housing depends on the income that stopped; illness removes the capacity to job-hunt; each problem removes the resource needed to solve the others.	Freeze the bleeding in order: (1) housing — apply for eviction protection/assistance the same day (deadlines are short); (2) income — file unemployment and disability claims day one; (3) health — confirm coverage continuity before any treatment gap.	Stack every available support simultaneously (benefits, charity care, food programs, utility arrears plans); one case worker or one organized folder with every claim number; creditors notified in writing with the plan.	Rebuild in sequence: stabilized housing, treated health, then re-employment at sustainable load; rebuild the 3-month reserve before any lifestyle restoration.	71%	Household-scale problems are the most controllable in this file (+8 scale rule) and the solutions are administrative, not technological; the joint penalty comes from deadline collisions, which sequencing removes.	Act inside the first 72 hours while every program is still open; keep documents in one folder (ID, lease, medical, employment) — claim speed is a documents problem; tell every creditor and agency the whole picture in writing: coordinated honesty unlocks discretion that silence forfeits.
4	Cyberattack on payments + bank panic + synthetic-media hoax	Payment failure feels like insolvency; a forged official video 'confirming' collapse converts a technical outage into a bank run within hours; cash demand then creates the real shortage.	Banks and the central bank publish signed, verifiable statements through pre-registered channels within 1 hour; cash distribution points announced with visible stock; deposit-insurance reminder with the exact coverage figure.	Forensic restoration in stages with public status percentages; platforms label and throttle the forged media under the provenance rules; prosecutions announced for the hoax's amplifiers if coordinated.	Provenance infrastructure completed (capture-signing default for official media); payment systems segmented with tested analogue fallback (offline card limits, cash logistics).	58%	Technical recovery is high-probability (~78%), but panic is a belief event, not a technical one; success is capped by how much verifiable trust infrastructure existed before the attack.	Pre-register the official channels and teach the public where they are in peacetime; hold the cash-logistics reserve (physical notes, distribution points) as seriously as the digital one; the first statement must be fast, signed, and numerically specific — vagueness is what the hoax feeds on.
5	Multi-year drought + food price spike + civil unrest	Drought cuts supply; prices transfer the pain to the urban poor; unrest disrupts the transport that moves food; governments fall, which removes the coordinator.	Release reserves and open subsidized staple channels in the cities first (unrest is urban and price-driven); protect transport corridors as critical infrastructure; publish the reserve position and import schedule to kill rumour pricing.	Rationing by lifeline entitlement (guaranteed staple quantity per person at fixed price) rather than by price; emergency employment on water infrastructure (wells, canals, soil works) paying partly in food.	Structural: extraction caps to recharge, soil regeneration, diversified import contracts, and the 90-day reserve rebuilt — with the drought plan written into law.	49%	The physics (drought) is not solvable by policy, only blunted; success means preventing famine and state collapse, not preventing hardship — hence below 50% for the full compound, while famine prevention alone rates ~75%.	Spend on the cities' staple price first — that is where the political cascade starts; never let the reserve position be a secret (secrecy converts shortage into panic); begin the structural water program during the crisis, with crisis legitimacy, instead of after it, when attention dies.
6	Major earthquake + urban fire + communications outage	Shaking ignites fires as it breaks the water mains needed to fight them; comms loss prevents coordination; aftershocks hit responders during rescue.	Search and rescue in the pre-mapped highest-risk building classes first (Pareto: the deadliest 20% of stock holds most victims); fire crews to water-independent tactics (firebreaks, demolitions) where mains are dead; emergency broadcast by battery radio and vehicle loudspeaker.	Shelter capacity for 10% of population activated; mutual-aid and military logistics under civilian coordination; damage map published daily so aid follows need, not cameras.	Retrofit program for the deadliest building classes with deadlines; water system segmented so breaks cannot drain a district; comms fallback (municipal broadcast, amateur radio) made permanent.	66%	Rescue and shelter are drilled, logistical problems with high success rates; the coinciding fire-without-water element is the irreducible loss driver, which is why the joint figure stays below the ~81% for the shelter solution alone.	The work is 90% before the event: retrofits, pre-sited shelters, mapped risk. In the event: one coordination authority, one public map, one broadcast channel — parallel uncoordinated heroics waste the scarcest resource, which is organized hours.
7	Automation displacement wave + fiscal crisis + legitimacy collapse	Displacement cuts income tax as demand for support rises; deficits force austerity into the downturn; both parties of the social contract (work pays, state protects) appear broken simultaneously.	Automatic stabilizers fire by formula (no emergency vote needed): benefits expand, wage insurance starts, public works rosters open in the hardest-hit sectors and regions first.	Citizen-fund dividends declared from automation rents; standard hours reduced in the most displaced sectors; a published transition dashboard (displacement, re-employment, wage recovery) updated monthly.	Ownership structures deepened while the economy recovers (the funds buy the dip); education rebuilt around unaided competence plus AI collaboration; legitimacy rebuilt through verified local service performance.	41%	This is the slowest compound in the file: every lever works on a 5–20 year lag, and legitimacy — the binding constraint — cannot be purchased; the honest figure reflects generational, not electoral, timescales.	Build the ownership and benefit mechanisms during the preceding growth years — starting them inside the crisis halves their effect; protect the stabilizers' automaticity against emergency politics; measure and publish relentlessly, because in a legitimacy crisis only verified performance compounds.
8	River flood + power loss + drinking-water contamination	Floodwater carries sewage and chemicals into wells and treatment intakes while power loss stops pumps and treatment; the disaster manufactures the contamination that outlasts it.	Boil-water and no-drink orders within the first hour via every channel including door-to-door in the flooded zone; bottled and tanker water to pre-mapped distribution points; power restoration sequenced water-plant-first.	Well testing across the flooded area with published results per street; temporary treatment (mobile units) until plants are certified; mould and contamination remediation standards enforced before reoccupation.	Floodplain buyouts executed while memory and funding last; treatment plants hardened and dual-powered; address-level risk attached to every property record.	68%	Each element has a mature playbook (water 85%, energy 76%, adaptation 69% at best) and they share logistics rather than competing for them; the deduction is for the contamination tail, which is slow and unevenly distributed.	Sequence publicly and keep the sequence: water before general power, testing before reassurance, buyouts before rebuilding-in-place; never declare water safe on a schedule — declare it per tested zone, with the numbers published.

Coinciding Scenario (Master file)	Joint Probabili	Extra Harm If All C	If NOT Addressed — Short-Term Implication	If NOT Addressed — Mid-Term Implication	If NOT Addressed — Long-Term Implication	If Addressed ASAP — What Early Action Buys
Grid failure + heatwave + hospital surge	62	70	Cooling, power, and care fail in the same hours; the prepared household (battery, cool room, water) is an island for 48-72 hours	Heat illness and spoiled medication concentrate in the unprepared; hospitals ration by arrival, not need	Each event raises insurance and lowers trust in infrastructure; the unprepared districts age faster	The Energy + Environment + Health builds together cover this almost entire scenario for under $800 of the $8,179
Pandemic + supply-chain break + misinformation	55	65	Household hygiene, records, and the two-source rule are the only defences that do not depend on shelves	Shortages reward the buffered and punish the just-in-time household monthly	Trust in medicine and information, lost in this combination, takes a generation to regrow	Health-Past + Food-Now + Information builds interlock: hygiene stock, 30-day buffer, verification habit
Household triple: job loss + illness + housing loss together	71	60	This is the standard biography of a bad year: the emergency fund and benefits file decide whether it is a bad year or a bad decade	Arrears, treatment gaps, and address instability feed each other; children change schools mid-year	An eviction record plus a health record plus a gap record price into every application for years	Economy-Now (fund, filing) + Shelter-Now (rights, rent banks) + Health (records, prevention) is the cheapest complete insurance in this file
Cyberattack + bank panic + synthetic-media hoax	58	55	Withdrawals and transfers made in the first hoax hours are the losses; the no-transfer rule and code word freeze the household out of the panic	Institutions ration trust; the unbanked-by-panic pay fees and lose places in queues	Each successful hoax funds better hoaxes; the defended population share decides the fraud economy	Technology-Now stack + paper vault + code word: under $100 total, all free-tier except one drive
Drought + food-price spike + unrest	49	60	Stored water and staples convert civil stress into a logistics problem the household has already solved	Gardens and soil decide who eats from their own ground while prices peak	Regions that feed themselves keep their politics; regions that cannot, import both food and disorder	Water + Food builds, especially perennials and soil - the slowest to build, therefore the most urgent to start
Earthquake + fire + communications loss	66	65	The go-bag, the plan, and the radio are the household government for the first days	Rebuilding queues reward the documented; the undocumented rebuild last and least	Neighbourhoods with rosters recover as communities; those without, as competitors	Shelter-Past + Communication-Past builds cost under $150 combined and one rehearsal
Automation displacement + fiscal strain + legitimacy crisis	41	50	No household item prevents this; ownership, a second skill, and collective terms decide who lands	Retraining markets flood; early movers buy courses at calm prices with calm minds	A displaced generation that lands badly becomes a permanent political fact	Economy-Future + Work-Future: contributions and skills bought during employment, at employment prices
Flood + power loss + water contamination	68	70	The valve, the battery, and stored water answer all three components with three items	Mould, displacement, and advisory weeks compound; documentation decides the insurance outcome	Repeated triple-events reprice the district itself - the stranded-asset mechanism in miniature	Environment-Now valve + Energy-Now battery + Water-Now reserve: the file's model triple, about $760 together

Statistic	Value
Total problems	162
Total solutions listed	162
Fields of discipline	18
Tier 1 (survival fields) solutions	90
Tier 2 solutions	72
Past problems	54
Now problems	54
Future problems	54
Root causes with solutions (5 Whys x 18 fields)	90
Coinciding scenarios analysed	8
Average success probability — all	62.5%
Average — Past	71.2%
Average — Now	67.0%
Average — Future	49.3%
Top 20% of solutions (32) — share of weighted harm	40.2%

Field	Solutions	Average Success %	AvgProbNum
Water	9	71.3%	71.3
Food & Agriculture	9	69.3%	69.3
Health & Medicine	9	62.0%	62
Energy	9	68.7%	68.7
Shelter & Housing	9	64.3%	64.3
Security & Defense	9	57.0%	57
Governance & Law	9	57.0%	57
Economy & Finance	9	62.0%	62
Technology & AI	9	60.7%	60.7
Environment & Climate	9	58.0%	58
Information & Media	9	57.3%	57.3
Psychology & Mind	9	58.0%	58
Education & Knowledge	9	70.3%	70.3
Work & Livelihood	9	60.7%	60.7
Transport & Mobility	9	65.7%	65.7
Communication	9	64.0%	64
Demographics & Population	9	55.3%	55.3
Justice & Ethics	9	63.7%	63.7



Field	Timeframe	Costs on PEOPLE	Costs on ANIMALS	Costs on PLANTS	Costs on THINGS & PROPERTY	Costs on the ENVIRONMENT	Costs on COMMUNITY & INSTITUTIONS	Non-Money Cost Account (health & years of life | mind | social | freedom | opportunity)	How to Mitigate These Costs
Water	Past	$2,000–15,000 — waterborne disease treatment, lost labour, funerals in severe outbreaks	$200–3,000 — livestock water hauling, animal deaths in drought-collapse zones	$500–10,000 — crop failure from irrigation loss at farm scale	$1,000–20,000 — wells, pumps and cisterns ruined by silt/salt; land value collapse	Regional: aquifer depletion and salinization take decades to reverse; lake/river ecosystems collapse	$10,000s per village — relief, trucking, and resettlement costs borne collectively	Health: waterborne disease, child mortality, years of life lost to hauling labour | Mind: constant anxiety over the next source | Social: water queues and source disputes corrode neighbour trust | Freedom: dependence on whoever controls the well | Opportunity: hours daily to hauling instead of school and work	Share storage and treatment at street level (one filter kit serves a courtyard); treat water communally to split fuel costs; prioritize children, elderly, and the sick in allocation — the order that saves the most life per litre.
Water	Now	$300–4,000 — bottled water at spike prices, illness, missed work	$50–500 — pet and small-livestock water purchase and hauling	$100–1,500 — garden and greenhouse losses during restrictions	$200–5,000 — appliance and plumbing damage from contaminated or sedimented supply	Localized: contamination plumes can persist in soil and sediment for years	$100,000s–millions per city event — emergency distribution, testing, litigation	Health: gastrointestinal illness risk, hygiene disease | Mind: advisory confusion and distrust | Social: bottled-water scrambles pit neighbours against each other | Freedom: minor | Opportunity: work and school hours lost to queues and closures	Never panic-buy: the stored reserve exists precisely to skip spike pricing; share filtration with neighbours (one unit, many households, scheduled); claim every utility credit and distribution point before spending privately.
Water	Future	$1,500–12,000 — chronic shortage pricing, hygiene disease, lost work	$500–5,000 — herd reductions and animal welfare crises in dry regions	$1,000–25,000 — farmland value and yield collapse in water-short zones	$5,000–60,000 — property value decline in day-zero-risk cities	Civilizational: dead aquifers and salinized deltas are effectively permanent on human timescales	Regional: migration pressure, utility bankruptcy, inter-jurisdiction conflict costs	Health: chronic hygiene disease in shortage zones | Mind: climate grief tied to place | Social: inter-community conflict over allocations | Freedom: rationing regimes govern daily life | Opportunity: careers and farms abandoned with the water table	Buy resilience once, collectively: co-op storage and wells divide the capital cost by every member household; phase demand-cutting fixtures by payback order (aerators first — weeks to repay).
Food & Agriculture	Past	$5,000–40,000 — famine: distress asset sales, malnutrition care, lost earning years; mortality beyond price	$1,000–20,000 — livestock starvation and distress slaughter at farm scale	Total crop loss — the originating cost; seed stock eaten, ending next year's capacity	$2,000–30,000 — land sold at famine prices; tools and stores consumed	Soil mined and eroded under desperation farming; recovery measured in decades	State-level: relief, unrest suppression, and the demographic scar of a famine generation	Health: stunting, famine mortality, developmental harm to children that no later wealth repays | Mind: trauma and food obsession transmitted across generations | Social: hoarding and theft dissolve community bonds | Freedom: dependence on whoever holds grain | Opportunity: a generation's schooling and growth, gone	Pool stores communally: pooled grain loses less to pests, theft, and spoilage than scattered stores, and rationing by need order saves the most life per calorie.
Food & Agriculture	Now	$1,200–6,000 — grocery inflation and spike pricing across a year; diet quality decline	$100–2,000 — feed price spikes passed into pet and livestock care costs	$200–5,000 — garden and small-farm input cost spikes (fertilizer, seed)	$500–10,000 — small food businesses bankrupted by input whiplash	Fertilizer runoff and panic land-use (marginal land ploughed) during spikes	$100,000s per community — food-bank demand surges and program top-ups	Health: diet quality decline — cheap calories, expensive nutrients | Mind: grocery anxiety as a monthly ritual | Social: food-bank use carries stigma that delays it | Freedom: minor | Opportunity: hours to bargain-hunting; small food businesses lost	The buffer itself is the cost mitigation — it converts spike prices into last month's prices; buying clubs cut the recurring cost of maintaining it; food-bank use frees cash before debt does the job worse.
Food & Agriculture	Future	$2,500–20,000 — structural food inflation; chronic disease from degraded diets	$1,000–15,000 — livestock systems failing under heat and feed scarcity	$5,000–100,000 — yield decline and farmland repricing across breadbaskets	$2,000–40,000 — food-processing and storage infrastructure stranded or overwhelmed	Soil loss and pollinator decline — the slowest, costliest repairs in this file	National: import dependence, strategic reserve drawdowns, political instability	Health: chronic disease from structurally degraded diets | Mind: food insecurity as a permanent background stress | Social: urban-rural political fracture over food prices | Freedom: dependence on distant suppliers and states | Opportunity: farming careers and regions abandoned	Perennials and soil are cost curves that fall every year — plant once, harvest for decades; cooperative land removes rent from the food cost stack permanently.
Health & Medicine	Past	$10,000–100,000 — pandemic mortality and morbidity at household scale; orphanhood and lost earners	$500–10,000 — livestock epidemics compounding human ones; animal culls	$200–5,000 — labour loss at planting/harvest spreading into food costs	$5,000–50,000 — businesses and practices destroyed; property abandoned in plague cities	Limited direct, but carrion and sanitation collapse poison water and land	Civilizational: demographic holes that echo for a century	Health: the cost IS health — mortality and disability at scale | Mind: mass grief and survivor trauma | Social: quarantine stigma, blame of the sick | Freedom: emergency powers normalize | Opportunity: orphaned children's entire trajectories	Household hygiene and isolation cost almost nothing and historically cut transmission more than any treatment then available — spend effort there before money anywhere.
Health & Medicine	Now	$1,500–25,000 — resistant infections: longer stays, stronger drugs, lost work; surgical risk repriced	$200–5,000 — veterinary antibiotic resistance; farm biosecurity costs	$100–2,000 — plant disease management costs rise in parallel (same chemistry treadmill)	$1,000–20,000 — hospital-acquired infection liability and remediation	Pharmaceutical residues in water systems accumulate	Health-system: stewardship, surveillance, and pipeline subsidies are permanent new budget lines	Health: resistant infections undo surgical safety | Mind: chronic uncertainty for patients and clinicians | Social: trust in medicine erodes | Freedom: minor | Opportunity: clinical years spent re-treating the treatable	Prevention arithmetic: vaccinations ($0), dental prevention ($120 vs. $3,000 emergencies), and medication optimization (pharmacist review, $0) are the three cheapest cost-cutters in the file.
Health & Medicine	Future	$3,000–50,000 — overload: delayed care converts treatable conditions into permanent disability	$500–10,000 — zoonotic spillover risk grows with system stress; culls and movement bans	$500–10,000 — food-safety failures multiply when inspection thins	$5,000–100,000 — private-care price exploitation during public-system failure	Engineered-organism release risk is the tail event priced here	National: reserve workforces, regional manufacturing, air infrastructure — decade-scale programs	Health: treatable becomes untreatable again | Mind: learned helplessness toward the care system | Social: triage resentment between groups | Freedom: health status as a gate to work and travel | Opportunity: a generation of clinicians burning out early	Air quality at home is bought once (one HEPA unit) and protects every season; early detection (BP cuff $45) intercepts the costliest chronic cascades at their cheapest point.
Energy	Past	$1,500–10,000 — cold and heat illness, lost work, embargo inflation on everything	$200–3,000 — livestock shelter heating/cooling failures	$300–5,000 — greenhouse and stored-crop losses without fuel	$5,000–40,000 — burst pipes, frozen systems, idled machinery	Woodland stripped around cities for fuel — a historical constant of energy crises	National: rationing administration, industrial shutdown losses	Health: cold and smoke illness | Mind: embargo anxiety | Social: fuel queues and black markets corrode fairness | Freedom: ration coupons as a second government | Opportunity: industry and study hours lost to darkness	Heat one room, not the house; insulate people (clothing) before spaces; communal cooking splits fuel three ways.
Energy	Now	$800–8,000 — spoiled food, lost remote work, hotel nights, medical-device risk	$100–1,500 — aquarium, reptile, and livestock deaths in outages; farm downtime	$100–2,000 — greenhouse and garden pump/heat losses	$2,000–30,000 — business interruption; frozen pipes in cold-climate outages	Generator emissions and fuel spills concentrate during outages	$Millions per regional event — utility restoration, mutual-aid crews, public shelters	Health: medical-device and temperature risk in outages | Mind: outage helplessness | Social: check-in burdens on the connected for the isolated | Freedom: minor | Opportunity: work-from-home and small-business days lost	The battery station's cost is mitigated by sizing to the survival core only (fridge, light, devices, medical) — not whole-home backup; DIY 12V builds cut the price by more than half.
Energy	Future	$2,000–30,000 — months-long outage: relocation, unemployment, health harm	$1,000–10,000 — industrial farming is energy-bound; mass livestock losses in long outages	$2,000–20,000 — cold-chain collapse across the food system	$10,000–150,000 — unheated/uncooled buildings degrade; electronics and vehicles idle	Grid-scale events drive emergency diesel burning at massive scale	Civilizational: transformer-class losses take years to manufacture back	Health: long-outage exposure deaths | Mind: civilizational insecurity | Social: energy hoarding vs. sharing conflicts | Freedom: movement and communication priced by the kilowatt | Opportunity: education and enterprise dark for seasons	Insulation is the only energy purchase whose cost falls every year forever (savings compound); buy it before generation, always.
Shelter & Housing	Past	$20,000–150,000 — the home itself, plus injury and displacement trauma	$200–5,000 — animals lost in fires, floods, and abandonment during flight	$500–10,000 — gardens, orchards, and stored harvests destroyed with the house	The property category IS this row: total loss of structure and contents	Debris, ash, and contamination loads on land and water	City-level: reconstruction programs measured in years and billions	Health: exposure, injury, displacement disease | Mind: homelessness trauma, especially in children | Social: diaspora scattering of entire communities | Freedom: shelter at the price of obedience to whoever provides it | Opportunity: schooling and work interrupted for years	Documentation is the cheapest cost mitigation in the file: $18 of copies routinely multiplies insurance and relief payouts several-fold.
Shelter & Housing	Now	$3,000–30,000 — eviction/forced move cascade: deposits, movers, premiums, lost work	$100–1,000 — pet surrender and emergency boarding during housing loss — a leading cause of shelter intake	$50–500 — houseplants and gardens lost in rushed moves (small money, real grief)	$2,000–25,000 — furniture and goods lost to storage fees and distress sales	Abandoned-unit decay and illegal dumping follow displacement waves	Municipal: shelter systems, encampment responses, court and enforcement time	Health: mould, crowding, and stress illness in bad housing | Mind: housing anxiety as the dominant adult stress | Social: displacement breaks the networks children and elderly live on | Freedom: tenancy insecurity disciplines every complaint you don't file | Opportunity: careers declined for location; commutes eating family hours	Knowledge before money: rights and rent banks cost $0 and interrupt the $3,000+ eviction cascade at its first step; shared housing halves the largest cost a household has.
Shelter & Housing	Future	$10,000–200,000 — stranded-asset loss: the family's main wealth store to zero	$500–5,000 — livestock and pet displacement in retreating zones	$2,000–50,000 — agricultural land lost to sea, fire, and salinization	The asset class itself reprices: whole districts to uninsurable	Abandoned zones: contamination, invasive species, unmanaged fire load	National: buyout programs are among the largest climate line items projected	Health: climate-zone illness burdens | Mind: grief for a home place becoming unlivable | Social: receiving-city resentment toward climate movers | Freedom: mobility priced beyond ordinary families | Opportunity: a lifetime of housing wealth, stranded	The risk audit costs $0 and prevents the only six-figure mistake in the file; portability (not holding all wealth in one exposed asset) costs nothing but a decision.
Security & Defense	Past	$5,000–100,000+ — persecution and war: property, livelihood, health, life	$1,000–20,000 — livestock looted or abandoned; farm animal populations crash in war zones	$2,000–30,000 — crops burned, orchards cut, fields mined — recovery in decades	$10,000–200,000 — homes and businesses destroyed or confiscated	War contamination: mines, ordnance, and chemicals in soil and water for generations	State-level: armies, occupation, and post-war reconstruction dominate national budgets	Health: injury and death — the ledger is bodies | Mind: persecution trauma across generations | Social: trust between groups destroyed for a century | Freedom: the cost IS freedom — its total loss | Opportunity: entire lives redirected to survival	Early action is the cost mitigation: leaving at the first trigger costs property; leaving at the last costs everything else. Pre-decided triggers remove the delay that history prices highest.
Security & Defense	Now	$500–20,000 — victimization: medical, lost work, trauma care	$100–2,000 — pet theft and cruelty associated with property crime	$50–1,000 — garden and allotment theft/vandalism (minor but corrosive)	$1,000–25,000 — break-in and vehicle losses net of insurance	Localized: fear changes land use — abandoned parks, dead high streets	Municipal: policing, courts, and victim services are permanent budget lines	Health: injury and chronic hypervigilance | Mind: fear repricing every route and hour | Social: suspicion between neighbours; emptied public spaces | Freedom: surveillance accepted as the price of safety | Opportunity: businesses and evenings not ventured	The first $25 of hardening (screws, lighting) captures most of the deterrence; evidence habits cost $0 and convert losses into recoveries.
Security & Defense	Future	$1,000–50,000 — mass-incident injury and liability chaos	$200–3,000 — animals in incident zones; farm biosecurity under drone threat	$500–10,000 — precision attacks on crops and irrigation become possible	$5,000–100,000 — critical-site damage; event and venue economics repriced	Autonomous-weapons proliferation contaminates battlefields with unexploded systems	National: counter-drone and hardening programs across all critical infrastructure	Health: mass-casualty risk at gatherings | Mind: ambient dread of invisible threats | Social: securitized suspicion of everyone | Freedom: countermeasures normalize total monitoring | Opportunity: public life retreating indoors and online	Demand-side discipline (venue standards, footprint reduction) costs nothing; regulation is collectively cheap compared to private countermeasures.
Governance & Law	Past	$5,000–100,000 — confiscation, imprisonment, exile; earnings and pensions annulled	$500–10,000 — livestock and land seized with the farms attached to them	$1,000–30,000 — estates, orchards, and vineyards confiscated or destroyed	$10,000–200,000 — homes, businesses, and bank accounts absorbed by the state	War and terror regimes poison land and water deliberately	The state itself is the cost: institutions converted from protection to extraction	Health: terror's direct toll | Mind: learned silence | Social: denunciation destroys the fabric between families | Freedom: total | Opportunity: a society's talent exiled or buried	Copies held abroad cost postage and prevent the total-loss scenario; networks cost dues and prevent isolation, the condition in which all other costs maximize.
Governance & Law	Now	$300–8,000 — bribes, rigged fees, and the time tax of corrupt systems	$50–1,000 — permits and inspections as extortion points for animal keepers	$100–3,000 — agricultural permits and land papers as rent-extraction points	$1,000–30,000 — procurement inflation passed into every public building and road	Permits for pollution are the classic captured commodity	Democracy's operating cost rises: audits, litigation, and duplicated private services	Health: services rationed by connection, not need | Mind: cynicism as civic mood | Social: trust in every institution discounted | Freedom: rights existing on paper only | Opportunity: talent spent on navigating capture instead of building	Written complaints are nearly free and are the unit from which all accountability is built; collective filing multiplies effect without multiplying cost.
Governance & Law	Future	$1,000–20,000 — parallel systems (private security, private water, black markets) replace failed public ones at premium prices	$200–3,000 — animal services collapse with general administration	$500–10,000 — land records unreliable: planting and investment freeze	$5,000–50,000 — property rights unenforceable: asset values discount by trust, not by brick	Unenforced environmental law compounds every other row in this file	Legitimacy is infrastructure: its failure taxes every transaction in the society	Health: unverifiable systems fail the sick first | Mind: epistemic helplessness — believing nothing | Social: coordination itself collapses | Freedom: no institution left to appeal to | Opportunity: collective projects become impossible	Local verification costs time, not money, and prevents the most expensive failure — a household governed by rumour.
Economy & Finance	Past	$5,000–80,000 — savings destroyed, wages unpaid, a decade of earnings lost to depression	$500–8,000 — farm animals sold or starved in credit collapses	$1,000–20,000 — unplanted or unharvested crops when credit dies mid-season	$10,000–100,000 — homes and businesses lost to deflation-era foreclosure	Deflation farming mines soil; dust-bowl dynamics are an economic product	Banking and relief systems rebuilt at public cost for a generation	Health: depression-era malnutrition and suicide | Mind: shame and despair as mass conditions | Social: family dissolution under debt | Freedom: debt as a cage | Opportunity: a decade of enterprise unbuilt	Deposit splitting and debt refusal are free decisions whose absence historically cost everything; durable value (tools, skills) is bought once.
Economy & Finance	Now	$2,000–25,000 — interest spirals, insolvency costs, credit damage priced into every future loan	$100–2,000 — pets surrendered and livestock sold early under debt pressure	$100–3,000 — gardens and small plots lost to arrears sales	$5,000–60,000 — homes cross-collateralized into business and consumer debt losses	Debt-driven consumption raises waste and extraction pressure	Consumer insolvency and credit systems carry permanent administrative cost	Health: debt stress is a measured cardiovascular and mental-health risk | Mind: money anxiety as the background of daily life | Social: relationships strained by lending and owing | Freedom: creditors' claims on your future time | Opportunity: risks not taken, moves not made, children deferred	The emergency fund is self-mitigating: each use avoids 20%+ interest; benefits accumulate as a free money already allocated — the cost of NOT claiming is the largest line most households never see.
Economy & Finance	Future	$10,000–100,000 — lifetime earnings loss for displaced workers without ownership	$500–5,000 — rural animal economies collapse with their human ones	$2,000–30,000 — farmland consolidation accelerates as family operations fail	$20,000–200,000 — housing in displaced regions reprices to abandonment	Abandoned industrial and urban land becomes a public remediation liability	Fiscal systems lose their tax base exactly as demand on them peaks	Health: displacement depression epidemics | Mind: status collapse for whole professions | Social: generational resentment politics | Freedom: dependence on whoever owns the machines | Opportunity: human potential wasted/unused	Employer pension matching is an instant 100% return — claim it before any other investing; skills in human-irreducible work are bought with time during employment, not with debt after displacement.
Technology & AI	Past	$1,000–50,000 — latent disease: lead and asbestos illness, lost working years	$200–5,000 — livestock exposure in industrial and agricultural settings	$200–5,000 — crop contamination near industrial sites	$5,000–80,000 — property remediation: lead and asbestos abatement per building	Contaminated sites are century-scale liabilities (smelters, fuel depots, asbestos mines)	Compensation funds and public remediation programs run for decades	Health: latent disease with 20–40 year fuses | Mind: betrayal by trusted products | Social: industrial communities divided over blame | Freedom: minor | Opportunity: working years lost to illness	Testing before purchase/renovation costs under $200 and stands against five-figure remediation; records cost a folder and decide compensation decades later.
Technology & AI	Now	$500–15,000 — fraud, ransomware, identity theft: money plus months of administration	$50–1,000 — farm and veterinary systems hit by the same intrusions	$100–3,000 — greenhouse and irrigation controllers as attack surface	$2,000–50,000 — business interruption, hospital diversions, pipeline shutdowns priced per day	E-waste and emergency diesel burning spike during incidents	National incident-response and recovery costs are now permanent budget lines	Health: fraud and breach stress; care diverted in attacks | Mind: vigilance fatigue | Social: trust in digital life discounted | Freedom: accounts and identities held hostage | Opportunity: months of administrative recovery per incident	The entire household cyber stack is free-tier: manager, MFA, updates, backups discipline. The only purchased item (a drive) costs ~$0.05/day across its life.
Technology & AI	Future	$2,000–40,000 — exclusion from money, records, and transport when systems fail or decide wrongly	$500–10,000 — precision livestock and wildlife systems under automated mismanagement	$1,000–20,000 — automated agriculture making correlated errors at scale	$10,000–150,000 — infrastructure under opaque automated control failing in correlated ways	Autonomous systems interacting with ecosystems is an unpriced frontier risk	Verification and fallback infrastructure becomes a standing civilizational cost	Health: automated error in care and transport | Mind: learned dependence and deskilling anxiety | Social: humans sorted by opaque scores | Freedom: the master cost — autonomy itself | Opportunity: unaided competence unpractised for a generation	Paper and exports are near-free and stand against lockout losses in the thousands; unaided competence costs practice time only.
Environment & Climate	Past	$3,000–60,000 — subsistence collapse: hunger, migration, lost livelihoods	$1,000–25,000 — herd and flock collapse with forage and water	The originating loss: soil and forest consumed by the civilization standing on them	$5,000–80,000 — farms, wills, and settlements abandoned	The cost IS environmental: salinization, erosion, deforestation — measured in centuries	Successor states inherit degraded land as their founding balance sheet	Health: subsistence disease and famine | Mind: ecological grief before the word existed | Social: resource wars between neighbours | Freedom: landlessness | Opportunity: civilizations' accumulated capital consumed	Soil care is labour converted directly into avoided loss; communal rules (grazing, fallowing) cost governance, not money.
Environment & Climate	Now	$1,000–45,000 — heat illness, flood injury, smoke exposure; displacement	$200–5,000 — pet and livestock heat deaths; farm animal losses in floods and fires	$500–20,000 — gardens to farmland: heat, flood, and smoke losses per event	$10,000–150,000 — flood and fire property damage per serious event, uninsured share	Each event loads water with sediment, sewage, and chemicals	Insurers withdraw; municipalities absorb roads, bridges, and emergency costs	Health: heat, smoke, and flood illness burdens rising yearly | Mind: climate anxiety, especially in the young | Social: blame politics after every event | Freedom: insurance and risk maps governing where life is possible | Opportunity: rebuilding years replacing building years	The backwater valve is the model: $250 against a $15,000 loss — buy mitigations in ratio order (loss ÷ cost), and claim municipal subsidies that cover 50–100% of exactly these items.
Environment & Climate	Future	$20,000–250,000 — relocation, health, and livelihood loss in uninhabitable zones	$5,000–50,000 — livestock geography ends in the hottest regions	$10,000–200,000 — breadbasket shifts: some regions gain, current ones lose catastrophically	$50,000–500,000 — coastal and fire-zone property systems repriced at civilizational scale	Tipping elements: ice, permafrost carbon, ocean circulation — the largest numbers in this file, denominated in ecosystems	Migration, defence, and adaptation budgets restructure national finances permanently	Health: uninhabitable-zone physiology | Mind: civilizational grief | Social: migration conflict as the default politics | Freedom: movement governed by climate | Opportunity: the largest wealth destruction in the file, in years as much as dollars	Audits and portability are decisions, not purchases; the costliest climate error is made at the property-purchase moment, where information is free.
Information & Media	Past	$1,000–30,000 — incitement's victims and participants: injury, imprisonment, flight	$100–3,000 — animals as early targets in communal violence	$200–5,000 — crops and stores destroyed in incited unrest	$5,000–100,000 — pogrom and riot property destruction	Propaganda precedes the burning of land and villages	Truth institutions (press, courts, archives) are rebuilt over generations	Health: incitement's physical toll | Mind: populations taught to hate on schedule | Social: neighbour trust destroyed | Freedom: thought itself policed | Opportunity: generations educated into falsehood	Source discipline is free and was the difference between families who believed incitement and families who survived it.
Information & Media	Now	$200–5,000 — fraud conversions, health misinformation harm, family rupture	$50–1,000 — animals harmed by viral challenges and misinformation-driven neglect scares	$50–2,000 — crop scares and food misinformation destroying farm incomes overnight	$1,000–30,000 — business reputations destroyed by coordinated falsehood	Misinformation targets environmental policy with industrial funding	Election and public-health administration carry a permanent verification overhead	Health: misinformation-driven treatment refusal and delay | Mind: outrage fatigue and attention damage | Social: families split by incompatible realities | Freedom: manipulation as governance | Opportunity: hours daily to the feed	The two-source rule and library access make the full defence $0; paid subscriptions are an upgrade, not a requirement.
Information & Media	Future	$500–20,000 — forgery fraud and wrongful accusation at personal scale	$100–2,000 — fabricated animal-cruelty and farm evidence in trade and activism wars	$200–5,000 — synthetic market-moving claims against harvests and food brands	$5,000–100,000 — property transactions and evidence systems repriced by universal doubt	Environmental data itself becomes contestable — monitoring loses authority	Courts and registries rebuild around provenance: a permanent institutional cost	Health: forged evidence in medical and legal records | Mind: radical doubt — believing nothing as a survival trait | Social: shared reality gone; coordination by tribe | Freedom: the liar's dividend for every powerful liar | Opportunity: science and journalism repriced by verification overhead	The family code word is a five-minute conversation standing against the most probable future fraud; paper originals cost one notary visit per decade.
Psychology & Mind	Past	$3,000–30,000 — lost working years, addiction, family breakdown transmitted across generations	$100–2,000 — animals neglected in traumatized households	$50–1,000 — gardens and farms untended in depression's grip	$2,000–30,000 — households and businesses dissolved by untreated illness	Land stewardship lapses when communities break	Post-war societies pay for trauma in crime, health, and welfare budgets for 50+ years	Health: trauma's somatic legacy (cardiovascular, immune) | Mind: the cost itself | Social: parenting under damage | Freedom: inner unfreedom — hypervigilance as a life sentence | Opportunity: lives at half capacity	Rhythm, testimony, and protected childhood cost nothing but steadiness — and they are where the intergenerational bill is decided.
Psychology & Mind	Now	$1,500–20,000 — absenteeism, presenteeism, treatment at crisis prices	$100–1,500 — pet care as both cost and therapy in mental-health households	$50–1,000 — allotment and garden loss when capacity fails	$1,000–20,000 — relationship and housing instability cascades	Attention damage is an environmental condition of the information ecosystem	Health systems: mental care is the fastest-growing major budget line in most countries	Health: the mind-body ledger (sleep, immunity, pain) | Mind: the cost itself | Social: withdrawal from every bond that would heal it | Freedom: compulsion by feed and substance | Opportunity: presenteeism — present everywhere, effective nowhere	Sleep, movement, and belonging are free interventions with effect sizes rivalling paid treatment for the common conditions; claim covered care early, when it is shortest and cheapest.
Psychology & Mind	Future	$1,000–15,000 — subscription stacks and lost human-capital formation	$100–1,000 — pets as the last physical relationship in withdrawn households (cost and lifeline)	$50–500 — houseplants and gardens as residual nature contact	$2,000–30,000 — fertility and household-formation collapse repricing housing and care	Nature contact itself becomes a scarce, priced good	Care systems for an aging, unpartnered population dominate late-century budgets	Health: loneliness physiology at population scale | Mind: synthetic comfort replacing resilience | Social: unpartnered, unparented generations | Freedom: dependency engineered as a business model | Opportunity: human attachment itself underbuilt	Written budgets for synthetic contact cost nothing; real-world anchors (one club) cost less per year than one month of companion subscriptions.
Education & Knowledge	Past	$5,000–60,000 — a family's earning ceiling lowered for generations by one lost literacy	$50–500 — animal husbandry knowledge lost with the texts and teachers	$200–5,000 — agricultural knowledge (seed, soil, season) erased with its carriers	$2,000–30,000 — libraries, schools, and records as physical property destroyed	Traditional ecological knowledge loss is permanent and unmeasurable in dollars	Civilizations pay in centuries: dark ages are education costs	Health: healing knowledge lost with its texts | Mind: enforced ignorance as humiliation | Social: knowledge castes | Freedom: reading as a punishable act | Opportunity: generations capped	Copying and memorizing are labour, not money; a used-book shelf costs less than one month of the ignorance it prevents.
Education & Knowledge	Now	$2,000–30,000 — tuition for credentials the market does not reward; a decade misdirected	$50–1,000 — veterinary and animal-science training gaps priced into farm losses	$200–5,000 — agronomy knowledge gaps priced into yields	$5,000–50,000 — school infrastructure maintained while learning stagnates inside it	Curricula ignoring local ecology produce maladapted graduates	Public systems: the largest single budget line in most states, audited on attendance	Health: health-literacy gaps priced in illness | Mind: credential anxiety | Social: sorting resentment between the certified and the capable | Freedom: minor | Opportunity: a decade spent acquiring the wrong proof	The library converts a $20,000 tuition problem into a $0 learning system for most purposes; read to children — the 120 hours are the cheapest education purchase that exists.
Education & Knowledge	Future	$3,000–40,000 — professionals unable to verify or function unaided	$200–3,000 — animal care automated without human judgment to catch errors	$500–10,000 — farming by dashboard: catastrophic when the dashboard is wrong	$5,000–60,000 — knowledge infrastructure (archives, models) as single points of failure	Synthetic content floods the knowledge commons itself	Licensing and examination systems rebuilt around unaided competence	Health: patients unable to question automated care | Mind: outsourced thinking atrophying | Social: a verification class and a trusting class | Freedom: dependence on systems one cannot check | Opportunity: judgment — the scarce skill — unbuilt	Unaided practice costs 30 minutes a week and defends the most expensive asset in the file: a mind that works without permission from a network.
Work & Livelihood	Past	$3,000–40,000 — a working life of wages recycled to the employer; net accumulation zero	$200–3,000 — company-farm animals as part of the scrip economy	$500–10,000 — tied cottages and plots as instruments of control	$5,000–50,000 — housing bound to employment: lose the job, lose the roof the same week	Company towns strip-mined their surroundings and left the tailings	Labour-law systems built from these costs still carry their shape	Health: industrial disease uncompensated | Mind: total dependence as humiliation | Social: the company as your whole society | Freedom: near-total | Opportunity: children's labour instead of schooling	Refusing the bundle costs inconvenience and prevents total capture; dues are the cheapest insurance a worker has ever been offered.

Work & Livelihood	Now	$1,500–20,000 — unpaid wages, missing benefits, injury without coverage	$100–1,500 — gig-economy animal care (delivery-era pet neglect is a documented pattern)	$50–1,000 — precarious workers' gardens and plots lost to schedule chaos	$3,000–40,000 — housing insecurity priced into every precarious contract	Precarity externalizes waste: disposable everything, no maintenance culture	Employment-standards enforcement is a permanent public cost	Health: injury and exhaustion without coverage | Mind: permanent job-search anxiety | Social: status shame of 'contractor' castes | Freedom: schedule as someone else's property | Opportunity: skills unbuilt while surviving	Records and claims are free; the wage theft they reverse is measured in billions yearly — the mitigation is literally filing the form.
Work & Livelihood	Future	$10,000–120,000 — mid-career displacement: retraining plus a decade of reduced earnings	$500–5,000 — rural animal livelihoods ending with their regions	$2,000–30,000 — land stewardship ending where livelihoods end	$10,000–100,000 — work-tied housing and pensions unwinding badly	Unmaintained land and infrastructure decay together	Transition programs are the fiscal price of the automation century	Health: displacement depression | Mind: obsolescence dread | Social: work-based identity and community gone together | Freedom: income at platforms' pleasure | Opportunity: midlife mastery discarded	Claim transition programs at announcement (they ration by date); negotiate collectively — the same outcome costs an individual ten times the effort.
Transport & Mobility	Past	$2,000–25,000 — siege and isolation pricing on every necessity	$500–8,000 — animals as transport dying with the system they powered	$1,000–20,000 — harvests rotting unshipped	$5,000–60,000 — stranded vehicles, carts, and vessels; cities as traps	Blockade and siege scarring of land around cities	Relief logistics are among the oldest state functions for a reason	Health: siege starvation and unreachable care | Mind: isolation despair | Social: cut-off communities fragmenting | Freedom: movement as a privilege of armies | Opportunity: trade and marriage radii collapsing	A used bicycle (~$160) replaces the entire fuel-and-fare cost stack for most local movement; route knowledge is free.
Transport & Mobility	Now	$3,000–12,000/yr — full car-dependence cost per household (the documented Canadian all-in figure)	$100–1,500 — pet transport and vet access priced by distance	$100–2,000 — garden and farm supply runs priced by sprawl	$8,000–15,000/yr — the vehicle itself as a depreciating, insurable liability	Transport is the second-largest household emissions source; sprawl locks it in	Road systems consume municipal budgets at rates transit never has	Health: commute hours priced in sleep and activity | Mind: traffic as a daily humiliation | Social: carless citizens as second-class participants | Freedom: the licence, the loan, and the fuel price as governors | Opportunity: 500+ hours a year inside a vehicle for many households	The honest comparison: $156/month pass against $8,000–12,000/yr car cost. Where a car is necessary, the sinking fund prevents repair debt — the spiral, not the car, is the cost.
Transport & Mobility	Future	$1,000–15,000 — grounding events: surge pricing, stranded commuters, missed care	$200–3,000 — animal transport and farm logistics under platform failure	$500–10,000 — just-in-time food logistics failing in correlated ways	$10,000–100,000 — fleets as bricked assets after software or cyber events	Fleet electrification concentrates mining and grid loads	Certification and fallback systems become standing transport costs	Health: crash and grounding chaos | Mind: dependence anxiety | Social: mobility inequality by software tier | Freedom: movement revocable by update | Opportunity: manual skills unpractised	The bicycle and the licence are one-time costs that no software event can revoke; provider diversification is free.
Communication	Past	$500–20,000 — rumour-driven losses; families separated without news for years	$50–1,000 — animal sales and slaughter decided on rumour prices	$100–3,000 — planting decided on rumour weather and war news	$2,000–30,000 — property sold in panic on false reports	Wars of rumour burn land as surely as armies	Postal and telegraph systems were built as state survival infrastructure	Health: warnings unheard | Mind: rumour terror | Social: families in decades of silence | Freedom: speech monopolized | Opportunity: coordination impossible	Paper plans and memberships cost almost nothing and are the only infrastructure that cannot be switched off centrally.
Communication	Now	$200–8,000 — lost audiences, records, and livelihoods in single bans and outages	$50–500 — pet and farm coordination lost with accounts	$50–1,000 — market and weather coordination priced by platform access	$1,000–20,000 — businesses whose only storefront is a platform page	Attention pollution is this era's communication externality	Public communication infrastructure was allowed to wither — rebuilding is priced now	Health: emergency communication failures | Mind: platform anxiety and audience dependence | Social: relationships held on rented land | Freedom: deplatforming as exile | Opportunity: businesses built on unowned channels	Exports and an owned channel are free habits; the library anchor makes full digital capability $0 for the device-less.
Communication	Future	$500–15,000 — chilling-effect losses: unfiled complaints, unspoken warnings, blackmail	$100–2,000 — animal-welfare reporting chilled by surveillance	$200–5,000 — farm and land data harvested as an asset class	$5,000–50,000 — property and identity records inside surveilled systems	Environmental monitoring depends on communication freedom	Rights-preserving architecture is a permanent design cost, paid once per system	Health: chilled reporting of hazards | Mind: self-censorship as habit | Social: trust requiring encryption and verification everywhere | Freedom: the cost itself | Opportunity: dissent and error-correction unspoken	Encryption and compartmentation are free settings; the annual data-correction hour prevents errors compounding into decisions.
Demographics & Population	Past	$3,000–40,000 — lineage restart costs: skills, records, and networks rebuilt from zero	$500–8,000 — herds and flocks lost with their keepers' knowledge	$1,000–20,000 — terraces, orchards, and irrigation abandoned with their populations	$5,000–60,000 — empty cities and untilled farms as physical waste	Abandoned land reverts; reclearing costs fall on the grandchildren	States fall with their populations; successors inherit the bill	Health: catastrophe mortality patterns | Mind: survivor generations' outlook | Social: kinship systems snapped | Freedom: minor | Opportunity: lineages ended entire	Writing things down and registering events are the cheapest acts in the file relative to the claims they later unlock.
Demographics & Population	Now	$5,000–50,000 — the care collision: childcare plus eldercare at market rates exceeding a salary	$100–2,000 — multi-pet and farm-animal care inside stretched households	$100–3,000 — family land underused while heirs live elsewhere	$10,000–100,000 — empty-nest oversized housing vs. young families priced out: the same stock, misallocated	Depopulating regions lose services in a spiral (school, clinic, store)	Pension and health systems repriced by the dependency ratio, permanently	Health: caregiver burnout as an epidemic | Mind: the sandwich generation's dread | Social: elders isolated, children institutionalized by schedule | Freedom: life timing (children, moves) dictated by prices | Opportunity: careers paused for care, uncounted	Care pooling replaces market prices with reciprocity; full benefits claiming is thousands per year for the cost of forms.
Demographics & Population	Future	$10,000–100,000 — cohort traps: retiring into systems designed for a pyramid that inverted	$1,000–10,000 — animal agriculture regions ending with their workforces	$5,000–50,000 — farmland abandonment and consolidation at continental scale	$50,000–300,000 — infrastructure built for populations that halve or triple	Abandoned land is an ecological opportunity with a management price	Fifty-year planning becomes a standing function of competent states	Health: care deserts for the old | Mind: demographic dread as politics | Social: generational contract fraying | Freedom: mobility and family size governed by systems | Opportunity: regions and lineages ending by arithmetic	Pathways and audits are time purchases: started early they cost hours; started late they cost years.
Justice & Ethics	Past	$5,000–100,000+ — confiscation, imprisonment, exile: the full stack of state-inflicted loss	$500–10,000 — animals seized with farms and homes	$1,000–30,000 — land as the first confiscation target	$10,000–200,000 — property systems converted to reward loyalty over title	Show-trial regimes poison land registries for generations	Tribunals and restitution processes run for half a century after	Health: imprisonment's toll | Mind: terror as a teacher | Social: denunciation cultures | Freedom: total | Opportunity: lives consumed by process and punishment	Archives cost postage; the restitution and asylum outcomes they power are priceless in the literal sense.
Justice & Ethics	Now	$1,000–30,000 — winnable cases lost unrepresented; collateral costs of single convictions run decades	$100–2,000 — animals seized in enforcement actions, held at owners' cost	$100–3,000 — small land and garden disputes priced beyond ordinary reach	$5,000–80,000 — homes lost to process costs before merits are heard	Environmental cases are the most priced-out category of all	Legal aid and court capacity are the standing price of equal justice	Health: incarceration and debt stress | Mind: the humiliation of being unheard | Social: legitimacy draining from courts | Freedom: rights rationed by wealth | Opportunity: years to process, for the poor	Free counsel before any signature is the highest-value zero in this file; small-claims fees are waivable — ask.
Justice & Ethics	Future	$1,000–25,000 — wrongful automated flags: jobs, credit, bail, and liberty priced per error	$100–2,000 — automated welfare and animal-control decisions at scale	$200–5,000 — land and subsidy algorithms allocating wrongly at scale	$5,000–60,000 — property and tenancy screening by opaque score	Environmental enforcement automated toward the measurable, away from the important	Audit and appeal infrastructure is the new courthouse cost	Health: automated flags' stress and error burdens | Mind: being judged by the unjudgeable | Social: scored castes | Freedom: appeal to no one | Opportunity: lives optimized for the metric instead of the good	Score hygiene is three hours a year against errors that price themselves into every loan, lease, and job screen you ever face.

#	Field	Timeframe	Item	Line Cost CA	Prevention (stops the problem or your exposure to it)	Mitigation (reduces the damage when it happens)	Cost Per Use — With Measurement
1	Water	Past	Stored water reserve 56 L/person	48	Prevention — Medium: storage does not prevent drought, but catchment and source mapping prevent dependence on a single failing source	Mitigation — High: 56 L/person reserve covers the lethal window (first 2 weeks) of any supply collapse	4 containers ≈ 8 fills/yr each over 5 yrs → ~$0.30/fill
2	Water	Past	Cloth + sand filtration setup	15	Prevention — Low: filtration does not prevent contamination of sources	Mitigation — High: converts unsafe water to survivable water at household scale	Filters ~50 batches/yr for years → under $0.10/batch
3	Water	Past	Fuel for boiling (1 month reserve)	25	Prevention — Low	Mitigation — High: boiling capability is the final barrier; fuel reserve guarantees it for a month	Fuel is consumed per use by design: ~$0.80/day of boiling
4	Water	Past	Rain barrel 200 L	60	Prevention — Medium: catchment reduces draw on failing municipal/aquifer supplies	Mitigation — Medium: 200 L buffer for hygiene and garden through dry spells	200 L ≈ 100+ garden/hygiene uses per season → ~$0.50/use
5	Water	Now	Certified gravity/pump water filter	90	Prevention — Medium: a home barrier prevents exposure even when the plant fails	Mitigation — High: safe water continues uninterrupted through the advisory period	Element treats ~1,000+ L → under $0.10/L protected
6	Water	Now	72-hour stored water (12 L/person drinking)	16	Prevention — Low	Mitigation — High: the first 72 hours — when information is worst — are fully covered	Consumed per event; ~$0.65/L in an emergency vs. $2+/L panic prices
7	Water	Now	Bathtub water bladder (fills in minutes at first notice)	35	Prevention — Low	Mitigation — High: ~100 L within minutes of first notice, before queues form	~100 L per deployment; reusable across events → ~$1/event after first uses
8	Water	Now	Water test strips (bacteria indicators + chlorine)	20	Prevention — Medium: early home detection precedes official notice in many events	Mitigation — Medium: verification prevents both panic buying and false reassurance	Strips ≈ $1/test; a test replaces a guess worth hundreds
9	Water	Future	Low-flow fixtures set (aerators, showerhead, toilet kit)	45	Prevention — High (for your household): permanent 20–30% demand cut reduces your exposure to every future restriction	Mitigation — Medium: lower baseline need stretches any reserve further	Permanent: saves ~15 L/day → payback in months, then free
10	Water	Future	30-day deep storage (barrels)	130	Prevention — Low	Mitigation — High: a month of autonomy converts a crisis into an inconvenience	Each fill = a month of autonomy; 2 fills/yr → protection at ~$65/yr
11	Water	Future	Greywater diverter for garden (where legal)	55	Prevention — Medium: keeps food production alive under watering bans	Mitigation — Medium: garden survives restrictions that kill neighbours' plots	Seasonal use; pays back in one saved crop
12	Water	Future	Cooperative membership share (community well/storage)	100	Prevention — Medium: collective storage prevents community-level day-zero dynamics	Mitigation — High: shared deep storage outlasts any household stock	Share ÷ expected years of membership → tens of dollars per protected year
13	Food & Agriculture	Past	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	220	Prevention — Medium: stores do not prevent famine, but they prevent YOUR household entering the famine market at peak prices	Mitigation — High: 90 days is the historical bridge across failed harvests to relief or the next crop	Consumed once — the 'use' is 90 days of life per adult; ~$2.45/day at famine prices that reach 10x
14	Food & Agriculture	Past	Garden start: seeds + soil for 10 m²	80	Prevention — Medium: home production reduces market dependence by 20–40% of fresh food	Mitigation — High: micronutrients and morale that stored grain cannot give	10 m² yields across a season ≈ 100+ servings → under $1/serving, falling yearly
15	Food & Agriculture	Past	Preservation kit (jars, lids, salt, vinegar)	60	Prevention — Low	Mitigation — High: gluts become months of food instead of rot	Each jar reused for years; ~$0.50/batch preserved
16	Food & Agriculture	Past	Seed-saving envelopes + storage tin	12	Prevention — High (for next year): saved seed prevents next season's dependence on seed markets that may not exist	Mitigation — Medium: garden restarts without cash	One season's saving = every future season's seed → cost per year approaches $0
17	Food & Agriculture	Now	30-day staple buffer per adult (~25 kg grain-equivalent)	85	Prevention — Medium: buffer prevents forced buying at spike prices	Mitigation — High: one month of price immunity per fill	Consumed across the month it covers: ~$2.85/day vs. spike-market $6+/day
18	Food & Agriculture	Now	Buying-club share (bulk staples with 4–10 households)	40	Prevention — Medium: bulk contracts lock pre-spike prices	Mitigation — Medium: permanent 15–30% staple discount softens every future spike	Savings of 15–30% on every bulk purchase, indefinitely
19	Food & Agriculture	Now	Sprouting kit (jars + seeds)	18	Prevention — Low	Mitigation — Medium: fresh food in 4 days when shelves empty	Jars produce weekly batches for years → pennies per batch of fresh greens
20	Food & Agriculture	Now	Community fridge / food bank registration + food-handler basics	5	Prevention — Low	Mitigation — High: frees cash exactly when prices peak; registration before the crisis avoids queues at the worst moment	Programs are free; cost per visit is transport only
21	Food & Agriculture	Future	Perennial food plants (berry bushes, fruit tree, herb bed)	90	Prevention — Medium: perennial production is structurally immune to annual seed/fuel shocks	Mitigation — Medium: yields grow yearly while annual systems strain	Plants yield 10–25 years → cost per harvest year falls below $10
22	Food & Agriculture	Future	Soil building: compost system + cover crop seed	45	Prevention — High (long-run): living soil is the only durable yield insurance	Mitigation — Medium: soil holds water through drought weeks	Soil improvements persist 5+ years per application
23	Food & Agriculture	Future	Dehydrator (entry electric or solar build)	70	Prevention — Low	Mitigation — Medium: preservation converts seasonal abundance into year-round supply	~$1–2 electricity per batch vs. $5+ commercial dried equivalents
24	Food & Agriculture	Future	Co-op / land-trust membership contribution	75	Prevention — Medium: owned growing space cannot be repriced or evicted	Mitigation — Medium: collective scale funds wells, fencing, and storage no household can	Membership per year ÷ harvests shared → lowest-cost food in the file
25	Health & Medicine	Past	Household medical kit: thermometer, pulse oximeter, ORS packets, fever medication (2 weeks)	55	Prevention — Medium: oximetry and early care prevent deterioration spirals	Mitigation — High: most pandemic deaths at home are worsened by late detection — the kit attacks exactly that	Devices serve daily for years → under $0.05/day
26	Health & Medicine	Past	Isolation-room setup (separate bedding, basin, waste bag system)	30	Prevention — High: room isolation is the main household transmission brake	Mitigation — Medium: organized care prevents secondary collapse of caregivers	Linens reused across illnesses → ~$1/illness episode
27	Health & Medicine	Past	Vaccination catch-up (all free public programs)	0	Prevention — High: vaccination is the strongest preventive in this entire file	Mitigation — High: vaccinated households convert pandemics into illnesses	$0 per dose — the best cost-per-use in the file
28	Health & Medicine	Past	Paper medical records folder (vaccines, conditions, medications, allergies)	6	Prevention — Medium: records prevent treatment errors and delays	Mitigation — Medium: any facility can treat you correctly on arrival	Folder consulted at every encounter for years
29	Health & Medicine	Now	Drug coverage optimization: benefits audit + generic substitution review with pharmacist	0	Prevention — High: correct, affordable medication prevents the non-adherence that drives resistance and relapse	Mitigation — Medium: savings buffer against drug-cost shocks	$0 — one review, recurring savings
30	Health & Medicine	Now	Home infection-control basics (soap stock, disinfectant, masks x50, gloves)	35	Prevention — High: hygiene basics interrupt transmission chains daily	Mitigation — Low	Consumables ≈ $3/week of household hygiene
31	Health & Medicine	Now	Dental prevention kit + one hygiene visit/yr	120	Prevention — High: prevention converts $3,000 emergencies into $120 visits	Mitigation — Low	One visit/yr prevents emergencies 10–30x its price
32	Health & Medicine	Now	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy take-back)	0	Prevention — High (population): unused antibiotics returned protect everyone's future cures	Mitigation — Low	$0 — take-back is free
33	Health & Medicine	Future	Room HEPA unit for the sick room / bedroom	130	Prevention — High: filtered air prevents the infections that overload creates	Mitigation — Medium: the sick room stops seeding the household	Runs ~8 hrs/day → ~$0.15/day across filter life
34	Health & Medicine	Future	Ventilation upgrade: CO2 monitor + window/exhaust habits	110	Prevention — Medium: measured air prevents unknowing exposure	Mitigation — Medium: ventilation habits are adjustable daily	Monitor serves for years → ~$0.10/day
35	Health & Medicine	Future	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	45	Prevention — High: early detection of BP/glucose prevents the two costliest chronic cascades	Mitigation — Medium: managed conditions survive system overload far better	Cuff: ~$0.05/measurement across its life
36	Health & Medicine	Future	Reserve medication buffer (1 month extra of essentials, rotated, where legal)	40	Prevention — Low	Mitigation — High: medication continuity through the exact weeks the system cannot supply you	Rotation cost ≈ normal refill timing — near $0 marginal
37	Energy	Past	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blanket fort supplies	60	Prevention — Medium: one-room living prevents whole-house heat loss consequences	Mitigation — High: survivable warmth at a fraction of embargoed fuel	Reusable every cold night for years
38	Energy	Past	Fuel reserve (safe storage: gas canisters / charcoal / wood)	70	Prevention — Low	Mitigation — High: stored fuel is stored time	Consumed per use: ~$2–3/day of emergency heat/cooking vs. black-market multiples

39	Energy	Past	Layered clothing upgrade (wool base layers x2 per person)	90	Prevention — High: body insulation prevents hypothermia regardless of fuel markets	Mitigation — Medium: comfort and sleep preserved	~200+ wears per garment → under $0.25/wear
40	Energy	Past	Pipe protection: insulation sleeves + shutoff knowledge	25	Prevention — High: insulated pipes and known shutoffs prevent the costliest cold loss outright	Mitigation — Medium: fast shutoff caps damage when prevention fails	One installation protects every freeze for decades
41	Energy	Now	Battery power station 1 kWh (or 12V battery + inverter build)	420	Prevention — Low: batteries do not prevent grid failure	Mitigation — High: 48–72 h of light, cold food, communication, and medical power — the difference between an outage and an emergency	~3,000+ charge cycles (LiFePO4) → ~$0.15/cycle; per outage-hour ≈ $0.50
42	Energy	Now	LED lanterns x3 + headlamp	35	Prevention — Low	Mitigation — Medium: safe light prevents candle-fire and injury losses	Rechargeable: hundreds of uses → ~$0.10/use
43	Energy	Now	Safe backup cooking: single-burner butane stove + 4 canisters	45	Prevention — Low	Mitigation — High: hot food and boiled water sustain health and morale through any outage	Canister ≈ $3–4 per day of cooking — vs. $30+ restaurant fallback
44	Energy	Now	Fridge discipline kit: thermometer + cooler + ice packs	30	Prevention — Medium: temperature knowledge prevents premature food discard AND food poisoning gambles	Mitigation — Medium: saves most of the $200–500 spoilage loss	Cooler serves every outage and trip for years
45	Energy	Future	Solar trickle panel 100–200 W for the battery station	160	Prevention — Low	Mitigation — High: converts a finite battery into indefinite survival-core power	Panel generates ~25 yrs → cost per kWh falls below grid price within years
46	Energy	Future	Home insulation priority pack (attic hatch seal, window film, draft proofing)	85	Prevention — Medium: lower demand prevents the depth of cold/heat injury in long outages	Mitigation — Medium: every outage becomes survivable longer	Savings recur every heating/cooling season for decades
47	Energy	Future	Community energy co-op share	100	Prevention — Medium: local generation reduces dependence on the failing whole	Mitigation — Medium: community power points serve non-members in practice	Share per year of local generation ownership
48	Energy	Future	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserve small bills	55	Prevention — Medium: cash and information prevent transaction and panic losses	Mitigation — Medium: you can buy, travel, and decide while systems are down	Radio serves decades; cash is spent only in the emergency it was saved for
49	Shelter & Housing	Past	Documents go-bag (waterproof pouch: ID, property, insurance, photos of every room's contents)	18	Prevention — Medium: documents do not prevent destruction; they prevent the second loss — uncompensated destruction	Mitigation — High: claims paid in weeks instead of years; identity provable anywhere	One kit, consulted at every claim for life
50	Shelter & Housing	Past	72-hour go-kit per person (clothes, medication, water, light, food bars)	65	Prevention — Low	Mitigation — High: the first 72 hours after losing a home decide the next year; the kit owns them	Per evacuation: replaces $200+/night emergency lodging chaos
51	Shelter & Housing	Past	Home inventory video + off-site copy	0	Prevention — Low	Mitigation — High: documented losses are paid at multiples of undocumented ones	$0 — one afternoon, lifetime claim value
52	Shelter & Housing	Past	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	70	Prevention — High: alarms and extinguishers prevent the majority of total-loss fires	Mitigation — Medium: shutoff knowledge caps water/fire damage	Alarms last ~10 yrs → ~$7/yr against total-loss risk
53	Shelter & Housing	Now	Tenant rights pack: your jurisdiction's rules printed + legal-aid and tenant-union numbers	4	Prevention — High: known rights prevent most illegal evictions and deposit thefts from being attempted	Mitigation — Medium: when attempted, the written process reverses them	One printout, used at every dispute
54	Shelter & Housing	Now	Housing documentation habit: dated photos at move-in/out, all landlord communication in writing	0	Prevention — Medium: documented condition prevents false damage claims	Mitigation — High: wins the disputes that do occur	$0 — minutes per tenancy
55	Shelter & Housing	Now	Rent-bank / eviction-prevention applications pre-filed knowledge + emergency contact list	0	Prevention — Medium: pre-known funds prevent the arrears-to-eviction cascade	Mitigation — High: rent banks exist precisely to interrupt the cascade	$0 — knowledge claimed once, used whenever arrears threaten
56	Shelter & Housing	Now	Shared-housing option prepared (one identified co-housing/roommate arrangement as Plan B)	0	Prevention — Medium: a prepared Plan B prevents panic-priced emergency housing	Mitigation — High: cost floor halves before the crisis, not during	$0 — one arrangement, standing by for years
57	Shelter & Housing	Future	Property risk check before buying/renewing: flood, fire, insurance-availability maps (free public data)	0	Prevention — High: the audit prevents the six-figure stranded purchase itself	Mitigation — Low	$0 per property decision — against six-figure stakes
58	Shelter & Housing	Future	Full insurance review incl. overland flood / sewer backup riders	180	Prevention — Low	Mitigation — High: transfers the catastrophic tail while transfer is still offered	Premium per year ÷ tail risk transferred
59	Shelter & Housing	Future	Portability reserve: keep at least 50% of net worth out of any single exposed property (financial accounts, skills)	0	Prevention — High: portable wealth prevents total-loss exposure	Mitigation — Medium: you can leave at a time of your choosing	$0 — an allocation decision
60	Shelter & Housing	Future	Buyout/relocation program registration (where offered) — filed early	0	Prevention — Low	Mitigation — High: early registration captures the generous phase of buyout programs	$0 — one filing, early-position value
61	Security & Defense	Past	Early-warning habit: named indicators list + decided trigger points ('if X happens, we leave within 48 h')	0	Prevention — High: decided triggers prevent the fatal delay that history punishes most	Mitigation — Medium: early departure preserves life and portable assets	$0 — one decision, rehearsed
62	Security & Defense	Past	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	90	Prevention — Medium: portable documents prevent the statelessness trap on arrival	Mitigation — High: employability from week one instead of year two	Translations serve every border and employer for life
63	Security & Defense	Past	Exit-route plan: two routes, rendezvous points, transport costs pre-estimated and saved	0	Prevention — Medium: a pre-costed route prevents improvisation at the worst moment	Mitigation — High: the earmarked fund makes the plan executable	The earmark is spent once — on the day it saves everything
64	Security & Defense	Past	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in signals)	0	Prevention — Medium: networks warn earlier than media	Mitigation — High: collective transport, housing, and information in flight	$0 — roster maintained in hours per year
65	Security & Defense	Now	Home target-hardening basics: solid locks, strike plates with 3-inch screws, window locks, lighting	85	Prevention — High: hardening deters the opportunistic majority of break-ins	Mitigation — Medium: delays forced entry long enough for response	Hardware lasts decades → ~$4/yr of deterrence
66	Security & Defense	Now	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	60	Prevention — Medium: evidence awareness deters repeat targeting	Mitigation — High: investigated reports recover property and stop patterns	Camera serves daily; evidence value realized per incident
67	Security & Defense	Now	Route and routine discipline + check-in system with one contact	0	Prevention — High: unpredictability prevents targeted incidents	Mitigation — Medium: check-ins shorten every emergency's discovery time	$0 — habits, daily
68	Security & Defense	Now	Victim-compensation and protective-order applications knowledge (legal aid)	0	Prevention — Low	Mitigation — High: compensation and orders convert victimhood into remedy	$0 — knowledge claimed when needed
69	Security & Defense	Future	Venue standards habit: choose venues with visible security, exits known on entry (30-second rule)	0	Prevention — Medium: exit awareness is the main variable in crowd-incident survival	Mitigation — Medium: 30 seconds of orientation saves the minutes that matter	$0 — 30 seconds per venue
70	Security & Defense	Future	Personal data footprint reduction: data-broker removals, minimal public exposure	30	Prevention — High: smaller footprint prevents targeting and fraud selection	Mitigation — Low	Removal effort once; some services ~$30/yr
71	Security & Defense	Future	Support for regulation: specific, written asks to representatives (drone ID, autonomous weapons limits)	0	Prevention — Medium (collective): regulation prevents the threat class rather than your instance of it	Mitigation — Low	$0 — letters per year
72	Security & Defense	Future	Home/business camera + signage (deterrence + evidence)	90	Prevention — Medium: visible cameras deter opportunistic crime	Mitigation — Medium: evidence after the fact	Camera daily for years → ~$0.10/day
73	Governance & Law	Past	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	75	Prevention — Medium: an archive abroad prevents total identity loss when records are destroyed or weaponized	Mitigation — High: claims, asylum, and restitution all run on this paper	One-time setup; protects for life → under $2/use across a lifetime of official interactions
74	Governance & Law	Past	Qualifications in portable form (transcripts, licences, references sealed and copied)	40	Prevention — Medium: portable credentials prevent the exile earnings collapse	Mitigation — High: work at your level from arrival	One-time; pays out across an entire career
75	Governance & Law	Past	Quiet-network membership (congregation, union, association dues)	60	Prevention — Medium: networks warn and vouch before states act	Mitigation — High: exit help, housing, and work arrive through people, not programs	Annual dues ≈ one restaurant meal; the payout is existential
76	Governance & Law	Now	Access-to-information request kit (templates + filing fees where charged)	15	Prevention — Medium: filed requests prevent decisions being made about you without a record	Mitigation — Medium: documents reverse specific corrupt acts	A handful of requests per year; ~$3–5 each

77	Governance & Law	Now	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	5	Prevention — Medium: complaint patterns prevent repeat extraction	Mitigation — Medium: individual recoveries plus deterrence	Free; cost is ~1.5 hours per complaint
78	Governance & Law	Now	Civic membership: one watchdog, union, or ratepayers' organization	40	Prevention — Medium (collective): organized citizens prevent capture from being costless	Mitigation — Low	Annual dues; monthly meeting time
79	Governance & Law	Future	Local verification habit: attend/join one local body (assembly, board, co-op) and keep its records	0	Prevention — Medium: local verification prevents rumour-governance at your scale	Mitigation — Medium: you hold ground truth when national narratives fail	Time only: ~6 hrs to join, then monthly hours
80	Governance & Law	Future	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	0	Prevention — Low	Mitigation — High: personal records win appeals in failing systems	Free; minutes per interaction
81	Governance & Law	Future	Support independent local media (one subscription)	60	Prevention — Medium: funded local media prevents the information vacuum collapse fills	Mitigation — Low	~$60/yr → under $0.20/day
82	Economy & Finance	Past	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	0	Prevention — High: splitting prevents any single bank failure from reaching your core savings	Mitigation — Medium: surviving institutions keep payments running	Free; set once, checked yearly
83	Economy & Finance	Past	Durable-goods and skills reserve (tools, preserved food, tradeable skills)	150	Prevention — Medium: durable value prevents total dependence on failing currency and banks	Mitigation — High: tools, food, and skills remain spendable when money is not	One-time ~$150; serves for decades
84	Economy & Finance	Past	Debt minimalism in boom times (fixed-rate only, no margin, no speculation on credit)	0	Prevention — High: in deflations this rule alone decides who survives financially	Mitigation — Medium	Free — a refusal, maintained
85	Economy & Finance	Now	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	500	Prevention — High: the fund prevents emergencies becoming debt in the first place	Mitigation — High: converts job loss and breakdowns into inconveniences	Your own money, retained: cost per use is negative — each use saves 20%+ interest
86	Economy & Finance	Now	Debt audit + avalanche order (highest rate first) printed and posted	0	Prevention — Medium: a visible order prevents interest waste	Mitigation — High: cuts total interest 20–40% on existing debt	Free; 1.5 hours once, minutes monthly
87	Economy & Finance	Now	Free insolvency-trustee first consultation + credit-counselling session (non-profit)	0	Prevention — Medium: professional mapping prevents the wrong-door spiral (payday loans, bankruptcy by default)	Mitigation — High: structured options replace panic	Free consultations
88	Economy & Finance	Now	Benefits and credits claim check (tax filing + benefits finder)	0	Prevention — Medium: claimed benefits prevent the shortfalls that become debt	Mitigation — Medium: hundreds per year for typical households	Free; ~1.5 hours yearly
89	Economy & Finance	Future	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	100	Prevention — Medium: ownership prevents total dependence on wages	Mitigation — High: dividends cushion displacement years	Automatic monthly; cost per year of protection falls as the balance compounds
90	Economy & Finance	Future	Human-irreducible skill course (care, craft, repair, judgment trades)	250	Prevention — High: human-irreducible skill prevents displacement reaching you	Mitigation — Medium: premium income in the automated economy	~40 hours of course time; career-long payout
91	Economy & Finance	Future	Union/professional-body membership	120	Prevention — Medium (collective): negotiated transitions prevent cliff-edge displacement	Mitigation — Medium	Annual dues
92	Technology & AI	Past	Home hazard test when buying older property (lead paint, asbestos survey before renovation)	180	Prevention — High: testing prevents the exposure itself	Mitigation — Medium: findings enable remediation before renovation spreads fibres/dust	One test per property purchase
93	Technology & AI	Past	Exposure records file (employers, products, dates) kept permanently	4	Prevention — Low	Mitigation — High: records are the compensation claim, decades later	One folder; minutes per job change
94	Technology & AI	Past	Adoption delay rule for unstudied non-essentials (wait for the 2nd generation / safety data)	0	Prevention — High: delay prevents first-generation latent harms	Mitigation — Low	Free — a waiting rule
95	Technology & AI	Now	Password manager + MFA on email and banking (the two master keys)	0	Prevention — High: manager + MFA prevent the dominant takeover path	Mitigation — Medium: vault limits any single breach's blast radius	Free tier; ~1.5 hrs setup → pennies per login
96	Technology & AI	Now	3-2-1 backup: external drive + cloud copy of irreplaceables	95	Prevention — Medium: backups prevent data hostage situations	Mitigation — High: full recovery in hours instead of never	~$95 per drive per ~5 years → ~$0.05/day
97	Technology & AI	Now	Automatic updates ON + router firmware check (quarterly, calendared)	0	Prevention — High: patched systems prevent most commodity exploitation	Mitigation — Low	Free; minutes quarterly
98	Technology & AI	Now	Verification protocol card (callback numbers printed; 'urgent = verify by second channel' rule)	2	Prevention — High: the protocol prevents the fraud that technology cannot	Mitigation — Medium: verification calls stop transfers mid-script	One card; seconds per use
99	Technology & AI	Future	Analogue life-line kit: paper originals of ID/property/credentials, cash reserve, offline maps and contacts	40	Prevention — Medium: paper prevents lockout from becoming identity loss	Mitigation — High: you can prove, pay, and travel offline	One-time ~$40; protects for decades
100	Technology & AI	Future	Data portability routine: monthly export of contacts, photos, records from every platform	0	Prevention — Medium: exports prevent platform decisions from erasing your records	Mitigation — High: migration between services in hours	30 min/month
101	Technology & AI	Future	Offline competence maintenance: one unaided skill practised regularly (navigation, arithmetic, writing)	0	Prevention — Medium: unaided competence prevents helplessness in system failure	Mitigation — Medium: employability persists across platform shifts	Time only
102	Environment & Climate	Past	Soil care basics: compost return, cover, no bare soil over winter	45	Prevention — High: soil care prevents the degradation itself	Mitigation — Medium: living soil buffers drought and flood years	Yearly seed cost; labour is the input
103	Environment & Climate	Past	Water-wise planting (drought-matched species, mulch)	60	Prevention — High: matched planting prevents water-stress losses	Mitigation — Medium	One-time planting; years of yield
104	Environment & Climate	Past	Fuel efficiency: improved cookstove / insulation of heat	80	Prevention — Medium: efficiency prevents fuel exhaustion	Mitigation — Medium: halves the crisis fuel bill	One build; daily use for years
105	Environment & Climate	Now	Address risk check (flood/fire/heat maps, free) + insurer confirmation in writing	0	Prevention — Medium: knowledge prevents uninformed exposure decisions	Mitigation — Medium: written insurer confirmation prevents claim denial surprises	Free; 1.5 hours
106	Environment & Climate	Now	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	250	Prevention — High: the valve prevents the most common five-figure household flood loss outright	Mitigation — Low	~$250 once → under $25/yr across a decade of protection
107	Environment & Climate	Now	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback, water stock	65	Prevention — Medium: a cool room prevents heat illness at home	Mitigation — High: survivable indoor refuge through heat domes	~$65 kit; pennies per heat-day
108	Environment & Climate	Now	Go-bag + document pouch (shared with Shelter build — counted once there)	0	Prevention — Low	Mitigation — High: evacuation readiness (costed in Shelter build)	Shared item — see Shelter & Housing
109	Environment & Climate	Future	Location decision audit every 5 years (risk trend, insurance trend, water trend for your address)	0	Prevention — High: five-yearly audits prevent the stranded-asset endgame	Mitigation — Medium: early exits keep prices	Free; 3 hours per 5 years
110	Environment & Climate	Future	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	200	Prevention — Medium: portable income prevents region-bound ruin	Mitigation — High: you can leave before leaving is fleeing	Course-scale investment; career payout
111	Environment & Climate	Future	Collective relocation readiness: community/municipal plan participation	0	Prevention — Medium (collective): planned moves prevent chaotic ones	Mitigation — High: collective terms and cohesion preserved	Time only
112	Information & Media	Past	Source-tracing habit taught at home (claim → origin → evidence) — a dinner-table practice, not a purchase	0	Prevention — High: household source-tracing prevents incitement from recruiting your family	Mitigation — Medium: verified networks correct rumours locally	Free; a habit, practised at meals
113	Information & Media	Past	Independent media support: one exile/public/independent subscription or donation	50	Prevention — Medium: a living alternative channel prevents monopoly from being total	Mitigation — Medium	~$50/yr
114	Information & Media	Past	Battery shortwave/AM radio for foreign and emergency broadcasts	35	Prevention — Medium: foreign broadcasts prevent information siege	Mitigation — Medium: verified outside news in blackouts	One radio; decades of service

115	Information & Media	Now	Two-source rule card + one scheduled news window (a household protocol, posted)	2	Prevention — High: the two-source rule prevents most misinformation harm at zero cost	Mitigation — Medium: scheduled news prevents anxiety spirals	Free; seconds per claim
116	Information & Media	Now	Primary-source subscriptions: one quality outlet + official data sources bookmarked	90	Prevention — Medium: a quality floor prevents a pure-outrage diet	Mitigation — Low	~$90/yr or $0 via library
117	Information & Media	Now	Fact-check and provenance tools bookmarked; reverse image search learned	0	Prevention — High: 2-minute verification prevents forwarding-driven harm	Mitigation — Medium	Free; minutes per claim
118	Information & Media	Future	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	0	Prevention — High: the code word prevents voice-clone fraud — the highest-probability future fraud — outright	Mitigation — Medium: rehearsed doubt stops transfers mid-call	Free; one family conversation
119	Information & Media	Future	Paper originals vault (identity, property, credentials) + notarial copies of the critical five	75	Prevention — Medium: paper originals prevent identity collapse under universal forgery	Mitigation — High: provable identity in any proceeding	One-time ~$75; lifetime service
120	Information & Media	Future	Content-credential tools adopted (signed capture apps) for important records	0	Prevention — Medium: signed capture prevents your genuine evidence being dismissed	Mitigation — Medium	Free tiers; per-record seconds
121	Psychology & Mind	Past	Routine and ritual structure (fixed sleep, meals, work, worship/rest — a written household rhythm)	0	Prevention — Medium: rhythm prevents chaos from becoming the household's operating system	Mitigation — High: structure is the container that carried families through every recorded catastrophe	Free; daily discipline
122	Psychology & Mind	Past	Mutual storytelling / testimony practice (regular, structured, within family or congregation)	0	Prevention — Medium: shared narrative prevents trauma from isolating	Mitigation — High: testimony converts private damage into survivable story	Free; weekly hours
123	Psychology & Mind	Past	Children-first protection of sleep, schooling, and play (the intergenerational shield)	0	Prevention — High: protected children prevent the intergenerational transmission itself	Mitigation — Medium	Free; parental steadiness is the cost
124	Psychology & Mind	Now	Sleep protocol: fixed window, morning light (free), screens out of the bedroom, caffeine cutoff	25	Prevention — High: sleep is the largest single preventive in mental health	Mitigation — Medium: rested people weather crises measurably better	~$25 optional; the protocol is free
125	Psychology & Mind	Now	Movement prescription: 60 min/day walking built into commute/errands + one strength session weekly	0	Prevention — High: movement prevents a large share of depression/anxiety onset	Mitigation — Medium: effect sizes rival medication for mild-moderate illness	Free; ~7 hrs/week of time — the real price, stated
126	Psychology & Mind	Now	One weekly in-person group (see Pareto action #6 — costed once)	0	Prevention — High: belonging prevents isolation-driven illness	Mitigation — Medium	Shared item — see Citizen Pareto
127	Psychology & Mind	Now	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	0	Prevention — Medium: saved numbers prevent the worst hour becoming a research problem	Mitigation — High: help in minutes, free	Free; minutes
128	Psychology & Mind	Future	Synthetic-companionship budget rule (fixed hours + money cap, written, reviewed monthly)	0	Prevention — High: a written budget prevents dependency from forming by drift	Mitigation — Medium: monthly review catches escalation early	Free; 30 min/month
129	Psychology & Mind	Future	Real-world anchor subscriptions: one club/league/class with dues (skin in the game raises attendance)	80	Prevention — High: paid anchors prevent synthetic substitution from winning by default	Mitigation — Medium	~$80/yr dues; weekly attendance
130	Psychology & Mind	Future	Device architecture: companions/feeds off the phone's home screen, notifications batched	0	Prevention — Medium: friction prevents compulsive engagement loops	Mitigation — Low	Free; one setup session
131	Education & Knowledge	Past	Home copying kit: notebooks, and one body of knowledge memorized per child (the unconfiscatable copy)	10	Prevention — High: memorized and home-copied knowledge prevents total loss in any confiscation	Mitigation — High: the family's restart capital survives every fire	One-time family project; protects for generations → cost per use approaches zero
132	Education & Knowledge	Past	Printed core references shelf (first aid, agriculture, mathematics, local law — bought used)	40	Prevention — Medium: a printed shelf prevents dependence on any single institution	Mitigation — High: offline capability in collapse	One-time ~$40; decades of reference use
133	Education & Knowledge	Past	Literacy defence: every child taught to read early at home if school fails (phonics materials)	25	Prevention — High: home literacy prevents the master loss from which all others follow	Mitigation — High: a reading child can rebuild every other knowledge	~1 hr/day for a year; the highest-return hours in this file
134	Education & Knowledge	Now	Library-as-university setup: card + course platforms + a written self-curriculum	0	Prevention — High: self-directed learning prevents credential debt for unmarketable skills	Mitigation — Medium: skills remain yours regardless of institutions	Free; weekly hours
135	Education & Knowledge	Now	Reading-with-children 20 min/day (the highest-yield household education act)	0	Prevention — High: daily reading prevents literacy failure at its root	Mitigation — High: effects compound across schooling	~120 hrs/yr of time — stated plainly as the price
136	Education & Knowledge	Now	Real-task testing kit: past papers, project portfolio, skill demonstrations filmed	8	Prevention — Medium: real-task evidence prevents self-deception about readiness	Mitigation — Medium: portfolios convert to employment	One-time ~$8 plus project time
137	Education & Knowledge	Future	No-device practice routine (weekly: handwriting, mental arithmetic, map navigation)	12	Prevention — High: unaided practice prevents competence atrophy	Mitigation — Medium: you remain functional and examinable offline	30 min/week indefinitely
138	Education & Knowledge	Future	Verification drill set: source-tracing exercises on real claims (family practice)	0	Prevention — High: verification drills prevent machine-error pass-through	Mitigation — Medium: errors caught at household scale	Family practice sessions
139	Education & Knowledge	Future	Offline knowledge reserve: printed field guides + manuals for your trades and region	45	Prevention — Medium: printed reserves prevent knowledge blackouts	Mitigation — Medium	One-time ~$45
140	Work & Livelihood	Past	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	0	Prevention — High: refusing the bundle prevents the capture itself	Mitigation — Medium: exit remains physically possible	A decision rule; occasional inconvenience costs
141	Work & Livelihood	Past	Union or worker-association membership	100	Prevention — Medium (collective): organization prevents individual helplessness	Mitigation — High: the historical record is unambiguous on this	Annual dues
142	Work & Livelihood	Past	Portable insurance/contributions check (confirm what is yours regardless of employer)	0	Prevention — Medium: portable benefits prevent job-loss from becoming total loss	Mitigation — Medium	Free; one audit
143	Work & Livelihood	Now	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	0	Prevention — Medium: records prevent wage theft from being deniable	Mitigation — High: claims are won on exactly this folder	5 min/week
144	Work & Livelihood	Now	Labour-board claim filed for any unpaid wages (free) + worker-centre intake	0	Prevention — Medium: filed claims prevent the practice continuing against the next worker	Mitigation — High: recovery with penalties in many jurisdictions	Free; hours per claim
145	Work & Livelihood	Now	Portable proof-of-skills file (certificates, references, portfolio — copies in documents folder)	6	Prevention — Medium: proof in hand prevents the long unemployment gap	Mitigation — High: re-employment speed is the mitigation	One-time ~$6 plus assembly time
146	Work & Livelihood	Now	Emergency fund (shared item — costed once in Economy & Finance)	0	Prevention — High	Mitigation — High: shared item — costed in Economy & Finance	Shared item — see Economy & Finance
147	Work & Livelihood	Future	Ownership participation (shared with Economy build — costed once there)	0	Prevention — Medium	Mitigation — High: shared item — costed in Economy & Finance	Shared item — see Economy & Finance
148	Work & Livelihood	Future	Transition account: dedicated savings line for retraining (target: one course + 3 months)	300	Prevention — Medium: a pre-funded account prevents retraining-by-debt	Mitigation — High: transition without panic	Savings, retained; target reached over 1–3 years
149	Work & Livelihood	Future	Collective representation maintained (union/professional dues — see Economy)	0	Prevention — Medium (collective)	Mitigation — Medium: shared item — costed in Economy & Finance	Shared item — see Economy & Finance
150	Transport & Mobility	Past	Bicycle + repair kit + panniers (the independent-mobility baseline)	160	Prevention — Medium: your own wheels prevent total siege dependence	Mitigation — High: 50 km/day of supply and evacuation range, fuel-free	Used bike ~$160; cost per trip falls below $0.50 within a year of regular use
151	Transport & Mobility	Past	Two-routes knowledge: printed local map with foot routes out of the district marked	8	Prevention — Medium: known routes prevent entrapment	Mitigation — Medium	One-time; verified yearly
152	Transport & Mobility	Past	30-day staple store (shared with Food build — costed once there)	0	Prevention — High	Mitigation — High: shared item — costed in Food & Agriculture	Shared item — see Food & Agriculture

153	Transport & Mobility	Now	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	156	Prevention — High: the pass+bike pattern prevents the car-cost trap itself	Mitigation — Medium: thousands per year returned to the household budget	Monthly pass vs. $8,000–12,000/yr car cost — the arithmetic is the argument
154	Transport & Mobility	Now	Car reserve discipline if car-dependent: never below quarter tank + maintenance sinking fund	50	Prevention — Medium: fuel and maintenance discipline prevents the breakdown-debt spiral	Mitigation — Medium	A sinking fund; minutes monthly
155	Transport & Mobility	Now	Evacuation-assistance registration for vulnerable members (municipal lists)	0	Prevention — Low	Mitigation — High: a guaranteed ride exists when private options fail	Free; one registration
156	Transport & Mobility	Future	The bicycle, maintained (see Past row — the only transport no update can brick)	40	Prevention — Medium: a maintained bicycle prevents platform dependence	Mitigation — High: the fallback that cannot be bricked	~$40/yr maintenance
157	Transport & Mobility	Future	Manual-driving competence maintained (licence + occasional practice) as vehicles automate	60	Prevention — Medium: manual competence prevents helplessness in automated failure	Mitigation — Medium: you can drive whatever still runs	Occasional practice
158	Transport & Mobility	Future	Provider diversification: accounts with 2+ mobility platforms + cash for taxis	0	Prevention — Medium: diversification prevents single-platform stranding	Mitigation — Medium	Free; minutes
159	Communication	Past	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	3	Prevention — Medium: a written plan prevents coordination collapse	Mitigation — High: rendezvous and drops work when nothing else does	One-time; rehearsed yearly
160	Communication	Past	Congregation/association membership (the historical message network)	50	Prevention — Medium: membership prevents isolation from information	Mitigation — High: the network carries warnings and help	Annual contributions
161	Communication	Past	Letter/postal discipline: addresses memorized + written copies held by two relatives	10	Prevention — Low	Mitigation — High: the channel that cannot be switched off centrally	Postage-scale costs
162	Communication	Now	Monthly export routine: contacts, photos, records downloaded to your own drive	0	Prevention — High: exports prevent platform loss from becoming record loss	Mitigation — High: restoration in hours	30 min/month
163	Communication	Now	One owned channel: personal email list / phone tree / radio net — set up and tested	0	Prevention — High: an owned channel prevents total silencing	Mitigation — High: your people remain reachable	Free tiers; setup hours
164	Communication	Now	Printed essential numbers card (family, doctors, banks, legal aid) in wallet + home	2	Prevention — Medium: the printed card prevents a dead phone erasing your network	Mitigation — High: any stranger's phone can reconnect you	One card
165	Communication	Now	Library digital-access anchor (card + weekly session if no home internet)	0	Prevention — Medium: an anchor prevents digital exclusion from becoming total	Mitigation — Medium	Free; weekly session
166	Communication	Future	Encrypted messaging as default for private communication (free apps) + device updates	0	Prevention — High: encryption prevents casual and mass surveillance of private life	Mitigation — Medium: compartmentation limits any breach	Free; setup hour
167	Communication	Future	Data-rights requests filed (access + correction) for your major files annually	10	Prevention — Medium: annual corrections prevent error-compounding in state files	Mitigation — Medium	Yearly hours
168	Communication	Future	Compartmentation basics: separate emails/aliases for finance, public, and private life	0	Prevention — High: compartments prevent one failure cascading across your whole life	Mitigation — Medium	Free; setup hours
169	Demographics & Population	Past	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	12	Prevention — Medium: the written book prevents knowledge dying with its holders	Mitigation — High: restart capital for the next generation	One family project; ~15 hours
170	Demographics & Population	Past	Skills duplication roster (each critical skill taught to 2+ people, tracked on paper)	0	Prevention — High: duplicated skills prevent single deaths disabling the household	Mitigation — Medium	Teaching hours across a year
171	Demographics & Population	Past	Civil registration discipline: every birth/marriage/death registered and copied, even in chaos	30	Prevention — Medium: registration prevents descendants' claims from being unprovable	Mitigation — High: documents unlock every future program	Small fees; one office visit per event
172	Demographics & Population	Now	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	0	Prevention — High: pooling prevents the care-cost collision itself	Mitigation — High: reciprocity absorbs shocks markets price cruelly	Coordination hours; no cash
173	Demographics & Population	Now	Benefits stack claim: child benefits, caregiver credits, parental top-ups — filed completely	0	Prevention — High: full claiming prevents the shortfalls that become crises	Mitigation — Medium: thousands per year for eligible families	Free; ~2 hours yearly
174	Demographics & Population	Now	Waitlist registrations early (childcare, family housing, seniors' housing for parents)	0	Prevention — Medium: early queue positions prevent crisis-time absence of places	Mitigation — Medium	Free; applications yearly
175	Demographics & Population	Future	50-year family location audit (where will care, work, and community exist for your household's ages)	0	Prevention — High: the audit prevents retiring into a collapsing map	Mitigation — Medium: early moves are cheap moves	4 hours per 5 years
176	Demographics & Population	Future	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	150	Prevention — Medium: lawful pathways prevent forced immobility	Mitigation — High: optionality for the whole lineage	Fees vary; timelines in years — start early
177	Demographics & Population	Future	Credential recognition pre-filing for any planned move (12-month target)	120	Prevention — Medium: pre-recognition prevents the year-three earnings penalty	Mitigation — Medium	Assessment fees; weeks of paperwork
178	Justice & Ethics	Past	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	60	Prevention — Low: archives do not prevent terror	Mitigation — High: they are the entire basis of asylum, restitution, and tribunal outcomes afterward	One-time; held abroad
179	Justice & Ethics	Past	Rights knowledge card for your system (arrest, search, detention limits) memorized + printed	3	Prevention — Medium: known limits prevent some checkpoint escalations	Mitigation — Medium: the record of violations is shaped by what you can cite	One card; memorized
180	Justice & Ethics	Past	Exit-readiness (shared with Security build — documents and routes)	0	Prevention — Medium	Mitigation — High: shared item — costed in Security & Defense	Shared item — see Security & Defense
181	Justice & Ethics	Now	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signature, or statement	0	Prevention — High: counsel before signatures prevents the irreversible errors	Mitigation — High: free advice changes outcomes in the majority of matters it touches	Free; hours per matter
182	Justice & Ethics	Now	Rights card + 'no signature under pressure' household rule (posted, rehearsed)	2	Prevention — High: the no-signature rule prevents coerced-consent losses	Mitigation — Medium	One printed card
183	Justice & Ethics	Now	Small-claims/tribunal filing for winnable civil matters (fees modest, waivable)	35	Prevention — Medium: functioning small-claims routes prevent grievances festering into feuds	Mitigation — High: recovery at citizen scale	~$35 fee (waivable); your hours are the price
184	Justice & Ethics	Now	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	0	Prevention — Medium: contemporaneous records prevent 'no evidence' dismissals	Mitigation — High: converts grievances into cases	Minutes per incident
185	Justice & Ethics	Future	Score hygiene routine: annual checks of credit, tenancy, and data-broker files; errors disputed in writing within windows	0	Prevention — High: clean files prevent automated flags from ever attaching	Mitigation — Medium: documented corrections shorten disputes	~3 hours yearly
186	Justice & Ethics	Future	Human-review demand template (written request for human review of any automated decision, kept ready)	2	Prevention — Medium: a ready template prevents the opaque 'no' from being final	Mitigation — High: human review reopens decided cases	One template, kept ready
187	Justice & Ethics	Future	Case documentation for test-case organizations (your documented case can become the precedent)	0	Prevention — Medium (collective): test cases prevent bad rules from settling	Mitigation — Medium	Free; your documented case is the contribution

Effects in Percentages — Definitions, Method, and a Worked Example. Read This First.

What "effect" means here	Every action in this system (all 187 items) is scored on two different effects, because prevention and mitigation are different mechanisms and confusing them produces bad decisions.
PREVENTION % (precise definition)	The estimated share of the problem's expected household harm that never happens because the action stops the event or the exposure from reaching this household. 100% would mean: with this action done, this problem's harm cannot reach you at all. 0% means: the action changes nothing about whether the harm arrives.
MITIGATION % (precise definition)	The estimated share of the harm that REMAINS after prevention, which the action removes when the event does reach the household. It is a percentage of the remainder, not of the original — that is why the two numbers must never simply be added.
COMBINED EFFECT % (the formula)	Combined = Prevention + (100 − Prevention) × Mitigation ÷ 100. Equivalently: harm remaining = (100 − Prevention) × (100 − Mitigation) ÷ 100, and Combined = 100 − harm remaining. Every Combined figure in this file is computed by this formula, not estimated separately.
Worked example (backwater valve)	Prevention 80%: in about 4 of 5 municipal sewer-backup events, the valve keeps sewage out entirely, so 80% of the expected harm never occurs. Mitigation 55%: in the 1 of 5 events where water still enters (other routes, extreme volume), the sump check and preparation remove about 55% of that remaining damage. Combined = 80 + 20 × 0.55 = 91%. Harm remaining: 9% of the unprotected expectation.
What kind of numbers these are	Structured expert estimates, made by the same method and standard as the Master file's success percentages: each figure was set by asking "in what share of realistic cases does this mechanism operate, and how completely?" Treat each as the centre of a range of roughly ±10 points, not as a measured trial result. Where a mechanism genuinely cannot prevent (insurance, archives, documentation), the Prevention figure is near zero ON PURPOSE — their power is mitigation, and the sheets show that honestly instead of inflating them.
How the basis column works	Each action's Basis states, in one sentence, the mechanism and its cap — why the number is what it is, and what limits it. If you disagree with a figure, the basis tells you exactly which assumption to change.
Positive and negative implications	Effects measure the problem the action answers. Implications are everything else the action does: co-benefits (positive — skills, savings, community, health beyond the target problem) and burdens (negative — upkeep, hazards, dependencies, false confidence). An action can have a modest effect and excellent implications (the documents folder), or a strong effect and real burdens (the battery station). Decide on both columns, not on effect alone.
Rollups (EFFECTS - PER SOLUTION)	Per problem, the sheet shows the simple mean of its actions' Prevention, Mitigation, and Combined percentages (unweighted — every action counts once), plus the strongest single action. Means are for comparison between problems, not a promise about any single action.

#	Action Taken (Item)	Field	Timeframe	Line Cost CA	Prevention E	Mitigation Effec	Combined Effe	Harm Rema	Basis — Why These Numbers, and What Caps Them	Positive Implications (beyond the problem)	Negative Implications (burdens the action itself creates)
1	Stored water reserve 56 L/person	Water	Past	48	55	70	86.5	13.5	Stored water removes hauling trips and queue exposure in most shortage days; when sources fail entirely it only rations the loss	Positive: stored water also covers fire first-response, cleaning, and guests; the habit teaches household logistics	Negative: weight and space burden; stale water risk if rotation lapses; false confidence beyond the stored days
2	Cloth + sand filtration setup	Water	Past	15	50	45	72.5	27.5	Improvised filtration stops most sediment and many pathogens, not dissolved contamination	Positive: filtration skill transfers to travel and camping; improves everyday taste/quality habits	Negative: slow throughput in high demand; does not remove dissolved chemicals - limits must be known
3	Fuel for boiling (1 month reserve)	Water	Past	25	30	55	68.5	31.5	Fuel guarantees the boiling option; it cannot create water that is not there	Positive: fuel stock doubles for cooking and light; warmth co-benefit	Negative: fire and CO hazard if stored or burned carelessly; fuel degrades
4	Rain barrel 200 L	Water	Past	60	45	30	61.5	38.5	Catchment adds a source in rain-fed shortages; useless in true drought	Positive: gardens watered free; roof runoff diverted from foundations	Negative: mosquito breeding if unscreened; legality varies; freeze damage
5	Certified gravity/pump water filter	Water	Now	90	80	40	88	12	A certified element blocks the contaminant class advisories are issued for; it does nothing for supply loss itself	Positive: daily drinking quality improves, not just emergency safety; reduces bottled-water plastic	Negative: element cost recurs; counterfeit elements are a real market risk
6	72-hour stored water (12 L/person drinking)	Water	Now	16	60	65	86	14	Stored volume bridges the standard 1-3 day advisory window in most events	Positive: doubles as outage and plumbing-repair reserve	Negative: takes shelf space; needs rotation discipline forever
7	Bathtub water bladder (fills in minutes at first notice)	Water	Now	35	35	60	74	26	Rapid fill converts warning time into stored supply; depends on hearing the warning	Positive: also fills for neighbours - social capital in an advisory	Negative: heavy to move when full; single-point storage
8	Water test strips (bacteria indicators + chlorine)	Water	Now	20	25	50	62.5	37.5	Testing prevents drinking contaminated water unknowingly; it detects, it does not clean	Positive: builds a testing habit that extends to wells and travel	Negative: strips test a limited panel - a pass is not a guarantee
9	Low-flow fixtures set (aerators, showerhead, toilet kit)	Water	Future	45	40	20	52	48	Demand cuts prevent shortage thresholds being reached at household scale; no effect on regional scarcity	Positive: bills fall immediately; fixtures raise home value	Negative: very low-flow fittings can disappoint; savings invisible without metering
10	30-day deep storage (barrels)	Water	Future	130	30	75	82.5	17.5	A month of autonomy converts rationing crises into inconveniences; the crisis still happens	Positive: resilience asset visible to insurers and buyers	Negative: capital locked in a fixed asset; pump/parts dependency
11	Greywater diverter for garden (where legal)	Water	Future	55	35	25	51.2	48.8	Greywater reuse prevents potable demand, mainly for gardens and flushing	Positive: garden thrives on free water; sewer load falls	Negative: detergent discipline required; some jurisdictions restrict reuse
12	Cooperative membership share (community well/storage)	Water	Future	100	45	55	75.2	24.8	Collective storage multiplies volume per dollar and adds governance in allocation	Positive: builds durable local institution and skills	Negative: governance time; collective decisions can be slow in a fast crisis
13	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	Food & Agriculture	Past	220	60	75	90	10	A 90-day store prevents hunger exposure in most single-season failures; in multi-year famine it only delays and softens	Positive: price smoothing in normal times too; cooking skills deepen	Negative: capital tied up in food; pest/spoilage losses if storage is poor; diet monotony risk
14	Garden start: seeds + soil for 10 m²	Food & Agriculture	Past	80	50	60	80	20	Gardens prevent dependence on failed markets for vegetables; staple calories still exposed	Positive: exercise, soil, seed autonomy, food quality in normal years	Negative: labour-heavy; first-year yields often disappoint; land/time competition
15	Preservation kit (jars, lids, salt, vinegar)	Food & Agriculture	Past	60	45	65	80.8	19.2	Preserved harvests prevent the winter gap; spoilage and error cap the effect	Positive: harvest gluts become assets; gifting builds community	Negative: preservation errors carry real food-safety risk; jars and space accumulate
16	Seed-saving envelopes + storage tin	Food & Agriculture	Past	12	70	30	79	21	Seed saving prevents the second-year collapse (no seed, no crop) almost entirely; first-year harm untouched	Positive: genetic diversity preserved; exchange networks form	Negative: viability bookkeeping; cross-pollination can spoil varieties
17	30-day staple buffer per adult (~25 kg grain-equivalent)	Food & Agriculture	Now	85	55	70	86.5	13.5	A buffer prevents spike-price exposure for a month; longer spikes exhaust it	Positive: fewer shopping trips; calmer monthly budgeting	Negative: rotation must be eaten through - waste if tastes change
18	Buying-club share (bulk staples with 4–10 households)	Food & Agriculture	Now	40	35	45	64.2	35.8	Bulk prices prevent retail-margin exposure on staples; availability crises unaffected	Positive: the club becomes a mutual-aid network beyond food	Negative: coordination friction; storage split across homes complicates access
19	Sprouting kit (jars + seeds)	Food & Agriculture	Now	18	30	50	65	35	Sprouts prevent fresh-food gaps of days; calorie contribution small	Positive: daily fresh nutrition for pennies; children learn food production	Negative: bacterial risk if rinsing lapses; small calorie share
20	Community fridge / food bank registration + food-handler basics	Food & Agriculture	Now	5	20	65	72	28	Food programs do not prevent price rises; they remove most of the hunger consequence	Positive: dignity-preserving routes stabilize nutrition for everything else	Negative: queues and limited choice; stigma if services are poorly designed
21	Perennial food plants (berry bushes, fruit tree, herb bed)	Food & Agriculture	Future	90	55	60	82	18	Perennial production prevents purchased-food dependence for its share of the diet, compounding yearly	Positive: property value and ecology rise; harvests gift and trade	Negative: years to first yield; pruning skill needed; wildlife shares the harvest
22	Soil building: compost system + cover crop seed	Food & Agriculture	Future	45	40	55	73	27	Soil water-holding prevents drought-year crop failure in many seasons; extreme years still fail	Positive: input bills fall yearly; soil becomes an appreciating asset	Negative: slow payoff tests patience; cover crops take beds out of rotation
23	Dehydrator (entry electric or solar build)	Food & Agriculture	Future	70	35	60	74	26	Drying prevents glut waste and bridges gaps; it creates no new calories	Positive: preserves any glut (garden, sales, gifts); no fridge space or power	Negative: texture/taste limits; electricity for electric models
24	Co-op / land-trust membership contribution	Food & Agriculture	Future	75	40	55	73	27	Shared land removes rent from the food stack and keeps production local under price shocks	Positive: land access without ownership; deep community roots	Negative: dues and workdays; collective governance overhead
25	Household medical kit: thermometer, pulse oximeter, ORS packets, fever medication (2 weeks)	Health & Medicine	Past	55	35	60	74	26	A stocked kit treats the household illness burden that is treatable at home; epidemics exceed it	Positive: everyday injuries and illnesses handled without clinics	Negative: expiry management; a kit invites overconfidence beyond its scope
26	Isolation-room setup (separate bedding, basin, waste bag system)	Health & Medicine	Past	30	55	40	73	27	Separation and dedicated linens cut household transmission roughly in half in respiratory/fecal illness	Positive: habits persist into ordinary flu seasons	Negative: space and privacy strain in small homes
27	Vaccination catch-up (all free public programs)	Health & Medicine	Past	0	75	30	82.5	17.5	Vaccination prevents the named diseases at high efficacy; other illness untouched	Positive: protects the whole community (herd effect), not only the household	Negative: none material at household level beyond appointment time
28	Paper medical records folder (vaccines, conditions, medications, allergies)	Health & Medicine	Past	6	15	70	74.5	25.5	Records prevent almost no illness; they remove most of the chaos cost when illness strikes	Positive: the folder serves every other system in this file (evacuation, legal, benefits)	Negative: sensitive data concentrated - must be guarded
29	Drug coverage optimization: benefits audit + generic substitution review with pharmacist	Health & Medicine	Now	0	45	35	64.2	35.8	Optimization removes duplicate/interacting prescriptions - a leading avoidable-harm class	Positive: fewer side effects; lower pharmacy bills permanently	Negative: requires bringing everything, every time - small recurring effort
30	Home infection-control basics (soap stock, disinfectant, masks x50, gloves)	Health & Medicine	Now	35	50	45	72.5	27.5	Household hygiene stock prevents a large share of everyday transmission	Positive: fewer sick days across the whole household yearly	Negative: consumable stream creates waste; stock rotation
31	Dental prevention kit + one hygiene visit/yr	Health & Medicine	Now	120	40	70	82	18	Annual dental prevention removes most emergency-class dental cost before it forms	Positive: pain, sleep, and eating improve beyond the avoided emergencies	Negative: annual cost is certain while the emergency is probabilistic - feels expensive until it doesn't
32	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy take-back)	Health & Medicine	Now	0	30	20	44	56	Take-back prevents leftover misuse and resistance selection pressure at household scale	Positive: protects water systems and everyone's future antibiotics	Negative: none material
33	Room HEPA unit for the sick room / bedroom	Health & Medicine	Future	130	35	55	70.8	29.2	Clean air prevents a share of respiratory/cardiovascular burden; exposure elsewhere remains	Positive: sleep and cognition improve with cleaner air daily	Negative: filter cost recurs forever; noise on high settings
34	Ventilation upgrade: CO2 monitor + window/exhaust habits	Health & Medicine	Future	110	30	50	65	35	Monitoring prevents silent bad-air exposure at home; it informs, others must act	Positive: data improves ventilation habits and housing choices	Negative: numbers can feed anxiety if read without context
35	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	Health & Medicine	Future	45	45	65	80.8	19.2	Home BP control prevents the silent cascade to stroke/heart failure in detected cases	Positive: detected risks improve insurability and clinical care	Negative: measurement anxiety in some people; cuff accuracy depends on fit
36	Reserve medication buffer (1 month extra of essentials, rotated, where legal)	Health & Medicine	Future	40	10	55	59.5	40.5	Rotation prevents none of the shortage; it removes over half the personal interruption harm	Positive: prescriber relationship strengthens through honest planning	Negative: strictly bounded by law and safety - never beyond prescribed amounts
37	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blanket fort supplies	Energy	Past	60	45	55	75.2	24.8	Sealing and one-room heating prevent cold-illness exposure in most cold snaps	Positive: comfort and bills improve in normal winters too	Negative: sealing too tight without ventilation harms air quality
38	Fuel reserve (safe storage: gas canisters / charcoal / wood)	Energy	Past	70	35	60	74	26	Fuel stock prevents the no-heat day; consumption caps total duration	Positive: cooking flexibility year-round	Negative: storage hazard; degradation bookkeeping
39	Layered clothing upgrade (wool base layers x2 per person)	Energy	Past	90	50	40	70	30	Insulating the body prevents most cold harm regardless of the building	Positive: outdoor work and sport benefit from the same layers	Negative: quality wool costs upfront; care requirements
40	Pipe protection: insulation sleeves + shutoff knowledge	Energy	Past	25	30	75	82.5	17.5	Pipe insulation prevents the burst-pipe loss class almost entirely where fitted	Positive: prevents the single most expensive household water loss	Negative: none material beyond the fitting hour
41	Battery power station 1 kWh (or 12V battery + inverter build)	Energy	Now	420	25	80	85	15	Backup power prevents nothing about the outage; it removes most of its household damage (food, medical, light, comms)	Positive: doubles for camping, work sites, and off-grid projects; silent vs. generators	Negative: the file's priciest item; capacity tempts oversizing; battery ages even unused
42	LED lanterns x3 + headlamp	Energy	Now	35	20	60	68	32	Reliable light prevents accident and security losses during outages	Positive: night work and reading improve in normal life too	Negative: cell management; cheap cells disappoint
43	Safe backup cooking: single-burner butane stove + 4 canisters	Energy	Now	45	15	65	70.2	29.8	A safe stove prevents the restaurant/takeaway cost spiral and cold-food days	Positive: outdoor cooking in summer keeps heat out of the kitchen	Negative: indoor misuse is a CO danger - discipline is the product
44	Fridge discipline kit: thermometer + cooler + ice packs	Energy	Now	30	20	75	80	20	Ice retention prevents most food-spoilage loss in the standard 24-48h outage	Positive: picnic and travel utility	Negative: cooler space in small kitchens
45	Solar trickle panel 100–200 W for the battery station	Energy	Future	160	35	55	70.8	29.2	Generation prevents grid dependence for its capacity share, sun permitting	Positive: bills fall in normal times; pairs with the battery for full autonomy	Negative: roof/orientation dependent; generation is weather's decision
46	Home insulation priority pack (attic hatch seal, window film, draft proofing)	Energy	Future	85	45	50	72.5	27.5	Insulation prevents the energy-price harm class permanently, in all weathers	Positive: comfort rises and bills fall in all seasons, crisis or not	Negative: disruptive install weekend; quality varies by installer
47	Community energy co-op share	Energy	Future	100	40	50	70	30	Ownership shares prevent retail-price exposure for the co-op share of supply	Positive: local jobs and democratic ownership of infrastructure	Negative: slow institutional returns; shares are illiquid
48	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserve small bills	Energy	Future	55	15	70	74.5	25.5	The outage kit prevents no outage; it removes most of the scramble, darkness, and silence costs	Positive: the radio habit reconnects the household to local information	Negative: cash float must be defended from everyday spending

49	Documents go-bag (waterproof pouch: ID, property, insurance, photos of every room's contents)	Shelter & Housing	Past	18	10	85	86.5	13.5	Documentation prevents almost no disaster; it removes most of the uncompensated-loss share afterward	Positive: the same folder serves benefits, legal, and medical systems	Negative: concentrated sensitive data - theft/loss risk if unguarded
50	72-hour go-kit per person (clothes, medication, water, light, food bars)	Shelter & Housing	Past	65	15	75	78.8	21.2	A packed bag prevents no evacuation; it removes most of the first-72-hours harm	Positive: doubles as travel/camping readiness	Negative: kits for children need yearly resizing; stored food tempts everyday use
51	Home inventory video + off-site copy	Shelter & Housing	Past	0	10	80	82	18	An inventory prevents no loss; documented claims are paid at multiples of undocumented ones	Positive: resale and warranty value in normal times	Negative: off-site copy must be maintained
52	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	Shelter & Housing	Past	70	70	60	88	12	Alarms and extinguishers prevent fire deaths at very high rates; property harm reduced, not removed	Positive: everyday fire readiness, not just disaster	Negative: extinguisher confidence can delay evacuation if misjudged - alarm first, always
53	Tenant rights pack: your jurisdiction's rules printed + legal-aid and tenant-union numbers	Shelter & Housing	Now	4	55	45	75.2	24.8	Rights knowledge prevents most unlawful-eviction and illegal-charge outcomes before they happen	Positive: deposits returned and illegal charges refused in ordinary tenancies too	Negative: asserting rights can strain landlord relations - documentation is the shield
54	Housing documentation habit: dated photos at move-in/out, all landlord communication in writing	Shelter & Housing	Now	0	30	70	79	21	Contemporaneous evidence prevents few disputes; it decides most of the ones that occur	Positive: the evidence habit serves work, consumer, and legal life	Negative: small recurring time; storage organization
55	Rent-bank / eviction-prevention applications pre-filed knowledge + emergency contact list	Shelter & Housing	Now	0	50	60	80	20	Pre-known rent relief prevents the arrears-to-eviction cascade at its first step in most cases	Positive: knowledge transfers to friends and family facing arrears	Negative: programs are rationed; knowing is necessary, not sufficient
56	Shared-housing option prepared (one identified co-housing/roommate arrangement as Plan B)	Shelter & Housing	Now	0	35	65	77.2	22.8	A standing Plan B prevents homelessness in most tenancy failures where it exists	Positive: deepens the friendships it rests on	Negative: depends on others' capacity in the same crisis
57	Property risk check before buying/renewing: flood, fire, insurance-availability maps (free public data)	Shelter & Housing	Future	0	60	30	72	28	The audit prevents the stranded-asset purchase - the single largest avoidable loss in this field	Positive: every future purchase decision improves, not only this one	Negative: analysis can delay necessary moves if thresholds aren't pre-set
58	Full insurance review incl. overland flood / sewer backup riders	Shelter & Housing	Future	180	5	80	81	19	Insurance prevents no flood; it transfers most of the financial loss when terms are right	Positive: contents and liability cover ride along in bundled policies	Negative: premiums are certain, payouts conditional; exclusions must be read
59	Portability reserve: keep at least 50% of net worth out of any single exposed property (financial accounts, skills)	Shelter & Housing	Future	0	45	45	69.8	30.2	Portability prevents lock-in harm; it cannot prevent the move itself	Positive: freedom compounds - careers and family choices widen	Negative: property wealth accumulates slower - a real trade-off, priced openly
60	Buyout/relocation program registration (where offered) — filed early	Shelter & Housing	Future	0	40	50	70	30	Early registration prevents the no-offer outcome when buyouts/relocations are funded	Positive: early registrants also receive information first	Negative: programs may never fund - position costs nothing, expectations must stay low
61	Early-warning habit: named indicators list + decided trigger points ('if X happens, we leave within 48 h')	Security & Defense	Past	0	55	40	73	27	Pre-decided triggers prevent the worst timing error - leaving too late - in most escalations	Positive: decision discipline transfers to every other emergency in this file	Negative: leaving early has real costs when the threat passes - accepted as insurance premium
62	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	Security & Defense	Past	90	25	75	81.2	18.8	Translated credentials prevent few persecutions; they remove most of the restart-from-zero cost	Positive: credentials also speed ordinary employment abroad	Negative: translation costs recur as documents renew
63	Exit-route plan: two routes, rendezvous points, transport costs pre-estimated and saved	Security & Defense	Past	0	40	65	79	21	A carried reserve prevents destitution at arrival in most flight paths	Positive: the reserve habit builds general financial resilience	Negative: cash carried is cash at risk - forms matter (small bills, split storage)
64	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in signals)	Security & Defense	Past	0	45	55	75.2	24.8	An organized roster prevents isolation harm and coordinates most mutual protection	Positive: the roster IS the community - it pays socially in peacetime	Negative: organizing time; roster data must be guarded
65	Home target-hardening basics: solid locks, strike plates with 3-inch screws, window locks, lighting	Security & Defense	Now	85	60	30	72	28	Hardening prevents most opportunistic entry attempts (offenders divert to softer targets)	Positive: insurance premiums and sleep both improve	Negative: hardening can look forbidding; locks fail if keys are unmanaged
66	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	Security & Defense	Now	60	25	70	77.5	22.5	Recording prevents few incidents; it converts most property loss into recovery/prosecution	Positive: package-theft and dispute evidence in ordinary life	Negative: privacy discipline required - aim at your own property only
67	Route and routine discipline + check-in system with one contact	Security & Defense	Now	0	55	35	70.8	29.2	Routine discipline prevents a large share of opportunistic street crime exposure	Positive: walking more improves health as a side effect	Negative: routine variation costs convenience
68	Victim-compensation and protective-order applications knowledge (legal aid)	Security & Defense	Now	0	15	65	70.2	29.8	Knowing compensation routes prevents no crime; it recovers most of the financial aftermath	Positive: the knowledge serves anyone you help afterward	Negative: compensation processes are slow - recovery is not immediate
69	Venue standards habit: choose venues with visible security, exits known on entry (30-second rule)	Security & Defense	Future	0	40	45	67	33	Situational awareness prevents a share of crowd-incident exposure; no effect on the incident itself	Positive: calm in crowds generally; the habit reassures children	Negative: hypervigilance risk if the habit becomes anxiety - 30 seconds, then enjoy the event
70	Personal data footprint reduction: data-broker removals, minimal public exposure	Security & Defense	Future	30	50	40	70	30	Footprint reduction prevents targeting data existing; determined actors remain	Positive: less spam, fewer scams, cleaner digital life daily	Negative: some services become harder to use; brokers respawn - annual sweep needed
71	Support for regulation: specific, written asks to representatives (drone ID, autonomous weapons limits)	Security & Defense	Future	0	35	30	54.5	45.5	Collective advocacy prevents some surveillance measures from being adopted at all	Positive: civic muscle built for every other cause	Negative: slow, uncertain payoffs; campaign fatigue
72	Home/business camera + signage (deterrence + evidence)	Security & Defense	Future	90	30	65	75.5	24.5	Visible recording deters a share of incidents and evidences most of the rest	Positive: deters package theft immediately	Negative: neighbour-relations care (aim and signage); storage upkeep
73	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	Governance & Law	Past	75	20	80	84	16	Archives prevent few persecutions; they decide most restitution, asylum, and tribunal outcomes afterward	Positive: archives become history itself - the community's memory survives	Negative: archive holders can be targeted for holding them; copies must be discreet
74	Qualifications in portable form (transcripts, licences, references sealed and copied)	Governance & Law	Past	40	25	75	81.2	18.8	Verified copies prevent few losses; they restore employability and identity in exile	Positive: credentials serve ordinary careers too	Negative: certification fees at each renewal
75	Quiet-network membership (congregation, union, association dues)	Governance & Law	Past	60	35	55	70.8	29.2	Networks prevent isolation harms; coordinated flight and shelter reduce losses when purges come	Positive: networks double as cultural and mutual-aid life in peacetime	Negative: dues and time; infiltration risk in true terror states
76	Access-to-information request kit (templates + filing fees where charged)	Governance & Law	Now	15	45	40	67	33	Records requests prevent decisions being made on hidden files; they correct a share of them	Positive: transparency habits improve every dealing with institutions	Negative: slow responses; requests can be stonewalled - persistence is the price
77	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	Governance & Law	Now	5	40	50	70	30	Written complaints prevent repeat violations against you; patterns from many complaints change practice	Positive: each upheld complaint raises the standard for everyone after you	Negative: time in process; occasional adversarial relations with offices you still need
78	Civic membership: one watchdog, union, or ratepayers' organization	Governance & Law	Now	40	35	45	64.2	35.8	Sustained membership prevents capture of the bodies that decide local matters, proportionally to participation	Positive: real influence over local decisions; civic skill compounds	Negative: meetings are the tax; burnout attrition is the failure mode
79	Local verification habit: attend/join one local body (assembly, board, co-op) and keep its records	Governance & Law	Future	0	40	40	64	36	Local verification prevents rumour-based household decisions; it cannot restore national trust alone	Positive: local trust networks strengthen every other system	Negative: verification labour is unpaid and endless
80	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	Governance & Law	Future	0	20	65	72	28	A filed record prevents little manipulation; it decides later disputes about what happened	Positive: your file becomes evidence others can cite	Negative: record-keeping minutes, forever
81	Support independent local media (one subscription)	Governance & Law	Future	60	35	40	61	39	Independent channels prevent total information monopoly for their subscribers	Positive: independent media survives because of exactly these subscriptions	Negative: paywalls fragment access - library cards are the equalizer
82	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	Economy & Finance	Past	0	65	45	80.8	19.2	Deposit splitting prevents total loss in any single bank failure, up to insurance limits	Positive: diversification discipline serves ordinary banking safety too	Negative: more accounts to track; small admin overhead
83	Durable-goods and skills reserve (tools, preserved food, tradeable skills)	Economy & Finance	Past	150	55	50	77.5	22.5	Durable value prevents cash evaporation under currency collapse for the converted share	Positive: tools and staples serve daily life, not only collapse	Negative: storage space; durable goods need maintenance
84	Debt minimalism in boom times (fixed-rate only, no margin, no speculation on credit)	Economy & Finance	Past	0	70	35	80.5	19.5	Refusing mania assets prevents the classic wipeout; the crash still harms wages and work	Positive: the rule also prevents ordinary speculative losses	Negative: you will watch some manias profit others - the emotional cost is real
85	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	Economy & Finance	Now	500	30	75	82.5	17.5	The fund prevents few emergencies; it removes most of their interest-and-debt damage	Positive: calm - the fund buys sleep before it buys anything else	Negative: building it feels slow; the first $500 is the hardest
86	Debt audit + avalanche order (highest rate first) printed and posted	Economy & Finance	Now	0	35	55	70.8	29.2	The debt map prevents minimum-payment traps; avalanche order cuts total interest substantially	Positive: credit score and options improve as balances fall	Negative: avalanche discipline cuts discretionary spending during the climb
87	Free insolvency-trustee first consultation + credit-counselling session (non-profit)	Economy & Finance	Now	0	25	60	70	30	Professional mapping prevents wrong-order moves; execution remains yours	Positive: a map you trust ends advice-shopping	Negative: free advice is general, not personal - verify against your numbers
88	Benefits and credits claim check (tax filing + benefits finder)	Economy & Finance	Now	0	55	50	77.5	22.5	Filing prevents benefit loss entirely for eligible credits - the harm here is mostly non-claiming	Positive: benefits claimed also stabilize the local services funded by uptake	Negative: annual filing discipline; audits possible - records are the answer
89	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	Economy & Finance	Future	100	60	45	78	22	Ownership converts displacement risk into asset income for the accumulated share	Positive: ownership income compounds into retirement security	Negative: market risk is real; the long view is the price of the return
90	Human-irreducible skill course (care, craft, repair, judgment trades)	Economy & Finance	Future	250	50	55	77.5	22.5	A second trade-level skill prevents total income loss in most displacement paths	Positive: the second skill enriches ordinary years (weekend income, options)	Negative: 40 hours is a real investment before any return
91	Union/professional-body membership	Economy & Finance	Future	120	45	50	72.5	27.5	Collective terms prevent the worst displacement terms; they rarely prevent displacement itself	Positive: solidarity protects coworkers, not only you	Negative: dues; collective outcomes are compromises
92	Home hazard test when buying older property (lead paint, asbestos survey before renovation)	Technology & AI	Past	180	60	35	74	26	Testing prevents exposure in untested materials almost entirely when acted on; installed exposure remains	Positive: findings raise property value and family health knowledge	Negative: positive findings can force expensive decisions - knowledge has a price
93	Exposure records file (employers, products, dates) kept permanently	Technology & AI	Past	4	15	70	74.5	25.5	Records prevent no past exposure; they decide compensation and care decades later	Positive: the folder serves insurance and medical systems too	Negative: none material beyond upkeep minutes
94	Adoption delay rule for unstudied non-essentials (wait for the 2nd generation / safety data)	Technology & AI	Past	0	55	30	68.5	31.5	The waiting rule prevents first-generation defect exposure for most purchases	Positive: second-generation products are cheaper AND better - the rule pays	Negative: waiting costs convenience and occasionally a needed capability

95	Password manager + MFA on email and banking (the two master keys)	Technology & AI	Now	0	75	50	87.5	12.5	A password manager + MFA prevents the large majority of account-takeover paths	Positive: digital life becomes calmer everywhere, not only safer	Negative: vault is a single point - backup codes discipline required
96	3-2-1 backup: external drive + cloud copy of irreplaceables	Technology & AI	Now	95	30	80	86	14	Backups prevent almost no attacks; tested backups remove most of the data-loss damage	Positive: photos and records gain permanence beyond emergencies	Negative: restore drills take an hour a year; untested backups are theatre
97	Automatic updates ON + router firmware check (quarterly, calendared)	Technology & AI	Now	0	65	45	80.8	19.2	Prompt updates prevent exploitation of known vulnerabilities - the majority of commodity attacks	Positive: devices run faster and longer on current software	Negative: rare update breakage; router admin learning curve
98	Verification protocol card (callback numbers printed; 'urgent = verify by second channel' rule)	Technology & AI	Now	2	60	55	82	18	The no-transfer rule prevents the dominant household fraud path (urgent impersonation)	Positive: the rule protects elderly relatives you teach it to	Negative: genuine urgent requests face friction too - the code word solves it
99	Analogue life-line kit: paper originals of ID/property/credentials, cash reserve, offline maps and contacts	Technology & AI	Future	40	20	75	80	20	A paper vault prevents no lockout scenario; it restores identity and function in most of them	Positive: paper records impress in legal and estate matters too	Negative: paper can burn or flood - two locations rule applies
100	Data portability routine: monthly export of contacts, photos, records from every platform	Technology & AI	Future	0	25	70	77.5	22.5	Exports prevent no platform failure; they remove most of the record-loss damage	Positive: migration freedom between services forever	Negative: 30 minutes a month, forever
101	Offline competence maintenance: one unaided skill practised regularly (navigation, arithmetic, writing)	Technology & AI	Future	0	55	40	73	27	Unaided competence prevents dependence harm in automated-failure scenarios	Positive: unaided skill is confidence in every outage and exam	Negative: practice time competes with tool convenience
102	Soil care basics: compost return, cover, no bare soil over winter	Environment & Climate	Past	45	50	55	77.5	22.5	Soil care prevents a large share of yield collapse in bad years; multi-year drought still wins	Positive: soil pays in every ordinary year, not only bad ones	Negative: labour is the input; results take seasons
103	Water-wise planting (drought-matched species, mulch)	Environment & Climate	Past	60	50	45	72.5	27.5	Matched planting prevents water-stress losses in most dry seasons	Positive: biodiversity and beauty rise with matched planting	Negative: establishment watering in year one
104	Fuel efficiency: improved cookstove / insulation of heat	Environment & Climate	Past	80	40	50	70	30	Efficiency prevents fuel scarcity from becoming cold and hunger for the saved share	Positive: fuel savings are permanent and immediate	Negative: stove quality varies; flue maintenance is safety-critical
105	Address risk check (flood/fire/heat maps, free) + insurer confirmation in writing	Environment & Climate	Now	0	55	35	70.8	29.2	The risk audit prevents buying or staying blind into mapped hazards for most households	Positive: the audit improves insurance negotiations immediately	Negative: unwelcome answers can force hard decisions early - that is their value
106	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	Environment & Climate	Now	250	80	55	91	9	A backwater valve prevents interior sewer damage in the large majority of municipal backup events	Positive: insurers and municipalities reward it; basements stay dry in ordinary storms too	Negative: valve needs yearly testing; failure mode is silent if untested
107	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback, water stock	Environment & Climate	Now	65	35	65	77.2	22.8	A prepared cool room prevents heat illness at home; outdoor and work exposure remains	Positive: comfort in every heatwave, not only emergencies	Negative: film/blinds slightly darken rooms
108	Go-bag + document pouch (shared with Shelter build — counted once there)	Environment & Climate	Now	0	15	75	78.8	21.2	Shared item - effect counted in Shelter & Housing (go-bag + document pouch)	Positive: shared item - see Shelter & Housing	Negative: shared item - see Shelter & Housing
109	Location decision audit every 5 years (risk trend, insurance trend, water trend for your address)	Environment & Climate	Future	0	55	40	73	27	Five-year audits prevent stranded-asset lock-in when acted on early	Positive: decisions improve across all assets, not only property	Negative: five-yearly discipline; data can be gamed by sellers - use authority sources
110	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	Environment & Climate	Future	200	50	50	75	25	Portable income prevents location-bound livelihood collapse	Positive: portable skills enrich the life you have now	Negative: investment competes with present consumption
111	Collective relocation readiness: community/municipal plan participation	Environment & Climate	Future	0	35	55	70.8	29.2	Collective plans prevent chaotic individual flight; they cannot prevent the move	Positive: collective plans protect neighbours who cannot plan alone	Negative: meetings; plans can be captured by the loudest voices
112	Source-tracing habit taught at home (claim → origin → evidence) — a dinner-table practice, not a purchase	Information & Media	Past	0	50	35	67.5	32.5	The two-question habit prevents a large share of rumour-driven decisions	Positive: the habit makes citizens formidable in ordinary life too	Negative: slower decisions; social friction with the credulous
113	Independent media support: one exile/public/independent subscription or donation	Information & Media	Past	50	40	45	67	33	Funded channels prevent information monopoly for their audiences in most crackdowns	Positive: journalism exists in peacetime because of these funders	Negative: steady cost; capture risk exists everywhere - diversify sources
114	Battery shortwave/AM radio for foreign and emergency broadcasts	Information & Media	Past	35	30	60	72	28	A battery radio prevents total information blackout in infrastructure loss	Positive: local news and weather in ordinary life	Negative: batteries and preset upkeep
115	Two-source rule card + one scheduled news window (a household protocol, posted)	Information & Media	Now	2	55	40	73	27	Cross-checking prevents most single-source manipulation of household decisions	Positive: family decisions improve across money, health, and safety	Negative: seconds-to-minutes per claim, daily
116	Primary-source subscriptions: one quality outlet + official data sources bookmarked	Information & Media	Now	90	40	50	70	30	One quality source prevents the outrage-diet harm for its share of attention	Positive: depth reading is pleasure, not only defence	Negative: subscription cost or library waits
117	Fact-check and provenance tools bookmarked; reverse image search learned	Information & Media	Now	0	45	50	72.5	27.5	Verification tools prevent most image/context deceptions that reach the household	Positive: you become the household's verifier - a valued role	Negative: tools change; the skill needs refreshing
118	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	Information & Media	Future	0	70	45	83.5	16.5	A family code word prevents the voice-clone fraud path almost entirely	Positive: the rule also stops ordinary phone scams against grandparents	Negative: code words must be refreshed as family changes
119	Paper originals vault (identity, property, credentials) + notarial copies of the critical five	Information & Media	Future	75	25	75	81.2	18.8	Notarized originals prevent few forgeries; they decide most authenticity disputes	Positive: notarized sets simplify estates and disputes in normal life	Negative: notary fees per set; renewal bookkeeping
120	Content-credential tools adopted (signed capture apps) for important records	Information & Media	Future	0	40	55	73	27	Provenance tools prevent unattributed synthetic content being trusted as yours	Positive: creators gain a durable authenticity asset	Negative: extra steps in every publishing workflow
121	Routine and ritual structure (fixed sleep, meals, work, worship/rest — a written household rhythm)	Psychology & Mind	Past	0	45	55	75.2	24.8	A protected rhythm prevents a large share of breakdown under sustained stress	Positive: rhythm improves ordinary days, not only crises	Negative: rigidity risk - the rhythm must bend or it breaks
122	Mutual storytelling / testimony practice (regular, structured, within family or congregation)	Psychology & Mind	Past	0	30	60	72	28	Testimony prevents little of the original harm; it removes much of its isolation and transmission	Positive: tellers report relief; listeners gain the map of what happened	Negative: retelling can retraumatize without pacing - stop rules are the safeguard
123	Children-first protection of sleep, schooling, and play (the intergenerational shield)	Psychology & Mind	Past	0	50	55	77.5	22.5	Protected childhood stability prevents a large share of intergenerational transmission	Positive: protected children outperform in every later measure this file tracks	Negative: adults carry the regulation load - support for them is not optional
124	Sleep protocol: fixed window, morning light (free), screens out of the bedroom, caffeine cutoff	Psychology & Mind	Now	25	55	45	75.2	24.8	A defended sleep window prevents much of the cognitive/mood cascade of modern schedules	Positive: sleep pays in cognition, mood, immunity, and safety daily	Negative: two-week adjustment; social schedules resist
125	Movement prescription: 60 min/day walking built into commute/errands + one strength session weekly	Psychology & Mind	Now	0	50	50	75	25	Daily movement prevents a large share of the common mood/energy burden	Positive: movement pays in every physical system simultaneously	Negative: time - 7 hrs/week stated plainly; injury risk if ramped carelessly
126	One weekly in-person group (see Pareto action #6 — costed once)	Psychology & Mind	Now	0	45	55	75.2	24.8	Belonging prevents loneliness harms; the group also catches crises early (shared item)	Positive: belonging is also fun, work leads, and care in ordinary weeks	Negative: attendance discipline; groups vary in quality - eight sessions before judging
127	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	Psychology & Mind	Now	0	20	70	76	24	A crisis plan prevents few crises; it removes most of the wrong-door, wrong-hour damage	Positive: knowing the doors lowers daily anxiety, not only crisis harm	Negative: none material
128	Synthetic-companionship budget rule (fixed hours + money cap, written, reviewed monthly)	Psychology & Mind	Future	0	55	40	73	27	Budgets for synthetic contact prevent dependency formation at the design stage	Positive: money and hours return to real life monthly	Negative: synthetic comforts are genuinely pleasant - the budget is a real sacrifice
129	Real-world anchor subscriptions: one club/league/class with dues (skin in the game raises attendance)	Psychology & Mind	Future	80	50	50	75	25	A real-world anchor prevents synthetic substitution for its share of social need	Positive: the anchor becomes friendship, skill, and standing	Negative: dues; weekly showing-up is the price
130	Device architecture: companions/feeds off the phone's home screen, notifications batched	Psychology & Mind	Future	0	45	45	69.8	30.2	Attention friction prevents a large share of compulsive-use harm	Positive: reclaimed attention funds reading, family, and sleep	Negative: a week of itch; some serendipity is lost with the feed
131	Home copying kit: notebooks, and one body of knowledge memorized per child (the unconfiscatable copy)	Education & Knowledge	Past	10	45	55	75.2	24.8	A copied canon prevents knowledge loss in most confiscations of the originals	Positive: copied texts seed schools after crises	Negative: copying labour is heavy; errors propagate if unchecked
132	Printed core references shelf (first aid, agriculture, mathematics, local law — bought used)	Education & Knowledge	Past	40	40	50	70	30	A home shelf prevents education blackout for its subjects; depth is limited	Positive: the shelf serves neighbours and children beyond the household	Negative: space; books age - reference editions date
133	Literacy defence: every child taught to read early at home if school fails (phonics materials)	Education & Knowledge	Past	25	65	45	80.8	19.2	Home literacy prevents the master loss - illiteracy multiplies every other problem in this file	Positive: literacy compounds into every other row of this file	Negative: adult time daily for years - the real price, stated
134	Library-as-university setup: card + course platforms + a written self-curriculum	Education & Knowledge	Now	0	40	50	70	30	Library-anchored learning prevents credential debt for most self-directed goals	Positive: skills and portfolios outlast any credential cycle	Negative: self-direction demands discipline institutions usually supply
135	Reading-with-children 20 min/day (the highest-yield household education act)	Education & Knowledge	Now	0	60	40	76	24	Daily reading prevents reading failure trajectories in most children so read to	Positive: reading households raise readers - the effect is generational	Negative: the daily hour is the price; screens compete for it
136	Real-task testing kit: past papers, project portfolio, skill demonstrations filmed	Education & Knowledge	Now	8	35	55	70.8	29.2	Demonstrated proof prevents credential-mismatch harm at hiring; it cannot create vacancies	Positive: proof portfolios open doors credentials cannot	Negative: assessment quality varies - external judges matter
137	No-device practice routine (weekly: handwriting, mental arithmetic, map navigation)	Education & Knowledge	Future	12	55	40	73	27	Unaided practice prevents competence atrophy in automated environments	Positive: confidence in outages and exams alike	Negative: 30 min/week forever - small but perpetual
138	Verification drill set: source-tracing exercises on real claims (family practice)	Education & Knowledge	Future	0	50	45	72.5	27.5	Verification drills prevent machine errors passing into household decisions	Positive: verifiers catch errors for employers and family too	Negative: verification friction slows genuine conveniences
139	Offline knowledge reserve: printed field guides + manuals for your trades and region	Education & Knowledge	Future	45	25	65	73.8	26.2	Printed reserves prevent knowledge blackout damage in connectivity loss	Positive: printed volumes are gifts and inheritance	Negative: storage; editions date slowly
140	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	Work & Livelihood	Past	0	60	35	74	26	Refusing the bundle prevents company-town capture almost entirely where exit exists	Positive: independence improves bargaining power in all jobs	Negative: unbundled living can cost more cash month to month

141	Union or worker-association membership	Work & Livelihood	Past	100	45	50	72.5	27.5	Organization prevents individual helplessness in wage and hour disputes	Positive: collective power improves terms for non-members too	Negative: dues; internal politics are real work
142	Portable insurance/contributions check (confirm what is yours regardless of employer)	Work & Livelihood	Past	0	40	55	73	27	Portable benefits prevent job loss becoming total loss for the covered share	Positive: portable coverage stabilizes family life across jobs	Negative: portable plans can be thinner than the best employer plans
143	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	Work & Livelihood	Now	0	35	65	77.2	22.8	Records prevent few wage thefts; they win most of the claims they support	Positive: records also support EI, tax, and pension claims	Negative: 5 minutes weekly, forever; employer friction possible
144	Labour-board claim filed for any unpaid wages (free) + worker-centre intake	Work & Livelihood	Now	0	30	60	72	28	Filing prevents repeat violations against the next worker; recovery is the mitigation	Positive: filed claims raise standards for the whole workplace	Negative: process time; retaliation risk managed by documentation
145	Portable proof-of-skills file (certificates, references, portfolio — copies in documents folder)	Work & Livelihood	Now	6	35	60	74	26	Proof in hand prevents the long-gap outcome; it shortens most re-employment searches	Positive: the portfolio serves promotions, not only exits	Negative: assembly hours up front
146	Emergency fund (shared item — costed once in Economy & Finance)	Work & Livelihood	Now	0	30	60	72	28	Shared item - effect counted in Economy & Finance (benefits audit/filing)	Positive: shared item - see Economy & Finance	Negative: shared item - see Economy & Finance
147	Ownership participation (shared with Economy build — costed once there)	Work & Livelihood	Future	0	55	50	77.5	22.5	Shared item - effect counted in Economy & Finance (ownership accumulation)	Positive: shared item - see Economy & Finance	Negative: shared item - see Economy & Finance
148	Transition account: dedicated savings line for retraining (target: one course + 3 months)	Work & Livelihood	Future	300	50	55	77.5	22.5	A transition fund prevents retraining-by-debt in most displacement paths	Positive: the fund buys choice of next work, not just survival	Negative: years of contributions before it matters
149	Collective representation maintained (union/professional dues — see Economy)	Work & Livelihood	Future	0	45	50	72.5	27.5	Shared item - effect counted in Economy & Finance (collective terms)	Positive: shared item - see Economy & Finance	Negative: shared item - see Economy & Finance
150	Bicycle + repair kit + panniers (the independent-mobility baseline)	Transport & Mobility	Past	160	45	55	75.2	24.8	Your own wheels prevent siege/isolation dependence for most local movement	Positive: health, savings, and local knowledge in ordinary times	Negative: weather and distance limits; theft risk
151	Two-routes knowledge: printed local map with foot routes out of the district marked	Transport & Mobility	Past	8	40	45	67	33	Known routes prevent entrapment in most disruptions	Positive: route knowledge serves visitors, children, emergencies	Negative: re-verification as the city changes
152	30-day staple store (shared with Food build — costed once there)	Transport & Mobility	Past	0	50	60	80	20	Shared item - effect counted in Food & Agriculture (household food store)	Positive: shared item - see Food & Agriculture	Negative: shared item - see Food & Agriculture
153	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	Transport & Mobility	Now	156	65	40	79	21	The pass+bike pattern prevents the car-cost trap for most urban households	Positive: thousands saved yearly fund the rest of this file	Negative: weather, cargo, and multi-stop days are genuinely harder
154	Car reserve discipline if car-dependent: never below quarter tank + maintenance sinking fund	Transport & Mobility	Now	50	35	60	74	26	Fuel/maintenance discipline prevents the breakdown-debt spiral in most car households	Positive: car lasts longer; surprises shrink	Negative: the fund requires honest arithmetic about true car costs
155	Evacuation-assistance registration for vulnerable members (municipal lists)	Transport & Mobility	Now	0	25	70	77.5	22.5	Registration prevents few transport failures; it removes most of the stranded consequence	Positive: registration also maps your eligibility for other programs	Negative: services are rationed; booking lead times
156	The bicycle, maintained (see Past row — the only transport no update can brick)	Transport & Mobility	Future	40	30	70	79	21	A maintained bicycle prevents platform-stranding damage almost entirely when needed	Positive: the bike serves daily life, not only failures	Negative: standby maintenance discipline
157	Manual-driving competence maintained (licence + occasional practice) as vehicles automate	Transport & Mobility	Future	60	35	55	70.8	29.2	Manual competence prevents helplessness in automated failure	Positive: licence and skill keep every vehicle option open	Negative: renewal fees; practice time
158	Provider diversification: accounts with 2+ mobility platforms + cash for taxis	Transport & Mobility	Future	0	40	45	67	33	Provider diversification prevents single-platform stranding in most events	Positive: competition among your providers improves prices	Negative: multiple apps/accounts to maintain
159	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	Communication	Past	3	30	70	79	21	A written plan prevents few communication losses; it removes most coordination damage	Positive: the plan also organizes ordinary reunions and emergencies of every kind	Negative: rehearsal discipline; plans age with every move
160	Congregation/association membership (the historical message network)	Communication	Past	50	40	55	73	27	Membership circles prevent information isolation in most disruptions	Positive: circles are friendship and culture, not only insurance	Negative: contribution expectations; circles can exclude if unguarded
161	Letter/postal discipline: addresses memorized + written copies held by two relatives	Communication	Past	10	25	65	73.8	26.2	Ink letters prevent no cutoff; they preserve the relationships and records through it	Positive: letters become family archives	Negative: slow - days, not seconds
162	Monthly export routine: contacts, photos, records downloaded to your own drive	Communication	Now	0	30	75	82.5	17.5	Exports prevent few bans/outages; they remove most of the record-loss damage	Positive: exports double as estate and migration records	Negative: 30 min/month forever
163	One owned channel: personal email list / phone tree / radio net — set up and tested	Communication	Now	0	35	65	77.2	22.8	An owned channel prevents total silencing in most platform actions	Positive: owned audiences are business assets in normal times	Negative: list upkeep; deliverability management
164	Printed essential numbers card (family, doctors, banks, legal aid) in wallet + home	Communication	Now	2	20	70	76	24	A printed card prevents no phone loss; it removes most of the network-loss damage	Positive: the card also helps a finder return your phone	Negative: replace at every number change
165	Library digital-access anchor (card + weekly session if no home internet)	Communication	Now	0	35	55	70.8	29.2	A public anchor prevents digital exclusion becoming total for its users	Positive: library anchors build broad digital skill	Negative: travel and hours limits
166	Encrypted messaging as default for private communication (free apps) + device updates	Communication	Future	0	60	40	76	24	Encryption prevents mass and casual surveillance of private communication	Positive: privacy improves every dealing, not only threatened ones	Negative: setup friction; some contacts won't follow you there
167	Data-rights requests filed (access + correction) for your major files annually	Communication	Future	10	30	60	72	28	Annual corrections prevent state-file errors compounding into decisions	Positive: clean files also improve credit and insurance outcomes	Negative: agencies are slow; corrections need follow-up
168	Compartmentation basics: separate emails/aliases for finance, public, and private life	Communication	Future	0	55	40	73	27	Compartmentation prevents one breach cascading across a whole life	Positive: breach damage stays small in ordinary hacks too	Negative: daily friction is the visible price
169	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	Demographics & Population	Past	12	20	75	80	20	A family book prevents few losses; it preserves most restart knowledge across generations	Positive: the book becomes the family's identity and inheritance	Negative: 15 hours of interviewing; elders' memories are perishable - start now
170	Skills duplication roster (each critical skill taught to 2+ people, tracked on paper)	Demographics & Population	Past	0	45	50	72.5	27.5	Duplicated skills prevent single-death household collapse in most cases	Positive: teaching deepens the teacher's mastery	Negative: teaching hours across a year
171	Civil registration discipline: every birth/marriage/death registered and copied, even in chaos	Demographics & Population	Past	30	30	70	79	21	Registration prevents few shocks; it unlocks most claims descendants will ever make	Positive: registered events simplify every future bureaucracy	Negative: fees and office visits; late registration is harder
172	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	Demographics & Population	Now	0	55	50	77.5	22.5	Pooling prevents the market-price care collision for the pooled share	Positive: pooled care IS the social fabric - it pays in belonging	Negative: rota friction; free-rider management
173	Benefits stack claim: child benefits, caregiver credits, parental top-ups — filed completely	Demographics & Population	Now	0	55	50	77.5	22.5	Shared item - effect counted in Economy & Finance (full benefits claiming)	Positive: shared item - see Economy & Finance	Negative: shared item - see Economy & Finance
174	Waitlist registrations early (childcare, family housing, seniors' housing for parents)	Demographics & Population	Now	0	40	60	76	24	Early queue positions prevent crisis-time absence of care places in most cases	Positive: applications reveal other programs you qualify for	Negative: waitlists can still fail - position is odds, not certainty
175	50-year family location audit (where will care, work, and community exist for your household's ages)	Demographics & Population	Future	0	50	40	70	30	The audit prevents retiring into a collapsing service map, when acted on early	Positive: the audit improves housing and savings decisions immediately	Negative: unwelcome arithmetic can force early hard choices
176	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	Demographics & Population	Future	150	45	55	75.2	24.8	Lawful pathways prevent forced immobility for the eligible	Positive: pathways also serve education and family options in calm times	Negative: fees vary widely; timelines in years
177	Credential recognition pre-filing for any planned move (12-month target)	Demographics & Population	Future	120	40	55	73	27	Pre-recognition prevents the year-three earnings penalty for most regulated trades	Positive: recognized credentials raise earnings from day one anywhere	Negative: assessment fees; some professions gatekeep beyond the published map
178	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	Justice & Ethics	Past	60	20	80	84	16	Shared item - effect counted in Governance & Law (protected archive)	Positive: shared item - see Governance & Law	Negative: shared item - see Governance & Law
179	Rights knowledge card for your system (arrest, search, detention limits) memorized + printed	Justice & Ethics	Past	3	35	45	64.2	35.8	Known limits prevent some checkpoint escalations	Positive: rights knowledge dignifies every official encounter	Negative: knowledge has limits against lawless actors - it shapes records, not outcomes, there
180	Exit-readiness (shared with Security build — documents and routes)	Justice & Ethics	Past	0	45	55	75.2	24.8	Shared item - effect counted in Security & Defense (exit triggers/roster)	Positive: shared item - see Security & Defense	Negative: shared item - see Security & Defense
181	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signature, or statement	Justice & Ethics	Now	0	45	60	78	22	Early counsel prevents most irreversible procedural errors	Positive: advice prevents the next ten problems, not only this one	Negative: free services are rationed; waits in peak times
182	Rights card + 'no signature under pressure' household rule (posted, rehearsed)	Justice & Ethics	Now	2	60	35	74	26	The no-signature rule prevents most coerced-consent losses	Positive: the rule protects family members you teach it to	Negative: legitimate urgent signings face delay - accepted as premium
183	Small-claims/tribunal filing for winnable civil matters (fees modest, waivable)	Justice & Ethics	Now	35	30	65	75.5	24.5	Small-claims routes prevent grievances festering; they recover most citizen-scale losses	Positive: a functioning small-claims habit lowers conflict in the whole community	Negative: your hours are the real fee; outcomes are never guaranteed
184	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	Justice & Ethics	Now	0	25	75	81.2	18.8	Contemporaneous logs prevent few violations; they decide most complaints they support	Positive: logs also serve insurance, employment, and tax matters	Negative: minutes per incident, forever
185	Score hygiene routine: annual checks of credit, tenancy, and data-broker files; errors disputed in writing within windows	Justice & Ethics	Future	0	50	45	72.5	27.5	Clean files prevent most automated-flag attachments	Positive: clean records price into cheaper credit and easier housing now	Negative: 3 hours yearly; agencies err in correcting too
186	Human-review demand template (written request for human review of any automated decision, kept ready)	Justice & Ethics	Future	2	20	70	76	24	A ready appeal prevents few flags; it reopens most wrongly decided cases	Positive: successful appeals improve the system for the next person	Negative: deadlines are short; not every review reverses
187	Case documentation for test-case organizations (your documented case can become the precedent)	Justice & Ethics	Future	0	40	40	64	36	Test cases prevent bad automated rules settling into permanence	Positive: clarified rights protect everyone, permanently	Negative: slow; your case may not be the chosen vehicle

Field	Timeframe	Actions in This	Mean Preventi	Mean Mitigati	Mean Combinec	Strongest Single Action (by combined effect)	Its Combine	Reading
Water	Past	4	45	50	72.2	Stored water reserve 56 L/person	86.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Water	Now	4	50	53.8	77.6	Certified gravity/pump water filter	88	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Water	Future	4	37.5	43.8	65.2	30-day deep storage (barrels)	82.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Food & Agriculture	Past	4	56.2	57.5	82.5	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	90	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Food & Agriculture	Now	4	35	57.5	71.9	30-day staple buffer per adult (~25 kg grain-equivalent)	86.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Food & Agriculture	Future	4	42.5	57.5	75.5	Perennial food plants (berry bushes, fruit tree, herb bed)	82	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Health & Medicine	Past	4	45	50	76	Vaccination catch-up (all free public programs)	82.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Health & Medicine	Now	4	41.2	42.5	65.7	Dental prevention kit + one hygiene visit/yr	82	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Health & Medicine	Future	4	30	56.2	69	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	80.8	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Energy	Past	4	40	57.5	75.4	Pipe protection: insulation sleeves + shutoff knowledge	82.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Energy	Now	4	20	70	75.8	Battery power station 1 kWh (or 12V battery + inverter build)	85	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Energy	Future	4	33.8	56.2	72	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserve small bills	74.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Shelter & Housing	Past	4	26.2	75	83.8	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	88	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Shelter & Housing	Now	4	42.5	60	77.8	Rent-bank / eviction-prevention applications pre-filed knowledge + emergency contact list	80	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Shelter & Housing	Future	4	37.5	51.2	73.2	Full insurance review incl. overland flood / sewer backup riders	81	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Security & Defense	Past	4	41.2	58.8	77.1	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	81.2	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Security & Defense	Now	4	38.8	50	72.6	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	77.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Security & Defense	Future	4	38.8	45	66.8	Home/business camera + signage (deterrence + evidence)	75.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Governance & Law	Past	3	26.7	70	78.7	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	84	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Governance & Law	Now	3	40	45	67.1	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	70	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Governance & Law	Future	3	31.7	48.3	65.7	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	72	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Economy & Finance	Past	3	63.3	43.3	79.6	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	80.8	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Economy & Finance	Now	4	36.2	60	75.2	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	82.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Economy & Finance	Future	3	51.7	50	76	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	78	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Technology & AI	Past	3	43.3	45	72.3	Exposure records file (employers, products, dates) kept permanently	74.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Technology & AI	Now	4	57.5	57.5	84.1	Password manager + MFA on email and banking (the two master keys)	87.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Technology & AI	Future	3	33.3	61.7	76.8	Analogue life-line kit: paper originals of ID/property/credentials, cash reserve, offline maps and contacts	80	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.

Environment & Climate	Past	3	46.7	50	73.3	Soil care basics: compost return, cover, no bare soil over winter	77.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Environment & Climate	Now	4	46.2	57.5	79.5	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	91	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Environment & Climate	Future	3	46.7	48.3	72.9	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	75	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Information & Media	Past	3	40	46.7	68.8	Battery shortwave/AM radio for foreign and emergency broadcasts	72	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Information & Media	Now	3	46.7	46.7	71.8	Two-source rule card + one scheduled news window (a household protocol, posted)	73	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Information & Media	Future	3	45	58.3	79.2	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	83.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Psychology & Mind	Past	3	41.7	56.7	74.9	Children-first protection of sleep, schooling, and play (the intergenerational shield)	77.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Psychology & Mind	Now	4	42.5	55	75.3	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	76	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Psychology & Mind	Future	3	50	45	72.6	Real-world anchor subscriptions: one club/league/class with dues (skin in the game raises attendance)	75	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Education & Knowledge	Past	3	50	50	75.3	Literacy defence: every child taught to read early at home if school fails (phonics materials)	80.8	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Education & Knowledge	Now	3	45	48.3	72.3	Reading-with-children 20 min/day (the highest-yield household education act)	76	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Education & Knowledge	Future	3	43.3	50	73.1	Offline knowledge reserve: printed field guides + manuals for your trades and region	73.8	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Work & Livelihood	Past	3	48.3	46.7	73.2	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	74	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Work & Livelihood	Now	4	32.5	61.2	73.8	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	77.2	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Work & Livelihood	Future	3	50	51.7	75.8	Ownership participation (shared with Economy build — costed once there)	77.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Transport & Mobility	Past	3	45	53.3	74.1	30-day staple store (shared with Food build — costed once there)	80	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Transport & Mobility	Now	3	41.7	56.7	76.8	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	79	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Transport & Mobility	Future	3	35	56.7	72.3	The bicycle, maintained (see Past row — the only transport no update can brick)	79	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Communication	Past	3	31.7	63.3	75.3	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	79	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Communication	Now	4	30	66.2	76.6	Monthly export routine: contacts, photos, records downloaded to your own drive	82.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Communication	Future	3	48.3	46.7	73.7	Encrypted messaging as default for private communication (free apps) + device updates	76	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Demographics & Population	Past	3	31.7	65	77.2	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	80	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Demographics & Population	Now	3	50	53.3	77	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	77.5	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Demographics & Population	Future	3	45	50	72.7	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	75.2	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Justice & Ethics	Past	3	33.3	60	74.5	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	84	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Justice & Ethics	Now	4	40	58.8	77.2	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	81.2	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.
Justice & Ethics	Future	3	36.7	51.7	70.8	Human-review demand template (written request for human review of any automated decision, kept ready)	76	Means are unweighted across the solution's actions — for comparing solutions, not a promise about any one action. See EFFECTS - HOW TO READ.

#	Action Taken (Item)	Field	Timeframe	Prevention	Mitigation	Combined Ef	Effect Realized	Effect Realized	Effect Realized	Effect Lost If	Effect Preserv	If NOT Addressed — Short-Term Implication (the window)	If NOT Addressed — Mid-Term Implication (1-5 yrs)	If NOT Addressed — Long-Term Implication (5+ yrs)	If Addressed ASAP — The Gain (positive implication)
1	Stored water reserve 56 L/person	Water	Past	55	70	86.5	70	25	5	40	60	Containers can be bought late, but filling them late means filling from the failing source itself	Mid: every season unstored is hauled at full labour price	Long: source knowledge and rights consolidate to others permanently	Positive: stored water also covers fire first-response, cleaning, and guests; the habit teaches household logistics
2	Cloth + sand filtration setup	Water	Past	50	45	72.5	65	25	10	30	70	The build works whenever made; the lost share is the exposure during the unfiltered interval	Mid: illness burden runs unfiltered through the household	Long: chronic exposure effects accumulate silently	Positive: filtration skill transfers to travel and camping; improves everyday taste/quality habits
3	Fuel for boiling (1 month reserve)	Water	Past	30	55	68.5	70	25	5	45	55	Fuel bought in shortage is bought at shortage prices, if it exists at all	Mid: fuel bought per-crisis at crisis prices, repeatedly	Long: the boiling habit never forms; each generation restarts unready	Positive: fuel stock doubles for cooking and light; warmth co-benefit
4	Rain barrel 200 L	Water	Past	45	30	61.5	55	30	15	55	45	Catchment must exist BEFORE the rains that fill it; a barrel built in drought stores nothing	Mid: rains fall on roofs with nothing under them	Long: water rights and practice settle to those who built early	Positive: gardens watered free; roof runoff diverted from foundations
5	Certified gravity/pump water filter	Water	Now	80	40	88	75	20	5	35	65	A filter bought mid-advisory still protects the remaining days; the first exposure days are unrecoverable	Mid: each advisory is met unfiltered, at bottled prices	Long: repeated low-dose exposure is the unmeasured ledger	Positive: daily drinking quality improves, not just emergency safety; reduces bottled-water plastic
6	72-hour stored water (12 L/person drinking)	Water	Now	60	65	86	80	15	5	85	15	Stored water only exists if stored before the advisory; during the event, shops empty within hours	Mid: every advisory repeats the scramble and the empty shelves	Long: the household is known - to itself - as one that scrambles; habit is destiny	Positive: doubles as outage and plumbing-repair reserve
7	Bathtub water bladder (fills in minutes at first notice)	Water	Now	35	60	74	85	10	5	90	10	The bathtub can only be filled while pressure lasts - the window is the first hours of the notice	Mid: warnings keep arriving to a house that cannot use them	Long: none beyond repeated losses - the window reopens each event, smaller	Positive: also fills for neighbours - social capital in an advisory
8	Water test strips (bacteria indicators + chlorine)	Water	Now	25	50	62.5	60	30	10	25	75	Testing works whenever started; late testing leaves the unknown-exposure interval unknowable	Mid: contamination scares are endured blind each time	Long: untested years of water are an unknown in every health file	Positive: builds a testing habit that extends to wells and travel
9	Low-flow fixtures set (aerators, showerhead, toilet kit)	Water	Future	40	20	52	35	40	25	20	80	Fixtures save from installation onward; delay loses only the interim savings	Mid: bills and demand stay at full rate through the tightening years	Long: fixtures fitted last save least; the savings curve is front-loaded	Positive: bills fall immediately; fixtures raise home value
10	30-day deep storage (barrels)	Water	Future	30	75	82.5	55	30	15	45	55	Storage bought at scarcity prices costs more and arrives after the first rationing season	Mid: the first rationing seasons are met with buckets	Long: storage bought last is bought at the era's highest prices	Positive: resilience asset visible to insurers and buyers
11	Greywater diverter for garden (where legal)	Water	Future	35	25	51.2	40	35	25	30	70	Greywater pays from installation; gardens planted meanwhile are a season behind	Mid: potable water keeps doing garden work at potable prices	Long: soil built on greywater years is years ahead	Positive: garden thrives on free water; sewer load falls
12	Cooperative membership share (community well/storage)	Water	Future	45	55	75.2	30	40	30	40	60	Co-op shares and governance seats fill early; late joiners inherit rules they did not shape	Mid: co-op terms are set; you are a customer, not a founder	Long: collective assets appreciate to their early members	Positive: builds durable local institution and skills
13	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	Food & Agriculture	Past	60	75	90	75	20	5	50	50	A store built after prices rise costs famine prices for the same kilograms - if they exist	Mid: the second hungry season arrives with no seed and no store - worse than the first	Long: the famine becomes the family's permanent economic starting line	Positive: price smoothing in normal times too; cooking skills deepen
14	Garden start: seeds + soil for 10 m^2	Food & Agriculture	Past	50	60	80	45	35	20	60	40	A garden planted late misses the season entirely; the year does not repeat	Mid: each season missed is a year of soil unbuilt	Long: land skill either compounds or is lost with the generation	Positive: exercise, soil, seed autonomy, food quality in normal years
15	Preservation kit (jars, lids, salt, vinegar)	Food & Agriculture	Past	45	65	80.8	50	35	15	55	45	Preservation only works on a harvest that exists; after the glut rots, there is nothing to preserve	Mid: gluts keep rotting at full value every year	Long: preservation knowledge dies unpractised	Positive: harvest gluts become assets; gifting builds community
16	Seed-saving envelopes + storage tin	Food & Agriculture	Past	70	30	79	25	40	35	65	35	Seed not saved in year one means no crop in year two - the loss compounds by a full cycle	Mid: purchased seed at scarcity prices, of varieties unsuited to your soil	Long: local varieties vanish - genetic wealth is not reprintable	Positive: genetic diversity preserved; exchange networks form
17	30-day staple buffer per adult (~25 kg grain-equivalent)	Food & Agriculture	Now	55	70	86.5	75	20	5	45	55	A buffer built during a spike is built AT the spike; the first month of exposure is already paid	Mid: spikes keep landing at full retail, monthly	Long: the price-shock years are the savings years of buffered households - the gap widens	Positive: fewer shopping trips; calmer monthly budgeting
18	Buying-club share (bulk staples with 4–10 households)	Food & Agriculture	Now	35	45	64.2	60	30	10	25	75	Clubs can form anytime; the lost share is the discounts of the unformed months	Mid: retail margin keeps being paid on every staple	Long: the club that never formed is a network that never exists	Positive: the club becomes a mutual-aid network beyond food
19	Sprouting kit (jars + seeds)	Food & Agriculture	Now	30	50	65	70	20	10	20	80	Sprouts start in a day whenever begun; delay loses only the gap days meanwhile	Mid: fresh gaps keep being filled at convenience-store prices	Long: minor - the habit is startable in a day, forever	Positive: daily fresh nutrition for pennies; children learn food production
20	Community fridge / food bank registration + food-handler basics	Food & Agriculture	Now	20	65	72	65	25	10	30	70	Programs can be claimed whenever needed; earlier registration skips the crisis queue	Mid: crisis queues lengthen; registration seniority accrues to others	Long: unclaimed years of support are simply gone	Positive: dignity-preserving routes stabilize nutrition for everything else
21	Perennial food plants (berry bushes, fruit tree, herb bed)	Food & Agriculture	Future	55	60	82	15	35	50	70	30	A tree not planted this year is a harvest missing in every future year - the most perishable item in this field	Mid: establishment years pass unplanted; purchased food continues at full share	Long: a decade of harvests is the price of a decade of hesitation - per tree	
22	Soil building: compost system + cover crop seed	Food & Agriculture	Future	40	55	73	30	40	30	55	45	Soil built a season late holds a season less water; the drought does not wait for your compost	Mid: droughts keep meeting unprepared soil	Long: soil age is wealth; late soil is permanently younger	Positive: input bills fall yearly; soil becomes an appreciating asset
23	Dehydrator (entry electric or solar build)	Food & Agriculture	Future	35	60	74	55	30	15	40	60	Drying capacity bought after the glut has nothing to dry until the next one	Mid: each glut is a missed winter	Long: minor - capacity is buyable in any year	Positive: preserves any glut (garden, sales, gifts); no fridge space or power
24	Co-op / land-trust membership contribution	Food & Agriculture	Future	40	55	73	30	40	30	45	55	Trust land and shares allocate early; late entry buys a smaller voice at a higher price	Mid: land prices and trust allocations move on without you	Long: shared land lost to other uses rarely returns	Positive: land access without ownership; deep community roots
25	Household medical kit: thermometer, pulse oximeter, ORS packets, fever medication (2 weeks)	Health & Medicine	Past	35	60	74	70	25	5	30	70	A kit assembled during illness is assembled at illness prices and illness speed	Mid: treatable illness keeps becoming complications	Long: household care knowledge never forms; dependence on scarce clinics hardens	Positive: everyday injuries and illnesses handled without clinics
26	Isolation-room setup (separate bedding, basin, waste bag system)	Health & Medicine	Past	55	40	73	70	25	5	50	50	Separation started after transmission has already seeded the household's secondary cases	Mid: secondary cases keep seeding inside the home	Long: chronic post-infection burdens accumulate member by member	Positive: habits persist into ordinary flu seasons
27	Vaccination catch-up (all free public programs)	Health & Medicine	Past	75	30	82.5	60	30	10	60	40	Vaccination after exposure is too late for that disease cycle; schedules exist because windows do	Mid: preventable diseases keep their appointment with the unvaccinated	Long: herd protection weakens community-wide	Positive: protects the whole community (herd effect), not only the household
28	Paper medical records folder (vaccines, conditions, medications, allergies)	Health & Medicine	Past	15	70	74.5	50	35	15	35	65	Records assembled mid-crisis are assembled from memory, under stress, with gaps that cost	Mid: every encounter restarts from confusion	Long: unrecorded histories corrupt every later diagnosis	Positive: the folder serves every other system in this file (evacuation, legal, benefits)
29	Drug coverage optimization: benefits audit + generic substitution review with pharmacist	Health & Medicine	Now	45	35	64.2	60	30	10	30	70	Optimization works whenever done; the lost share is the months of interactions meanwhile	Mid: interactions and duplicates keep billing and harming monthly	Long: adverse-event risk compounds with every added medication	Positive: fewer side effects; lower pharmacy bills permanently
30	Home infection-control basics (soap stock, disinfectant, masks x50, gloves)	Health & Medicine	Now	50	45	72.5	65	25	10	35	65	Hygiene stock bought in an outbreak is bought at outbreak prices and queues	Mid: transmission seasons repeat at full rate	Long: habit-free homes raise habit-free children	Positive: fewer sick days across the whole household yearly
31	Dental prevention kit + one hygiene visit/yr	Health & Medicine	Now	40	70	82	35	40	25	55	45	Dental delay converts $120 prevention into root canals and extractions on a known schedule	Mid: the $120 problem becomes the $1,200 problem on schedule	Long: tooth loss is forever; reconstruction is the five-figure endpoint	Positive: pain, sleep, and eating improve beyond the avoided emergencies
32	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy take-back)	Health & Medicine	Now	30	20	44	30	40	30	20	80	Take-back works any day; the lost share is diffuse and population-level	Mid: leftovers keep circulating in medicine cabinets	Long: resistance is a commons - your share of its loss is everyone's	Positive: protects water systems and everyone's future antibiotics
33	Room HEPA unit for the sick room / bedroom	Health & Medicine	Future	35	55	70.8	50	35	15	25	75	Filtration helps from installation onward; delay loses only interim exposure	Mid: respiratory burden accumulates season by season	Long: lung and vessel damage is substantially irreversible	Positive: sleep and cognition improve with cleaner air daily
34	Ventilation upgrade: CO2 monitor + window/exhaust habits	Health & Medicine	Future	30	50	65	55	30	15	20	80	Monitoring informs whenever started; unmonitored years are simply unknown	Mid: bad-air years pass unmeasured and unargued	Long: housing decisions keep being made blind	Positive: data improves ventilation habits and housing choices
35	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	Health & Medicine	Future	45	65	80.8	30	40	30	40	60	Pressure found late has already spent years on vessels; the cuff only helps early	Mid: silent pressure spends its years on vessels	Long: the stroke/heart-failure endpoint arrives unheralded and largely unpaid-back	Positive: detected risks improve insurability and clinical care
36	Reserve medication buffer (1 month extra of essentials, rotated, where legal)	Health & Medicine	Future	10	55	59.5	55	30	15	35	65	Rotation built during a shortage cannot be built - pharmacies ration precisely then	Mid: each shortage re-runs the interruption lottery	Long: trust in the care system erodes one shortage at a time	Positive: prescriber relationship strengthens through honest planning
37	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blanket fort supplies	Energy	Past	45	55	75.2	70	25	5	55	45	Sealing done in autumn protects the winter; done in January it protects half of one	Mid: cold seasons keep billing health and pipes at full rate	Long: cold-damaged health and housing stock deteriorate together	Positive: comfort and bills improve in normal winters too
38	Fuel reserve (safe storage: gas canisters / charcoal / wood)	Energy	Past	35	60	74	65	25	10	50	50	Fuel stocked in shortage is stocked at shortage prices, in shortage queues	Mid: per-crisis fuel purchases at per-crisis prices	Long: no stock discipline forms; scarcity stays personal	Positive: cooking flexibility year-round
39	Layered clothing upgrade (wool base layers x2 per person)	Energy	Past	50	40	70	70	25	5	25	75	Layers help from the day they are worn; delay loses only the cold days meanwhile	Mid: cold harm keeps landing on the least-layered members first	Long: minor - layers are startable any day	Positive: outdoor work and sport benefit from the same layers
40	Pipe protection: insulation sleeves + shutoff knowledge	Energy	Past	30	75	82.5	70	25	5	70	30	Insulation fitted after the first freeze protects the pipes that have not burst yet	Mid: each winter is a fresh lottery on the pipes	Long: one burst rewrites the house's finances for years	Positive: prevents the single most expensive household water loss
41	Battery power station 1 kWh (or 12V battery + inverter build)	Energy	Now	25	80	85	75	20	5	75	25	A battery bought during an outage arrives uncharged and untested - its worst state at the worst moment	Mid: outages keep taxing food, work, and sleep at full rate	Long: battery prices fall slowly; the losses do not wait for them	Positive: doubles for camping, work sites, and off-grid projects; silent vs. generators
42	LED lanterns x3 + headlamp	Energy	Now	20	60	68	70	25	5	40	60	Lights bought mid-outage are bought in the dark, at surge prices	Mid: dark hours keep producing accidents and anxiety	Long: minor - lights are buyable in any week	Positive: night work and reading improve in normal life too
43	Safe backup cooking: single-burner butane stove + 4 canisters	Energy	Now	15	65	70.2	70	25	5	45	55	A stove bought during an outage faces empty canister shelves - everyone had the same idea	Mid: outage eating keeps costing restaurant prices	Long: minor beyond repetition	Positive: outdoor cooking in summer keeps heat out of the kitchen
44	Fridge discipline kit: thermometer + cooler + ice packs	Energy	Now	20	75	80	75	20	5	50	50	Ice retention only works if the packs were frozen before the power failed	Mid: freezers keep failing the 24-hour test	Long: food-waste habit becomes the household norm	Positive: picnic and travel utility
45	Solar trickle panel 100–200 W for the battery station	Energy	Future	35	55	70.8	40	35	25	30	70	Panels produce from installation; delay loses the sun meanwhile, nothing more	Mid: grid prices are paid in full through the rising years	Long: 25-year assets bought last produce least	Positive: bills fall in normal times; pairs with the battery for full autonomy
46	Home insulation priority pack (attic hatch seal, window film, draft proofing)	Energy	Future	45	50	72.5	35	40	25	35	65	Insulation saves from the first season after fitting; each unfitted season is spent at full price	Mid: every season is paid at the uninsulated rate	Long: insulation's payback window closes one season at a time	Positive: comfort rises and bills fall in all seasons, crisis or not
47	Community energy co-op share	Energy	Future	40	50	70	30	40	30	40	60	Co-op allocations and founding terms go to early members	Mid: co-op dividends accrue to members, not waiters	Long: ownership shares bought last cost most	Positive: local jobs and democratic ownership of infrastructure
48	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserve small bills	Energy	Future	15	70	74.5	70	25	5	60	40	The kit's value concentrates in the first blackout hours; assembled after, it missed its event	Mid: each blackout re-runs the scramble at full cost	Long: the kit's absence becomes a family story told after each event	Positive: the radio habit reconnects the household to local information
49	Documents go-bag (waterproof pouch: ID, property, insurance, photos of every room's contents)	Shelter & Housing	Past	10	85	86.5	60	30	10	45	55	Copies made after the disaster are made from ashes and memory; the claim window also runs	Mid: claims and aid are fought from weakness, slowly	Long: uncompensated losses set the family's wealth line for a generation	Positive: the same folder serves benefits, legal, and medical systems
50	72-hour go-kit per person (clothes, medication, water, light, food bars)	Shelter & Housing	Past	15	75	78.8	70	25	5	65	35	A go-bag packed during evacuation is a fraction of the bag packed in calm - and the calm items are the ones forgotten	Mid: displacement scrambles work, school, and care simultaneously	Long: the first-72-hours harm (medication, documents, children) echoes longest	Positive: doubles as travel/camping readiness
51	Home inventory video + off-site copy	Shelter & Housing	Past	10	80	82	55	30	15	50	50	An inventory filmed after the loss documents an empty room	Mid: insurers' numbers, unchallenged, become the settlement	Long: underpaid claims cannot be re-fought after windows close	Positive: resale and warranty value in normal times
52	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	Shelter & Housing	Past	70	60	88	65	25	10	55	45	Alarms fitted after the first fire are fitted for the second one	Mid: fire risk runs unmitigated nightly	Long: the loss, when it comes, is total in the way few losses are	Positive: everyday fire readiness, not just disaster
53	Tenant rights pack: your jurisdiction's rules printed + legal-aid and tenant-union numbers	Shelter & Housing	Now	55	45	75.2	65	25	10	50	50	Rights learned after signing or after the notice arrives lose the prevention half of their power	Mid: arrears and charges compound under rules you learn too late	Long: an eviction record prices into every future tenancy for years	Positive: deposits returned and illegal charges refused in ordinary tenancies too
54	Housing documentation habit: dated photos at move-in/out, all landlord communication in writing	Shelter & Housing	Now	30	70	79	60	30	10	55	45	Evidence created after the fact is testimony; created at the time, it is proof	Mid: disputes keep being decided by the other side's paperwork	Long: a reputation as an undocumented tenant follows the file, not the person	Positive: the evidence habit serves work, consumer, and legal life
55	Rent-bank / eviction-prevention applications pre-filed knowledge + emergency contact list	Shelter & Housing	Now	50	60	80	55	30	15	60	40	Rent banks and relief are claimed at arrears month one or fought for at month four	Mid: the cascade reaches the hearing stage where outcomes harden	Long: housing instability writes itself into children's schooling records	Positive: knowledge transfers to friends and family facing arrears
56	Shared-housing option prepared (one identified co-housing/roommate arrangement as Plan B)	Shelter & Housing	Now	35	65	77.2	55	30	15	45	55	A Plan B arranged mid-crisis is arranged from weakness, at crisis prices	Mid: crisis moves are made at crisis prices, into worse housing	Long: each forced move breaks networks that took years to build	Positive: deepens the friendships it rests on
57	Property risk check before buying/renewing: flood, fire, insurance-availability maps (free public data)	Shelter & Housing	Future	60	30	72	50	30	20	65	35	The audit only has value BEFORE the purchase; after, it is an obituary of the decision	Mid: exposed purchases keep being made at yesterday's prices	Long: the stranded asset is the retirement - its loss is the long term	Positive: every future purchase decision improves, not only this one
58	Full insurance review incl. overland flood / sewer backup riders	Shelter & Housing	Future	5	80	81	45	35	20	80	20	Insurance cannot be bought for a flood already forecast, let alone underway - the window is absolute	Mid: each renewal is a fresh chance to be uninsurable	Long: uninsured losses in the withdrawal era are unrecoverable by definition	Positive: contents and liability cover ride along in bundled policies

59	Portability reserve: keep at least 50% of net worth out of any single exposed property (financial accounts, skills)	Shelter & Housing	Future	45	45	69.8	35	40	25	30	70	Portability can be built anytime; delay loses the compounding of the portable years	Mid: lock-in deepens as equity and roots grow in the exposed place	Long: mobility, once lost, is rarely repurchased	Positive: freedom compounds - careers and family choices widen
60	Buyout/relocation program registration (where offered) — filed early	Shelter & Housing	Future	40	50	70	30	40	30	55	45	Programs fund early registrants; late registration queues behind the funded	Mid: funded offers go to the registered; you read about them	Long: relocation help is a one-generation window	Positive: early registrants also receive information first
61	Early-warning habit: named indicators list + decided trigger points ("if X happens, we leave within 48 h")	Security & Defense	Past	55	40	73	65	25	10	85	15	Triggers decided late are decided under fire, which is to say barely decided at all	Mid: escalation finds the household mid-deliberation each time	Long: the late-exit biography is written once and inherited	Positive: decision discipline transfers to every other emergency in this file
62	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	Security & Defense	Past	25	75	81.2	45	35	20	60	40	Translations done after flight are done in exile, at exile prices, without the originals to hand	Mid: exile employment runs below qualification for years	Long: unrecognized credentials cap a working life	Positive: credentials also speed ordinary employment abroad
63	Exit-route plan: two routes, rendezvous points, transport costs pre-estimated and saved	Security & Defense	Past	40	65	79	55	30	15	55	45	A reserve accumulated late is smaller at the exact moment size decides the route	Mid: arrival poverty compounds into housing and work penalties	Long: the reserve never rebuilt is the business never started	Positive: the reserve habit builds general financial resilience
64	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in signals)	Security & Defense	Past	45	55	75.2	55	30	15	50	50	A roster built during the crisis competes with the crisis for everyone's hours	Mid: isolation removes warning, shelter, and witness simultaneously	Long: scattered communities do not reassemble	Positive: the roster IS the community - it pays socially in peacetime
65	Home target-hardening basics: solid locks, strike plates with 3-inch screws, window locks, lighting	Security & Defense	Now	60	30	72	65	25	10	35	65	Hardening works from the night it is fitted; delay loses the exposure nights meanwhile	Mid: opportunistic crime keeps selecting the soft target	Long: a victimization record prices into insurance and peace of mind	Positive: insurance premiums and sleep both improve
66	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	Security & Defense	Now	25	70	77.5	60	30	10	40	60	Recording installed after an incident documents the next one	Mid: incidents stay unprovable; recovery stays theoretical	Long: repeat targeting follows unrecorded first successes	Positive: package-theft and dispute evidence in ordinary life
67	Route and routine discipline + check-in system with one contact	Security & Defense	Now	55	35	70.8	65	25	10	30	70	Habits protect from the week they form; the unformed weeks are at full exposure	Mid: exposure hours accumulate at full rate	Long: minor - habits form in weeks at any age	Positive: walking more improves health as a side effect
68	Victim-compensation and protective-order applications knowledge (legal aid)	Security & Defense	Now	15	65	70.2	55	30	15	45	55	Compensation deadlines run from the incident, not from when you learned of them	Mid: deadlines expire per incident, quietly	Long: unrecovered losses compound into the household's risk posture	Positive: the knowledge serves anyone you help afterward
69	Venue standards habit: choose venues with visible security, exits known on entry (30-second rule)	Security & Defense	Future	40	45	67	60	30	10	45	55	Awareness learned after an incident is learned at the highest tuition	Mid: crowds keep being read by others' instincts, not yours	Long: minor - the habit is learnable at any point	Positive: calm in crowds generally; the habit reassures children
70	Personal data footprint reduction: data-broker removals, minimal public exposure	Security & Defense	Future	50	40	70	50	35	15	55	45	Data collected meanwhile cannot be un-collected; removal only stops the flow, not the stock	Mid: the data stock grows monthly; removal chases a moving total	Long: profiles assembled now are the targeting files of later decades	Positive: less spam, fewer scams, cleaner digital life daily
71	Support for regulation: specific, written asks to representatives (drone ID, autonomous weapons limits)	Security & Defense	Future	35	30	54.5	30	40	30	50	50	Rules harden early in a technology's life; late advocacy litigates what early advocacy could have prevented	Mid: each rule hardens unopposed in its first version	Long: surveillance infrastructure, once built, is a permanent resident	Positive: civic muscle built for every other cause
72	Home/business camera + signage (deterrence + evidence)	Security & Defense	Future	30	65	75.5	60	30	10	35	65	Cameras deter from installation; delay loses the unwitnessed interval	Mid: unwitnessed incidents repeat against the same address	Long: minor beyond repetition	Positive: deters package theft immediately
73	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	Governance & Law	Past	20	80	84	35	40	25	55	45	Archives assembled after searches begin are assembled from what the search left behind	Mid: restitution windows and witnesses narrow yearly	Long: unarchived history becomes the victor's version	Positive: archives become history itself - the community's memory survives
74	Qualifications in portable form (transcripts, licences, references sealed and copied)	Governance & Law	Past	25	75	81.2	40	35	25	45	55	Copies verified after exile are verified without access to the originals	Mid: exile years run below credential, compounding	Long: identity gaps harden into statelessness risks for descendants	Positive: credentials serve ordinary careers too
75	Quiet-network membership (congregation, union, association dues)	Governance & Law	Past	35	55	70.8	50	35	15	50	50	Networks formed during purges form under surveillance, at purge prices	Mid: unnetworked exiles absorb shocks alone	Long: communities that scattered do not reassemble	Positive: networks double as cultural and mutual-aid life in peacetime
76	Access-to-information request kit (templates + filing fees where charged)	Governance & Law	Now	45	40	67	55	30	15	50	50	Requests filed after decisions are appeals; before, they are inputs - the same paper, half the power	Mid: decisions made on hidden files become precedents	Long: unrequested records are destroyed on schedule - permanently	Positive: transparency habits improve every dealing with institutions
77	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	Governance & Law	Now	40	50	70	55	30	15	40	60	Complaints filed after limitation periods are history lessons, not claims	Mid: violations repeat against the silent at full rate	Long: unchallenged practice becomes the de facto rule	Positive: each upheld complaint raises the standard for everyone after you
78	Civic membership: one watchdog, union, or ratepayers' organization	Governance & Law	Now	35	45	64.2	40	35	25	35	65	Bodies captured meanwhile are captured for their whole term, not until you join	Mid: captured bodies serve full terms	Long: institutional memory of capture outlives the term	Positive: real influence over local decisions; civic skill compounds
79	Local verification habit: attend/join one local body (assembly, board, co-op) and keep its records	Governance & Law	Future	40	40	64	45	35	20	30	70	Local proof habits can start anytime; the lost share is the rumour-governed years meanwhile	Mid: rumour-governed households make rumour-priced decisions	Long: verification capacity, unbuilt, is a generation's project later	Positive: local trust networks strengthen every other system
80	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	Governance & Law	Future	20	65	72	50	30	20	45	55	Records begun late start from the late date; the unrecorded years stay unrecorded	Mid: the unwritten years stay unwritten	Long: your file is the only history your grandchildren get	Positive: your file becomes evidence others can cite
81	Support independent local media (one subscription)	Governance & Law	Future	35	40	61	50	30	20	40	60	Subscriptions begun in a crisis begin after the crisis's first, decisive information battle	Mid: information monopoly consolidates channel by channel	Long: independent channels, once dead, take decades to regrow	Positive: independent media survives because of exactly these subscriptions
82	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	Economy & Finance	Past	65	45	80.8	70	20	10	60	40	Splitting after the failure is announced queues behind everyone who split before	Mid: failures cascade through unsplit households completely	Long: wiped-out savings rewrite the family's starting line for decades	Positive: diversification discipline serves ordinary banking safety too
83	Durable-goods and skills reserve (tools, preserved food, tradeable skills)	Economy & Finance	Past	55	50	77.5	60	25	15	45	55	Durable value bought in a panic is bought at panic prices	Mid: cash keeps evaporating at inflation's rate	Long: durable skills, unbought, are careers unbuilt	Positive: tools and staples serve daily life, not only collapse
84	Debt minimalism in boom times (fixed-rate only, no margin, no speculation on credit)	Economy & Finance	Past	70	35	80.5	65	25	10	65	35	Refusal decided during the mania is decided against the loudest social pressure there is	Mid: each mania finds the same unruled money	Long: the pattern repeats per generation that never writes the rule	Positive: the rule also prevents ordinary speculative losses
85	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	Economy & Finance	Now	30	75	82.5	45	35	20	50	50	The fund's power is its age: a fund started at the emergency is a loan application	Mid: emergencies keep converting to 20%+ debt, repeatedly	Long: debt years displace ownership years - the swap is the long term	Positive: calm - the fund buys sleep before it buys anything else
86	Debt audit + avalanche order (highest rate first) printed and posted	Economy & Finance	Now	35	55	70.8	55	30	15	55	45	Every month unmapped is a month of minimum payments compounding at card rates	Mid: interest keeps being paid on the unmapped balance	Long: credit damage prices into housing and work for years	Positive: credit score and options improve as balances fall
87	Free insolvency-trustee first consultation + credit-counselling session (non-profit)	Economy & Finance	Now	25	60	70	60	25	15	35	65	Advice sought mid-crisis is sought in a queue of the equally desperate	Mid: wrong-order moves keep being made confidently	Long: minor - a map is drawable in any year	Positive: a map you trust ends advice-shopping
88	Benefits and credits claim check (tax filing + benefits finder)	Economy & Finance	Now	55	50	77.5	55	30	15	45	55	Benefits unclaimed for a tax year are mostly unrecoverable after it - July recalculations do not wait	Mid: each unclaimed July resets the loss, not the entitlement	Long: unclaimed benefits across a childhood are a house deposit, gone	Positive: benefits claimed also stabilize the local services funded by uptake
89	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	Economy & Finance	Future	60	45	78	20	35	45	40	60	Contributions delayed lose the compounding years, which are the majority of the ending value	Mid: displacement finds savings where contributions should be	Long: compounding's lost decade is the retirement's largest line	Positive: ownership income compounds into retirement security
90	Human-irreducible skill course (care, craft, repair, judgment trades)	Economy & Finance	Future	50	55	77.5	40	35	25	45	55	Courses taken after displacement compete with job search for the same hours	Mid: the course stays untaken while the window is cheap	Long: the second skill unbought is the second career unhad	Positive: the second skill enriches ordinary years (weekend income, options)
91	Union/professional-body membership	Economy & Finance	Future	45	50	72.5	35	40	25	55	45	Terms negotiated after displacement is announced are negotiated from the announcement	Mid: terms are set by whoever organized first	Long: weak terms normalize into the industry standard	Positive: solidarity protects coworkers, not only you
92	Home hazard test when buying older property (lead paint, asbestos survey before renovation)	Technology & AI	Past	60	35	74	55	30	15	50	50	Testing after symptoms begin answers a harder question with worse evidence	Mid: exposure continues daily in untested rooms	Long: latency delivers the diagnosis decades after the cheap answer	Positive: findings raise property value and family health knowledge
93	Exposure records file (employers, products, dates) kept permanently	Technology & AI	Past	15	70	74.5	30	40	30	40	60	Records kept from now forward cannot retroactively document the unrecorded years	Mid: cases weaken as records scatter	Long: unfiled exposures are unprovable exposures	Positive: the folder serves insurance and medical systems too
94	Adoption delay rule for unstudied non-essentials (wait for the 2nd generation / safety data)	Technology & AI	Past	55	30	68.5	65	25	10	30	70	The rule can start anytime; the lost share is the first-generation products already bought	Mid: first-generation defects keep being bought at launch prices	Long: minor - the rule is adoptable in any year	Positive: second-generation products are cheaper AND better - the rule pays
95	Password manager + MFA on email and banking (the two master keys)	Technology & AI	Now	75	50	87.5	70	20	10	55	45	MFA enabled after the takeover is enabled for the attacker's next visit, not yours	Mid: takeover attempts keep arriving at unlocked doors	Long: a breached identity echoes through credit and accounts for years	Positive: digital life becomes calmer everywhere, not only safer
96	3-2-1 backup: external drive + cloud copy of irreplaceables	Technology & AI	Now	30	80	86	65	25	10	45	55	Backups begun after the ransomware are begun for the next incident	Mid: drives fail on their own schedule, unbacked	Long: lost archives (photos, records) are lost in the permanent tense	Positive: photos and records gain permanence beyond emergencies
97	Automatic updates ON + router firmware check (quarterly, calendared)	Technology & AI	Now	65	45	80.8	70	20	10	50	50	Updates deferred are vulnerabilities published - attackers read the same notes	Mid: known-vulnerability exploits are automated and continuous	Long: an unpatched device becomes the household's permanent open window	Positive: devices run faster and longer on current software
98	Verification protocol card (callback numbers printed; 'urgent = verify by second channel' rule)	Technology & AI	Now	60	55	82	70	20	10	60	40	The rule learned from the first successful fraud costs the amount of the first successful fraud	Mid: fraud scripts improve monthly against the unruled	Long: one success funds the fraudster's next hundred attempts on your family	Positive: the rule protects elderly relatives you teach it to
99	Analogue life-line kit: paper originals of ID/property/credentials, cash reserve, offline maps and contacts	Technology & AI	Future	20	75	80	55	30	15	50	50	A paper vault assembled during a lockout is assembled without access to what it should contain	Mid: each lockout is improvised from nothing	Long: paper unmaintained decays into the same dependence it insured against	Positive: paper records impress in legal and estate matters too
100	Data portability routine: monthly export of contacts, photos, records from every platform	Technology & AI	Future	25	70	77.5	55	30	15	45	55	Exports begun after a ban appeal the ban, not restore the data	Mid: platform dependence deepens with every unexported month	Long: a decade of records inside one company is a decade hostage	Positive: migration freedom between services forever
101	Offline competence maintenance: one unaided skill practised regularly (navigation, arithmetic, writing)	Technology & AI	Future	55	40	73	50	35	15	35	65	Practice begun in the failure is begun under the failure's time pressure	Mid: deskilling compounds quietly	Long: unaided competence, unpractised for a generation, is a rebuild, not a refresh	Positive: unaided skill is confidence in every outage and exam
102	Soil care basics: compost return, cover, no bare soil over winter	Environment & Climate	Past	50	55	77.5	35	40	25	55	45	Soil care begun in the bad year begins on soil the bad year is eating	Mid: yields keep discounting the lost topsoil	Long: soil age is the farm's true capital - late soil is poor soil	Positive: soil pays in every ordinary year, not only bad ones
103	Water-wise planting (drought-matched species, mulch)	Environment & Climate	Past	50	45	72.5	40	35	25	60	40	Planting matched species late misses the rains they establish on	Mid: dry seasons keep punishing mismatched planting	Long: water tables fall to those who adapted first	Positive: biodiversity and beauty rise with matched planting
104	Fuel efficiency: improved cookstove / insulation of heat	Environment & Climate	Past	40	50	70	55	30	15	40	60	Efficiency built in scarcity is built at scarcity prices	Mid: fuel poverty repeats each winter at full price	Long: minor beyond repetition and smoke exposure	Positive: fuel savings are permanent and immediate
105	Address risk check (flood/fire/heat maps, free) + insurer confirmation in writing	Environment & Climate	Now	55	35	70.8	50	35	15	50	50	Risk discovered by the flood itself arrives with the water	Mid: hazard prices keep being discovered by events	Long: insurability, once lost in a zone, does not return	Positive: the audit improves insurance negotiations immediately
106	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	Environment & Climate	Now	80	55	91	70	20	10	75	25	The valve installed after the first backup prevents the second one	Mid: every storm season re-rolls the backup dice	Long: repeated small floods quietly consume the house's value	Positive: insurers and municipalities reward it; basements stay dry in ordinary storms too
107	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback, water stock	Environment & Climate	Now	35	65	77.2	60	30	10	55	45	A cool room prepared during the heat dome is prepared by people already heat-stressed	Mid: heat seasons lengthen; the unprepared room stays unprepared	Long: heat illness burden accumulates in kidneys and hearts	Positive: comfort in every heatwave, not only emergencies
108	Go-bag + document pouch (shared with Shelter build — counted once there)	Environment & Climate	Now	15	75	78.8	70	25	5	65	35	Shared item - delay counted in Shelter & Housing (a go-bag packed during evacuation is a fraction of the bag packed in calm)	Mid: shared - see Shelter & Housing	Long: shared - see Shelter & Housing	Positive: shared item - see Shelter & Housing
109	Location decision audit every 5 years (risk trend, insurance trend, water trend for your address)	Environment & Climate	Future	55	40	73	45	35	20	55	45	Audits done after prices move audit a decision the market already made for you	Mid: the exit discount widens yearly	Long: stranded assets are the definition of a long-term loss	Positive: decisions improve across all assets, not only property
110	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	Environment & Climate	Future	50	50	75	40	35	25	45	55	Portable skills built after the region declines are built without the region's income to fund them	Mid: location-bound income stays location-bound	Long: portable income unbought is the move unmade	Positive: portable skills enrich the life you have now
111	Collective relocation readiness: community/municipal plan participation	Environment & Climate	Future	35	55	70.8	35	40	25	50	50	Plans joined late are plans designed by those who came early	Mid: plans finalize without your constraints in them	Long: collective relocations protect their members first, by design	Positive: collective plans protect neighbours who cannot plan alone
112	Source-tracing habit taught at home (claim → origin → evidence) — a dinner-table practice, not a purchase	Information & Media	Past	50	35	67.5	60	30	10	45	55	Habits formed under propaganda form against resistance the calm never applies	Mid: each rumour cycle finds the same undefended household	Long: children raised on incitement carry it for life	Positive: the habit makes citizens formidable in ordinary life too
113	Independent media support: one exile/public/independent subscription or donation	Information & Media	Past	40	45	67	50	35	15	50	50	Channels funded after the crackdown are funded through the crackdown's filters	Mid: monopoly deepens per unsubscribed year	Long: dead channels do not resurrect for the crisis that needs them	Positive: journalism exists in peacetime because of these funders

114	Battery shortwave/AM radio for foreign and emergency broadcasts	Information & Media	Past	30	60	72	65	25	10	55	45	A radio bought in the blackout is bought at blackout prices, in the dark	Mid: blackouts keep being total	Long: minor beyond repetition	Positive: local news and weather in ordinary life
115	Two-source rule card + one scheduled news window (a household protocol, posted)	Information & Media	Now	55	40	73	65	25	10	35	65	The rule works from adoption; the lost share is the manipulations already acted on	Mid: manipulation keeps pricing household decisions monthly	Long: a poisoned information diet compounds into a worldview	Positive: family decisions improve across money, health, and safety
116	Primary-source subscriptions: one quality outlet + official data sources bookmarked	Information & Media	Now	40	50	70	55	30	15	25	75	A quality source improves the diet from the first issue; delay loses only the meanwhile	Mid: the outrage diet keeps billing attention daily	Long: reading depth, unpractised, attenuates	Positive: depth reading is pleasure, not only defence
117	Fact-check and provenance tools bookmarked; reverse image search learned	Information & Media	Now	45	50	72.5	60	30	10	35	65	Tools learned after sharing the hoax retract to a fraction of its audience	Mid: hoaxes keep being shared first and checked never	Long: your credibility, spent repeatedly, stops being believed when it matters	Positive: you become the household's verifier - a valued role
118	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	Information & Media	Future	70	45	83.5	70	20	10	80	20	A code word created after the first clone call authenticates nothing retroactively - the first call is the test it had to exist before	Mid: clone attempts multiply against the unworded family	Long: the first success teaches the fraudster your family's full shape	Positive: the rule also stops ordinary phone scams against grandparents
119	Paper originals vault (identity, property, credentials) + notarial copies of the critical five	Information & Media	Future	25	75	81.2	50	35	15	45	55	Originals notarized after a forgery dispute begins are exhibits, not anchors	Mid: disputes keep being lost for want of an anchor document	Long: a forgeable paper trail is the estate your heirs inherit	Positive: notarized sets simplify estates and disputes in normal life
120	Content-credential tools adopted (signed capture apps) for important records	Information & Media	Future	40	55	73	50	35	15	40	60	Provenance adopted late leaves the earlier archive permanently doubtful	Mid: synthetic content keeps being mistaken for yours	Long: authorship itself becomes doubtful by default	Positive: creators gain a durable authenticity asset
121	Routine and ritual structure (fixed sleep, meals, work, worship/rest — a written household rhythm)	Psychology & Mind	Past	45	55	75.2	45	35	20	40	60	Rhythm begun late still works; the lost share is the years the household ran without it	Mid: stress keeps billing the body untreated	Long: the unprocessed years become the children's baseline	Positive: rhythm improves ordinary days, not only crises
122	Mutual storytelling / testimony practice (regular, structured, within family or congregation)	Psychology & Mind	Past	30	60	72	35	40	25	45	55	Testimony begun after witnesses die is begun with the sources gone	Mid: witnesses age; tellings get thinner	Long: untold history becomes family mythology - or silence	Positive: tellers report relief; listeners gain the map of what happened
123	Children-first protection of sleep, schooling, and play (the intergenerational shield)	Psychology & Mind	Past	50	55	77.5	40	35	25	55	45	Stability protected after the damage years protect the remaining ones	Mid: damage years accumulate in development, where they are most expensive	Long: transmission completes: the pattern becomes the next generation's own	Positive: protected children outperform in every later measure this file tracks
124	Sleep protocol: fixed window, morning light (free), screens out of the bedroom, caffeine cutoff	Psychology & Mind	Now	55	45	75.2	60	30	10	40	60	Sleep defended after the cascade has spent months compounding interest in every system	Mid: sleep debt compounds into mood, immunity, and error	Long: chronic short sleep is a dementia-era risk factor, not a lifestyle footnote	Positive: sleep pays in cognition, mood, immunity, and safety daily
125	Movement prescription: 60 min/day walking built into commute/errands + one strength session weekly	Psychology & Mind	Now	50	50	75	60	30	10	35	65	Movement helps from the first week; the lost share is the deconditioning meanwhile	Mid: deconditioning makes each restart harder than the last	Long: fitness unbought in midlife is frailty bought for old age	Positive: movement pays in every physical system simultaneously
126	One weekly in-person group (see Pareto action #6 — costed once)	Psychology & Mind	Now	45	55	75.2	50	35	15	30	70	Groups can be joined anytime; crisis-time joining queues behind crisis-time demand (shared item)	Mid: isolation deepens; groups fill and close (shared item)	Long: unpartnered, unfriended decades are the endpoint loneliness plans for	Positive: belonging is also fun, work leads, and care in ordinary weeks
127	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	Psychology & Mind	Now	20	70	76	65	25	10	65	35	A crisis plan written during the crisis is written by the crisis	Mid: crises keep arriving at wrong doors at wrong hours	Long: each mishandled crisis teaches the household that help does not work	Positive: knowing the doors lowers daily anxiety, not only crisis harm
128	Synthetic-companionship budget rule (fixed hours + money cap, written, reviewed monthly)	Psychology & Mind	Future	55	40	73	55	30	15	55	45	Budgets set after dependency forms are withdrawals, not budgets	Mid: dependency deepens as products improve at holding	Long: synthetic-first attachment becomes the generational baseline	Positive: money and hours return to real life monthly
129	Real-world anchor subscriptions: one club/league/class with dues (skin in the game raises attendance)	Psychology & Mind	Future	50	50	75	50	35	15	45	55	Anchors joined after substitution is total must compete with a perfect product, not a habit	Mid: the anchor's seats fill with earlier joiners	Long: real-world community, unjoined, thins past viability in some places	Positive: the anchor becomes friendship, skill, and standing
130	Device architecture: companions/feeds off the phone's home screen, notifications batched	Psychology & Mind	Future	45	45	69.8	55	30	15	35	65	Friction installed late still works; the lost share is the compulsive years meanwhile	Mid: compulsive hours keep displacing sleep and reading	Long: attention is the mind's soil - erosion is slow and then total	Positive: reclaimed attention funds reading, family, and sleep
131	Home copying kit: notebooks, and one body of knowledge memorized per child (the unconfiscatable copy)	Education & Knowledge	Past	45	55	75.2	50	35	15	35	65	Copies made anytime preserve what still exists; the lost share is what burns meanwhile	Mid: uncopied knowledge keeps dying with its holders	Long: the canon, once gone, is archaeology - partial and expensive	Positive: copied texts seed schools after crises
132	Printed core references shelf (first aid, agriculture, mathematics, local law — bought used)	Education & Knowledge	Past	40	50	70	55	30	15	25	75	A shelf built late still teaches; delay loses only the unschooled interval	Mid: unschooled years accumulate at home	Long: minor - shelves teach at any age	Positive: the shelf serves neighbours and children beyond the household
133	Literacy defence: every child taught to read early at home if school fails (phonics materials)	Education & Knowledge	Past	65	45	80.8	45	35	20	45	55	Literacy taught late still transforms; the lost share is the child's early window, which is the cheapest one	Mid: the child's window narrows grade by grade	Long: adult literacy is learnable and slower - the tuition is years	Positive: literacy compounds into every other row of this file
134	Library-as-university setup: card + course platforms + a written self-curriculum	Education & Knowledge	Now	40	50	70	50	35	15	25	75	Learning can start any week; the lost share is the credential-debt paths entered meanwhile	Mid: credential paths keep being paid for unexamined	Long: skills unbought in the cheap years are bought in the dear ones	Positive: skills and portfolios outlast any credential cycle
135	Reading-with-children 20 min/day (the highest-yield household education act)	Education & Knowledge	Now	60	40	76	45	35	20	50	50	Reading begun at eight still helps - at a fraction of the effect of reading begun at three	Mid: reading gaps widen measurably per school year	Long: the non-reader's ceiling sets in across every subject	Positive: reading households raise readers - the effect is generational
136	Real-task testing kit: past papers, project portfolio, skill demonstrations filmed	Education & Knowledge	Now	35	55	70.8	50	35	15	30	70	Proof can be built anytime; the lost share is the hiring cycles missed meanwhile	Mid: hiring cycles pass the unproven by	Long: minor - portfolios compound from any start	Positive: proof portfolios open doors credentials cannot
137	No-device practice routine (weekly: handwriting, mental arithmetic, map navigation)	Education & Knowledge	Future	55	40	73	45	35	20	35	65	Practice begun in the failure still rebuilds; slower, under pressure	Mid: dependence deepens per automated year	Long: competence gaps become class gaps	Positive: confidence in outages and exams alike
138	Verification drill set: source-tracing exercises on real claims (family practice)	Education & Knowledge	Future	50	45	72.5	55	30	15	40	60	Drills begun after the first machine error audit the errors already absorbed	Mid: machine errors keep passing into decisions uncounted	Long: verification, unbuilt, becomes a profession you must buy	Positive: verifiers catch errors for employers and family too
139	Offline knowledge reserve: printed field guides + manuals for your trades and region	Education & Knowledge	Future	25	65	73.8	60	30	10	30	70	Reserves printed during an outage are printed from memory - choose the five volumes now	Mid: each outage re-runs the knowledge blackout	Long: printed reserves, unprinted, are a library that never existed	Positive: printed volumes are gifts and inheritance
140	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	Work & Livelihood	Past	60	35	74	60	30	10	40	60	Refusal decided after bundling must unwind housing, credit, and schooling at once	Mid: bundling deepens; exit prices rise	Long: capture becomes the family's normal - children are raised inside it	Positive: independence improves bargaining power in all jobs
141	Union or worker-association membership	Work & Livelihood	Past	45	50	72.5	55	30	15	35	65	Organization begun mid-dispute still wins some; begun before, it deters	Mid: disputes keep being lost individually	Long: workplaces without organization stay without it	Positive: collective power improves terms for non-members too
142	Portable insurance/contributions check (confirm what is yours regardless of employer)	Work & Livelihood	Past	40	55	73	55	30	15	40	60	Portability audited after signing audits a signed contract	Mid: each job loss re-runs the total-loss script	Long: benefit gaps compound into retirement gaps	Positive: portable coverage stabilizes family life across jobs
143	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	Work & Livelihood	Now	35	65	77.2	55	30	15	55	45	Records begun after the theft document the thefts after this one	Mid: theft continues monthly, unrecorded and therefore legal in practice	Long: unrecovered wages are the fund, the course, and the deposit - gone in instalments	Positive: records also support EI, tax, and pension claims
144	Labour-board claim filed for any unpaid wages (free) + worker-centre intake	Work & Livelihood	Now	30	60	72	55	30	15	50	50	Claims filed late meet limitation periods - some windows are already closed	Mid: violations normalize; new hires inherit them as 'the job'	Long: workplaces are trained by what goes unfiled	Positive: filed claims raise standards for the whole workplace
145	Portable proof-of-skills file (certificates, references, portfolio — copies in documents folder)	Work & Livelihood	Now	35	60	74	55	30	15	35	65	Proof assembled mid-search assembles under time pressure, thinner	Mid: gaps lengthen; explanations get harder	Long: career capital decays in long gaps, measurably	Positive: the portfolio serves promotions, not only exits
146	Emergency fund (shared item — costed once in Economy & Finance)	Work & Livelihood	Now	30	60	72	55	30	15	45	55	Shared item - delay counted in Economy & Finance	Mid: shared - see Economy & Finance	Long: shared - see Economy & Finance	Positive: shared item - see Economy & Finance
147	Ownership participation (shared with Economy build — costed once there)	Work & Livelihood	Future	55	50	77.5	50	30	20	40	60	Shared item - delay counted in Economy & Finance	Mid: shared - see Economy & Finance	Long: shared - see Economy & Finance	Positive: shared item - see Economy & Finance
148	Transition account: dedicated savings line for retraining (target: one course + 3 months)	Work & Livelihood	Future	50	55	77.5	45	35	20	60	40	A fund started at the announcement is an application for credit, not a fund	Mid: the announcement arrives to an empty account	Long: debt-funded transitions echo for a decade	Positive: the fund buys choice of next work, not just survival
149	Collective representation maintained (union/professional dues — see Economy)	Work & Livelihood	Future	45	50	72.5	45	35	20	50	50	Shared item - delay counted in Economy & Finance	Mid: shared - see Economy & Finance	Long: shared - see Economy & Finance	Positive: shared item - see Economy & Finance
150	Bicycle + repair kit + panniers (the independent-mobility baseline)	Transport & Mobility	Past	45	55	75.2	65	25	10	45	55	A bike bought in the disruption is bought at disruption prices, if the shops are open	Mid: isolation prices every necessity, monthly	Long: cut-off households relocate or wither - both are costs	Positive: health, savings, and local knowledge in ordinary times
151	Two-routes knowledge: printed local map with foot routes out of the district marked	Transport & Mobility	Past	40	45	67	60	30	10	35	65	Routes scouted under pressure are scouted at the worst time to learn them	Mid: disruptions keep improvising routes at the worst hours	Long: minor - knowledge is startable in any calm week	Positive: route knowledge serves visitors, children, emergencies
152	30-day staple store (shared with Food build — costed once there)	Transport & Mobility	Past	50	60	80	70	25	5	50	50	Shared item - delay counted in Food & Agriculture	Mid: shared - see Food & Agriculture	Long: shared - see Food & Agriculture	Positive: shared item - see Food & Agriculture
153	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	Transport & Mobility	Now	65	40	79	60	30	10	30	70	The pattern can be adopted any month; the lost share is the car-cost months meanwhile	Mid: car costs keep consuming the file's budget line	Long: a decade of car-dependence is six figures redirected from every other defence	Positive: thousands saved yearly fund the rest of this file
154	Car reserve discipline if car-dependent: never below quarter tank + maintenance sinking fund	Transport & Mobility	Now	35	60	74	55	30	15	45	55	Discipline begun at the breakdown begins with a repair bill already due	Mid: repairs keep arriving as emergencies	Long: the spiral, once started, ends in repossession statistics	Positive: car lasts longer; surprises shrink
155	Evacuation-assistance registration for vulnerable members (municipal lists)	Transport & Mobility	Now	25	70	77.5	55	30	15	55	45	Registration during the crisis queues behind the crisis	Mid: each transport failure is improvised alone	Long: unregistered need stays uncounted in service planning	Positive: registration also maps your eligibility for other programs
156	The bicycle, maintained (see Past row — the only transport no update can brick)	Transport & Mobility	Future	30	70	79	65	25	10	40	60	A standby bike unmaintained fails on its first emergency ride	Mid: platform dependence deepens per update	Long: the unmaintained fallback is a museum piece	Positive: the bike serves daily life, not only failures
157	Manual-driving competence maintained (licence + occasional practice) as vehicles automate	Transport & Mobility	Future	35	55	70.8	55	30	15	35	65	Licences lapsed long enough must be re-earned by examination	Mid: skills fade measurably per unpractised year	Long: re-licensing in later life is harder, dearer, sometimes barred	Positive: licence and skill keep every vehicle option open
158	Provider diversification: accounts with 2+ mobility platforms + cash for taxis	Transport & Mobility	Future	40	45	67	55	30	15	25	75	Diversification can be set up anytime; the lost share is the stranding events meanwhile	Mid: single points of failure keep failing singly	Long: minor beyond repetition	Positive: competition among your providers improves prices
159	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	Communication	Past	30	70	79	55	30	15	60	40	A plan written during the cutoff is written without the information it exists to carry	Mid: cutoffs keep finding the household planless	Long: separated families' stories diverge permanently	Positive: the plan also organizes ordinary reunions and emergencies of every kind
160	Congregation/association membership (the historical message network)	Communication	Past	40	55	73	55	30	15	40	60	Circles joined in crisis are joined as a supplicant, not a member	Mid: isolation persists per disruption	Long: circles unjoined in calm rarely exist in crisis	Positive: circles are friendship and culture, not only insurance
161	Letter/postal discipline: addresses memorized + written copies held by two relatives	Communication	Past	25	65	73.8	45	35	20	30	70	Letters work whenever sent; the lost share is the silent years meanwhile	Mid: silence accumulates between the separated	Long: unwritten years are unrecoverable years	Positive: letters become family archives
162	Monthly export routine: contacts, photos, records downloaded to your own drive	Communication	Now	30	75	82.5	60	30	10	40	60	Exports begun after the ban are appeals, not backups	Mid: the hostage archive grows monthly	Long: a decade of records inside platforms is a decade unownable	Positive: exports double as estate and migration records
163	One owned channel: personal email list / phone tree / radio net — set up and tested	Communication	Now	35	65	77.2	55	30	15	40	60	A channel built after silencing starts from zero audience, announced to no one	Mid: audience stays rented; rent rises	Long: channels unbuilt are followings unhad	Positive: owned audiences are business assets in normal times
164	Printed essential numbers card (family, doctors, banks, legal aid) in wallet + home	Communication	Now	20	70	76	60	30	10	25	75	A card printed anytime works; the lost share is the phones lost meanwhile	Mid: each lost phone re-runs the network loss	Long: minor beyond repetition	Positive: the card also helps a finder return your phone
165	Library digital-access anchor (card + weekly session if no home internet)	Communication	Now	35	55	70.8	55	30	15	30	70	Anchors serve walk-ins daily; delay loses the unconnected months	Mid: digital exclusion compounds as services go digital-only	Long: the excluded fall a full service-generation behind	Positive: library anchors build broad digital skill
166	Encrypted messaging as default for private communication (free apps) + device updates	Communication	Future	60	40	76	60	25	15	45	55	Encryption adopted after surveillance begins protects the future, not the archive	Mid: the unprotected archive grows monthly	Long: retroactive privacy is a contradiction - the stock is permanent	Positive: privacy improves every dealing, not only threatened ones
167	Data-rights requests filed (access + correction) for your major files annually	Communication	Future	30	60	72	55	30	15	50	50	Corrections filed after decisions are appeals against decided matters	Mid: errors compound into credit, tenancy, and work decisions	Long: a corrupted state file is a biography you must disprove forever	Positive: clean files also improve credit and insurance outcomes
168	Compartmentation basics: separate emails/aliases for finance, public, and private life	Communication	Future	55	40	73	55	30	15	40	60	Compartments built after a breach organize the next life, not the breached one	Mid: each account added uncompartmented enlarges the blast radius	Long: one late breach can be a whole-life event	Positive: breach damage stays small in ordinary hacks too
169	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	Demographics & Population	Past	20	75	80	40	35	25	65	35	The book begun after the elders die is begun from gravestones and guesses	Mid: sources keep dying; the book shortens	Long: what the last elder knew is the definition of gone	Positive: the book becomes the family's identity and inheritance

170	Skills duplication roster (each critical skill taught to 2+ people, tracked on paper)	Demographics & Population	Past	45	50	72.5	55	30	15	40	60	Skills taught after the holder's death are researched, not taught	Mid: single points of failure keep being single	Long: untaught skills become research projects, not inheritances	Positive: teaching deepens the teacher's mastery
171	Civil registration discipline: every birth/marriage/death registered and copied, even in chaos	Demographics & Population	Past	30	70	79	40	35	25	45	55	Late registration is possible and punitive - fees and proof multiply with every year	Mid: unregistered lives keep proving everything from zero	Long: descendants inherit the proof burden	Positive: registered events simplify every future bureaucracy
172	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	Demographics & Population	Now	55	50	77.5	55	30	15	50	50	Pooling arranged during the care crisis is arranged by exhausted people at market prices	Mid: care keeps being bought at crisis retail	Long: caregiver health, spent unpooled, is the second retirement lost	Positive: pooled care IS the social fabric - it pays in belonging
173	Benefits stack claim: child benefits, caregiver credits, parental top-ups — filed completely	Demographics & Population	Now	55	50	77.5	55	30	15	45	55	Shared item - delay counted in Economy & Finance	Mid: shared - see Economy & Finance	Long: shared - see Economy & Finance	Positive: shared item - see Economy & Finance
174	Waitlist registrations early (childcare, family housing, seniors' housing for parents)	Demographics & Population	Now	40	60	76	45	35	20	55	45	Queues joined in crisis start at the back, during the surge	Mid: queues keep being joined at the back	Long: unplaced elders and unplaced children are the system's long shadow	Positive: applications reveal other programs you qualify for
175	50-year family location audit (where will care, work, and community exist for your household's ages)	Demographics & Population	Future	50	40	70	45	35	20	45	55	Audits after the map has collapsed audit your own address in it	Mid: the service map keeps thinning unexamined	Long: retiring into a collapsed map is a one-way door	Positive: the audit improves housing and savings decisions immediately
176	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	Demographics & Population	Future	45	55	75.2	40	35	25	55	45	Pathways begun late meet age cutoffs and closed cohorts - some doors are birthdays	Mid: windows close on schedule - age cutoffs do not extend	Long: immobility in a declining region is the definition of trapped	Positive: pathways also serve education and family options in calm times
177	Credential recognition pre-filing for any planned move (12-month target)	Demographics & Population	Future	40	55	73	45	35	20	50	50	Recognition sought after migrating runs from abroad, slower and dearer	Mid: earnings penalties run monthly at the unrecognized rate	Long: a career priced at the wrong level compounds into pension	Positive: recognized credentials raise earnings from day one anywhere
178	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	Justice & Ethics	Past	20	80	84	35	40	25	55	45	Shared item - delay counted in Governance & Law	Mid: shared - see Governance & Law	Long: shared - see Governance & Law	Positive: shared item - see Governance & Law
179	Rights knowledge card for your system (arrest, search, detention limits) memorized + printed	Justice & Ethics	Past	35	45	64.2	55	30	15	40	60	Limits learned after the checkpoint are learned for the archive, not the encounter	Mid: encounters keep being navigated blind	Long: unrecorded violations are unprosecutable violations	Positive: rights knowledge dignifies every official encounter
180	Exit-readiness (shared with Security build — documents and routes)	Justice & Ethics	Past	45	55	75.2	55	30	15	50	50	Shared item - delay counted in Security & Defense	Mid: shared - see Security & Defense	Long: shared - see Security & Defense	Positive: shared item - see Security & Defense
181	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signature, or statement	Justice & Ethics	Now	45	60	78	55	30	15	55	45	Counsel after the signature advises on a signed world	Mid: signatures keep preceding understanding	Long: bad contracts echo for their full terms	Positive: advice prevents the next ten problems, not only this one
182	Rights card + 'no signature under pressure' household rule (posted, rehearsed)	Justice & Ethics	Now	60	35	74	60	30	10	60	40	The rule adopted after the first pressured signature protects the second contract, not the first	Mid: pressure keeps working on the unruled	Long: one coerced contract can price for a decade	Positive: the rule protects family members you teach it to
183	Small-claims/tribunal filing for winnable civil matters (fees modest, waivable)	Justice & Ethics	Now	30	65	75.5	50	35	15	40	60	Small claims filed late meet limitation periods and scattered evidence	Mid: grievances keep festering into feuds or silence	Long: unrecovered small losses are a regressive tax on the unassertive	Positive: a functioning small-claims habit lowers conflict in the whole community
184	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	Justice & Ethics	Now	25	75	81.2	50	35	15	55	45	Logs begun after the pattern is established reconstruct from memory - weaker paper	Mid: patterns keep being undocumentable	Long: the unlogged years are the defendant's best evidence - absence	Positive: logs also serve insurance, employment, and tax matters
185	Score hygiene routine: annual checks of credit, tenancy, and data-broker files; errors disputed in writing within windows	Justice & Ethics	Future	50	45	72.5	55	30	15	50	50	Hygiene begun after the flag cleans the record going forward; the flag's decisions stand meanwhile	Mid: flags keep pricing into decisions monthly	Long: scored lives optimize for the score; the good is crowded out	Positive: clean records price into cheaper credit and easier housing now
186	Human-review demand template (written request for human review of any automated decision, kept ready)	Justice & Ethics	Future	20	70	76	50	35	15	60	40	Appeals filed after their deadlines are correspondence, not appeals	Mid: wrong decisions settle into records and precedents	Long: unappealed errors become the training data of the next system	Positive: successful appeals improve the system for the next person
187	Case documentation for test-case organizations (your documented case can become the precedent)	Justice & Ethics	Future	40	40	64	35	40	25	35	65	Test cases can be joined anytime; bad rules settle deeper meanwhile	Mid: bad rules normalize per unchallenged year	Long: settled bad rules take movements, not letters, to move	Positive: clarified rights protect everyone, permanently

Field	Timeframe	Mean Preven	Mean Mitigat	Mean Combine	Mean Effect Los	PROBLEM If Not	PROBLEM — Shar	PROBLEM — Eff	PROBLEM — Effe	If Addressed ASAP — Positive Implication	If NOT Addressed ASAP — Negative Implication	How to Read the Timing
Water	Past	45	50	72.2	42.5	12	55	50	50	Acting early keeps every source mapped and every container filled while water still flows - the cheap season of this problem	Each season unaddressed, wells are claimed, knowledge of sources dies with elders, and the household enters scarcity already dependent	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Water	Now	50	53.8	77.6	58.8	8	30	65	35	A filter and reserve bought in calm times cost budget prices and work from the first advisory hour	Bought during an advisory, the same items cost spike prices - if shelves have any - and the first 24-72 hours of exposure are already spent	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Water	Future	37.5	43.8	65.2	33.8	6	45	40	60	Fixtures and storage installed early compound savings for decades and precede the scarcity pricing era	Demand keeps rising against a falling table; late movers buy storage at scarcity prices and inherit rationing designed by others	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Food & Agriculture	Past	56.2	57.5	82.5	57.5	15	65	55	45	A store built at harvest prices is the difference between a bad season and a famine biography	Unaddressed, the first failed season consumes the seed, the animals, and the children's growth - in that order, and the last does not come back	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Food & Agriculture	Now	35	57.5	71.9	30	7	25	45	55	A buffer built at normal prices converts every future spike into a non-event for a month at a time	Every month without the buffer is paid at retail spike prices; the household enters each crisis one paycheque thinner	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Food & Agriculture	Future	42.5	57.5	75.5	52.5	5	60	50	50	Perennials planted this year are producing in year three; soil built now holds water in the drought years that are coming	Soil and trees cannot be backdated. A decade unaddressed is a decade of yield that no later money repurchases	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Health & Medicine	Past	45	50	76	43.8	14	70	60	40	Hygiene, records, and vaccination done in calm times are nearly free and stop transmission chains at the first link	Unaddressed epidemics write themselves into orphanhoods and lifelong disability; the dead and the damaged do not return when the medicine arrives	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Health & Medicine	Now	41.2	42.5	65.7	35	9	50	45	55	Stewardship and prevention now keep the antibiotic era alive and dental costs at the $120 level	Resistance compounds silently; each year of delay retires drugs and multiplies the dental bill toward the five-figure reconstruction class	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Health & Medicine	Future	30	56.2	69	30	7	55	40	60	Monitoring and rotation habits built now intercept cascades while they are still cheap physiology, not yet events	Silent burdens (pressure, air, resistance) accumulate below symptom level; by the time they announce themselves the cheap decade has closed	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Energy	Past	40	57.5	75.4	50	10	40	55	45	Sealing, layers, and fuel laid in during summer cost off-season prices and are ready before the first cold night	The first unheated winter spends health, pipes, and schooling that later fuel cannot buy back; burst pipes are forever-costs	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Energy	Now	20	70	75.8	52.5	6	30	70	30	A battery bought in calm times is sized correctly, charged, and tested before it is ever needed	Bought during an outage - if stock exists - it arrives uncharged, untested, and at surge prices; the spoiled food and the dark nights are already spent	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Energy	Future	33.8	56.2	72	41.2	5	35	45	55	Insulation installed now pays back every season from this year; generation bought early produces through the price-rise decades	Energy poverty compounds like debt: cold homes damage health, health damage cuts income, cut income buys less energy	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Shelter & Housing	Past	26.2	75	83.8	53.8	11	50	60	40	Documents copied and bags packed in calm times take an afternoon and decide outcomes for decades	After the disaster, the undocumented compete for the same aid with no proof; uncompensated losses become permanent poverty	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Shelter & Housing	Now	42.5	60	77.8	52.5	13	35	55	45	Rights learned and rent relief claimed at the first missed payment interrupt the cascade while it is still a conversation	Arrears compound monthly, filings harden positions, and the same tenancy that was savable in month one is a hearing in month four	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Shelter & Housing	Future	37.5	51.2	73.2	57.5	6	60	50	50	Risk audited before purchase, the six-figure stranded asset is simply never bought	Insurance withdrawal and price discovery move one way; each year unaddressed, the exit value of exposed property falls and the cost of leaving rises	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Security & Defense	Past	41.2	58.8	77.1	62.5	16	75	70	30	Triggers decided in calm times fire on time; documents move before borders close	History prices late exits in lives. The window in which leaving is possible is the whole asset, and it closes without announcement	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Security & Defense	Now	38.8	50	72.6	37.5	8	35	50	50	Hardening and habits installed early cost budget prices and deter from the first night	Each unaddressed month is exposure at full rate; after an incident, the same measures cost more and prove less	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Security & Defense	Future	38.8	45	66.8	46.2	7	45	45	55	Footprint reduced now cannot be collected later; advocacy started early shapes the rules before they harden	Data once collected is never un-collected; surveillance normalized in one decade is litigated for three	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.

Governance & Law	Past	26.7	70	78.7	50	13	70	55	45 Archives copied abroad in calm times cost postage and are ready before the first knock	Under way, archives are seized with the houses that hold them; what is not already duplicated is already lost	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Governance & Law	Now	40	45	67.1	41.7	9	40	45	55 Requests filed while records still exist and officials still answer create the paper trail everything later stands on	Delay lets retention schedules destroy records lawfully; the file you needed is shredded on schedule, not on malice	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Governance & Law	Future	31.7	48.3	65.7	38.3	8	50	40	60 Local verification habits built now are the seed stock if national trust fails	Epistemic habits harden with practice; a population that waits to learn verification learns it under fire	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Economy & Finance	Past	63.3	43.3	79.6	56.7	14	55	50	50 Deposits split and debt refused before the crash are the whole defence; both are free decisions made in boring times	In the crash, every exit is crowded; the undiversified discover that 'later' was the position	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Economy & Finance	Now	36.2	60	75.2	46.2	18	40	55	45 A fund started today is small in month one and decisive by month twelve; the audit takes one afternoon	Debt compounds at 20%+ while the fund does not exist; every month unaddressed is a month the interest works and you do not	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Economy & Finance	Future	51.7	50	76	46.7	7	45	45	55 Ownership and skills accumulated during employment convert to income during displacement	Retraining begun after displacement is funded by debt and desperation; begun during, by surplus and choice	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Technology & AI	Past	43.3	45	72.3	40	10	65	40	60 Testing before purchase or renovation costs $180 and settles the question while answers are cheap	Latency is the trap: exposure continues daily while untested, and symptoms arrive decades after the evidence was fresh	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Technology & AI	Now	57.5	57.5	84.1	52.5	12	35	55	45 The free security stack takes one evening now and closes the majority of attack paths from tonight	Every unpatched month is exposure at full rate; after a breach, the same evening of work buys recovery measured in months instead	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Technology & AI	Future	33.3	61.7	76.8	43.3	6	40	45	55 Paper and exports maintained monthly mean no platform event can strand you	Lock-in compounds: each year of unexported data raises the cost of ever leaving, until leaving is theoretical	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Environment & Climate	Past	46.7	50	73.3	51.7	9	60	50	50 Soil and planting done in ordinary years are the only versions that exist in the bad year	Soil erodes on its own schedule; a decade of neglect is a decade of topsoil that no budget rebuys on any timeline	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Environment & Climate	Now	46.2	57.5	79.5	61.2	15	45	65	35 The valve and the audit cost a weekend in the dry season, at booked prices	Sewer backups do not schedule; the first storm after you meant to do it is the tuition - and installers book out for months after a flood	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Environment & Climate	Future	46.7	48.3	72.9	50	8	65	55	45 Audits and portable skills bought early are bought at calm prices with full choice of destination	Receiving regions price in the movers; each year, the good exits cost more and the forced ones multiply	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Information & Media	Past	40	46.7	68.8	50	14	60	50	50 Source habits formed in peacetime are the ones available in wartime	Incitement works on the unpractised first; by the time the lesson is obvious, the harm it teaches has already happened	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Information & Media	Now	46.7	46.7	71.8	31.7	10	30	45	55 The two-source rule costs seconds from the first day it is adopted	Manipulation compounds through your own shares; your network's trust, once spent on a hoax, does not fully refill	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Information & Media	Future	45	58.3	79.2	55	11	45	60	40 The code word takes five minutes today and must exist BEFORE the first clone call, by definition	The first synthetic fraud against an unprepared family is the most likely to succeed, and it teaches the fraudster your family's shape	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Psychology & Mind	Past	41.7	56.7	74.9	46.7	10	65	45	55 Rhythm and testimony begun early shorten the shadow by years	Unaddressed trauma transmits; the delay is not paid by the delayer alone but by the next generation's nervous systems	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Psychology & Mind	Now	42.5	55	75.3	42.5	9	45	40	60 Sleep, movement, and belonging started now intercept the common cascades while they are habits, not diagnoses	Insomnia, inactivity, and isolation entrench neurologically and socially; each season makes the same intervention slower, not impossible	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Psychology & Mind	Future	50	45	72.6	45	8	40	50	50 Budgets and anchors set before the products mature are set by you; set after, they are negotiated against an industry	Dependency engineered early becomes the baseline the person defends; unwinding costs more than prevention by design	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Education & Knowledge	Past	50	50	75.3	35	12	60	45	55 Copies and literacy secured early survive every later confiscation	Knowledge not copied dies with its holders; illiteracy, once a generation is raised in it, takes a generation to undo	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.

Education & Knowledge	Now	45	48.3	72.3	35	8	35	40	60	Library-anchored learning and daily reading started now compound through every school year	Reading failure hardens by grade level; the child not read to at five is the struggling student at ten and the capped earner at thirty	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Education & Knowledge	Future	43.3	50	73.1	35	6	40	45	55	Unaided competence practised now is the skill the automated age cannot devalue	Deskilling is invisible while systems work; it announces itself the first day they do not	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Work & Livelihood	Past	48.3	46.7	73.2	38.3	11	45	45	55	Independence kept and dues paid in calm times are the worker's whole negotiating position	Company-town capture deepens with every bundled year; exit gets more expensive as roots grow in company soil	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Work & Livelihood	Now	32.5	61.2	73.8	46.2	10	30	50	50	Records kept from this week and claims filed inside their windows recover money that is simply owed	Wage theft unrecorded is wage theft completed; limitation periods close claims monthly, silently	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Work & Livelihood	Future	50	51.7	75.8	50	7	40	50	50	The transition fund and collective terms built during employment are ready the day the announcement comes	Displacement met without a fund is met with debt; the fund cannot be backdated	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Transport & Mobility	Past	45	53.3	74.1	43.3	9	35	50	50	Your own wheels and known routes, ready before the disruption, are freedom of movement itself	Isolation compounds: missed supplies become missed medicine become missed rescue, in sequence	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Transport & Mobility	Now	41.7	56.7	76.8	43.3	8	30	40	60	The pass-and-bike pattern adopted now frees thousands per year from this month	Every year of car-dependence at $8,000-12,000 is a year the rest of this file goes unfunded	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Transport & Mobility	Future	35	56.7	72.3	33.3	6	35	45	55	A maintained bike and a live licence are options no software event can revoke	Manual skills and fallback vehicles, unmaintained, are theoretical options - they fail the day they are needed	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Communication	Past	31.7	63.3	75.3	43.3	12	50	50	50	A written plan and a living circle, made in calm times, coordinate the household through any cutoff	Rumour fills every silence; families separated without a plan stay separated for years, not weeks	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Communication	Now	30	66.2	76.6	33.8	8	35	45	55	Exports and an owned channel set up this month mean no platform action can erase you	Records accumulate inside platforms monthly; the unexported archive grows more hostage, not less	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Communication	Future	48.3	46.7	73.7	45	9	45	55	45	Encryption and compartments adopted now protect every conversation from tonight forward	The archive of unprotected communication only grows; you cannot retroactively encrypt a decade	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Demographics & Population	Past	31.7	65	77.2	50	11	65	45	55	The family book written while elders live is the lineage's permanent capital	Elders are perishable sources; each year unaddressed, the book gets shorter and the restart harder	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Demographics & Population	Now	50	53.3	77	50	10	35	50	50	Pooling arranged and benefits claimed now convert the care collision into a rota and a budget	Unpooled care is bought at market panic prices during the crisis it creates; caregiver burnout is the second casualty	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Demographics & Population	Future	45	50	72.7	50	7	50	45	55	Audits and pathways started early buy options at their cheapest - sometimes free	Demographic arithmetic does not negotiate; the options expire on schedule whether examined or not	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Justice & Ethics	Past	33.3	60	74.5	48.3	13	70	55	45	Archives and rights knowledge ready early decide outcomes after the terror ends	The record not made during the events cannot be made from the aftermath; impunity is built from missing paper	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Justice & Ethics	Now	40	58.8	77.2	52.5	9	35	50	50	Counsel before the first signature and logs from the first incident decide most matters before they become cases	Process hardens positions monthly; the unrepresented sign first and understand later, in that order, permanently	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.
Justice & Ethics	Future	36.7	51.7	70.8	48.3	8	40	50	50	Clean files and ready appeals maintained yearly keep automated errors small and reversible	Scores compound their errors: one uncorrected flag prices into every later decision the file feeds	Short term is governed by the Effect Lost If Late %: that share of the solution's power exists only inside the early window. Mid term is governed by Harm Growth %/yr: the problem compounds at that rate while unaddressed. Long term is governed by Irreversibility %: that share of the harm no later action, at any price, buys back.

#	Item	Field	Timeframe	Pareto Build	Short-Term (0–1 year) — Cost & Mitigation	Mid-Term (1–5 years) — Cost & Mitigation	Long-Term (5+ years) — Cost & Mitigation
1	Stored water reserve 56 L/person	Water	Past	76	Cost: purchase + first fill labour | Mitigation: food-grade containers only; date every fill	Cost: rotation labour ~1 hr/yr; container replacement ~$10/yr | Mitigation: rotate on a calendar date (spring/fall); inspect for cracks	Cost: full replacement cycle every ~5 yrs (~$48) | Mitigation: store containers out of sunlight — UV is what kills them
2	Cloth + sand filtration setup	Water	Past	77	Cost: setup labour 2 hrs | Mitigation: layer order matters (gravel, sand, cloth) — build it once, correctly	Cost: sand/cloth renewal ~$5/yr | Mitigation: pre-filter through cloth to extend sand life	Cost: negligible | Mitigation: teach the build to a second person — knowledge is the long-term asset
3	Fuel for boiling (1 month reserve)	Water	Past	78	Cost: first month stock ~$25 | Mitigation: dry, ventilated, CO-safe storage from day one	Cost: stock rotation; fuel price exposure | Mitigation: buy fuel in summer (off-season prices)	Cost: ongoing by design | Mitigation: the boiling habit itself is the permanent mitigation
4	Rain barrel 200 L	Water	Past	79	Cost: install 2 hrs | Mitigation: screen fitted at install (keeps animals and debris out)	Cost: cleaning 1 hr/season; tap seals ~$5 | Mitigation: annual scrub prevents biofilm	Cost: barrel life 10+ yrs → ~$6/yr | Mitigation: winter drain or freeze-proof placement
5	Certified gravity/pump water filter	Water	Now	1	Cost: $90 once + setup | Mitigation: register purchase date; elements age by volume, not time	Cost: elements $30–60/yr | Mitigation: pre-filter tap water through cloth in sediment events to double element life	Cost: ~$45/yr averaged across a decade | Mitigation: keep one spare element — supply chains fail exactly when filters matter
6	72-hour stored water (12 L/person drinking)	Water	Now	2	Cost: $8–16 per fill cycle | Mitigation: fill from tap at first advisory rumour, not confirmation	Cost: rotation every 6–12 months (water used in cooking, replaced free) | Mitigation: FIFO labels	Cost: near $0/yr if rotation uses the water | Mitigation: never let the reserve become a forgotten, stale asset
7	Bathtub water bladder (fills in minutes at first notice)	Water	Now	3	Cost: $35 once | Mitigation: rinse and dry fully after each use (mould is the failure mode)	Cost: negligible | Mitigation: store with the hose adapter attached — seconds count	Cost: bladder material life ~5 yrs → ~$7/yr | Mitigation: inspect seams annually
8	Water test strips (bacteria indicators + chlorine)	Water	Now	4	Cost: $20 per kit (~20 tests) | Mitigation: test on a schedule, not on suspicion	Cost: $20/yr restock | Mitigation: batch-test with neighbours and share results	Cost: $20/yr ongoing | Mitigation: a testing habit outlives any single kit
9	Low-flow fixtures set (aerators, showerhead, toilet kit)	Water	Future	40	Cost: $45 + 1.5 hrs fitting | Mitigation: fit aerators first (fastest payback)	Cost: $0 — savings recur (~15 L/day) | Mitigation: check washers yearly	Cost: fitting replacement ~10 yrs | Mitigation: the demand cut is permanent once installed
10	30-day deep storage (barrels)	Water	Future	41	Cost: $130 + floor check | Mitigation: distribute weight; fill in place, never move full	Cost: pump seals ~$10/5 yrs; rotation labour | Mitigation: rotate one barrel per season into household use	Cost: barrels 10+ yrs → ~$13/yr | Mitigation: a second household sharing halves both cost and risk
11	Greywater diverter for garden (where legal)	Water	Future	42	Cost: $55 + 3 hrs | Mitigation: soap discipline (plant-safe detergents only)	Cost: hose replacement ~$10/3 yrs | Mitigation: mulch basin renewed seasonally	Cost: negligible recurring | Mitigation: legality checked at each tenancy/bylaw change
12	Cooperative membership share (community well/storage)	Water	Future	43	Cost: $100 share | Mitigation: attend the co-op's meetings — governance is the product	Cost: annual dues vary (~$25–50) | Mitigation: labour shares substitute for cash in most co-ops	Cost: dues ongoing | Mitigation: the asset (well + storage) appreciates in scarcity — your share is the hedge
13	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil + salt)	Food & Agriculture	Past	80	Cost: $220 staged over weeks | Mitigation: buy in harvest season when staples are cheapest	Cost: rotation labour; pest losses 2–5%/yr if unsealed | Mitigation: sealed bins + bay leaves; FIFO	Cost: restock as consumed (~$220/cycle per adult) | Mitigation: the store is eaten and rebuilt — it is a flow, not a monument
14	Garden start: seeds + soil for 10 m²	Food & Agriculture	Past	81	Cost: $80 + heavy first-season labour | Mitigation: start with 4 proven crops, not 20 experiments	Cost: seed and compost ~$20/yr; labour 2–4 hrs/week in season | Mitigation: perennials and mulch cut labour yearly	Cost: soil improves — yields rise on the same spend | Mitigation: skills compound; teach a second gardener
15	Preservation kit (jars, lids, salt, vinegar)	Food & Agriculture	Past	82	Cost: $60 kit | Mitigation: tested recipes only — preservation errors are the risk	Cost: lids are single-use (~$10/yr); jars permanent | Mitigation: buy lids in bulk off-season	Cost: ~$10/yr | Mitigation: fermentation skills need no equipment at all once learned
16	Seed-saving envelopes + storage tin	Food & Agriculture	Past	83	Cost: $12 | Mitigation: label variety + year on every envelope	Cost: germination falls yearly (onion 1 yr, tomato 4+) | Mitigation: grow out each variety before its viability window closes	Cost: $0 once the cycle runs | Mitigation: seed exchanged with neighbours diversifies genetics free
17	30-day staple buffer per adult (~25 kg grain-equivalent)	Food & Agriculture	Now	5	Cost: $85 | Mitigation: build at $10/week of staples per shop until complete	Cost: rotation only | Mitigation: eat from the buffer weekly so rotation is automatic	Cost: flow cost = your normal grocery spend, redirected | Mitigation: the buffer is a pricing shield, renewed by eating
18	Buying-club share (bulk staples with 4–10 households)	Food & Agriculture	Now	6	Cost: $40 float | Mitigation: written split rules before the first bulk buy	Cost: coordination ~1 hr/month | Mitigation: rotate the organizer role	Cost: savings 15–30% recur indefinitely | Mitigation: the club is also a mutual-aid network — protect it
19	Sprouting kit (jars + seeds)	Food & Agriculture	Now	7	Cost: $18 | Mitigation: sterile rinse routine from day one	Cost: seed ~$10/yr | Mitigation: mung beans are the cheapest reliable sprout — standardize on them	Cost: ~$10/yr | Mitigation: jars last indefinitely
20	Community fridge / food bank registration + food-handler basics	Food & Agriculture	Now	8	Cost: transport ~$5/visit | Mitigation: register BEFORE crisis weeks (queues double then)	Cost: time per visit | Mitigation: combine with other errands; community fridges need no registration at all	Cost: $0 structural | Mitigation: giving back (volunteering) when able keeps the system you may need again
21	Perennial food plants (berry bushes, fruit tree, herb bed)	Food & Agriculture	Future	44	Cost: $90 + planting labour | Mitigation: plant in autumn (cheaper bare-root, better establishment)	Cost: pruning/mulch ~2 hrs/yr per plant | Mitigation: choose disease-resistant varieties at purchase — a one-time decision with decade effects	Cost: yields rise for 10–25 yrs → cost per harvest approaches $0 | Mitigation: propagate cuttings to replace and share
22	Soil building: compost system + cover crop seed	Food & Agriculture	Future	45	Cost: $45 + setup labour | Mitigation: site the pile once, correctly (drainage, shade)	Cost: turning labour; cover seed ~$15/yr | Mitigation: municipal free compost tops up your own	Cost: soil capital compounds — this is an appreciating asset | Mitigation: never sell or pave your soil
23	Dehydrator (entry electric or solar build)	Food & Agriculture	Future	46	Cost: $70 | Mitigation: solar build removes electricity cost entirely	Cost: electricity ~$1–2/batch (electric model); screens cleaned | Mitigation: batch-dry at harvest glut when produce is cheapest	Cost: unit life 5–10 yrs → ~$10/yr | Mitigation: dehydrated stores need no fridge space or power
24	Co-op / land-trust membership contribution	Food & Agriculture	Future	47	Cost: $75 | Mitigation: choose co-ops with transparent books	Cost: workdays ~4/yr | Mitigation: labour shares over cash where offered	Cost: land held in trust cannot be sold out from under the garden | Mitigation: governance participation is the premium
25	Household medical kit: thermometer, pulse oximeter, ORS packets, fever medication (2 weeks)	Health & Medicine	Past	84	Cost: $55 kit | Mitigation: check expiry dates every 6 months (calendar it)	Cost: consumable refills ~$15/yr | Mitigation: buy refills with seasonal pharmacy sales	Cost: devices last 5+ yrs → ~$11/yr | Mitigation: the kit's real long-term product is the household's care habit
26	Isolation-room setup (separate bedding, basin, waste bag system)	Health & Medicine	Past	85	Cost: $30 | Mitigation: designate the isolation room in advance, not mid-illness	Cost: laundry and disinfectant per episode | Mitigation: washable everything; avoid disposables	Cost: negligible | Mitigation: the protocol is remembered — rehearse it once with the household
27	Vaccination catch-up (all free public programs)	Health & Medicine	Past	86	Cost: $0 + appointment time | Mitigation: keep the record card with the health documents	Cost: boosters per schedule — $0 in public programs | Mitigation: set phone reminders; records in the folder	Cost: $0 | Mitigation: herd protection compounds — your vaccination protects the vulnerable in your own home
28	Paper medical records folder (vaccines, conditions, medications, allergies)	Health & Medicine	Past	87	Cost: $6 + 1 hr assembly | Mitigation: one folder, one place, told to all adults	Cost: 15 min/yr updates | Mitigation: update at every prescription change	Cost: near $0 | Mitigation: photograph the folder into the phone as backup
29	Drug coverage optimization: benefits audit + generic substitution review with pharmacist	Health & Medicine	Now	9	Cost: $0 + 1.5 hrs | Mitigation: bring every medication bottle to the review	Cost: repeat review yearly (free) | Mitigation: formularies change — last year's waste returns	Cost: $0 recurring; savings recur | Mitigation: teach the audit to family — it transfers
30	Home infection-control basics (soap stock, disinfectant, masks x50, gloves)	Health & Medicine	Now	10	Cost: $35 stock | Mitigation: warehouse sizes split with neighbours	Cost: ~$35/yr consumables | Mitigation: soap is the active ingredient — stock it deepest	Cost: ~$35/yr | Mitigation: habits persist after stocks lapse — teach the why
31	Dental prevention kit + one hygiene visit/yr	Health & Medicine	Now	11	Cost: $120/yr visit | Mitigation: dental-school clinics halve it	Cost: $120/yr steady | Mitigation: prevention is the mitigation — the ratio is 1:10–30 against emergencies	Cost: steady $120/yr vs. five-figure late-life dental reconstruction | Mitigation: start now; dental debt compounds like financial debt
32	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy take-back)	Health & Medicine	Now	12	Cost: $0 | Mitigation: tell your prescriber you return leftovers — it changes prescribing	Cost: $0 | Mitigation: pharmacy take-back is permanent infrastructure	Cost: $0 | Mitigation: population-level: your returned drugs protect everyone's future cures
33	Room HEPA unit for the sick room / bedroom	Health & Medicine	Future	48	Cost: $130 | Mitigation: size to the room (oversized units waste filters)	Cost: filters $40–70/yr | Mitigation: DIY Corsi-Rosenthal refills at ~$15	Cost: ~$55/yr averaged | Mitigation: in overload years this is the household's hospital-grade room
34	Ventilation upgrade: CO2 monitor + window/exhaust habits	Health & Medicine	Future	49	Cost: $110 | Mitigation: place at breathing height, away from the stove's false readings	Cost: battery/sensor life ~10 yrs | Mitigation: ventilation habits cost $0 once learned	Cost: ~$11/yr | Mitigation: the number makes air a managed resource permanently
35	Chronic-disease prevention stack: BP cuff + glucose basics + activity routine	Health & Medicine	Future	50	Cost: $45 | Mitigation: upper-arm cuff, correct size — accuracy is the product	Cost: weekly measurements (time) | Mitigation: log readings; trends beat single numbers	Cost: ~$9/yr across device life | Mitigation: early-found hypertension is the cheapest save in medicine
36	Reserve medication buffer (1 month extra of essentials, rotated, where legal)	Health & Medicine	Future	51	Cost: ~$40 to establish rotation | Mitigation: earliest-expiry at front; prescriber informed	Cost: rotation discipline | Mitigation: never exceed one extra month — hoarding degrades the system for everyone	Cost: near $0 steady-state | Mitigation: legal and safe only within prescribed quantities
37	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blanket fort supplies	Energy	Past	88	Cost: $60 | Mitigation: fit seals before the first cold snap, not during	Cost: curtain/seal wear ~$10/yr | Mitigation: the one-room doctrine is rehearsed, not improvised	Cost: ~$10/yr | Mitigation: children who learn it keep it for life
38	Fuel reserve (safe storage: gas canisters / charcoal / wood)	Energy	Past	89	Cost: $70 stock | Mitigation: approved containers only; dated fuel	Cost: fuel degrades (gasoline ~6–12 months) | Mitigation: rotate into vehicle/lawn use; charcoal and wood store for years	Cost: ongoing by design | Mitigation: dry wood is the only multi-year fuel — favour it
39	Layered clothing upgrade (wool base layers x2 per person)	Energy	Past	90	Cost: $45–90 per person | Mitigation: buy wool in summer sales and thrift first	Cost: wool care (gentle wash) | Mitigation: base layers are worn ~200 days/yr in cold homes — cost per wear is pennies	Cost: quality wool lasts a decade → ~$5/yr | Mitigation: repair, don't replace (darning is a 10-minute skill)
40	Pipe protection: insulation sleeves + shutoff knowledge	Energy	Past	91	Cost: $25 + 1 hr | Mitigation: tag the shutoffs while calm	Cost: foam degrades ~10 yrs | Mitigation: re-inspect each autumn (one calendar entry)	Cost: ~$3/yr | Mitigation: prevented burst pipes are the largest avoided loss in the energy rows
41	Battery power station 1 kWh (or 12V battery + inverter build)	Energy	Now	13	Cost: $420 (or $120–180 DIY) | Mitigation: LiFePO4 chemistry — cheaper cells die in 2 years and cost more by year 5	Cost: keep charge between 20–90%; one full cycle test monthly | Mitigation: a battery never exercised fails when needed	Cost: LiFePO4 life 10 yrs / 3,000+ cycles → ~$0.15/cycle, ~$42/yr | Mitigation: this is the file's model buy-once item
42	LED lanterns x3 + headlamp	Energy	Now	14	Cost: $35 | Mitigation: rechargeables only — disposable batteries are the expensive path	Cost: cell replacement ~3–5 yrs (~$10) | Mitigation: store charged; rotate monthly	Cost: ~$9/yr | Mitigation: headlamps keep hands free — the safety feature that matters
43	Safe backup cooking: single-burner butane stove + 4 canisters	Energy	Now	15	Cost: $45 | Mitigation: CO alarm in the home BEFORE the stove is ever lit indoors (see Shelter alarms)	Cost: canisters ~$3–4 each, shelf life years | Mitigation: buy canisters in summer	Cost: ~$15/yr stock rotation | Mitigation: outdoor-rated use keeps the option safe for decades
44	Fridge discipline kit: thermometer + cooler + ice packs	Energy	Now	16	Cost: $30 | Mitigation: thermometer lives in the fridge permanently (knowledge, not gear)	Cost: gel packs refrozen free | Mitigation: a full freezer holds 48 hrs; half-full, 24 — keep it full of water bottles	Cost: ~$6/yr | Mitigation: the saved $200–500 per outage repays the kit many times over

45	Solar trickle panel 100–200 W for the battery station	Energy	Future	52	Cost: $160 (or $90 for 100 W) | Mitigation: tilt and shade checked at install	Cost: panel cleaning 2x/yr | Mitigation: partial shade kills output — trim once, benefit for years	Cost: panels degrade ~0.5%/yr over 25 yrs → cost per kWh below grid | Mitigation: pairs with the battery to make protection indefinite
46	Home insulation priority pack (attic hatch seal, window film, draft proofing)	Energy	Future	53	Cost: $85 + a weekend | Mitigation: attic hatch and rim joists first (biggest leaks)	Cost: savings recur every season | Mitigation: re-caulk at paint cycles	Cost: measures last 15–30 yrs → ~$4/yr | Mitigation: insulation never needs a subscription
47	Community energy co-op share	Energy	Future	54	Cost: $100 share | Mitigation: read the co-op's annual report — ownership means oversight	Cost: dues vary | Mitigation: dividends offset dues in mature co-ops	Cost: generation assets last 25+ yrs | Mitigation: your share is a hedge denominated in electricity
48	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserve small bills	Energy	Future	55	Cost: $55 (incl. starter cash float) | Mitigation: small bills only; replenish after any use	Cost: radio batteries/crank check yearly | Mitigation: paper contacts updated at every number change	Cost: near $0/yr | Mitigation: the kit's value is realized in hours, once — price accordingly
49	Documents go-bag (waterproof pouch: ID, property, insurance, photos of every room's contents)	Shelter & Housing	Past	92	Cost: $18 | Mitigation: waterproof pouch; copies in TWO places (bag + trusted person)	Cost: 30 min/yr refresh (new IDs, policies) | Mitigation: refresh on a fixed date — birthday or tax day	Cost: ~$4/yr of copies | Mitigation: the folder's value peaks exactly when everything else is lost
50	72-hour go-kit per person (clothes, medication, water, light, food bars)	Shelter & Housing	Past	93	Cost: $65/person | Mitigation: pack for the climate you would evacuate INTO	Cost: food/medication rotation 2x/yr (~$10) | Mitigation: eat the rotated items; nothing is wasted	Cost: ~$13/person/yr | Mitigation: children's kits resized yearly — put it on the school-year calendar
51	Home inventory video + off-site copy	Shelter & Housing	Past	94	Cost: $0 + 1 hr filming | Mitigation: narrate purchases and serial numbers aloud	Cost: re-film after major purchases (minutes) | Mitigation: store the copy OFF the property (cloud + relative)	Cost: $0 | Mitigation: claims paid on documented losses run multiples higher
52	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (water/gas/power)	Shelter & Housing	Past	95	Cost: $70 | Mitigation: test alarms monthly (the 1st of the month rule)	Cost: batteries/yr ~$8; extinguisher check | Mitigation: fire departments often test/replace free for seniors	Cost: alarms replaced at 10 yrs → ~$7/yr | Mitigation: prevention items are the cheapest insurance ever sold
53	Tenant rights pack: your jurisdiction's rules printed + legal-aid and tenant-union numbers	Shelter & Housing	Now	17	Cost: $4 printing | Mitigation: verify against the current statute — rules change (Ontario changed in 2026)	Cost: re-check yearly (30 min) | Mitigation: law newsletters from tenant unions are free	Cost: ~$1/yr | Mitigation: knowledge compounds — tenants who know, teach
54	Housing documentation habit: dated photos at move-in/out, all landlord communication in writing	Shelter & Housing	Now	18	Cost: $0 habit | Mitigation: photos on day one, before furniture	Cost: minutes per tenancy event | Mitigation: email creates timestamps you cannot fake later	Cost: $0 | Mitigation: the habit transfers to every future tenancy and purchase
55	Rent-bank / eviction-prevention applications pre-filed knowledge + emergency contact list	Shelter & Housing	Now	19	Cost: $0 + 1 hr research | Mitigation: write the numbers down NOW — in crisis you will not research	Cost: refresh contacts yearly | Mitigation: programs change names; the need doesn't	Cost: $0 | Mitigation: pre-knowledge is the entire product
56	Shared-housing option prepared (one identified co-housing/roommate arrangement as Plan B)	Shelter & Housing	Now	20	Cost: $0 + social time | Mitigation: agree terms in writing while relations are good	Cost: relationship maintenance | Mitigation: a Plan B unused is still cheaper than a panic move	Cost: $0 | Mitigation: shared housing experience now makes later co-housing easy
57	Property risk check before buying/renewing: flood, fire, insurance-availability maps (free public data)	Shelter & Housing	Future	56	Cost: $0 + 2 hrs per property decision | Mitigation: check insurance AVAILABILITY, not just price — withdrawal is the leading indicator	Cost: re-check at each renewal | Mitigation: risk maps update; so must you	Cost: $0 | Mitigation: the audit's value is the purchase it prevents
58	Full insurance review incl. overland flood / sewer backup riders	Shelter & Housing	Future	57	Cost: ~$180/yr premium delta | Mitigation: raise the deductible — insure the catastrophe, self-insure the nuisance	Cost: annual premium | Mitigation: document mitigation work (valves, clearing) — insurers discount it	Cost: premiums rise with zone risk — the trend itself is your exit signal | Mitigation: exit while insurance still exists to transfer value to a buyer
59	Portability reserve: keep at least 50% of net worth out of any single exposed property (financial accounts, skills)	Shelter & Housing	Future	58	Cost: $0 (allocation) | Mitigation: skills and accounts are the portable half	Cost: opportunity cost of slower property accumulation | Mitigation: revisit the ratio yearly	Cost: $0 | Mitigation: mobility compounds like money
60	Buyout/relocation program registration (where offered) — filed early	Shelter & Housing	Future	59	Cost: $0 filing | Mitigation: register the moment a program opens	Cost: update the file yearly | Mitigation: early registrants historically receive the funded offers	Cost: $0 | Mitigation: queue position is the asset
61	Early-warning habit: named indicators list + decided trigger points ('if X happens, we leave within 48 h')	Security & Defense	Past	96	Cost: $0 + 1 hr decision | Mitigation: write the triggers down — in the moment, unwritten triggers move	Cost: review when conditions change | Mitigation: share the list with one person outside the household	Cost: $0 | Mitigation: the discipline, once learned, is permanent
62	Portable wealth & documents kit (copies abroad, small high-value items, qualifications translated)	Security & Defense	Past	97	Cost: $90 translations/copies | Mitigation: translate QUALIFICATIONS first (they monetize fastest)	Cost: keep copies current | Mitigation: digital copies with two holders on two continents	Cost: ~$10/yr | Mitigation: documents appreciate — they are your identity's title deed
63	Exit-route plan: two routes, rendezvous points, transport costs pre-estimated and saved	Security & Defense	Past	98	Cost: the earmarked savings themselves | Mitigation: hold the fund in a form you can carry or access from anywhere	Cost: fund competes with daily needs — defend it | Mitigation: even a partial fund buys the first, decisive leg	Cost: $0 carrying | Mitigation: routes re-verified when borders shift
64	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in signals)	Security & Defense	Past	99	Cost: $0 + 2 hrs organizing | Mitigation: written roster — memory fails under stress	Cost: one gathering per season | Mitigation: rosters decay; refresh contacts quarterly	Cost: time only | Mitigation: the roster IS the community's immune system
65	Home target-hardening basics: solid locks, strike plates with 3-inch screws, window locks, lighting	Security & Defense	Now	21	Cost: $85 hardware | Mitigation: 3-inch screws in strike plates are the $5 core of this item — do those first	Cost: locks rekeyed at tenancy changes (~$40) | Mitigation: lighting bulbs ~$5/yr	Cost: hardware life 20+ yrs → ~$5/yr | Mitigation: hardening is bought once per home, forever
66	Evidence habit kit: phone backup on, cloud copies, incident log notebook, dash/door camera where lawful	Security & Defense	Now	22	Cost: $60 camera optional | Mitigation: aim at your own property only — lawful and neighbour-safe	Cost: cloud storage free tiers suffice | Mitigation: incident log kept contemporaneously (same-day notes are evidence; later notes are memories)	Cost: ~$12/yr | Mitigation: backups tested quarterly
67	Route and routine discipline + check-in system with one contact	Security & Defense	Now	23	Cost: $0 | Mitigation: vary times, not just routes	Cost: minor inconvenience | Mitigation: check-in contact rotated if one is unavailable	Cost: $0 | Mitigation: habits persist across addresses and decades
68	Victim-compensation and protective-order applications knowledge (legal aid)	Security & Defense	Now	24	Cost: $0 + 1 hr | Mitigation: compensation deadlines are short — learn them before you need them	Cost: refresh yearly | Mitigation: legal-aid numbers change; verify	Cost: $0 | Mitigation: knowledge transfers to helping others — teach it
69	Venue standards habit: choose venues with visible security, exits known on entry (30-second rule)	Security & Defense	Future	60	Cost: $0 habit | Mitigation: exits first, belongings never	Cost: seconds per venue | Mitigation: teach children the 30-second rule as a game	Cost: $0 | Mitigation: the habit is lifelong and free
70	Personal data footprint reduction: data-broker removals, minimal public exposure	Security & Defense	Future	61	Cost: $30 + 2 hrs | Mitigation: start with the free removals (search engines, brokers' opt-outs)	Cost: new brokers appear — sweep yearly (1 hr) | Mitigation: a secondary email/phone for public life caps the work	Cost: ~$10/yr | Mitigation: footprint discipline gets easier as it becomes default
71	Support for regulation: specific, written asks to representatives (drone ID, autonomous weapons limits)	Security & Defense	Future	62	Cost: $0 letters | Mitigation: specific asks (ID mandates, geofencing) outperform general concern	Cost: minutes per campaign | Mitigation: join existing campaigns — signatures aggregate	Cost: $0 | Mitigation: laws, once passed, protect everyone permanently
72	Home/business camera + signage (deterrence + evidence)	Security & Defense	Future	63	Cost: $90 | Mitigation: signage alone deters a share — post it day one	Cost: storage/batteries ~$15/yr | Mitigation: aim and lighting checked seasonally	Cost: ~$25/yr | Mitigation: evidence value realized per incident — one incident repays years
73	Personal archive abroad (certified copies of identity, property, qualifications with trusted person/institution in another jurisdiction)	Governance & Law	Past	100	Cost: $75 assembly | Mitigation: two holders, two jurisdictions — never one basket	Cost: re-certification as documents expire | Mitigation: a contents list travels with the archive so gaps are visible	Cost: ~$10/yr | Mitigation: the archive is inherited — tell the next generation where it is
74	Qualifications in portable form (transcripts, licences, references sealed and copied)	Governance & Law	Past	101	Cost: $40 | Mitigation: sealed copies stay sealed (tamper-evident)	Cost: refresh at career changes | Mitigation: digital scans bridge paper loss	Cost: ~$5/yr | Mitigation: credentials are the exile-proof estate
75	Quiet-network membership (congregation, union, association dues)	Governance & Law	Past	102	Cost: $60/yr dues | Mitigation: choose organizations with real member networks, not mailing lists	Cost: dues + participation | Mitigation: attend — value lives in the room	Cost: dues steady | Mitigation: membership compounds: ten years in a network is a different asset than one
76	Access-to-information request kit (templates + filing fees where charged)	Governance & Law	Now	25	Cost: $15 fees | Mitigation: request fee waivers in the same letter	Cost: a few requests/yr | Mitigation: calendar follow-ups — silence is a tactic used on you	Cost: ~$10/yr | Mitigation: your filed history makes each next request stronger
77	Complaints filed in writing to ombudsman/auditor (free) with evidence folder	Governance & Law	Now	26	Cost: $5 copies | Mitigation: everything in writing; verbal answers are not answers	Cost: persistence hours | Mitigation: file collectively wherever possible — patterns compel	Cost: near $0 | Mitigation: published outcomes teach the next complainant
78	Civic membership: one watchdog, union, or ratepayers' organization	Governance & Law	Now	27	Cost: $40/yr | Mitigation: pick ONE body and actually participate	Cost: dues + meetings | Mitigation: committee seats are where citizens touch power	Cost: steady | Mitigation: institutional memory accrues to members
79	Local verification habit: attend/join one local body (assembly, board, co-op) and keep its records	Governance & Law	Future	64	Cost: time (6 hrs to join) | Mitigation: choose bodies with published minutes — verifiability is the point	Cost: monthly hours | Mitigation: rotate with a partner to prevent burnout attrition	Cost: time only | Mitigation: local proof is the seed stock of wider trust
80	Personal service records (your own file of wait times, outcomes, correspondence with agencies)	Governance & Law	Future	65	Cost: $0 habit | Mitigation: log outcomes the same day	Cost: minutes per interaction | Mitigation: the file grows into evidence	Cost: $0 | Mitigation: your records outlive administrations
81	Support independent local media (one subscription)	Governance & Law	Future	66	Cost: $60/yr | Mitigation: library access is the free substitute — subscribe only if you can	Cost: steady | Mitigation: independent outlets die without subscribers in peacetime	Cost: $60/yr | Mitigation: information infrastructure is bought before it is needed or not at all
82	Deposit splitting rule: never above the insured limit in one institution (CDIC $100k/category in Canada)	Economy & Finance	Past	103	Cost: $0 (account admin time) | Mitigation: automate the split — willpower is not a system	Cost: tracking overhead | Mitigation: one statement day per month for all accounts	Cost: $0 | Mitigation: insurance limits change — re-check at each renewal of the rules
83	Durable-goods and skills reserve (tools, preserved food, tradeable skills)	Economy & Finance	Past	104	Cost: $150 tools/staples | Mitigation: buy tools used; skills appreciate, goods depreciate	Cost: maintenance of tools ~$10/yr | Mitigation: tool libraries cover the rare-use items	Cost: ~$10/yr | Mitigation: durable value survives every currency regime
84	Debt minimalism in boom times (fixed-rate only, no margin, no speculation on credit)	Economy & Finance	Past	105	Cost: $0 (a refusal) | Mitigation: write the rule down; bubbles argue, paper doesn't	Cost: social/opportunity friction during booms | Mitigation: judge the rule at 10-year marks, not at party conversation	Cost: $0 | Mitigation: the rule's payoff arrives in the worst decade, which is the point
85	Emergency fund build: $500 starter then 1 month, then 3 (automatic transfer on payday)	Economy & Finance	Now	28	Cost: the saving itself (retained, not spent) | Mitigation: automate on payday — before spending, not after	Cost: discipline during the build (months) | Mitigation: milestones celebrated at $500 / 1 month / 3 months	Cost: the fund's 'cost' is negative — each emergency use saves 20%+ interest | Mitigation: refill rule: any use triggers automatic rebuild
86	Debt audit + avalanche order (highest rate first) printed and posted	Economy & Finance	Now	29	Cost: $0 + 1.5 hrs | Mitigation: list EVERY balance — partial audits lie	Cost: 15 min/month review | Mitigation: avalanche order recalculated as balances fall	Cost: $0 | Mitigation: interest saved compounds into the emergency fund
87	Free insolvency-trustee first consultation + credit-counselling session (non-profit)	Economy & Finance	Now	30	Cost: $0 | Mitigation: book before crisis peaks (their calendars fill too)	Cost: revisit at each major change | Mitigation: one professional map, updated, beats ten opinions	Cost: $0 | Mitigation: the consultation's value decays — re-run it every few years

88	Benefits and credits claim check (tax filing + benefits finder)	Economy & Finance	Now	31	Cost: $0 + filing time | Mitigation: file even at zero income — the return IS the application	Cost: yearly filing | Mitigation: benefits recalculate each July from the return	Cost: $0 | Mitigation: unclaimed benefits are the file's largest recoverable sum
89	Broad index investing habit (small automatic monthly amount in a low-fee diversified fund, inside tax shelters first)	Economy & Finance	Future	67	Cost: monthly contributions (retained, growing) | Mitigation: tax shelters first (TFSA/RRSP-class), low-fee index only	Cost: market volatility endured | Mitigation: never sell in the displacement panic — the fund exists FOR that decade	Cost: negative (compounding) | Mitigation: ownership is the long answer; time in it is the mechanism
90	Human-irreducible skill course (care, craft, repair, judgment trades)	Economy & Finance	Future	68	Cost: $250 + ~40 hrs | Mitigation: choose by employment outcomes data, not advertising	Cost: practice hours to earning level | Mitigation: first clients through your network (see Pareto action #6)	Cost: the skill appreciates with automation | Mitigation: refresh tools, not the trade
91	Union/professional-body membership	Economy & Finance	Future	69	Cost: $120/yr dues | Mitigation: attend the bargaining briefings — information is the member product	Cost: dues steady | Mitigation: collective terms are negotiated before displacement, or not at all	Cost: dues | Mitigation: the contract outlives the crisis
92	Home hazard test when buying older property (lead paint, asbestos survey before renovation)	Technology & AI	Past	106	Cost: $180 survey (or $40 kits) | Mitigation: test BEFORE renovation disturbs materials	Cost: findings filed permanently | Mitigation: disclose honestly on resale — liability follows concealment	Cost: one test per property → ~$6/yr across a 30-year occupancy | Mitigation: abatement only when needed, by licensed removal
93	Exposure records file (employers, products, dates) kept permanently	Technology & AI	Past	107	Cost: $4 folder | Mitigation: add employers/products as life happens (2 min each)	Cost: minutes per year | Mitigation: photograph the folder (cloud + paper)	Cost: ~$0 | Mitigation: the file matures into evidence exactly when latency expires
94	Adoption delay rule for unstudied non-essentials (wait for the 2nd generation / safety data)	Technology & AI	Past	108	Cost: $0 (waiting) | Mitigation: exceptions for genuine necessities, decided in advance	Cost: novelty lag | Mitigation: second-generation products are also cheaper — the rule pays you	Cost: negative | Mitigation: a lifetime rule, not a purchase
95	Password manager + MFA on email and banking (the two master keys)	Technology & AI	Now	32	Cost: $0 free tier + 1.5 hrs | Mitigation: secure EMAIL first, then banking (email resets everything)	Cost: vault hygiene 1 hr/quarter | Mitigation: backup codes printed and stored with documents	Cost: $0–40/yr if premium | Mitigation: per-login cost is fractions of a cent; the protected value is everything
96	3-2-1 backup: external drive + cloud copy of irreplaceables	Technology & AI	Now	33	Cost: $95 drive | Mitigation: automatic backup software — manual backups don't happen	Cost: one restore TEST per year (1 hr) | Mitigation: an untested backup is a rumour (drill it)	Cost: drive every ~5 yrs → ~$19/yr | Mitigation: 3-2-1 means any single failure is boring
97	Automatic updates ON + router firmware check (quarterly, calendared)	Technology & AI	Now	34	Cost: $0 + minutes | Mitigation: auto-update ON for OS and apps, router checked quarterly	Cost: occasional update breakage (rare, fixable) | Mitigation: update promptly — delay is the vulnerability	Cost: $0 | Mitigation: the habit is the patch
98	Verification protocol card (callback numbers printed; 'urgent = verify by second channel' rule)	Technology & AI	Now	35	Cost: $2 card | Mitigation: rehearse the script once with family (under stress, unrehearsed rules fail)	Cost: card replaced when numbers change | Mitigation: the rule is 'urgent = verify', posted where calls are taken	Cost: ~$0 | Mitigation: one stopped transfer repays a lifetime of cards
99	Analogue life-line kit: paper originals of ID/property/credentials, cash reserve, offline maps and contacts	Technology & AI	Future	70	Cost: $40 incl. cash float | Mitigation: laminate the critical sheets	Cost: refresh copies at document renewals | Mitigation: cash float replenished after any use	Cost: ~$5/yr | Mitigation: paper does not expire, get breached, or change terms
100	Data portability routine: monthly export of contacts, photos, records from every platform	Technology & AI	Future	71	Cost: $0 + 30 min/month | Mitigation: same day monthly — attach it to a bill-paying day	Cost: 6 hrs/yr | Mitigation: exports are also your migration kit between services	Cost: $0 | Mitigation: no platform can hold you hostage with your own past
101	Offline competence maintenance: one unaided skill practised regularly (navigation, arithmetic, writing)	Technology & AI	Future	72	Cost: time (practice) | Mitigation: tie practice to existing routines (mental arithmetic while shopping)	Cost: weekly minutes | Mitigation: skills maintained are skills kept	Cost: $0 | Mitigation: competence is the only asset class no system can revoke
102	Soil care basics: compost return, cover, no bare soil over winter	Environment & Climate	Past	109	Cost: $45 + labour | Mitigation: start with the worst square metre, not the whole plot	Cost: seasonal inputs ~$15/yr | Mitigation: municipal compost/leaf programs cut cash to near zero	Cost: soil gains compound | Mitigation: your soil in year 10 outperforms any purchased input
103	Water-wise planting (drought-matched species, mulch)	Environment & Climate	Past	110	Cost: $60 | Mitigation: plant for the climate arriving, not the one remembered	Cost: mulch renewal yearly | Mitigation: neighbour plant-swaps diversify free	Cost: established plants are near-free producers | Mitigation: perennials again — the climate answer repeats
104	Fuel efficiency: improved cookstove / insulation of heat	Environment & Climate	Past	111	Cost: $80 (or $20 DIY) | Mitigation: build quality decides fuel savings — follow a tested design	Cost: fuel halved permanently | Mitigation: chimney/flue kept clean (safety + efficiency)	Cost: stove life 10+ yrs → ~$8/yr | Mitigation: efficiency is the only fuel that never runs out
105	Address risk check (flood/fire/heat maps, free) + insurer confirmation in writing	Environment & Climate	Now	36	Cost: $0 + 1.5 hrs | Mitigation: get the insurer's answer IN WRITING	Cost: re-check at renewal | Mitigation: risk maps improve yearly — use the newest	Cost: $0 | Mitigation: knowledge positions every later dollar
106	Backwater valve + sump check (the $250 that prevents the $15,000 basement flood)	Environment & Climate	Now	37	Cost: $250 (often 50–100% subsidized) | Mitigation: claim the municipal subsidy FIRST — retroactive claims usually fail	Cost: test yearly (bucket of water) — minutes | Mitigation: sump battery backup if outages coincide with storms	Cost: ~$25/yr amortized | Mitigation: against a $15,000 loss class, this is the file's best ratio
107	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback, water stock	Environment & Climate	Now	38	Cost: $65 | Mitigation: shade OUTSIDE the glass first (film and blinds second)	Cost: fan electricity minimal | Mitigation: pre-cool at night; rehearse the room plan before the heat dome	Cost: ~$10/yr | Mitigation: pairs with the battery station for power-free cooling (fan on battery)
108	Go-bag + document pouch (shared with Shelter build — counted once there)	Environment & Climate	Now	39	Cost: shared item	Cost: shared — costed in Shelter & Housing	Cost: shared — see Shelter & Housing
109	Location decision audit every 5 years (risk trend, insurance trend, water trend for your address)	Environment & Climate	Future	73	Cost: $0 + 3 hrs per 5 yrs | Mitigation: track three numbers — insurance, water, services — trends, not headlines	Cost: the audit repeats | Mitigation: exit thresholds decided in advance, in writing	Cost: $0 | Mitigation: early movers sell to buyers; late movers sell to no one
110	Portable livelihood investment (skills and savings not tied to local climate-sensitive industry)	Environment & Climate	Future	74	Cost: $200 + 20 hrs | Mitigation: choose skills by portability AND local demand in receiving regions	Cost: practice to earning level | Mitigation: clients and reputation are portable assets	Cost: career-long returns | Mitigation: income that can move is the true climate insurance
111	Collective relocation readiness: community/municipal plan participation	Environment & Climate	Future	75	Cost: time | Mitigation: join the municipal process while it is still planning, not evacuation	Cost: meetings | Mitigation: collective plans protect the vulnerable first — be inside the plan	Cost: time only | Mitigation: communities that plan together move together
112	Source-tracing habit taught at home (claim → origin → evidence) — a dinner-table practice, not a purchase	Information & Media	Past	164	Cost: $0 habit | Mitigation: practise on small claims so the skill holds on big ones	Cost: dinner-table minutes | Mitigation: make the question ('who says, and how would they know?') a family reflex	Cost: $0 | Mitigation: the habit is inheritable — it outlives every platform
113	Independent media support: one exile/public/independent subscription or donation	Information & Media	Past	165	Cost: $50/yr | Mitigation: fund in PEACETIME — subscriptions raised during crises arrive after them	Cost: steady | Mitigation: share access within the household	Cost: $50/yr | Mitigation: independent channels are infrastructure bought by the year
114	Battery shortwave/AM radio for foreign and emergency broadcasts	Information & Media	Past	166	Cost: $35 | Mitigation: hand-crank or battery — power-independent by design	Cost: batteries ~$5/yr | Mitigation: preset the stations that matter	Cost: ~$5/yr | Mitigation: radios outlive networks
115	Two-source rule card + one scheduled news window (a household protocol, posted)	Information & Media	Now	112	Cost: $2 card | Mitigation: post it where the phone is answered	Cost: seconds per claim | Mitigation: the rule applies to 'good' news too — flattery is also a vector	Cost: ~$0 | Mitigation: attention returned is time earned back daily
116	Primary-source subscriptions: one quality outlet + official data sources bookmarked	Information & Media	Now	113	Cost: $90/yr or $0 via library | Mitigation: one quality source, deeply read, beats ten skimmed	Cost: steady | Mitigation: renew deliberately each year — audit the diet	Cost: ≤$90/yr | Mitigation: your floor source sets your ceiling of error
117	Fact-check and provenance tools bookmarked; reverse image search learned	Information & Media	Now	114	Cost: $0 + 1 hr learning | Mitigation: bookmark the tools on the phone itself	Cost: minutes per claim | Mitigation: screenshot before sharing — evidence first	Cost: $0 | Mitigation: verification speed compounds with practice
118	Family verification code word + callback rule for money/urgent requests (agreed, written, rehearsed)	Information & Media	Future	140	Cost: $0 + one conversation | Mitigation: rehearse it — a code word never used in calm will fail in panic	Cost: refresh the word yearly (holidays work) | Mitigation: extend the rule to texts and video calls, not just voice	Cost: $0 | Mitigation: one stopped clone-fraud repays forever
119	Paper originals vault (identity, property, credentials) + notarial copies of the critical five	Information & Media	Future	141	Cost: $75 notary | Mitigation: the critical five documents only — don't notarize the trivial	Cost: renew copies as documents renew | Mitigation: copies held by the trusted person abroad (see Governance)	Cost: ~$8/yr amortized | Mitigation: provable identity is priceless exactly when forgery is free
120	Content-credential tools adopted (signed capture apps) for important records	Information & Media	Future	142	Cost: $0 tiers | Mitigation: use signed capture for records that may be disputed (contracts, incidents, property state)	Cost: seconds per record | Mitigation: originals archived off-platform monthly	Cost: $0 | Mitigation: the provenance habit future-proofs your evidence
121	Routine and ritual structure (fixed sleep, meals, work, worship/rest — a written household rhythm)	Psychology & Mind	Past	167	Cost: $0 structure | Mitigation: protect ONE anchor first (sleep or meals) when everything shakes	Cost: steadiness labour | Mitigation: the rhythm bends in crises and returns after — that return is the skill	Cost: $0 | Mitigation: children raised inside rhythm carry it into their own households
122	Mutual storytelling / testimony practice (regular, structured, within family or congregation)	Psychology & Mind	Past	168	Cost: $0 | Mitigation: go gently — testimony at the teller's pace, never extracted	Cost: weekly time | Mitigation: stop rules agreed in advance (anyone may pause)	Cost: $0 | Mitigation: stories told become stories survivable
123	Children-first protection of sleep, schooling, and play (the intergenerational shield)	Psychology & Mind	Past	169	Cost: $0 | Mitigation: boring predictability is the intervention — same wake, same meals, same school	Cost: adult self-regulation is the hidden price — budget support for the adults too	Cost: $0 | Mitigation: the intergenerational dividend is the largest in this file
124	Sleep protocol: fixed window, morning light (free), screens out of the bedroom, caffeine cutoff	Psychology & Mind	Now	115	Cost: $25 optional curtain | Mitigation: light before phone in the morning (the order matters)	Cost: two-week adjustment discomfort | Mitigation: shift the window by 15 minutes, not an hour	Cost: $0 steady | Mitigation: sleep defends every other system in this file simultaneously
125	Movement prescription: 60 min/day walking built into commute/errands + one strength session weekly	Psychology & Mind	Now	116	Cost: $0 + ~7 hrs/wk time (stated plainly) | Mitigation: build it into commutes and errands — transport walking is free movement	Cost: time is the price, permanently | Mitigation: pair it with the weekly group (walk together) — two interventions, one slot	Cost: time | Mitigation: fitness is the only intervention here that also pays in every other row
126	One weekly in-person group (see Pareto action #6 — costed once)	Psychology & Mind	Now	117	Cost: shared item | Mitigation: shared — costed in Citizen Pareto	Cost: attendance | Mitigation: eight consecutive attendances before judging	Cost: $0–dues | Mitigation: see Citizen Pareto action #6
127	Crisis lines saved in phone + EAP/benefits check (free session entitlements)	Psychology & Mind	Now	118	Cost: $0 | Mitigation: save lines under plain names in the phone (no searching mid-crisis)	Cost: refresh numbers yearly | Mitigation: tell one person where your crisis plan lives	Cost: $0 | Mitigation: EAP entitlements renew annually — use them or lose them
128	Synthetic-companionship budget rule (fixed hours + money cap, written, reviewed monthly)	Psychology & Mind	Future	143	Cost: $0 rule | Mitigation: write the cap down; review with the statement monthly	Cost: honesty in review | Mitigation: treat breaches as data, not failure — tighten the design, not the guilt	Cost: $0 | Mitigation: the budget rule appreciates as the products get better at hooking
129	Real-world anchor subscriptions: one club/league/class with dues (skin in the game raises attendance)	Psychology & Mind	Future	144	Cost: $80/yr dues | Mitigation: pay dues deliberately — paid memberships attend at multiples of free ones	Cost: weekly attendance | Mitigation: choose the anchor you would miss if it vanished	Cost: ~$80/yr | Mitigation: real rooms, real people — the competitor synthetic comfort cannot copy
130	Device architecture: companions/feeds off the phone's home screen, notifications batched	Psychology & Mind	Future	145	Cost: $0 setup | Mitigation: batch notifications to two windows daily	Cost: a week of itch | Mitigation: keep ONE deliberate joy-channel — total abstinence designs fail	Cost: $0 | Mitigation: friction is self-regulation you install once

131	Home copying kit: notebooks, and one body of knowledge memorized per child (the unconfiscatable copy)	Education & Knowledge	Past	170	Cost: $0 + copying labour | Mitigation: two copies, two places (pair with the documents folder)	Cost: refresh as knowledge updates | Mitigation: margins stay open — a finished notebook stops learning	Cost: ~$0 | Mitigation: the notebook is taught forward — that is its permanence
132	Printed core references shelf (first aid, agriculture, mathematics, local law — bought used)	Education & Knowledge	Past	171	Cost: ~$40/yr at used prices | Mitigation: buy reference, not novelty (the shelf's spine is dictionaries, atlas, manuals)	Cost: steady small spend | Mitigation: library discards and swaps feed the shelf free	Cost: falling (a complete shelf stops growing) | Mitigation: the shelf's cost per consultation approaches zero for life
133	Literacy defence: every child taught to read early at home if school fails (phonics materials)	Education & Knowledge	Past	172	Cost: adults' daily hour (the real price, stated) | Mitigation: read-alouds count; story IS literacy's engine	Cost: the hour continues through childhood | Mitigation: the child reads alone by year 3–4 and the adult hour ends	Cost: $0 after | Mitigation: one literate generation teaches the next — the cost line ends
134	Library-as-university setup: card + course platforms + a written self-curriculum	Education & Knowledge	Now	119	Cost: $0 + weekly hours | Mitigation: one project at a time, finished publicly (portfolio)	Cost: hours steady | Mitigation: credentials audited yearly against job postings — prune dead paths fast	Cost: time | Mitigation: skills are the only tuition that cannot be refunded by regret
135	Reading-with-children 20 min/day (the highest-yield household education act)	Education & Knowledge	Now	120	Cost: ~120 hrs/yr of adult time | Mitigation: library story-times and school reading programs share the load	Cost: the hour shrinks as the child takes over (year 3+) | Mitigation: protect the habit through the middle-school dip — that is where it dies	Cost: $0 | Mitigation: reading children become teaching adults; the line terminates
136	Real-task testing kit: past papers, project portfolio, skill demonstrations filmed	Education & Knowledge	Now	121	Cost: $8 + project time | Mitigation: pick assessments with external judges (exams, clients, contests)	Cost: one new proof per year | Mitigation: proofs stack into a portfolio employers actually open	Cost: ~$8/yr | Mitigation: evidence outlives certificates
137	No-device practice routine (weekly: handwriting, mental arithmetic, map navigation)	Education & Knowledge	Future	146	Cost: 30 min/week | Mitigation: attach practice to an existing habit (coffee, commute)	Cost: time steady | Mitigation: teach a family member weekly — teaching is practice with witnesses	Cost: $0 | Mitigation: unaided competence is a wasting asset only if unpractised
138	Verification drill set: source-tracing exercises on real claims (family practice)	Education & Knowledge	Future	147	Cost: $0 (drills) | Mitigation: verify ONE important output daily as routine, not paranoia	Cost: minutes daily | Mitigation: error logs reveal your personal failure modes	Cost: $0 | Mitigation: the verifier's habit scales from homework to hospital bills
139	Offline knowledge reserve: printed field guides + manuals for your trades and region	Education & Knowledge	Future	148	Cost: $45 | Mitigation: choose the five reference volumes your life actually queries	Cost: refresh at decade marks | Mitigation: printed knowledge doesn't change terms or vanish	Cost: ~$5/yr | Mitigation: pairs with the archive (Governance)
140	Cash-wage rule + independent housing/food (never bundle more than 2 of job/housing/food/credit with one master)	Work & Livelihood	Past	173	Cost: $0 (a decision) | Mitigation: decide BEFORE the offer arrives, in writing	Cost: occasional higher cash costs for unbundled living | Mitigation: unbundle gradually (housing first)	Cost: the premium of independence, small and steady | Mitigation: against company-town capture, it is the cheapest line in this file
141	Union or worker-association membership	Work & Livelihood	Past	174	Cost: $50/yr dues | Mitigation: attend one meeting per quarter minimum — presence is the membership	Cost: dues steady | Mitigation: union legal and insurance benefits typically exceed dues	Cost: dues | Mitigation: seniority and solidarity both compound
142	Portable insurance/contributions check (confirm what is yours regardless of employer)	Work & Livelihood	Past	175	Cost: $0 + one audit | Mitigation: request the portability answer in writing	Cost: re-audit at every job change | Mitigation: portable benefits checked BEFORE signing new contracts	Cost: $0 | Mitigation: the audit habit follows you across every employer
143	Personal work log: hours, schedules (photos), pay stubs — one folder, updated weekly	Work & Livelihood	Now	122	Cost: $0 + 5 min/week | Mitigation: photograph schedules the day they post	Cost: the weekly minutes | Mitigation: one folder, chronological — boards read chronology	Cost: $0 | Mitigation: records within limitation periods only — file older ones as history, current ones as claims
144	Labour-board claim filed for any unpaid wages (free) + worker-centre intake	Work & Livelihood	Now	123	Cost: $0 + hours per claim | Mitigation: file even small amounts — patterns are built from small filings	Cost: retaliation risk managed via documentation (item 1) | Mitigation: file collectively where coworkers share the practice	Cost: $0 | Mitigation: recovered wages typically exceed the fund's first milestone
145	Portable proof-of-skills file (certificates, references, portfolio — copies in documents folder)	Work & Livelihood	Now	124	Cost: $6 + 2 hrs | Mitigation: tailor per application FROM the master proof, never rewrite it	Cost: refresh at each credential | Mitigation: one afternoon yearly keeps it current	Cost: ~$2/yr | Mitigation: the proof portfolio is your portable reputation
146	Emergency fund (shared item — costed once in Economy & Finance)	Work & Livelihood	Now	125	Cost: shared item	Cost: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance
147	Ownership participation (shared with Economy build — costed once there)	Work & Livelihood	Future	149	Cost: shared item	Cost: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance
148	Transition account: dedicated savings line for retraining (target: one course + 3 months)	Work & Livelihood	Future	150	Cost: savings build (retained) | Mitigation: automate from the first month of employment, not the first crisis	Cost: the account defends itself — watch it quarterly only | Mitigation: hold it separate from the emergency fund (different wars)	Cost: negative (it earns while it waits) | Mitigation: time converts this from luxury to inevitability
149	Collective representation maintained (union/professional dues — see Economy)	Work & Livelihood	Future	151	Cost: shared item | Mitigation: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance	Cost: shared — see Economy & Finance
150	Bicycle + repair kit + panniers (the independent-mobility baseline)	Transport & Mobility	Past	176	Cost: $160 used | Mitigation: buy through a community bike shop (fit + a year of adjustments)	Cost: maintenance ~$40/yr; parts widely available | Mitigation: learn three repairs (tube, chain, brake pads) — YouTube + one evening	Cost: a good frame lasts decades → ~$10/yr | Mitigation: the bike is the only vehicle whose fuel is dinner
151	Two-routes knowledge: printed local map with foot routes out of the district marked	Transport & Mobility	Past	177	Cost: $0 + reconnaissance time | Mitigation: walk routes once in daylight before needing them at night	Cost: re-verify seasonally | Mitigation: routes shift with construction and seasons	Cost: $0 | Mitigation: knowledge shared with the household multiplies it
152	30-day staple store (shared with Food build — costed once there)	Transport & Mobility	Past	178	Cost: shared item	Cost: shared — costed in Food & Agriculture	Cost: shared — see Food & Agriculture
153	Transit pass + cycling combination replacing second-car or car-lite pattern (costed as monthly pass)	Transport & Mobility	Now	126	Cost: $156/month pass (or concession rate) | Mitigation: employer/university/benefit passes first — many riders overpay by default	Cost: monthly steady — against $8,000–12,000/yr car cost | Mitigation: the freed difference funds the rest of this file	Cost: steady | Mitigation: transit dependence is a budget feature, not a compromise
154	Car reserve discipline if car-dependent: never below quarter tank + maintenance sinking fund	Transport & Mobility	Now	127	Cost: the sinking fund contributions (retained) | Mitigation: calculate YOUR car's true monthly figure once, honestly	Cost: fund tops up monthly | Mitigation: repairs stop being debts the month the fund starts	Cost: the spiral never starts | Mitigation: sell the second car before the fund — arithmetic decides
155	Evacuation-assistance registration for vulnerable members (municipal lists)	Transport & Mobility	Now	128	Cost: $0 registration | Mitigation: register while healthy — crisis registration queues	Cost: eligibility renewals | Mitigation: keep the booking app AND phone number both	Cost: $0 | Mitigation: paratransit and volunteer-driver networks persist for decades — join early
156	The bicycle, maintained (see Past row — the only transport no update can brick)	Transport & Mobility	Future	152	Cost: $40/yr maintenance | Mitigation: store indoors — weather, not use, kills standby bikes	Cost: tires/tubes ~$20/2 yrs | Mitigation: one annual tune keeps it evacuation-ready	Cost: ~$25/yr | Mitigation: the fallback that cannot be bricked, priced honestly
157	Manual-driving competence maintained (licence + occasional practice) as vehicles automate	Transport & Mobility	Future	153	Cost: practice time | Mitigation: one manual drive per month keeps the skill	Cost: licence renewals | Mitigation: keep the licence even car-free — re-testing later costs more	Cost: renewal fees only | Mitigation: competence is retained optionality
158	Provider diversification: accounts with 2+ mobility platforms + cash for taxis	Transport & Mobility	Future	154	Cost: $0 + setup | Mitigation: register payment methods across providers in advance	Cost: minutes yearly | Mitigation: never let one app hold your only route home	Cost: $0 | Mitigation: diversification compounds quietly
159	Family communication plan on paper: rendezvous points, message drops, contact chain, agreed before crisis	Communication	Past	179	Cost: $6 print | Mitigation: laminate; one copy per adult + one at the rendezvous contact	Cost: rehearse yearly (15 min) | Mitigation: update at every move/number change	Cost: ~$1/yr | Mitigation: plans un-rehearsed are fiction — the rehearsal is the product
160	Congregation/association membership (the historical message network)	Communication	Past	180	Cost: $25/yr contributions | Mitigation: show up — networks are maintained in person	Cost: steady small | Mitigation: reciprocity balances itself over years	Cost: steady | Mitigation: the circle is also your emergency housing, job leads, and care rota
161	Letter/postal discipline: addresses memorized + written copies held by two relatives	Communication	Past	181	Cost: postage | Mitigation: addresses written in the plan's contacts	Cost: occasional | Mitigation: letters are archived by recipients — your words persist	Cost: ~$5/yr | Mitigation: the channel cannot be deprecated
162	Monthly export routine: contacts, photos, records downloaded to your own drive	Communication	Now	129	Cost: $0 + 30 min/month | Mitigation: calendar it with bill-paying	Cost: the monthly half hour | Mitigation: automate what can be (cloud sync) and hand-check the rest	Cost: drive space free tiers | Mitigation: two copies on two services — one ban can't erase you
163	One owned channel: personal email list / phone tree / radio net — set up and tested	Communication	Now	130	Cost: $0 (free tiers) | Mitigation: tell people the new channel once, in the old one	Cost: minutes of upkeep | Mitigation: the owned list exports with you forever	Cost: free tiers persist | Mitigation: audience you own is income insurance
164	Printed essential numbers card (family, doctors, banks, legal aid) in wallet + home	Communication	Now	131	Cost: $1 card | Mitigation: card in the phone case AND the wallet	Cost: replace at number changes | Mitigation: memorize the top two numbers as backup	Cost: ~$0 | Mitigation: the card outlives every phone
165	Library digital-access anchor (card + weekly session if no home internet)	Communication	Now	132	Cost: $0 + weekly 1 hr | Mitigation: book the library session like an appointment	Cost: the weekly hour | Mitigation: staff help is free — ask; it is literally their job	Cost: $0 | Mitigation: anchor sessions build skill until the anchor is needed less
166	Encrypted messaging as default for private communication (free apps) + device updates	Communication	Future	155	Cost: $0 + 1 hr setup | Mitigation: set up with ONE trusted contact first, together	Cost: app updates | Mitigation: safety numbers shared once, verified in person	Cost: $0 | Mitigation: encrypted-by-default becomes the household standard
167	Data-rights requests filed (access + correction) for your major files annually	Communication	Future	156	Cost: $0 + 2 hrs/yr | Mitigation: pull every file on a fixed date (birthday)	Cost: the yearly hours | Mitigation: dispute errors in writing the same week	Cost: $0 | Mitigation: clean state files compound — each year the baseline is truer
168	Compartmentation basics: separate emails/aliases for finance, public, and private life	Communication	Future	157	Cost: $0 + setup hours | Mitigation: financial and identity accounts in the strictest compartment, always	Cost: minor daily friction | Mitigation: friction is the visible price of an invisible protection	Cost: $0 | Mitigation: breach blast-radius stays small forever
169	Family knowledge book: recipes, remedies, trades, histories, contacts — written, copied, shared	Demographics & Population	Past	182	Cost: $15 + ~15 hrs family labour | Mitigation: interview the oldest first — the source list is mortal	Cost: additions at family events | Mitigation: bring the book TO gatherings; memories surface socially	Cost: ~$2/yr | Mitigation: copies to two households — the book outlives its house
170	Skills duplication roster (each critical skill taught to 2+ people, tracked on paper)	Demographics & Population	Past	183	Cost: $0 (teaching time) | Mitigation: teach under supervision in calm seasons, not crises	Cost: refresh drills yearly | Mitigation: skills taught to two survive one absence	Cost: $0 | Mitigation: redundancy is the family's continuity plan
171	Civil registration discipline: every birth/marriage/death registered and copied, even in chaos	Demographics & Population	Past	184	Cost: $20 fees | Mitigation: register immediately — late registration multiplies fees and difficulty	Cost: copies for each institution that asks | Mitigation: the documents folder holds masters; hand out copies only	Cost: ~$5/yr | Mitigation: registered lives claim everything; unregistered lives prove everything
172	Care-pooling arrangement: 2–3 families or multi-generational roster with written schedule	Demographics & Population	Now	133	Cost: $0 + coordination hours | Mitigation: written rota — goodwill without a rota burns out by month three	Cost: the rota's steady hours | Mitigation: rotate the heavy slots; nobody carries two in a row	Cost: time | Mitigation: pooled care outlasts purchased care because it is also the social fabric
173	Benefits stack claim: child benefits, caregiver credits, parental top-ups — filed completely	Demographics & Population	Now	134	Cost: shared item	Cost: shared — costed in Economy & Finance	Cost: shared — see Economy & Finance

174	Waitlist registrations early (childcare, family housing, seniors' housing for parents)	Demographics & Population	Now	135	Cost: $0 applications | Mitigation: apply to waitlists BEFORE need — positions are rationed by date	Cost: renewals yearly | Mitigation: keep files warm with annual confirmations	Cost: $0 | Mitigation: queue seniority compounds like interest
175	50-year family location audit (where will care, work, and community exist for your household's ages)	Demographics & Population	Future	158	Cost: $0 + 4 hrs per 5 yrs | Mitigation: use pension/health AUTHORITY projections, not headlines	Cost: the 5-yearly half day | Mitigation: thresholds and triggers written into the audit	Cost: $0 | Mitigation: the audit's product is never being surprised by arithmetic
176	Second-residency/citizenship pathway research + applications where lawfully entitled (descent, etc.)	Demographics & Population	Future	159	Cost: fees (vary widely) + years of timeline | Mitigation: start with the FREE eligibility assessments many states offer	Cost: document gathering | Mitigation: every document doubles for the destination's credential file	Cost: the option's price falls the earlier it is bought | Mitigation: optionality bought young is nearly free; bought late, it is nearly priceless
177	Credential recognition pre-filing for any planned move (12-month target)	Demographics & Population	Future	160	Cost: ~$200 assessments | Mitigation: assess BEFORE migrating — recognition from abroad is slower and dearer	Cost: bridging courses vary | Mitigation: regulated professions publish their pathways — follow the published map only	Cost: one-time | Mitigation: credentials recognized once earn forever
178	Evidence archive abroad (dated records of abuses, property, identity — duplicated outside the jurisdiction)	Justice & Ethics	Past	185	Cost: shared item	Cost: shared — costed in Governance & Law	Cost: shared — see Governance & Law
179	Rights knowledge card for your system (arrest, search, detention limits) memorized + printed	Justice & Ethics	Past	186	Cost: $2 card | Mitigation: memorize the five key phrases; the card is the backup	Cost: refresh at legal changes | Mitigation: rights known approximately beat rights unknown perfectly	Cost: ~$0 | Mitigation: knowledge is the only property that cannot be confiscated at the checkpoint
180	Exit-readiness (shared with Security build — documents and routes)	Justice & Ethics	Past	187	Cost: shared item	Cost: shared — costed in Security & Defense	Cost: shared — see Security & Defense
181	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signature, or statement	Justice & Ethics	Now	136	Cost: $0 + hours per matter | Mitigation: bring the documents folder to the FIRST consultation	Cost: duties counsel exist for arrests round the clock — save the number	Cost: $0 | Mitigation: advice sought early prevents the matters that cost
182	Rights card + 'no signature under pressure' household rule (posted, rehearsed)	Justice & Ethics	Now	137	Cost: $1 card | Mitigation: the rule covers family too — time pressure is the tell, whoever applies it	Cost: $0 | Mitigation: legitimate parties accept 'I need advice first' instantly	Cost: $0 | Mitigation: the habit protects every contract of your life
183	Small-claims/tribunal filing for winnable civil matters (fees modest, waivable)	Justice & Ethics	Now	138	Cost: ~$35 fee (waivable) + your hours | Mitigation: organize evidence chronologically — adjudicators reward it	Cost: hearing time | Mitigation: mediation first where offered (faster, free)	Cost: ~$35 per matter | Mitigation: recoverable when you win, in most tariffs
184	Record-keeping for discrimination/wrongdoing (contemporaneous notes, saved messages)	Justice & Ethics	Now	139	Cost: $3 notebook | Mitigation: date + time + names, same day, every time	Cost: minutes per incident | Mitigation: contemporaneous beats voluminous	Cost: ~$1/yr | Mitigation: your log is the plaintiff's exhibit and the defendant's shield
185	Score hygiene routine: annual checks of credit, tenancy, and data-broker files; errors disputed in writing within windows	Justice & Ethics	Future	161	Cost: $0 + ~3 hrs/yr | Mitigation: pull scores on a fixed date; correct in writing	Cost: the yearly hours | Mitigation: freeze credit when not borrowing (free) — prevention beats correction	Cost: $0 | Mitigation: clean files are a form of wealth — they price into every contract
186	Human-review demand template (written request for human review of any automated decision, kept ready)	Justice & Ethics	Future	162	Cost: $0 template | Mitigation: customize per decision; keep tone factual	Cost: minutes per dispute | Mitigation: deadlines for human review are short — file fast	Cost: $0 | Mitigation: templates improve with each use — keep your outcomes noted
187	Case documentation for test-case organizations (your documented case can become the precedent)	Justice & Ethics	Future	163	Cost: $0 + document-keeping | Mitigation: your documented case is the standing test cases need	Cost: time if your case is chosen | Mitigation: clinics carry the legal costs of test litigation	Cost: $0 | Mitigation: rights are clarified once and enjoyed by everyone after

Field	Timeframe	Problem (abbreviated)	Short-Term Cost	Mid-Term (Years 1–5, per year)	Long-Term (Years 5+, per year)	Mitigating the Costs — Together, Across All Horizons	Problem Cost Avo	Leverage Ratio (¢	Pareto Effici
Psychology & Mind	Past	Psychology & Mind — Untreated trauma: lost working years, family transmission, addiction self-medication costs	0	Mid: $0 (time only)	Long: $0	Together: rhythm and testimony cost steadiness, not money — the correct currency for the problem.	16500	Infinite ($0 build)	1
Demographics & Population	Now	Demographics & Population — Care-cost collision: childcare and eldercare purchased at market rates simultaneou	0	Mid: rota hours; benefits shared via Economy	Long: rota hours	Together: pooled care's currency is reciprocity hours, and it returns more than it costs.	27500	Infinite ($0 build)	2
Justice & Ethics	Future	Justice & Ethics — Algorithmic-injustice costs: wrongful flags costing jobs, credit, bail, and liberty — fough	2	Mid: 3 hrs/yr audits	Long: 3 hrs/yr	Together: score hygiene is rented in hours, and it prices into every future contract.	13000	6500	3
Shelter & Housing	Now	Shelter & Housing — Eviction or forced move: first/last deposits, movers, storage, rent premiums, job and scho	4	Mid: $0–5/yr (reprints, refresh)	Long: $0–5/yr	Together: the Now build is knowledge; its carrying cost is one hour a year.	16500	4125	4
Communication	Now	Communication — Platform-loss costs: businesses and families losing their only contact list, records, and audi	2	Mid: $0 (habits + free tiers)	Long: $0	Together: exports and lists are free forever once set; the cost was the setup hour.	4100	2050	5
Education & Knowledge	Now	Education & Knowledge — Credential-without-learning cost: $10,000s of tuition for skills the market does not r	8	Mid: time (~120 hrs/yr reading) + $8/yr proofs	Long: time, falling	Together: skills and reading carry no cash rent; the budget is hours, protected by scheduling them like bills.	16000	2000	6
Work & Livelihood	Now	Work & Livelihood — Precarity costs: unpaid wages never claimed ($100s–$1,000s per worker yearly), no benefits	6	Mid: $0 recurring (records habit)	Long: $0	Together: the Now build pays out (recovered wages) rather than costing forward.	10750	1791.7	7
Justice & Ethics	Past	Justice & Ethics — Weaponized-justice costs: confiscation, imprisonment, exile — defence impossible, so the co	63	Mid: shared via Governance/Security	Long: shared	Together: the Past build is carried by the Governance archive and Security roster — costed there, once.	52500	833.3	8
Communication	Future	Communication — Surveillance/chilling costs: self-censorship losses, wrongful suspicion from data errors, blac	10	Mid: 2 hrs/yr audits	Long: 2 hrs/yr	Together: privacy's rent is two hours a year.	7750	775	9
Environment & Climate	Future	Environment & Climate — Uninhabitable-zone stranding: property worthless, relocation at personal cost, livelih	200	Mid: $40–80/yr (course amortization, participation)	Long: falling as portable income establishes	Together: audits are free; the investment is in portability, which pays wages, not just protection.	135000	675	10
Health & Medicine	Past	Health & Medicine — Pandemic without preparation: income loss, funeral costs, lifelong disability from untreat	91	Mid: $15–30/yr (refills, boosters free)	Long: $15–30/yr	Together: the entire Past build costs less per year than one pharmacy run during an outbreak; keep the folder current and the rest maintains itself.	55000	604.4	11
Shelter & Housing	Future	Shelter & Housing — Climate-stranded asset: an unsellable, uninsurable home is a total loss of the family's ma	180	Mid: ~$180/yr (insurance delta)	Long: $180+/yr, rising with zone risk — the rise is the signal	Together: the free audits govern the one paid line; when the premium trend breaks your threshold, the portability reserve does its work.	105000	583.3	12
Security & Defense	Past	Security & Defense — Persecution and war: loss of property, livelihood, and in the worst cases life — costs be	90	Mid: $10–30/yr (copies, roster upkeep)	Long: $10–30/yr	Together: almost everything here is decisions and paper; the cash cost is trivial against the losses governed.	52500	583.3	13
Shelter & Housing	Past	Shelter & Housing — Loss of the home itself: the largest single financial event in a household's life; displac	153	Mid: $15–25/yr (kit rotation, alarm batteries)	Long: $15–25/yr	Together: under $25/yr maintains the entire Past build — the cheapest insurance portfolio in the file.	85000	555.6	14
Technology & AI	Future	Technology & AI — System-dependence failure: locked out of money, records, transport, and identity when platfo	40	Mid: $5–10/yr (copy refresh)	Long: $5–10/yr	Together: paper and exports — the cheapest rows in the file to maintain.	21000	525	15
Demographics & Population	Past	Demographics & Population — Lineage-collapse costs: skills and records dying with elders; families restarting	42	Mid: $5/yr (copies, registrations)	Long: $5/yr	Together: paper maintenance — the lineage's records cost less than a streaming subscription.	21500	511.9	16
Education & Knowledge	Past	Education & Knowledge — Knowledge loss at family scale: trades, languages, and literacies die in one generatio	75	Mid: $40/yr (shelf) + adults' hour through childhood	Long: ~$0 (self-teaching generation)	Together: the Past build's currency is one daily hour, and it terminates — literate children teach themselves.	32500	433.3	17
Psychology & Mind	Now	Psychology & Mind — Mental-health crisis costs: lost work ($1,000s/yr in absenteeism and presenteeism), relati	25	Mid: $25 one-time + time	Long: time only (~7 hrs/wk movement — the honest price, stated)	Together: the Now build's real budget is hours, and the file prices them openly instead of pretending otherwise.	10750	430	18
Justice & Ethics	Now	Justice & Ethics — Two-tier justice costs: the unrepresented plead, pay, and lose winnable cases; one convicti	37	Mid: $35/matter + $1/yr notebook	Long: same	Together: counsel free, process cheap, records priceless — the Now build costs less than one lost deposit.	15500	418.9	19
Education & Knowledge	Future	Education & Knowledge — Unaided-incompetence cost: professionals who cannot function or verify without systems	57	Mid: 30 min/week practice	Long: 30 min/week	Together: competence rent is half an hour a week — cheaper than any subscription it defends against.	21500	377.2	20
Governance & Law	Past	Governance & Law — Regime collapse costs: confiscation, forced labour, exile — money is the smallest part.	175	Mid: $15–30/yr (copies, dues)	Long: $15–30/yr	Together: paper and memberships — under $30/yr maintains the entire historical-governance build.	52500	300	21
Economy & Finance	Past	Economy & Finance — Deflation/depression: savings destroyed by bank failure, debts compounding in real terms,	150	Mid: $10–20/yr (tool upkeep)	Long: $10–20/yr	Together: the Past build is nearly free to maintain — its costs were decisions, and decisions don't invoice.	42500	283.3	22
Work & Livelihood	Future	Work & Livelihood — Full displacement of a mid-career worker: retraining at personal cost plus a decade of red	300	Mid: fund contributions (retained) via shared Economy build	Long: negative (fund earns)	Together: see Economy & Finance Future — one shared build carries both.	65000	216.7	23
Work & Livelihood	Past	Work & Livelihood — Company-town capture: wages recycled to the employer through scrip, store, and rent — a wo	100	Mid: $50/yr dues	Long: $50/yr	Together: dues are the only recurring line; audits and decisions are free.	21500	215	24
Security & Defense	Future	Security & Defense — Drone/autonomous attack or mass incident: injury, property loss, event liability chaos.	120	Mid: $35–50/yr (camera, removals)	Long: $30–45/yr	Together: habits and letters are free; the camera is the only subscription-class cost, and it is optional.	25500	212.5	25
Demographics & Population	Future	Demographics & Population — Demographic-trap costs: pensions and care systems repriced against your cohort; st	270	Mid: fees where pathways are pursued (varies)	Long: falling once options are secured	Together: buy optionality early and cheap; the audit itself is free.	55000	203.7	26
Information & Media	Past	Information & Media — Propaganda-driven harm: participation in or victimization by incited violence; property	85	Mid: $55/yr (subscription + batteries)	Long: $55/yr	Together: one subscription maintains the household's alternative channel.	15500	182.4	27
Governance & Law	Future	Governance & Law — Legitimacy collapse costs: services failing for lack of compliance and trust — longer queue	60	Mid: $60/yr (one subscription) + time	Long: $60/yr + time	Together: time is the currency here — 2 hrs/month, budgeted like money.	10500	175	28
Environment & Climate	Past	Environment & Climate — Land degradation at household scale: failed wells, eroded plots, fuelwood exhaustion —	185	Mid: $15–25/yr (seed, upkeep)	Long: falling (soil compounds)	Together: labour is the input; cash declines yearly as perennials and soil mature.	31500	170.3	29
Communication	Past	Communication — Isolation-by-monopoly costs: rumour-driven losses, missed warnings, family separation without	63	Mid: $30/yr (circle + postage)	Long: $30/yr	Together: circles and paper — $30/yr carries crisis communication.	10250	162.7	30
Technology & AI	Past	Technology & AI — Latent-exposure harm: lead/asbestos class injuries — medical costs and lost years, compensat	184	Mid: ~$0/yr (records kept)	Long: ~$6/yr amortized (tests per property)	Together: vigilance items: near-zero carrying cost, five-figure avoided losses.	25500	138.6	31
Information & Media	Future	Information & Media — Forgery-era losses: fraud via cloned voices/faces, legal evidence chaos, wrongful accusa	75	Mid: $8/yr (notary amortized)	Long: $8/yr	Together: a code word and paper vault — near-zero carrying cost against the most probable future fraud.	10250	136.7	32
Economy & Finance	Future	Economy & Finance — Displacement without ownership: a working life of wages ends with no claim on the machine	470	Mid: contributions (retained) + $120/yr dues	Long: contributions compound — the 'cost' becomes the income	Together: fund ownership monthly like a bill; the course is bought once; dues are the rent on collective power.	55000	117	33
Psychology & Mind	Future	Psychology & Mind — Synthetic-dependence costs: subscription stacks, lost human relationships, fertility and c	80	Mid: $80/yr (anchor dues)	Long: $80/yr	Together: one dues line defends against an entire product category engineered to extract.	8000	100	34
Health & Medicine	Now	Health & Medicine — One uninsured ER visit or a resistant infection: $1,000s in bills where uninsured, weeks o	155	Mid: $170–200/yr (dental visit + hygiene consumables)	Long: $170–220/yr	Together: the dental line looks like the cost and is actually the savings — budget it as insurance, which it is.	13250	85.5	35
Health & Medicine	Future	Health & Medicine — Care-system overload: delayed surgeries, untreated chronic disease progression, private-ca	325	Mid: $100–140/yr (filters, monitoring)	Long: $90–130/yr	Together: one HEPA unit and one cuff carry the build; DIY filter refills halve the recurring line.	26500	81.5	36
Transport & Mobility	Past	Transport & Mobility — Isolation costs: goods at siege prices, medical care unreachable, evacuation impossible	168	Mid: $40/yr (bike upkeep)	Long: ~$25/yr	Together: a bicycle and known routes — the entire Past build runs on $40/yr.	13500	80.4	37
Transport & Mobility	Future	Transport & Mobility — Platform-grounding cost: fleets bricked by software or cyber events; surge pricing in t	100	Mid: $40/yr (bike maintenance)	Long: $40/yr	Together: the manual fallback is maintained for the price of two tubes a year.	8000	80	38

Technology & AI	Now	Technology & AI — Ransomware/fraud loss: average consumer fraud losses run hundreds to thousands; identity the	97	Mid: $19–25/yr (drive amortized, batteries)	Long: $19–40/yr	Together: under $40/yr carries the whole household cyber build — less than one hour of fraud-recovery time.	7750	79.9	39
Environment & Climate	Now	Environment & Climate — One flood or fire event uninsured: $10,000s in repairs, displacement months, health ef	315	Mid: $25–40/yr (valve checks, kit upkeep)	Long: $25–40/yr	Together: one subsidized valve plus a $65 kit — the recurring line is smaller than a single insurance deductible.	23000	73	40
Security & Defense	Now	Security & Defense — Victimization costs: theft, injury, lost work, trauma care; a single break-in averages th	145	Mid: $45–70/yr (rekeying amortized, storage, bulbs)	Long: $35–55/yr	Together: hardware bought once per home; the recurring line is smaller than one insurance deductible.	10250	70.7	41
Governance & Law	Now	Governance & Law — Corruption tax: bribes, rigged procurement passed into prices, fines and fees extracted by	60	Mid: $40–60/yr (dues, fees)	Long: $40–60/yr	Together: the paid line is one membership; everything else is writing. Collective filing divides even the time.	4150	69.2	42
Food & Agriculture	Past	Food & Agriculture — Famine conditions: asset sales at distress prices, malnutrition, child development harm —	372	Mid: $250–350/yr (restock flow + garden inputs)	Long: $200–300/yr, falling as soil and seed mature	Together: buy staples at harvest-season lows; rotate communally to cut spoilage; each year the garden and seed cycle carry more of the load and the cash share falls.	22500	60.5	43
Water	Past	Water — Illness from unsafe water; days of labour hauling water; crop loss.	148	Mid (1–5 yrs): $30–80/yr rotation and parts	Long (5+ yrs): $50–100/yr steady state	Together: short term, pool purchases with neighbours (containers and barrels are cheaper by the pallet); mid term, one rotation calendar for the whole street; long term, the co-op well ends the per-household cost curve entirely.	8500	57.4	44
Food & Agriculture	Future	Food & Agriculture — Structural food inflation; diet-related disease costs; dependence on failing supply regio	280	Mid: $60–100/yr (plants, lids, dues)	Long: $40–80/yr, yields rising	Together: perennials invert the cost curve — spend early, harvest for decades; the dehydrator pays for itself each harvest glut.	11250	40.2	45
Energy	Future	Energy — Months-long outage scenario: relocation costs, lost employment, home damage from cold/heat without po	400	Mid: $30–60/yr (dues, batteries)	Long: negative-to-low (panel generation offsets)	Together: insulation first (it pays you), panel second (it pays you slower), co-op last (it pays everyone).	16000	40	46
Transport & Mobility	Now	Transport & Mobility — Car-dependence cost: $8,000–12,000/yr all-in per household car in Canada — the largest	206	Mid: pass cost monthly (the cheap option) or fund contributions	Long: steady — always below car cost by $5,000+/yr	Together: transport is where households overpay most; the Pareto answer is the pass first, the fund second, the second car never.	7500	36.4	47
Information & Media	Now	Information & Media — Misinformation losses: fraud conversions, health misinformation harm, family and civic c	92	Mid: $0–90/yr (one subscription or library $0)	Long: $0–90/yr	Together: the defence is free at library tier; the paid tier buys depth, not safety.	2600	28.3	48
Economy & Finance	Now	Economy & Finance — Debt spiral: interest at 20%+ on revolving credit converts a $3,000 emergency into years o	500	Mid: negative (interest saved + benefits claimed recur)	Long: negative and compounding	Together: the Now build PAYS the household — benefits claimed and interest avoided typically exceed $1,000/yr for eligible households.	13500	27	49
Food & Agriculture	Now	Food & Agriculture — Price-spike exposure: grocery inflation on a full budget runs $100–250/month extra; empty	148	Mid: $100–150/yr (buffer flow + sprouting seed)	Long: $85–120/yr	Together: the buffer is eaten, not warehoused — its carrying cost is near zero; the buying club's discount funds its own continuation.	3600	24.3	50
Energy	Past	Energy — Embargo winter: burst pipes ($5,000+), lost work, fuel at black-market prices.	245	Mid: $40–70/yr (fuel flow, seals)	Long: $35–60/yr	Together: insulate people, then one room, then pipes — in that cost order; every step funds the next from saved fuel.	5750	23.5	51
Water	Future	Water — Day-zero style shortages: water trucking prices, lost work days, garden and hygiene collapse.	330	Mid: $30–60/yr (dues, seals, rotation)	Long: $25–50/yr	Together: fixtures pay for themselves first (negative net cost), then fund the storage; co-op dues replace capital outlays; the household that cuts demand 30% needs 30% less of everything else.	6750	20.5	52
Water	Now	Water — Bottled-water panic buying at spike prices; missed work; illness risk if advisory ignored.	161	Mid: $40–80/yr (elements, strips, rotation)	Long: $45–70/yr	Together: short, fill everything at first rumour (free); mid, share one filter across households in rotation; long, the testing-and-rotation habit costs less than one bottled-water panic per year.	2150	13.4	53
Energy	Now	Energy — Blackout losses: spoiled food ($200–500), lost remote-work days, medical-device risk, hotel nights.	530	Mid: $25–45/yr (canisters, cells, ice packs)	Long: $50–70/yr averaged (battery amortized)	Together: the battery is the only capital item; everything else is consumables under $50/yr combined.	4400	8.3	54

Pareto Analysis of Term Costs

The vital few solutions	The top 11 of 54 solutions by leverage ratio (20%) carry $354,600 of the $1,359,250 total avoided problem cost = 26.1%, and cost only $386 of the $8,179 short-term build = 4.7% of the money. That is the Pareto pattern in this file, computed, not asserted: the highest-leverage fifth is also among the cheapest, because most of it is documents, rights, hygiene, and knowledge.
The vital few items	Items with Pareto Build Rank 1–37 (the top 20% of 187 items) are the first builds of the sequence: the $0 claims, documents, and rights tier plus the first water, food, and records hardware. In term terms they carry near-zero mid- and long-term rents — protection bought in this fifth does not invoice you later.
Term-cost shape of the whole build	Short term: $8,179 total (one-time). Mid term: most problems run $0–200/yr to maintain; the expensive mid-terms are time (Psychology and Education run on hours, stated in their rows). Long term: several builds turn negative (Economy Now pays the household; Energy Future generates; Food Future yields rise) — the long-term cost of the solutions is lower than the long-term cost of every problem they answer.
How to use this sheet	When money is shortest, buy in leverage order (TERM COSTS - PER PROBLEM, rank ascending). When time is shortest, skip to the $0-time items inside each rank (they are the ones whose Short-Term cell begins "Cost: $0"). When both are short, the first 11 ranks are the whole answer.

Optimal Budgeting — Rules and Computed Build Schedules

Rule 1 — Free before money	Complete the entire $0 tier first (claims, documents, rights, registrations, protocols). It carries the majority of protection and costs only time (~236 hours, schedulable at 2 hours/week across a year, or 30 minutes/day).
Rule 2 — Pareto order for every dollar	Paid items are bought strictly in the BUDGET TIERS order from the costed file: survival fields first, Now before Future before Past. Never buy a Future item while a Now item in the same field is unbought.
Rule 3 — Payback order inside tiers	Where two items compete, buy the one with the shorter payback first (aerators repay in weeks; the backwater valve in one avoided flood; insulation every season forever).
Rule 4 — The substitution ladder	For every item: claim it free (programs, subsidies) → borrow it (library, tool library, neighbours) → buy used → buy budget-new → buy premium only where failure is lethal (water filtration, alarms). The ladder is printed per item in the costed file's Free or Cheaper Alternative column.
Rule 5 — Sinking funds, not splurges	Four envelopes: Water & Food, Power & Shelter, Health & Records, Growth (skills/ownership). A fixed monthly amount lands in each on payday, before discretionary spending. The build completes itself; willpower is not part of the mechanism.
Rule 6 — Mitigate the cost of mitigating	Bulk and co-op buying cuts staples 15–30%; shared big items (one filter, one battery station per courtyard or floor) divide capital costs by every household sharing; subsidies for exactly these items (rain barrels, backwater valves, retrofits) run 50–100% in many municipalities — claim before buying, every time.

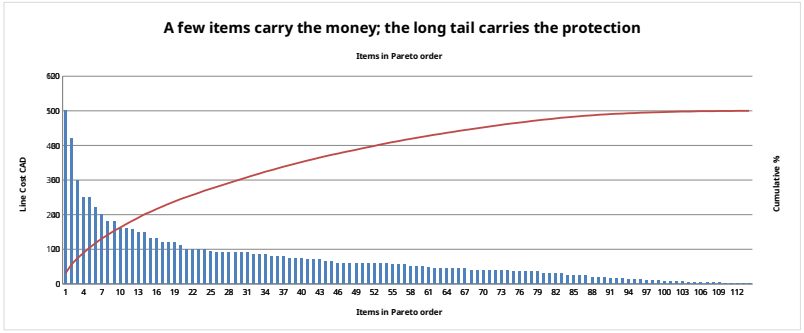
Monthly Budget	Months to Complete the $0 Tier (time only, at 8 hrs/month)	Months to Complete the FULL Paid Build	Cumulative Paid Spend at Month 3	Cumulative Paid Spend at Month 6	Cumulative Paid Spend at Month 12	What Is True at Month 12
$50/month	≈30 months of 8 hrs (or 12 months at ~20 hrs/month)	≈164 months	$141	$286	$464	At all three budgets the $0 tier plus the first survival hardware (water, food buffer start, go-bag, records) is complete within the first year — because Pareto order spends the first dollars on the highest-protection items.
$75/month	≈30 months of 8 hrs (or 12 months at ~20 hrs/month)	≈110 months	$161	$344	$884	At all three budgets the $0 tier plus the first survival hardware (water, food buffer start, go-bag, records) is complete within the first year — because Pareto order spends the first dollars on the highest-protection items.
$150/month	≈30 months of 8 hrs (or 12 months at ~20 hrs/month)	≈55 months	$344	$884	$1,800	At all three budgets the $0 tier plus the first survival hardware (water, food buffer start, go-bag, records) is complete within the first year — because Pareto order spends the first dollars on the highest-protection items.

Total paid build (all items, one household)	$8,179 CAD — the same total as the costed file; budgeting changes the ORDER and the timing, never the honesty of the total. Shared items counted once.

Applies To	Guide / Route	Hyperlink	Source (cited)	What It Provides
Water, Energy, Shelter — emergency situations	Municipal emergency preparedness guide (kit, plan, evacuation)	https://www.sylvanlake.ca/public-services/fire-department/emergency-preparedness/ev	Town of Sylvan Lake — Emergency Preparedness Guide (accessed Oct 19, 2026)	Kit contents, family plan, FireSmart home protection steps, apartment/condo preparedness
Water, Food, Shelter — household preparedness standards	Emergency preparedness standards (water 4.5 L/person/day, 3-day food, first aid, sanitation)	http://www.highlandseast.ca/en/live-here/emergency-preparedness.aspx	Municipality of Highlands East — Emergency Preparedness (accessed Oct 19, 2026)	The measured household standards used across the preparedness items
All fields — finding help in any situation (Canada)	211 — community, social, health, and government services, 24/7, 150+ languages	https://www.unitedwaygt.org/partnerships/helpstartshere-for-all-canadians/	United Way Greater Toronto — Help Starts Here / 211 (accessed Oct 19, 2026)	Food banks, income assistance, legal resources, mental health, housing and shelters — the master referral route
All fields — service navigation data	211 Day / 211 Canada national overview	https://www.unitedway.ca/february-11-is-211-day/	United Way Centraide Canada (accessed Oct 19, 2026)	What 211 connects to, unmet-needs data, how navigation works
Economy, Work, Demographics — claiming benefits	Benefits guide for newcomers and residents (CCB, EI, GIS, credits; filing unlocks benefits)	https://welcomeaide.com/en/benefits	WelcomeAide — Canadian Government Benefits Finder guide (accessed Oct 19, 2026)	Which benefits exist per status, and the rule that filing taxes activates them
Economy — full benefits catalogue	Complete Canadian benefits guide (amounts, how each is obtained)	https://www.canooq.ca/blog/complete-canadian-benefits-guide	Canooq — Complete Canadian Benefits Guide (accessed Oct 19, 2026)	Benefit amounts and application routes, incl. Canada Disability Benefit and Dental Care Plan
Conditions — disability benefits	Canada Disability Benefit (eligibility, documents, amounts)	https://benefitswayfinder.org/federal/canada-disability-benefit	Prosper Canada — Benefits Wayfinder (accessed Oct 19, 2026)	The disability benefit route in full, incl. DTC prerequisite
Security (Past), all displacement statuses — protection	UNHCR — who we protect (refugees, asylum seekers, IDPs, stateless people)	https://www.unhcr.org/us/about-unhcr/who-we-protect	UNHCR — Who We Protect (accessed Oct 19, 2026)	Protection categories, asylum as a right, what UNHCR provides
Security (Past), statelessness — protection operations	UNHCR Protection operations	https://www.unhcr.org/protection.html	UNHCR — Protection (accessed Oct 19, 2026)	Legal and physical protection, minimum shelter/food/water/medical standards after flight
Statuses in Canada — refugee and claimant help	UNHCR Canada via 211 BC agency listing (legal assistance, hearing observation, individual help)	https://bc.211.ca/agency-details/united-nations-high-commissioner-for-refugees-unhcr-c	211 British Columbia — UNHCR Canada agency details (accessed Oct 19, 2026)	UNHCR Canada's actual services and contact route
Technology & AI (Now) — cyber situations	Five practical ways to make yourself cybersafe (passwords, 2FA, backups, updates, phishing)	https://www.cyber.gc.ca/en/guidance/five-practical-ways-make-yourself-cybersafe	Canadian Centre for Cyber Security (accessed Oct 19, 2026)	The official household cyber standard behind the Technology items
Information, Communication — digital safety	Get Cyber Safe — Generation Cyber Safe guide (MFA, passphrases, device hygiene)	https://www.getcybersafe.gc.ca/en/resources/your-guide-generation-cyber-safe	Get Cyber Safe, Government of Canada (accessed Oct 19, 2026)	Plain-language digital safety for every generation
Shelter (Now), Justice — tenancy situations (Ontario)	Tenant Rights Ontario 2026 — RTA, rent guideline 2.1%, eviction process, Bill 60 changes	https://www.nihanglaw.ca/blog-tenant-rights-ontario-2026/	Nihang Law — Tenant Rights Ontario 2026 (accessed Oct 19, 2026)	Current Ontario tenant law in one guide, incl. hearing rights and notice periods
Shelter (Now), Justice — LTB forms and eviction help	Ontario LTB forms and tenant/landlord rights hub (T2, T5, T6 applications)	https://www.ontariotenancyadvisor.com/	Ontario Tenancy Advisor (accessed Oct 19, 2026)	The forms and routes behind the tenant items
Justice — tribunal process	Landlord and Tenant Board practice guides	https://www.pklawfirm.ca/category/landlord-and-tenant-board/	PK Law — LTB category (accessed Oct 19, 2026)	Eviction process step by step, entry and harassment rules
Health (Now) — the antimicrobial resistance situation	WHO Antimicrobial Resistance Fact Sheet (key facts, drivers, prevention)	https://cdn.who.int/media/docs/default-source/antimicrobial-resistance/amr-factsheet.p	World Health Organization — AMR Fact Sheet (accessed Oct 19, 2026)	The authoritative basis for the Health–Now problem and its stewardship solution
Health (Now) — AMR evidence library	Antimicrobial Resistance Fact Sheet (Harvard GHELI repository of the WHO sheet)	https://repository.gheli.harvard.edu/repository/11909/	Harvard Global Health Education and Learning Incubator (accessed Oct 19, 2026)	Prevention and control methods per sector (individuals, policymakers, agriculture)
Food (Now) — food access situations (Toronto example)	Community Fridges Toronto — free food, no questions, locations	https://communityfridgesto.org/	Community Fridges TO (accessed Oct 19, 2026)	A working model of the no-barrier food route
Food (Now) — food bank route (Toronto example)	Parkdale Community Food Bank — hours, registration, location	https://www.pcfb.ca/	Parkdale Community Food Bank (accessed Oct 19, 2026)	A concrete food-bank route of the type 211 refers to nationally
Psychology, Governance — crisis and rights routes (Canada)	Other support services: 988 Suicide Crisis Helpline, Hope for Wellness (1-855-242-3310), 211, human-rights routes	http://www.chrc-ccdp.gc.ca/find-help/find-help-other-agencies-and-support-services/oth	Canadian Human Rights Commission — Find Help (accessed Oct 19, 2026)	Crisis lines, Indigenous support, housing and rights complaint routes in one official page
Environment — home wildfire protection	FireSmart home protection (roof, vents, decks, defensible space) within the Sylvan Lake guide	https://www.sylvanlake.ca/public-services/fire-department/emergency-preparedness/ev	Town of Sylvan Lake / FireSmart Alberta (accessed Oct 19, 2026)	The property-level wildfire mitigation steps behind the Environment items
Preparedness — the classic family guide (PDF)	Emergency Preparedness family guide (kit, kids, special needs, sources: Public Safety Canada)	http://robinmarwick.ca/wp-content/uploads/2016/02/Emergency-preparedness.pdf	Toronto Office of Emergency Management guide, archived copy (accessed Oct 19, 2026)	Family preparedness incl. children and special-needs planning

Pareto Graph — Every Costed Item (all problems, all solutions)
How to Bars: each paid item's line cost, sorted largest to smallest. Line: cumulative share of the $8,179 paid build. The chart shows the Pareto shape of the whole build at a glance — a small number of items (battery station, water storage, food buffers) carry most of the money, while the long tail is dozens of items under $50 that carry most of the protection per dollar. The $0 items (the majority of the protection) cannot appear on a cost axis; they are listed beneath the chart data.

Rank	Item	Field	Timeframe	Line Cost CAD	Cumulative % of P
1	Emergency fund build: $500 starter then 1 month, then 3 (automatic tr	Economy & Finance	Now	500	6.1
2	Battery power station 1 kWh (or 12V battery + inverter build)	Energy	Now	420	11.2
3	Transition account: dedicated savings line for retraining (target: one c	Work & Livelihood	Future	300	14.9
4	Human-irreducible skill course (care, craft, repair, judgment trades)	Economy & Finance	Future	250	18
5	Backwater valve + sump check (the $250 that prevents the $15,000 bas	Environment & Climate	Now	250	21
6	90-day staple store per adult (~75 kg grain-equivalent + legumes + oil +	Food & Agriculture	Past	220	23.7
7	Portable livelihood investment (skills and savings not tied to local clima	Environment & Climate	Future	200	26.2
8	Full insurance review incl. overland flood / sewer backup riders	Shelter & Housing	Future	180	28.4
9	Home hazard test when buying older property (lead paint, asbestos su	Technology & AI	Past	180	30.6
10	Solar trickle panel 100–200 W for the battery station	Energy	Future	160	32.5
11	Bicycle + repair kit + panniers (the independent-mobility baseline)	Transport & Mobility	Past	160	34.5
12	Transit pass + cycling combination replacing second-car or car-lite patt	Transport & Mobility	Now	156	36.4
13	Durable-goods and skills reserve (tools, preserved food, tradeable skill	Economy & Finance	Past	150	38.2
14	Second-residency/citizenship pathway research + applications where la	Demographics & Populatio	Future	150	40.1
15	30-day deep storage (barrels)	Water	Future	130	41.6
16	Room HEPA unit for the sick room / bedroom	Health & Medicine	Future	130	43.2
17	Dental prevention kit + one hygiene visit/yr	Health & Medicine	Now	120	44.7
18	Union/professional-body membership	Economy & Finance	Future	120	46.2
19	Credential recognition pre-filing for any planned move (12-month targe	Demographics & Populatio	Future	120	47.6
20	Ventilation upgrade: CO2 monitor + window/exhaust habits	Health & Medicine	Future	110	49
21	Cooperative membership share (community well/storage)	Water	Future	100	50.2
22	Community energy co-op share	Energy	Future	100	51.4
23	Broad index investing habit (small automatic monthly amount in a low-	Economy & Finance	Future	100	52.6
24	Union or worker-association membership	Work & Livelihood	Past	100	53.9
25	3-2-1 backup: external drive + cloud copy of irreplaceables	Technology & AI	Now	95	55
26	Certified gravity/pump water filter	Water	Now	90	56.1
27	Perennial food plants (berry bushes, fruit tree, herb bed)	Food & Agriculture	Future	90	57.2
28	Layered clothing upgrade (wool base layers x2 per person)	Energy	Past	90	58.3
29	Portable wealth & documents kit (copies abroad, small high-value item	Security & Defense	Past	90	59.4
30	Home/business camera + signage (deterrence + evidence)	Security & Defense	Future	90	60.5
31	Primary-source subscriptions: one quality outlet + official data source	Information & Media	Now	90	61.6
32	30-day staple buffer per adult (~25 kg grain-equivalent)	Food & Agriculture	Now	85	62.7
33	Home insulation priority pack (attic hatch seal, window film, draft proo	Energy	Future	85	63.7
34	Home target-hardening basics: solid locks, strike plates with 3-inch scr	Security & Defense	Now	85	64.8
35	Garden start: seeds + soil for 10 m²	Food & Agriculture	Past	80	65.7
36	Fuel efficiency: improved cookstove / insulation of heat	Environment & Climate	Past	80	66.7
37	Real-world anchor subscriptions: one club/league/class with dues (skin	Psychology & Mind	Future	80	67.7
38	Co-op / land-trust membership contribution	Food & Agriculture	Future	75	68.6
39	Personal archive abroad (certified copies of identity, property, qualifica	Governance & Law	Past	75	69.5
40	Paper originals vault (identity, property, credentials) + notarial copies o	Information & Media	Future	75	70.4
41	Dehydrator (entry electric or solar build)	Food & Agriculture	Future	70	71.3
42	Fuel reserve (safe storage: gas canisters / charcoal / wood)	Energy	Past	70	72.1
43	Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags (	Shelter & Housing	Past	70	73
44	72-hour go-kit per person (clothes, medication, water, light, food bars)	Shelter & Housing	Past	65	73.8
45	Cool-room kit: external shading, reflective film, fan, wet-cloth fallback,	Environment & Climate	Now	65	74.6
46	Rain barrel 200 L	Water	Past	60	75.3
47	Preservation kit (jars, lids, salt, vinegar)	Food & Agriculture	Past	60	76.1
48	One-heated-room kit: door draft seals, heavy curtains, indoor tent/blan	Energy	Past	60	76.8
49	Evidence habit kit: phone backup on, cloud copies, incident log noteb	Security & Defense	Now	60	77.5
50	Quiet-network membership (congregation, union, association dues)	Governance & Law	Past	60	78.3
51	Support independent local media (one subscription)	Governance & Law	Future	60	79
52	Water-wise planting (drought-matched species, mulch)	Environment & Climate	Past	60	79.7
53	Manual-driving competence maintained (licence + occasional practice)	Transport & Mobility	Future	60	80.5
54	Evidence archive abroad (dated records of abuses, property, identity —	Justice & Ethics	Past	60	81.2
55	Greywater diverter for garden (where legal)	Water	Future	55	81.9
56	Household medical kit: thermometer, pulse oximeter, ORS packets, fev	Health & Medicine	Past	55	82.5
57	Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserv	Energy	Future	55	83.2
58	Independent media support: one exile/public/independent subscriptio	Information & Media	Past	50	83.8
59	Car reserve discipline if car-dependent: never below quarter tank + ma	Transport & Mobility	Now	50	84.4
60	Congregation/association membership (the historical message networ	Communication	Past	50	85
61	Stored water reserve 56 L/person	Water	Past	48	85.6
62	Low-flow fixtures set (aerators, showerhead, toilet kit)	Water	Future	45	86.2
63	Soil building: compost system + cover crop seed	Food & Agriculture	Future	45	86.7
64	Chronic-disease prevention stack: BP cuff + glucose basics + activity ro	Health & Medicine	Future	45	87.3
65	Safe backup cooking: single-burner butane stove + 4 canisters	Energy	Now	45	87.8
66	Soil care basics: compost return, cover, no bare soil over winter	Environment & Climate	Past	45	88.4
67	Offline knowledge reserve: printed field guides + manuals for your tra	Education & Knowledge	Future	45	88.9
68	Buying-club share (bulk staples with 4–10 households)	Food & Agriculture	Now	40	89.4
69	Reserve medication buffer (1 month extra of essentials, rotated, where	Health & Medicine	Future	40	89.9
70	Qualifications in portable form (transcripts, licences, references sealed	Governance & Law	Past	40	90.4
71	Civic membership: one watchdog, union, or ratepayers' organization	Governance & Law	Now	40	90.9
72	Analogue life-line kit: paper originals of ID/property/credentials, cash	Technology & AI	Future	40	91.4
73	Printed core references shelf (first aid, agriculture, mathematics, local	Education & Knowledge	Past	40	91.9
74	The bicycle, maintained (see Past row — the only transport no update c	Transport & Mobility	Future	40	92.4
75	Bathtub water bladder (fills in minutes at first notice)	Water	Now	35	92.8
76	Home infection-control basics (soap stock, disinfectant, masks x50, glo	Health & Medicine	Now	35	93.2
77	LED lanterns x3 + headlamp	Energy	Now	35	93.6
78	Battery shortwave/AM radio for foreign and emergency broadcasts	Information & Media	Past	35	94.1
79	Small-claims/tribunal filing for winnable civil matters (fees modest, wai	Justice & Ethics	Now	35	94.5
80	Isolation-room setup (separate bedding, basin, waste bag system)	Health & Medicine	Past	30	94.9
81	Fridge discipline kit: thermometer + cooler + ice packs	Energy	Now	30	95.2
82	Personal data footprint reduction: data-broker removals, minimal publ	Security & Defense	Future	30	95.6
83	Civil registration discipline: every birth/marriage/death registered and	Demographics & Populatio	Past	30	96
84	Fuel for boiling (1 month reserve)	Water	Past	25	96.3
85	Pipe protection: insulation sleeves + shutoff knowledge	Energy	Past	25	96.6
86	Sleep protocol: fixed window, morning light (free), screens out of the b	Psychology & Mind	Now	25	96.9
87	Literacy defence: every child taught to read early at home if school fails	Education & Knowledge	Past	25	97.2
88	Water test strips (bacteria indicators + chlorine)	Water	Now	20	97.4
89	Sprouting kit (jars + seeds)	Food & Agriculture	Now	18	97.7
90	Documents go-bag (waterproof pouch: ID, property, insurance, photos	Shelter & Housing	Past	18	97.9
91	72-hour stored water (12 L/person drinking)	Water	Now	16	98.1
92	Cloth + sand filtration setup	Water	Past	15	98.3
93	Access-to-information request kit (templates + filing fees where charge	Governance & Law	Now	15	98.4
94	Seed-saving envelopes + storage tin	Food & Agriculture	Past	12	98.6
95	No-device practice routine (weekly: handwriting, mental arithmetic, ma	Education & Knowledge	Future	12	98.7
96	Family knowledge book: recipes, remedies, trades, histories, contacts –	Demographics & Populatio	Past	12	98.9
97	Home copying kit: notebooks, and one body of knowledge memorized	Education & Knowledge	Past	10	99
98	Letter/postal discipline: addresses memorized + written copies held by	Communication	Past	10	99.1
99	Data-rights requests filed (access + correction) for your major files ann	Communication	Future	10	99.2
100	Real-task testing kit: past papers, project portfolio, skill demonstration	Education & Knowledge	Now	8	99.3
101	Two-routes knowledge: printed local map with foot routes out of the di	Transport & Mobility	Past	8	99.4
102	Paper medical records folder (vaccines, conditions, medications, allergi	Health & Medicine	Past	6	99.5
103	Portable proof-of-skills file (certificates, references, portfolio — copies	Work & Livelihood	Now	6	99.6
104	Community fridge / food bank registration + food-handler basics	Food & Agriculture	Now	5	99.6
105	Complaints filed in writing to ombudsman/auditor (free) with evidence	Governance & Law	Now	5	99.7
106	Tenant rights pack: your jurisdiction's rules printed + legal-aid and te	Shelter & Housing	Now	4	99.8
107	Exposure records file (employers, products, dates) kept permanently	Technology & AI	Past	4	99.8
108	Family communication plan on paper: rendezvous points, message dro	Communication	Past	3	99.8
109	Rights knowledge card for your system (arrest, search, detention limit	Justice & Ethics	Past	3	99.9
110	Verification protocol card (callback numbers printed; 'urgent = verify by	Technology & AI	Now	2	99.9
111	Two-source rule card + one scheduled news window (a household prot	Information & Media	Now	2	99.9



112	Printed essential numbers card (family, doctors, banks, legal aid) in wa	Communication	Now	2	100	
113	Rights card + 'no signature under pressure' household rule (posted, re	Justice & Ethics	Now	2	100	
114	Human-review demand template (written request for human review of	Justice & Ethics	Future	2	100	

$0 items (no cost bar — the protection majority)

	Vaccination catch-up (all free public programs)	Health & Medicine	Past
	Drug coverage optimization: benefits audit + generic substitution revi	Health & Medicine	Now
	Antibiotic discipline: none stockpiled; return leftovers (free pharmacy t	Health & Medicine	Now
	Home inventory video + off-site copy	Shelter & Housing	Past
	Housing documentation habit: dated photos at move-in/out, all landlor	Shelter & Housing	Now
	Rent-bank / eviction-prevention applications pre-filed knowledge + em	Shelter & Housing	Now
	Shared-housing option prepared (one identified co-housing/roommate	Shelter & Housing	Now
	Property risk check before buying/renewing: flood, fire, insurance-avail	Shelter & Housing	Future
	Portability reserve: keep at least 50% of net worth out of any single exp	Shelter & Housing	Future
	Buyout/relocation program registration (where offered) — filed early	Shelter & Housing	Future
	Early-warning habit: named indicators list + decided trigger points ('if	Security & Defense	Past
	Exit-route plan: two routes, rendezvous points, transport costs pre-est	Security & Defense	Past
	Mutual-aid roster (5–10 households: skills, vehicles, languages, check-in	Security & Defense	Past
	Route and routine discipline + check-in system with one contact	Security & Defense	Now
	Victim-compensation and protective-order applications knowledge (leg	Security & Defense	Now
	Venue standards habit: choose venues with visible security, exits know	Security & Defense	Future
	Support for regulation: specific, written asks to representatives (dron	Security & Defense	Future
	Local verification habit: attend/join one local body (assembly, board, co	Governance & Law	Future
	Personal service records (your own file of wait times, outcomes, corre	Governance & Law	Future
	Deposit splitting rule: never above the insured limit in one institution	Economy & Finance	Past
	Debt minimalism in boom times (fixed-rate only, no margin, no specula	Economy & Finance	Past
	Debt audit + avalanche order (highest rate first) printed and posted	Economy & Finance	Now
	Free insolvency-trustee first consultation + credit-counselling session (r	Economy & Finance	Now
	Benefits and credits claim check (tax filing + benefits finder)	Economy & Finance	Now
	Adoption delay rule for unstudied non-essentials (wait for the 2nd gen	Technology & AI	Past
	Password manager + MFA on email and banking (the two master keys)	Technology & AI	Now
	Automatic updates ON + router firmware check (quarterly, calendared)	Technology & AI	Now
	Data portability routine: monthly export of contacts, photos, records f	Technology & AI	Future
	Offline competence maintenance: one unaided skill practised regularly	Technology & AI	Future
	Address risk check (flood/fire/heat maps, free) + insurer confirmation ir	Environment & Climate	Now
	Location decision audit every 5 years (risk trend, insurance trend, wate	Environment & Climate	Future
	Collective relocation readiness: community/municipal plan participatio	Environment & Climate	Future
	Source-tracing habit taught at home (claim → origin → evidence) — a di	Information & Media	Past
	Fact-check and provenance tools bookmarked; reverse image search le	Information & Media	Now
	Family verification code word + callback rule for money/urgent request	Information & Media	Future
	Content-credential tools adopted (signed capture apps) for important r	Information & Media	Future
	Routine and ritual structure (fixed sleep, meals, work, worship/rest —	Psychology & Mind	Past
	Mutual storytelling / testimony practice (regular, structured, within fam	Psychology & Mind	Past
	Children-first protection of sleep, schooling, and play (the intergenerati	Psychology & Mind	Past
	Movement prescription: 60 min/day walking built into commute/errand	Psychology & Mind	Now
	Crisis lines saved in phone + EAP/benefits check (free session entitleme	Psychology & Mind	Now
	Synthetic-companionship budget rule (fixed hours + money cap, writte	Psychology & Mind	Future
	Device architecture: companions/feeds off the phone's home screen, n	Psychology & Mind	Future
	Library-as-university setup: card + course platforms + a written self-cur	Education & Knowledge	Now
	Reading-with-children 20 min/day (the highest-yield household educati	Education & Knowledge	Now
	Verification drill set: source-tracing exercises on real claims (family pra	Education & Knowledge	Future
	Cash-wage rule + independent housing/food (never bundle more than	Work & Livelihood	Past
	Portable insurance/contributions check (confirm what is yours regardle	Work & Livelihood	Past
	Personal work log: hours, schedules (photos), pay stubs — one folder,	Work & Livelihood	Now
	Labour-board claim filed for any unpaid wages (free) + worker-centre i	Work & Livelihood	Now
	Evacuation-assistance registration for vulnerable members (municipal l	Transport & Mobility	Now
	Provider diversification: accounts with 2+ mobility platforms + cash for	Transport & Mobility	Future
	Monthly export routine: contacts, photos, records downloaded to your	Communication	Now
	One owned channel: personal email list / phone tree / radio net — set	Communication	Now
	Library digital-access anchor (card + weekly session if no home internet	Communication	Now
	Encrypted messaging as default for private communication (free apps)	Communication	Future
	Compartmentation basics: separate emails/aliases for finance, public, a	Communication	Future
	Skills duplication roster (each critical skill taught to 2+ people, tracked	Demographics & Populatio	Past
	Care-pooling arrangement: 2–3 families or multi-generational roster wi	Demographics & Populatio	Now
	Benefits stack claim: child benefits, caregiver credits, parental top-ups	Demographics & Populatio	Now
	Waitlist registrations early (childcare, family housing, seniors' housing	Demographics & Populatio	Now
	50-year family location audit (where will care, work, and community exi	Demographics & Populatio	Future
	Legal-aid / duty-counsel / law-clinic intake (free) before any plea, signa	Justice & Ethics	Now
	Record-keeping for discrimination/wrongdoing (contemporaneous not	Justice & Ethics	Now
	Score hygiene routine: annual checks of credit, tenancy, and data-brok	Justice & Ethics	Future
	Case documentation for test-case organizations (your documented cas	Justice & Ethics	Future

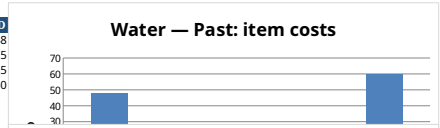
Graphs Per Problem — Items, Problem Cost, and Solution Cost

How to read them

Each problem has its own bar chart: one bar per solution item at its line cost, so the shape of every solution is visible (one big item vs. many small ones). Beside each chart, the problem cost range and the solution total are printed for the comparison the charts cannot show on the same axis (problem costs run to six figures).

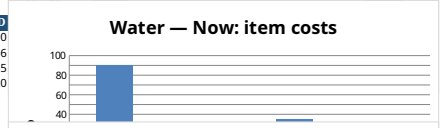
Water — Past

Item	Line Cost CAD
Stored water reserve 56 L/person	48
Cloth + sand filtration setup	15
Fuel for boiling (1 month reserve)	25
Rain barrel 200 L	60



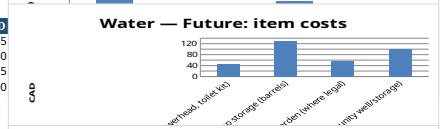
Water — Now

Item	Line Cost CAD
Certified gravity/pump water filter	90
72-hour stored water (12 L/person drinking)	16
Bathtub water bladder (fills in minutes at first notice)	35
Water test strips (bacteria indicators + chlorine)	20



Water — Future

Item	Line Cost CAD
Low-flow fixtures set (aerators, showerhead, toilet kit)	45
30-day deep storage (barrels)	130
Greywater diverter for garden (where legal)	55
Cooperative membership share (community well/storage)	100



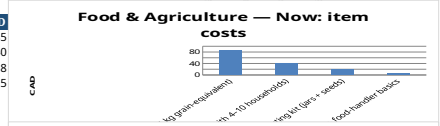
Food & Agriculture — Past

Item	Line Cost CAD
90-day staple store per adult (~75 kg grain-equivalent + legumes + oil +	220
Garden start: seeds + soil for 10 m²	80
Preservation kit (jars, lids, salt, vinegar)	60
Seed-saving envelopes + storage tin	12



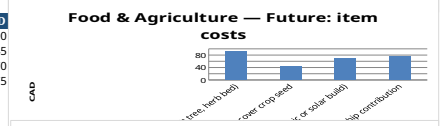
Food & Agriculture — Now

Item	Line Cost CAD
30-day staple buffer per adult (~25 kg grain-equivalent)	85
Buying-club share (bulk staples with 4–10 households)	40
Sprouting kit (jars + seeds)	18
Community fridge / food bank registration + food-handler basics	5



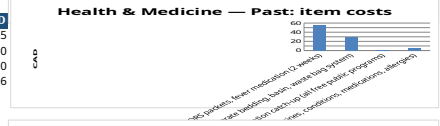
Food & Agriculture — Future

Item	Line Cost CAD
Perennial food plants (berry bushes, fruit tree, herb bed)	90
Soil building: compost system + cover crop seed	45
Dehydrator (entry electric or solar build)	70
Co-op / land-trust membership contribution	75



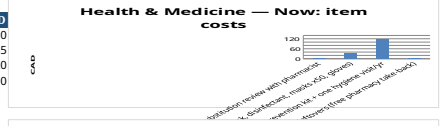
Health & Medicine — Past

Item	Line Cost CAD
Household medical kit: thermometer, pulse oximeter, ORS packets, fev	55
Isolation-room setup (separate bedding, basin, waste bag system)	30
Vaccination catch-up (all free public programs)	0
Paper medical records folder (vaccines, conditions, medications, allerg	6



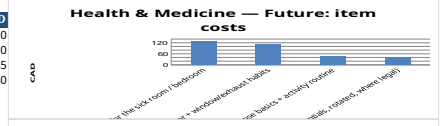
Health & Medicine — Now

Item	Line Cost CAD
Drug coverage optimization: benefits audit + generic substitution revi	0
Home infection-control basics (soap stock, disinfectant, masks x50, glo	35
Dental prevention kit + one hygiene visit/yr	120
Antibiotic discipline: none stockpiled; return leftovers (free pharmacy t	0



Health & Medicine — Future

Item	Line Cost CAD
Room HEPA unit for the sick room / bedroom	130
Ventilation upgrade: CO2 monitor + window/exhaust habits	110
Chronic-disease prevention stack: BP cuff + glucose basics + activity ro	45
Reserve medication buffer (1 month extra of essentials, rotated, where	40



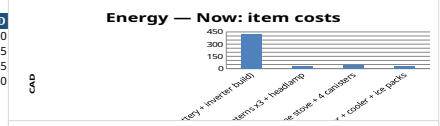
Energy — Past

Item	Line Cost CAD
One-heated-room kit: door draft seals, heavy curtains, indoor tent/blar	60
Fuel reserve (safe storage: gas canisters / charcoal / wood)	70
Layered clothing upgrade (wool base layers x2 per person)	90
Pipe protection: insulation sleeves + shutoff knowledge	25



Energy — Now

Item	Line Cost CAD
Battery power station 1 kWh (or 12V battery + inverter build)	420
LED lanterns x3 + headlamp	35
Safe backup cooking: single-burner butane stove + 4 canisters	45
Fridge discipline kit: thermometer + cooler + ice packs	30



Energy — Future

Item	Line Cost CAD
Solar trickle panel 100–200 W for the battery station	160



Home insulation priority pack (attic hatch seal, window film, draft proo	85
Community energy co-op share	100
Analogue fallback kit: wind-up radio, paper maps/contacts, cash reserv	55



Shelter & Housing — Past

Item	Line Cost CAD
Documents go-bag (waterproof pouch: ID, property, insurance, photos	18
72-hour go-kit per person (clothes, medication, water, light, food bars)	65
Home inventory video + off-site copy	0
Basic hardening: smoke/CO alarms tested, extinguisher, shutoff tags	70



Shelter & Housing — Now

Item	Line Cost CAD
Tenant rights pack: your jurisdiction's rules printed + legal-aid and t	4
Housing documentation habit: dated photos at move-in/out, all landlo	0
Rent-bank / eviction-prevention applications pre-filed knowledge + em	0
Shared-housing option prepared (one identified co-housing/roommate	0



Shelter & Housing — Future

Item	Line Cost CAD
Property risk check before buying/renewing: flood, fire, insurance-avai	0
Full insurance review incl. overland flood / sewer backup riders	180
Portability reserve: keep at least 50% of net worth out of any single exp	0
Buyout/relocation program registration (where offered) — filed early	0



Security & Defense — Past

Item	Line Cost CAD
Early-warning habit: named indicators list + decided trigger points ('if	0
Portable wealth & documents kit (copies abroad, small high-value item	90
Exit-route plan: two routes, rendezvous points, transport costs pre-es	0
Mutual-aid roster (5–10 households: skills, vehicles, languages, check-i	0



Security & Defense — Now

Item	Line Cost CAD
Home target-hardening basics: solid locks, strike plates with 3-inch scr	85
Evidence habit kit: phone backup on, cloud copies, incident log noteb	60
Route and routine discipline + check-in system with one contact	0
Victim-compensation and protective-order applications knowledge (leg	0



Security & Defense — Future

Item	Line Cost CAD
Venue standards habit: choose venues with visible security, exits know	0
Personal data footprint reduction: data-broker removals, minimal publ	30
Support for regulation: specific, written asks to representatives (dron	0
Home/business camera + signage (deterrence + evidence)	90



Governance & Law — Past

Item	Line Cost CAD
Personal archive abroad (certified copies of identity, property, qualifica	75
Qualifications in portable form (transcripts, licences, references sealed	40
Quiet-network membership (congregation, union, association dues)	60



Governance & Law — Now

Item	Line Cost CAD
Access-to-information request kit (templates + filing fees where charge	15
Complaints filed in writing to ombudsman/auditor (free) with evidence	5
Civic membership: one watchdog, union, or ratepayers' organization	40



Governance & Law — Future

Item	Line Cost CAD
Local verification habit: attend/join one local body (assembly, board, c	0
Personal service records (your own file of wait times, outcomes, corre	0
Support independent local media (one subscription)	60



Economy & Finance — Past

Item	Line Cost CAD
Deposit splitting rule: never above the insured limit in one institution	0
Durable-goods and skills reserve (tools, preserved food, tradeable skill	150
Debt minimalism in boom times (fixed-rate only, no margin, no specul	0



Economy & Finance — Now

Item	Line Cost CAD
Emergency fund build: $500 starter then 1 month, then 3 (automatic tr	500
Debt audit + avalanche order (highest rate first) printed and posted	0
Free insolvency-trustee first consultation + credit-counselling session (	0
Benefits and credits claim check (tax filing + benefits finder)	0



Economy & Finance — Future

Item	Line Cost CAD
Broad index investing habit (small automatic monthly amount in a low	100
Human-irreducible skill course (care, craft, repair, judgment trades)	250
Union/professional-body membership	120



Technology & AI — Past

Item	Line Cost CAD
Home hazard test when buying older property (lead paint, asbestos su	180
Exposure records file (employers, products, dates) kept permanently	4
Adoption delay rule for unstudied non-essentials (wait for the 2nd gen	0



Technology & AI — Now

Item	Line Cost CAD
Password manager + MFA on email and banking (the two master keys)	0
3-2-1 backup: external drive + cloud copy of irreplaceables	95
Automatic updates ON + router firmware check (quarterly, calendared	0
Verification protocol card (callback numbers printed; 'urgent = verify b	2



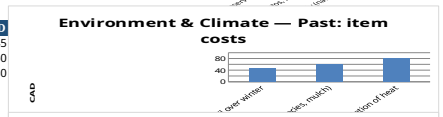
Technology & AI — Future

Item	Line Cost CAD
Analogue life-line kit: paper originals of ID/property/credentials, cash	40
Data portability routine: monthly export of contacts, photos, records	0
Offline competence maintenance: one unaided skill practised regularly	0



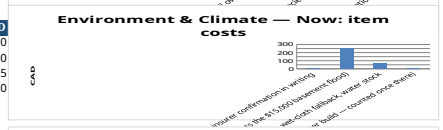
Environment & Climate — Past

Item	Line Cost CAD
Soil care basics: compost return, cover, no bare soil over winter	45
Water-wise planting (drought-matched species, mulch)	60
Fuel efficiency: improved cookstove / insulation of heat	80



Environment & Climate — Now

Item	Line Cost CAD
Address risk check (flood/fire/heat maps, free) + insurer confirmation i	0
Backwater valve + sump check (the $250 that prevents the $15,000 bas	250
Cool-room kit: external shading, reflective film, fan, wet-cloth fallback,	65
Go-bag + document pouch (shared with Shelter build — counted once	0



Environment & Climate — Future

Item	Line Cost CAD
Location decision audit every 5 years (risk trend, insurance trend, wate	0
Portable livelihood investment (skills and savings not tied to local clima	200
Collective relocation readiness: community/municipal plan participatio	0



Information & Media — Past

Item	Line Cost CAD
Source-tracing habit taught at home (claim → origin → evidence) — a d	0
Independent media support: one exile/public/independent subscriptio	50
Battery shortwave/AM radio for foreign and emergency broadcasts	35



Information & Media — Now

Item	Line Cost CAD
Two-source rule card + one scheduled news window (a household prot	2
Primary-source subscriptions: one quality outlet + official data sourc	90
Fact-check and provenance tools bookmarked; reverse image search l	0



Information & Media — Future

Item	Line Cost CAD
Family verification code word + callback rule for money/urgent reques	0
Paper originals vault (identity, property, credentials) + notarial copies c	75
Content-credential tools adopted (signed capture apps) for important	0



Psychology & Mind — Past

Item	Line Cost CAD
Routine and ritual structure (fixed sleep, meals, work, worship/rest —	0
Mutual storytelling / testimony practice (regular, structured, within fa	0
Children-first protection of sleep, schooling, and play (the intergenerat	0



Psychology & Mind — Now

Item	Line Cost CAD
Sleep protocol: fixed window, morning light (free), screens out of the	25
Movement prescription: 60 min/day walking built into commute/errar	0
One weekly in-person group (see Pareto action #6 — costed once)	0
Crisis lines saved in phone + EAP/benefits check (free session entitlem	0



Psychology & Mind — Future

Item	Line Cost CAD
Synthetic-companionship budget rule (fixed hours + money cap, writte	0
Real-world anchor subscriptions: one club/league/class with dues (skin	80
Device architecture: companions/feeds off the phone's home screen, r	0



Education & Knowledge — Past

Item	Line Cost CAD
Home copying kit: notebooks, and one body of knowledge memorized	10
Printed core references shelf (first aid, agriculture, mathematics, loca	40
Literacy defence: every child taught to read early at home if school fail	25



Education & Knowledge — Now

Item	Line Cost CAD
Library-as-university setup: card + course platforms + a written self-cu	0
Reading-with-children 20 min/day (the highest-yield household educat	0
Real-task testing kit: past papers, project portfolio, skill demonstration	8



Education & Knowledge — Future

Item	Line Cost CAD
No-device practice routine (weekly: handwriting, mental arithmetic, ma	12
Verification drill set: source-tracing exercises on real claims (family pra	0
Offline knowledge reserve: printed field guides + manuals for your tra	45



Work & Livelihood — Past

Item	Line Cost CAD
Cash-wage rule + independent housing/food (never bundle more than	0
Union or worker-association membership	100
Portable insurance/contributions check (confirm what is yours regardl	0



Work & Livelihood — Now

Item	Line Cost CAD
Personal work log: hours, schedules (photos), pay stubs — one folder,	0
Labour-board claim filed for any unpaid wages (free) + worker-centre i	0
Portable proof-of-skills file (certificates, references, portfolio — copie	6
Emergency fund (shared item — costed once in Economy & Finance)	0



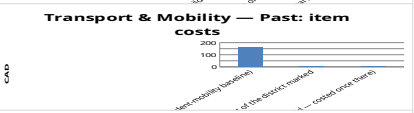
Work & Livelihood — Future

Item	Line Cost CAD
Ownership participation (shared with Economy build — costed once th	0
Transition account: dedicated savings line for retraining (target: one c	300
Collective representation maintained (union/professional dues — see	0



Transport & Mobility — Past

Item	Line Cost CAD
Bicycle + repair kit + panniers (the independent-mobility baseline)	160
Two-routes knowledge: printed local map with foot routes out of the d	8
30-day staple store (shared with Food build — costed once there)	0



Transport & Mobility — Now

Item	Line Cost CAD
Transit pass + cycling combination replacing second-car or car-lite pat	156
Car reserve discipline if car-dependent: never below quarter tank + ma	50
Evacuation-assistance registration for vulnerable members (municipal	0



Transport & Mobility — Future

Item	Line Cost CAD
The bicycle, maintained (see Past row — the only transport no update	40
Manual-driving competence maintained (licence + occasional practice)	60
Provider diversification: accounts with 2+ mobility platforms + cash for	0



Communication — Past

Item	Line Cost CAD
Family communication plan on paper: rendezvous points, message dro	3
Congregation/association membership (the historical message networ	50
Letter/postal discipline: addresses memorized + written copies held by	10



Communication — Now

Item	Line Cost CAD
Monthly export routine: contacts, photos, records downloaded to your	0
One owned channel: personal email list / phone tree / radio net — set	0
Printed essential numbers card (family, doctors, banks, legal aid) in wa	2
Library digital-access anchor (card + weekly session if no home interne	0



Communication — Future

Item	Line Cost CAD
Encrypted messaging as default for private communication (free apps)	0
Data-rights requests filed (access + correction) for your major files ann	10
Compartmentation basics: separate emails/aliases for finance, public,	0



Demographics & Population — Past

Item	Line Cost CAD
Family knowledge book: recipes, remedies, trades, histories, contacts	12
Skills duplication roster (each critical skill taught to 2+ people, tracked	0
Civil registration discipline: every birth/marriage/death registered and	30



Demographics & Population — Now

Item	Line Cost CAD
Care-pooling arrangement: 2–3 families or multi-generational roster w	0
Benefits stack claim: child benefits, caregiver credits, parental top-ups	0
Waitlist registrations early (childcare, family housing, seniors' housing	0



Demographics & Population — Future

Item	Line Cost CAD
50-year family location audit (where will care, work, and community ex	0
Second-residency/citizenship pathway research + applications where la	150
Credential recognition pre-filing for any planned move (12-month targ	120



Justice & Ethics — Past

Item	Line Cost CAD
Evidence archive abroad (dated records of abuses, property, identity –	60
Rights knowledge card for your system (arrest, search, detention limit	3
Exit-readiness (shared with Security build — documents and routes)	0



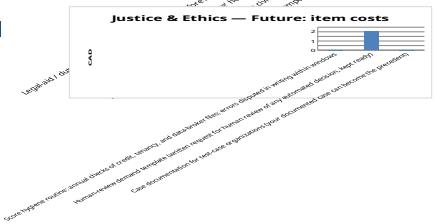
Justice & Ethics — Now

Item	Line Cost CAD
Legal-aid / duty-counsel / law-clinic intake (free) before any plea, sign	0
Rights card + 'no signature under pressure' household rule (posted, re	2
Small-claims/tribunal filing for winnable civil matters (fees modest, wai	35
Record-keeping for discrimination/wrongdoing (contemporaneous not	0



Justice & Ethics — Future

Item	Line Cost CAD
Score hygiene routine: annual checks of credit, tenancy, and data-brok	0
Human-review demand template (written request for human review of	2
Case documentation for test-case organizations (your documented cas	0

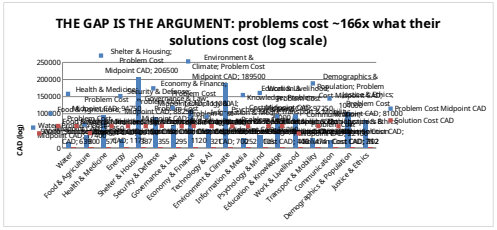


Visuals — The Whole System at a Glance

What is here — Six charts the other sheets cannot show: (1) problem cost vs. solution cost by field (log scale — the gap is the story), (2) solution hours by field (the time price), (3) the item-cost histogram (where the $8,179 actually sits), (4) citizen success without help vs. with help by timeframe (from the Citizen file), (5) the eight coinciding-problem scenarios and their joint probabilities (from the Master file), (6) solution totals by timeframe. All data blocks sit beside their charts so every visual can be audited against its numbers.

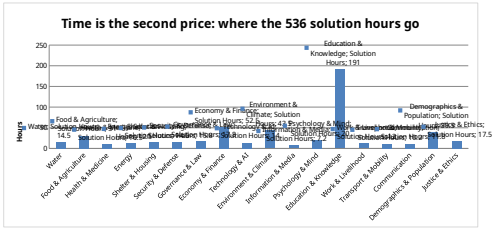
1 — Problem Cost (midpoint) vs. Solution Cost by Field — LOG SCALE

Field	Problem Cost Mid	Solution Cost CAD
Water	17400	639
Food & Agriculture	37350	800
Health & Medicine	94750	571
Energy	26150	1175
Shelter & Housing	206500	337
Security & Defense	88250	355
Governance & Law	67150	295
Economy & Finance	111000	1120
Technology & AI	54250	321
Environment & Climate	189500	700
Information & Media	28350	252
Psychology & Mind	35250	105
Education & Knowledge	70000	140
Work & Livelihood	97250	406
Transport & Mobility	29000	474
Communication	22100	75
Demographics & Population	104000	312
Justice & Ethics	81000	102



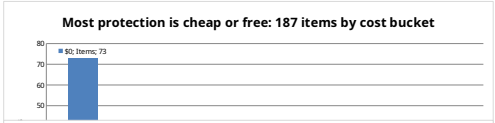
2 — Solution Time Investment by Field (hours)

Field	Solution Hours
Water	14.5
Food & Agriculture	31
Health & Medicine	10.5
Energy	12.5
Shelter & Housing	14.5
Security & Defense	15.8
Governance & Law	17.8
Economy & Finance	52.5
Technology & AI	14
Environment & Climate	43.5
Information & Media	7.2
Psychology & Mind	20
Education & Knowledge	191
Work & Livelihood	12
Transport & Mobility	10.2
Communication	11.8
Demographics & Population	39.5
Justice & Ethics	17.5



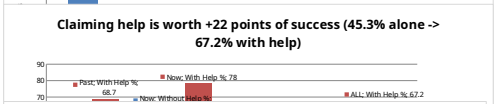
3 — Where the Money Sits: Items by Cost Bucket (all 187 items)

Cost Bucket	Items
$0	73
$1–25	31
$26–50	26
$51–100	37
$101–250	17
$251+	3



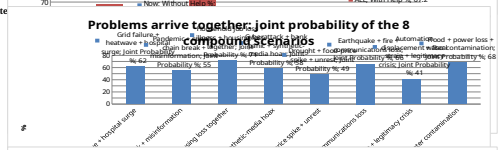
4 — Citizen Success: Without Help vs. With Help, by Timeframe (Citizen file)

Timeframe	Without Help %	With Help %
Past	44.8	68.7
Now	59.1	78
Future	32	54.8
ALL	45.3	67.2



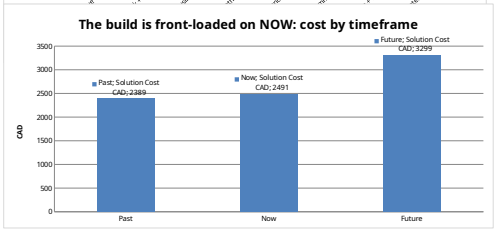
5 — Coinciding Problems: Joint Probability of the Compound Scenarios (Maste

Scenario	Joint Probability %
Grid failure + heatwave + hospil	62
Pandemic + supply-chain break	55
Household job loss + illness + h	71
Cyberattack + bank panic + syn	58
Drought + food-price spike + ur	49
Earthquake + fire + communica	66
Automation displacement + fisc	41
Flood + power loss + water con	68



6 — Solution Cost and Time by Timeframe

Timeframe	Solution Cost CAD	Solution Hours
Past	2389	148
Now	2491	202
Future	3299	185.8



The Persuasive Case — Ten Arguments, Every One of Them a Number From These Files

How this sheet persuades

Nothing here is inflated. Persuasion in this file is selection and framing of verified figures: each argument states its claim, the exact number behind it, what the number means in a citizen's life, and where the number comes from. Modelled figures are labelled modelled. Anyone may audit any line back to its sheet.

#	The Argument	The Figure	What It Means in a Citizen's Life	Source
1	Prevention is not slightly cheaper than disaster. It is two orders of magnitude cheaper.	$8,179 builds all 54 solutions. The same 54 problems unsolved cost $1,359,250 at their midpoints — a ratio of 166 : 1.	The entire build costs less than ONE mid-range problem landing once: one flooded basement, one resistant infection, one eviction cascade.	Costed file, PROBLEM COST vs SOLUTION COST (midpoints summed; modelled, labelled in-file)
2	The majority of protection costs nothing.	73 of 187 items are $0 — documents, rights, claims, registrations, protocols. The $0 tier alone is 235.8 of the 536 hours and the majority of the protection.	A citizen with no money at all can still complete most of this system. Poverty blocks the hardware; it does not block the protection.	Costed file, COSTED ITEMS + BUDGET TIERS (recount: 187 items)
3	Asking for help is the single highest-yield action in the file.	Citizen success rises from 45.3% alone to 67.2% with help claimed — +22 points, the largest single improvement available at any price, and it is free.	211, legal aid, benefits, rent banks: the routes exist and are unclaimed mostly from not knowing. Knowing is the whole cost.	Citizen file, CITIZEN PLAYBOOK (54 rows, both percentages per row)
4	Some ratios are absurd enough to repeat.	A $250 backwater valve stands against a $15,000 sewer-backup loss — 60 : 1 — and most municipalities subsidize 50–100% of the valve.	The best purchase in the file costs less than a month of groceries and is often free after subsidy.	Environment & Climate — Now (costed file); subsidy practice cited in GUIDE LINKS & SOURCES
5	The whole system costs less per day than coffee.	$8,179 amortized across a conservative 10-year protection life = $2.24/day, falling every year after, because most items are bought once.	Framed daily, the question stops being "can we afford prevention" and becomes "why are we paying for the alternative at 166x".	Derived: Costed file total / 3,650 days (arithmetic, shown)
6	The problems are not hypothetical. The world's health authority counts the bodies.	Bacterial antimicrobial resistance was directly responsible for 1.27 million deaths in 2019 and contributed to 4.95 million; the World Bank projects up to US$1 trillion in extra healthcare costs by 2050.	Health — Now is in the build because the threat is measured, current, and priced by the institutions that will pay for it.	WHO Antimicrobial Resistance Fact Sheet, via Harvard GHELI reposit
7	Preparedness has an official measure, and it is small.	The municipal standard is 4.5 litres of water per person per day and a 3-day household supply — the entire Water-Now defence for one person is 13.5 litres and a filter.	The gap between "prepared" and "not prepared" is not wealth. It is 13.5 litres and one afternoon.	Municipality of Highlands East, Emergency Preparedness — http://ww
8	You do not need everything. The vital fifth carries the system.	In the Master file's weighted ranking, the top 20% of solutions carry 40.2% of the harm. In term costs, the top 11 of 54 solutions by leverage are also among the cheapest.	A limited budget is not a failed budget. Spent in Pareto order, a small budget buys a disproportionate share of safety.	Master file, PARETO PRIORITY (computed); this file, PARETO - TERM VIEW (computed)
9	Problems do not arrive one at a time — and neither should preparation.	The eight compound scenarios carry joint probabilities of 41–71%. The likeliest single entry is an ordinary household triple — job loss + illness + housing loss together — at 71%.	The household triple is not exotic. It is the standard biography of a bad year, which is why the solutions are built to overlap.	Master file, COINCIDING PROBLEMS (structured expert estimates, labelled in-file)
10	Time is priced openly here — which is why the plan fits real lives.	The full build is ~536 hours. At 30 minutes a day, the $0 tier (235.8 hours) completes in about 16 months; at 8 hours a month, the paid build completes at $50–150/month inside the schedules in BUDGETING SOLUTIONS.	No step requires wealth, permission, or a free month. It requires half an hour and an order to do things in. This file is the order.	Costed file (hours per item); this file, BUDGETING SOLUTIONS (OPTIMAL) — schedules computed from the item sequence

Full Statistics — All Four Files Reconciled

Problems in the Master file	162 (18 fields x Past/Now/Future x Household/Society/Global scales)
Core problems fully costed (this system)	54 (18 fields x 3 timeframes, household-citizen scale)
Solution items costed	187 (180 in the cores + 7 shared items counted once)
Items by price	73 at $0 (incl. shared) ; 114 paid ; median paid item $55 ; largest item $420 (1 kWh LiFePO4 battery station)
Total solution build	$8,179 CAD + ~536 hours (all 54 solutions)
Total problem cost if unsolved (midpoints)	$1,359,250 CAD across the 54 cores
Modelled savings	$1,351,071 CAD (problem midpoints minus solution build)
Average leverage (problem / solution)	166 : 1 across the whole system
Master-file solution success (structured estimates)	62.5% average — Past ~68%, Now ~65%, Future ~54%
Citizen success (Citizen file)	Without help: 45.3% average ; with help claimed: 67.2% average — claiming help is worth roughly +22 points
Pareto findings (computed, both files)	Master: top 20% of solutions carry 40.2% of weighted harm. Term view: top 11 of 54 solutions by leverage carry 26.1% of avoided problem cost.
Coinciding scenarios	8 compound scenarios, joint probabilities 41–71% (charted in VISUALS)
Entity costing	54 problems x 6 entity columns (People, Animals, Plants, Things & Property, Environment, Community & Institutions) + 5 non-money dimensions each
Guide links	22 verbatim URLs with named sources, accessed Oct 19, 2026

Per-Field Table

Field	Items	Solution CAD	Solution Hours	Problem Midpoint CAD
Water	12	639	14.5	17400
Food & Agriculture	12	800	31	37350
Health & Medicine	12	571	10.5	94750
Energy	12	1175	12.5	26150
Shelter & Housing	12	337	14.5	206500
Security & Defense	12	355	15.8	88250
Governance & Law	9	295	17.8	67150
Economy & Finance	10	1120	52.5	111000
Technology & AI	10	321	14	54250
Environment & Climate	10	700	43.5	189500
Information & Media	9	252	7.2	28350
Psychology & Mind	10	105	20	35250
Education & Knowledge	9	140	191	70000
Work & Livelihood	10	406	12	97250
Transport & Mobility	9	474	10.2	29000
Communication	10	75	11.8	22100
Demographics & Population	9	312	39.5	104000
Justice & Ethics	10	102	17.5	81000
TOTAL	187	8179	535.8	1359250

SOURCE FOR EVERY HEADLINE FIGURE

$8,179 total build; 187 items; ~536 hours	Derived in-file: sum of item line costs and hours, Costed file COSTED ITEMS (recount-verified; shared items counted once)
$1,359,250 problem midpoints; $1.35M modelled savings	Costed file, PROBLEM COST vs SOLUTION COST — household-scale CAD loss ranges from documented loss patterns; modelled and labelled in-file, not quotes
166 : 1 average leverage	Arithmetic on the two figures above (shown in THE PERSUASIVE CASE #1)
62.5% master success; Past 68 / Now 65 / Future 54	Master file, PROBLEMS & SOLUTIONS — structured expert estimates with per-row basis, labelled in-file
40.2% — top 20% of solutions carry weighted harm	Master file, PARETO PRIORITY — computed from the file's priority scores
45.3% / 67.2% citizen success	Citizen file, CITIZEN PLAYBOOK — 54 rows; recomputed for the VISUALS chart from the row values (matches)
41–71% compound-scenario probabilities	Master file, COINCIDING PROBLEMS — structured expert estimates, labelled in-file
AMR: 1.27M deaths (2019); up to US$1T extra costs by 2050	WHO AMR Fact Sheet via Harvard GHELI — https://repository.gheli.harvard.edu/repository/11909/ ; WHO sheet — https://cdn.who.in
4.5 L/person/day; 3-day supply standard	Municipality of Highlands East — http://www.highlandseast.ca/en/live-here/emergency-preparedness.aspx (accessed Oct 19, 2026)

Ontario rent guideline 2.1% (2026); Bill 60 notice changes	Nihang Law, Tenant Rights Ontario 2026 — https://www.nihanglaw.ca/blog-tenant-rights-ontario-2026/ (third-party law-firm guide, a
Cyber household standard (passwords, MFA, backups, updates)	Canadian Centre for Cyber Security — https://www.cyber.gc.ca/en/guidance/five-practical-ways-make-yourself-cybersafe (accessed O
Benefits amounts/routes (CCB, EI, disability benefit)	WelcomeAide benefits guide — https://welcomeaide.com/en/benefits ; Canooq complete guide — https://www.canooq.ca/blog/comple
Protection categories and routes (refugees, stateless, IDPs)	UNHCR — https://www.unhcr.org/us/about-unhcr/who-we-protect ; https://www.unhcr.org/protection.html (accessed Oct 19, 2026)
211 national referral system	United Way — https://www.unitedwaygt.org/partnerships/helpstartshere-for-all-canadians/ ; https://www.unitedway.ca/february-11-
Crisis and rights lines (988; Hope for Wellness 1-855-242-3310)	Canadian Human Rights Commission, Find Help — http://www.chrc-ccdp.gc.ca/find-help/find-help-other-agencies-and-support-servic
Prices of items (Oct 2026)	Canadian budget-tier retail planning figures used consistently across the Costed file; labelled in-file as planning figures, not quotes